

卷五 · 治疗、用药与并发症管理

贫血、高血压与蛋白尿、恶心呕吐、食欲、磷结合剂、钾、新药（HIF-PHI、SGLT2、MSC、姜黄素、NAC、Renaltec）以及猫内科新旧药物更新。

#	文献（中文标题 / 原文标题 / 要点）	来源	页
1	猫CKD疗法证据系统综述 Therapies for Feline Chronic Kidney Disease: What is the Evidence? - PMC 有什么证据? Philip Roudebush 1,5, David J Polzin 2, Sheri J Ross 3, Todd L Towell 1, Larry G Adams 4, S Dru Forrester 1 实用相关性 成功治疗和预防宠物肾脏疾病需要多维方法, 以识别并消除原因或加重因素, 定期提供专业检查和护理, 并在必要时规划和实施全面的治疗方案。证据基础 多年来, 已经开发或倡导了许多慢性肾脏疾病 (CKD) 的治疗和预防干预措施, 但疗效或有效性的证据往往缺乏或高度不稳定。因此, 本系统评价的主要目标是识别并批判性地评估支持各种猫慢性肾病 (CKD) 管理方法的证据; 即液体治疗、钙三...	—	48
2	猫内科临床治疗学新旧药物更新 Over the past 10 years, feline medicine has significantly advanced through the addition of new pharmaceuticals and alternative 猫医学临床治疗学: 旧药与新药的更新 实际意义: 在过去的10年里, 猫科医学通过新增的用于猫的药物和替代制剂, 以及对现有药物知识的提高, 取得了显著进展。通过持续的药物开发, 猫的治疗选择正在不断增加。临床挑战: 在治疗猫科疾病时会遇到许多特定于物种的临床挑战。此外, 治疗选择可能因地域而异, 并随时间变化。证据基础: 本综述文章讨论了关于某些较新治疗药物的现有证据, 这些药物目前可供猫科医学从业者使用, 或正在开发中并进行临床试验。它们的近期引入意味着对其中一些药物的证据仍在不断出现。文章介绍了这些药物的临床使用情况、可用的药代动力学和药效学数据, 以及在猫中报告的不良反应。观众: 本文中的信息与所有从...	—	20
3	猫肾性贫血综述 Anemia of Renal Disease: What it is, what to do and what's new - PMC 肾脏疾病贫血: 是什么、该怎么办以及最新进展 Serge Chalhoub 1, Cathy Langston 2, Adam Eatroff 3 患者群体 据估计, 15–30%的老年猫会发展为慢性肾病 (CKD), 而在这些猫中, 随着肾脏疾病的加重, 30–65%会出现贫血。肾性贫血的发病机制是多因素的, 但主要原因是红细胞生成素 (肾脏激素) 产量减少, 它控制骨髓红细胞的生成。随着肾脏疾病的进展, 红细胞生成素的产生减少。实际意义 认识到肾性贫血的存在非常重要, 以便采取适当的治疗来改善生活质量和代谢功能。促红细胞生成剂 (ESAs), 如阿法依波汀 (epoetin alfa)、贝塔依波汀 (epoetin...	—	44
4	莫立司他治疗猫CKD贫血RCT Erythropoietic effects of molidustat, a hypoxia-inducible factor prolyl hydroxylase (HIF-PH) inhibitor, were 抑制剂, 在猫慢性肾病 (CKD) 相关性贫血中的应用 Molidustat (一种缺氧诱导因子脯氨酰羟化酶抑制剂, HIF-PH抑制剂) 的促红细胞生成作用在健康猫中已被验证。评估每日口服给予molidustat对慢性肾病 (CKD) 贫血猫的安全性和促红细胞生成作用。21只由客户拥有的CKD猫 (4-17岁) 伴有贫血。多中心现场研究; 随机、盲法、安慰剂对照。猫每天口服治疗28天, 给予对照产品 (CP; n=6) 或5 mg/kg molidustat (n=15) 的悬液。每周评估血细胞比容 (HCT)。个体猫治疗成功定义为HCT较基线增加≥4个百分点。对照组的平均血细胞比容 (HCT) 在整个研究期间保持较...	—	12
5	HIF-PHI治疗猫CKD贫血的新选择 Hypoxia-inducible factor prolyl hydroxylase inhibitor as a new therapeutic option for treating anemia in feline chronic kidney disease. 贫血是患有慢性肾病 (CKD) (CKD) 的猫常见的并发症, 主要由于肾脏红细胞生成素 (EPO) 产生减少, 这源于肾脏红细胞生成素产生 (REP) 细胞的丧失。贫血会通过引起嗜睡和厌食等临床症状影响生存和生活质量。通过监测HCT/压积 (Htc/PCV) 趋势早期发现贫血, 对于在进行性肾损伤限制治疗效果之前启动治疗至关重要。Molidustat 是一种缺氧诱导因子 (HIF) 脯氨酰羟化酶抑制剂 (HIF-PHI), 是一种新型口服药物, 可以刺激内源性红细胞生成素的合成。然而, 治疗必须在 REP 细胞仍然功能正常时开始, PCV 目标范围为...	Tierärztliche Praxis. Ausgabe K, K 2026	24
6	健康猫与CKD猫收缩压随时间变化 Hypertension is a common problem in older cats, most often associated with chronic kidney disease (CKD). Cross- 健康猫与慢性肾病 (CKD) 猫的收缩压 (SBP) 随时间变化 高血压是老年猫中常见的问题, 最常与慢性肾病 (CKD) 相关。横断面研究表明, 猫的血压会随着年龄的增长而升高。假设/目标 旨在确定猫的血压是否会随年龄增加, 以及这种情况是否独立于CKD的存在。同时调查患高血压的风险因素。回顾性地识别了265只患有CKD的猫和133只≥9岁的健康猫。根据初次评估的状态 (CKD或健康) 和最后一次随访时的血压 (正常血压[NT]或发展为高血压	—	13

[DH]), 创建了四个组: Healthy-NT、Healthy-DH、CKD-NT和CKD-DH。比较了随时间变化的收缩压 (SBP) 斜率与0的差异及组间差异。研究了高血压发...

7	替米沙坦vs贝那普利治疗猫CKD蛋白尿 The efficacy and benefits of telmisartan in cats with chronic kidney disease (CKD) have not previously been reported. 长期口服替米沙坦与贝那普利治疗慢性肾病 (CKD) 猫的疗效比较 替米沙坦在慢性肾病 (CKD) 猫中的疗效和益处此前未有报道。使用替米沙坦对慢性肾病 (CKD) 猫进行长期治疗, 可以降低尿蛋白与肌酐比值 (UPC) (UP/C), 效果与224只由客户拥有的成年CKD猫相似。前瞻性、多中心、对照、随机、平行组、盲法临床试验, 采用非劣效设计。猫按1:1的比例分配到替米沙坦组 (1 mg/kg; n = 112) 或贝那普利组 (0.5–1.0 mg/kg; n = 112), 每天口服一次 (q24 h)。主要终点预先定义为基于UP/C变化对数转换加权平均值 (AUC 0→t/t) 百分比变化的蛋白尿变化 (贝那普利: 替米沙坦), ...	—	17
8	非高血压CKD猫循环RAAS标志物 Circulating renin-angiotensin-aldosterone system markers in cats with non-hypertensive chronic kidney disease or systemic arterial hypertension. 背景关于非高血压慢性肾病 (NHT-CKD) 或系统性动脉高压 (SAH) 猫的经典和替代循环 RAAS (RAAS) 信息有限。年龄和饮食会影响RAAS结果。假设/目标比较健康猫与患有 NHT-CKD或末治疗SAH的猫的血清平衡血管紧张素肽和醛固酮浓度, 并评估服用氨氯地平的高血压猫的这些浓度变化。动物 客户拥有的猫患有NHT-CKD (血清肌酐≥1.6 mg/dL, 收缩压 (SBP) [SBP]<160 mmHg;n = 17)、SAH (SBP≥160 mmHg或≥150 mmHg 伴高血压眼部病变;n = 6), 或正常SBP (<1...	Journal of Veterinary Internal Med 2026	40
9	氮质血症猫犬离子低钙的严重病因与结局 Severity, etiology and outcomes of ionized hypocalcemia in azotemic cats and dogs. 目的 描述在患有各种原因血氮增多症的犬猫群体中, 离子化低钙血症的严重程度、发生时间、病因及结果。方法 回顾性分析在单一机构中10年期间 (2014年1月1日至2023年12月31日) 同时患有离子化低钙血症和血氮增多症且在血清生化检测24小时内测量离子化钙 (iCa) 的犬猫的病历。离子化低钙血症在猫定义为iCa < 1.21 mmol/L, 在犬定义为iCa < 1.30 mmol/L。中度低钙血症定义为iCa < 1.00 mmol/L, 重度低钙血症定义为iCa < 0.75 mmol/L。血氮增多症定义为肌酐浓度 ≥...	Frontiers in Veterinary Science 2026	8
10	长期使用马罗匹坦控制CKD猫呕吐 Chronic use of maropitant for the management of vomiting and inappetence in cats with chronic kidney disease: a blinded, placebo-controlled clinical t 慢性使用马罗匹坦 (Maropitant) 治疗患有慢性肾病 (CKD) 猫的呕吐和食欲不振: 一项盲法、安慰剂对照的临床试验 Jessica M Quimby 1,✉, William T Brock 1, Kelsey Moses 2, David Bolotin 3, Kayla Patricelli 3 马罗匹坦 (Maropitant) 通常用于急性呕吐。一项针对猫的药代动力学和毒性研究表明, 长期使用似乎是安全的。本研究的目的是评估马罗匹坦 (Maropitant) 在猫慢性肾病 (CKD) 相关的慢性呕吐和食欲不振管理中的疗效。入组本随机、安慰剂对照、盲法临床研究的共有41只猫, 这些猫具有稳定的国际肾脏...	—	15
11	脂肪来源同种异体MSC治疗猫CKD随机对照试验 Assessment of intravenous adipose-derived allogeneic mesenchymal stem cells for the treatment of feline chronic kidney disease: a randomized, placebo- 用于治疗猫慢性肾脏疾病 (CKD) 的干细胞: 在八只猫中进行的随机、安慰剂对照临床试验 Jessica M Quimby 1,✉, Tracy L Webb 1, Elissa Randall 2, Angela Marolf 2, Alex Valdes-Martinez 2,* , Steve W Dow 1,3 猫慢性肾脏疾病 (CKD) 的特点是慢性肾小管间质性肾炎, 炎症会促进肾纤维化的进展。间充质干细胞 (MSC) (MSCs) 在啮齿动物CKD模型中已显示出抗炎和抗纤维化作用。然而, 关于MSC疗法对伴侣动物疾病效果的随机试验报道很少。本研究的目的是使用随机、安慰剂对照试验评估异体MSC治疗猫CKD...	—	17
12	同种异体冻存MSC治疗猫CKD三项先导试验 R E S E A R C H Open Access Safety and efficacy of intravenous infusion of 研究 开放获取 静脉输注的安全性和有效性 异体冷冻间充质干细胞 (MSC) 用于治疗猫慢性肾病 (CKD) : 三项连续试点研究的结果 Jessica M Quimby 1*, Tracy L Webb 1, Lauren M Habenicht 1 和 Steven W Dow 1,2 引言: 间充质干细胞 (MSC) 的给药已被证明可改善慢性肾病 (CKD) 啮齿动物模型的肾功能, 部分机制是通过减少肾内炎症和抑制纤维化。猫的CKD以肾小管间质炎症和纤维化为特征, 因此用MSC治疗可能改善该疾病患者的肾功能和尿液炎症标志物。因此, 进行了一系列试点研究, 以评估静脉注射异体脂肪来源 MSC (aMSCs) 在自然发生CKD的...	—	12
13	2025同种异体脂肪源MSC治疗重症猫CKD病例报告 Case Report: Allogeneic adipose-derived mesenchymal stem cells for severe feline chronic kidney disease - PMC 病例报告: 异体脂肪来源间充质干细胞 (MSC) 治疗严重猫慢性肾病 (CKD) 宋怡哲 1,* , † , # , 刘裕阳 1, † , # , 余瑜 1, 王莹 1, 穆云鹏 1, 王思宇 1, 韩婉婷 2, 海玲 一只10岁的绝育雄性布偶猫出现食欲不振、恶心、呕吐、嗜睡和进行性体重下降的症状。实验室指标和临床表现提示其患有国际肾脏疾病学会 (IRIS) 慢性肾脏病 (CKD) III期。经过基础支持性治疗后, 临床症状缓解, 患者出院进行家庭管理。三个月后, 猫的病情加重, 经复查, 其符合IRIS CKD	—	27

IV期的诊断标准。在按照IRIS指南进行为期4个月的规范化支持性治疗后，临床症状和实验室指标均无明显变化，猫仍表现为消瘦...

14 间充质干细胞治疗伴侣动物肾病从机制到临床

Frontiers in Veterinary Science 2026 17

Advancing mesenchymal stem cell therapy for kidney diseases in companion animals: from mechanisms to clinical application
急性和慢性肾病（CKD）是伴侣动物中的主要临床挑战，但逆转已建立损伤的治疗选择仍然有限。间充质干细胞（MSC）（MSCs）因其免疫调节和组织修复特性而成为一种有前景的治疗方法。本综述综合了目前关于猫狗肾脏疾病MSC治疗的证据，重点关注其作用机制、预处理策略、给药途径和临床结果。关键发现表明，MSCs主要通过旁分泌介导的免疫调节发挥肾脏保护作用，而非直接分化。静脉给药虽然简单，但会导致超过80%的干细胞滞留在肺部，而肾脏归巢率不足5%；动脉或局部注射可将归巢率提高到15–20%，但存在操作风险且缺乏标准化剂量。预处理...

15 姜黄素对CKD猫的肾脏保护作用

Journal of Veterinary Internal Med 2026 36

Renoprotective effects of curcumin in cats with chronic kidney disease.
背景 猫的慢性肾病（CKD）以氧化应激、炎症和细胞凋亡为特征。姜黄素在实验模型中显示出肾脏保护作用，但在患有CKD的猫中临床证据仍然有限。假设/目的 评估口服姜黄素补充剂（80 mg/猫/天）在标准治疗下是否能有利地调节患有国际肾脏兴趣学会（IRIS）3期CKD的猫的肾功能、氧化、炎症和凋亡生物标志物。动物 对象为十六只IRIS 3期CKD的顾客拥有的猫。方法 前瞻性、随机试验。猫被分配为仅接受药物治疗组（MT；n = 8）或药物治疗加姜黄素组（MT+C；n = 8）。姜黄素按80 mg/猫/天口服，分2次给...

16 N-乙酰半胱氨酸降低肌酐BUN_SDMA与尿蛋白

BMC veterinary research 2026 7

N-acetylcysteine reduces serum creatinine, blood urea nitrogen, symmetric dimethylarginine and urine protein to creatinine ratio in cats with chronic
背景 氧化应激被认为是慢性肾病（CKD）的一个重要促成因素。迄今为止，文献中关于N-乙酰半胱氨酸（NAC）对自然发生CKD的猫的影响的临床数据仍然很少。本研究的目的是评估在治疗急性加重的CKD猫时，添加NAC是否能够比单纯静脉输液治疗更改善肾功能生物标志物。方法 本研究纳入了总计50只由客户拥有的猫。纳入标准包括先前被诊断为氮质血症性CKD（IRIS 2-4期）的猫，并且超声检查显示双侧肾脏体积减小、表面轮廓粗糙及肾皮质回声异常。所有猫都使用标准的临床程序进行检查，包括临床检查、血液分析、腹部超声检查、试纸尿液分...

17 Renaltec治疗IRIS2-3期猫CKD的效果评价

Journal of Feline Medicine and Sur 2026 34

Evaluation of the effect of Renaltec in cats with IRIS stage 2 and 3 chronic kidney disease
目标 慢性肾病（CKD）与血液和肾小管上皮细胞中尿素氮如靛基硫酸盐（IS）的积累有关，这些物质会促进CKD的进展。本前瞻性、随机、双中心临床试点研究的目的是评估口服球状碳质吸附剂Renaltec（Porus One）对IS血浆浓度、CKD进展、钙/磷稳态参数（总钙[tCa]、磷、FGF-2323 [FGF-23]）、尿蛋白:肌酐比值（UPC）以及粪便失衡指数（FDI）在2级和3级稳定CKD猫（根据国际肾脏兴趣学会IRIS分级）中的影响，观察期为6个月。方法 共19只患有慢性肾病（CKD）的猫被随机分配到Renalt...

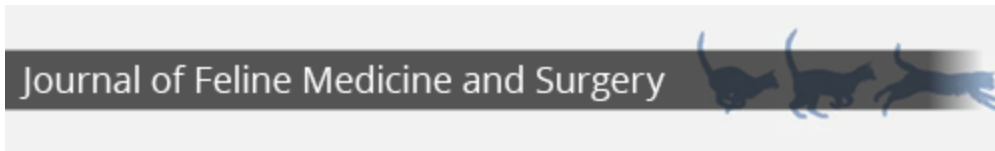
18 SGLT2抑制剂用于非理想候选糖尿病猫

Journal of Feline Medicine and Sur 2026 19

SGLT2 inhibitor therapy in diabetic cats: first clinical experiences with non-ideal candidates
病例系列总结 本回顾性病例系列描述了七只使用韦拉格列净（velagliflozin）治疗的糖尿病猫，这些猫被认为并非该疗法的理想候选者。这些更为复杂的糖尿病猫病例被转诊至比利时根特大学兽医学院小动物科。本病例系列的唯一纳入标准是在诊断为糖尿病后接受韦拉格列净（Senvelgo；勃林格殷格翰）治疗。所有猫的数据包括信号信息、病史、临床发现、诊断、治疗、反应及结果。所有猫的初次会诊时间均在2024年3月至2025年5月之间。相关性 与新颖信息 目前关于钠-葡萄糖共转运蛋白2（SGLT2）抑制剂在猫科患者中使用的现有文献仍...

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Therapies for Feline Chronic Kidney Disease

What is the Evidence?

[Philip Roudebush](#)^{1,5}, [David J Polzin](#)², [Sheri J Ross](#)³, [Todd L Towell](#)¹, [Larry G Adams](#)⁴, [S Dru Forrester](#)¹

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Abstract



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Practical relevance Successful treatment and prevention of kidney disease in pet animals requires a multidimensional approach to identify and eliminate causes or exacerbating factors, provide professional examination and care on a regular basis, and plan and implement a comprehensive treatment program when necessary.

Evidence base Over the years, many therapeutic and preventive interventions have been developed or advocated for chronic kidney disease (CKD), but evidence of efficacy or effectiveness is often lacking or highly variable. Accordingly, the main objective of this systematic review was to identify and critically appraise the evidence supporting various approaches to managing feline CKD; namely, fluid therapy, calcitriol therapy, antihypertensive therapy, ACE inhibitor therapy, erythropoietic hormone replacement therapy, potassium supplementation, antioxidant supplementation, alkalization therapy, dietary phosphorus restriction and intestinal phosphate binders, therapeutic renal foods, assisted feeding, dialysis and renal transplantation.

Chronic kidney disease defined and staged

It has recently been proposed that the term ‘renal failure’ be discarded in favor of ‘kidney disease’ for two reasons. First, pet owners know what a kidney is, but many are unfamiliar with the medical term ‘renal.’ Thus using the term ‘kidney’ facilitates communication between veterinarians and clients. Secondly, there has never been complete agreement on the specific meanings of renal insufficiency and renal failure. By using the term ‘kidney disease’ and staging the severity of disease, it is possible to greatly facilitate understanding, communication and application of management guidelines for patients.

Kidney disease is defined as the presence of functional or structural abnormalities in one or both kidneys.¹ It may be recognised by reduced kidney function or the presence of kidney damage. Kidney damage is defined as either: microscopic or macroscopic pathology detected by biopsy or direct visualization of the kidneys; or markers of damage detected by blood or urine tests or imaging studies.

The severity of kidney disease varies greatly depending on the magnitude of kidney involvement.

Patients with chronic kidney disease (CKD, as defined below) can be categorized into stages along a continuum of progressive disease.³ The International Renal Interest Society (IRIS) has proposed a four-tier system for staging CKD in dogs and cats (see box, page 196).⁴ The staging system was developed by a panel of experts in veterinary nephrology, rather than specific clinical studies, and its use has been advocated by the American and European Societies of Veterinary Nephrology and

Urology. The stage is based on the level of kidney function signified by at least two sequential determinations of serum creatinine concentration obtained in a well hydrated patient fasted for 12–24 h. Ideally, serum creatinine concentrations should be determined at least twice over 1–2 weeks to stage CKD. As a variety of factors affect serum creatinine concentrations, published reference ranges are often exceedingly broad. Using the IRIS staging system, some patients classified as having mild renal azotemia (stage 2) may have serum creatinine values within published reference ranges. As a consequence, the patient's overall clinical status should be considered when interpreting serum creatinine concentration and other laboratory tests and when planning patient management. In particular, the serum creatinine concentration must be interpreted in the light of a concurrently obtained measure of urine concentrating ability (urine specific gravity or osmolality). In the absence of urinary obstruction, serum creatinine values > 1.6 mg/dl (140 μ mol/l) are likely to indicate kidney disease when the patient's urine specific gravity is < 1.035. Daily production of creatinine in the body is determined largely by the muscle mass of the individual, so care must be taken in interpreting values obtained from animals with reduced muscle mass; for example, elderly cats with CKD and poor muscle mass may have lower than expected serum creatinine concentrations. [5](#)

CKD is defined as kidney damage present for at least 3 months, with or without decreased glomerular filtration rate (GFR), or a > 50% reduction in GFR persisting for at least 3 months. [2](#)

Stages of feline CKD

Stage 1 Non-azotemia

Markers of kidney damage present

Plasma creatinine < 140 $\mu\text{mol/l}$ (1.6 mg/dl)

Proteinuria: classify (P/NP/BP)*

Hypertension: classify (N/L/M/H/nc/c/RND)†

Stage 2 Mild renal azotemia

Markers of kidney damage present

Plasma creatinine 140–249 $\mu\text{mol/l}$ (1.6–2.8 mg/dl)

Proteinuria: classify (P/NP/BP)*

Hypertension: classify (N/L/M/H/nc/c/RND)†

stage 3 Moderate renal azotemia

Plasma creatinine 250–439 $\mu\text{mol/l}$ (2.9–5.0 mg/dl)

Proteinuria: classify (P/NP/BP)*

Hypertension: classify (N/L/M/H/nc/c/RND)†

Stage 4 Severe renal azotemia

Plasma creatinine ≥ 440 $\mu\text{mol/l}$ (5.0 mg/dl)

Proteinuria: classify (P/NP/BP)*

Hypertension: classify (N/LyM/H/nc/c/RND)†

For full details of the staging system, see the International Renal Interest Society's website:
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P = proteinuria (protein:creatinine ratio [UPC] > 0.4), NP = non-proteinuria (UPC < 0.2), BP
= borderline proteinuria (UPC 0.2–0.4]

N = minimal risk of complications, L = low risk of complications, M = moderate risk of complications, H = high risk of complications, nc = no complications, c = complications, RND = risk not determined (blood pressure not measured)

The IRIS has proposed to further subclassify patients with CKD according to their magnitude of proteinuria and systemic blood pressure. Proteinuria and hypertension appear to influence prognosis; however, the impact of therapeutic intervention to lessen proteinuria and hypertension on long-term prognosis is still uncertain.⁶⁻⁹ Classification of patients as proteinuric necessitates determining the urine protein:creatinine ratio (UPC) and eliminating the presence of hemorrhage, inflammation and/or infection. Feline patients are classified as proteinuric when their UPC > 0.4. Values for UPC between 0.2 and 0.4 are classified as borderline proteinuric, while UPC values < 0.2 are classified as non-proteinuric. Ideally, UPC values between 0.4 and 1.0 should be confirmed using several urine samples obtained over 2–4 weeks. With respect to blood pressure, the IRIS has proposed that patients be further staged according to the degree of risk of end-organ damage and whether there is current evidence of such damage or complications.⁴

Goals and limits of therapy for CKD

Conservative medical management of CKD consists of supportive and symptomatic therapy designed to correct abnormalities in fluid, electrolyte, acid-base, endocrine and nutritional balance. Therapy aims to minimize the clinical and pathophysiological consequences of reduced kidney function. In general, this type of management should not be expected to halt, reverse or eliminate renal lesions responsible for CKD. Therefore, management strategies are most beneficial when combined with specific therapy directed at correcting the primary cause of the kidney disease such as hypercalcemic nephropathy, bacterial urinary tract infections and obstructive uropathy. Efforts should be directed at detecting and treating active primary kidney diseases in addition to managing the various pathophysiological and clinical consequences of CKD.

Medical management guidelines provided here are intended for pets with compensated disease; not for patients unable to eat or accept oral medications because of severe uremia. Clinical signs of uremia must be corrected before instituting the general principles of medical management. The goals of medical management of CKD are to:

- ameliorate clinical signs of uremia;

minimize disturbances associated with excesses or losses of electrolytes, vitamins and minerals;
support adequate nutrition by supplying daily protein, calorie, mineral and other nutrient requirements; and
modify progression of kidney disease.²

Although evidence-based medicine does not always lead to a definitive answer, it does provide a framework for making decisions and understanding the risk-benefit relationship of various therapeutic or preventive plans.

These goals are best achieved when recommendations are individualized to the patient's needs on the basis of clinical and laboratory findings. Because CKD is progressive and dynamic, serial assessment of the patient and modification of therapy in response to changes are essential parts of the management strategy.

Evidence-based veterinary nephrology concepts

Evidence-based medicine (EBM) represents a major, although largely untested, intellectual advance when making clinical decisions and determining patient care.¹⁰⁻¹² EBM has been defined as the integration of the best research evidence with clinical expertise and patient values.¹⁰ Best research evidence means clinically relevant research, especially from patient-centered clinical studies. Clinical expertise refers to the use of clinical skills to identify each patient's unique health status, establish a diagnosis, and determine the risks and benefits of potential interventions. For veterinary medicine, the concept of patient values must include the unique preferences and expectations of each owner. In addition, if a therapeutic intervention is not readily available, then it is unlikely to benefit the patient. The basic tenet of EBM is that integration of these elements (clinically relevant research, clinical expertise, patient/owner preferences, and availability of resources) will result in the formation of a diagnostic and therapeutic alliance that optimizes clinical outcomes and quality of life.¹⁰

In the past, information based on pathophysiological rationale and data from models of kidney disease were often used to justify clinical recommendations in cats with CKD. The proliferation of randomized controlled clinical trials has led to an increase in the quantity and quality of clinically valid evidence. When possible, veterinarians should use information derived from systematically conducted, rigorously controlled clinical studies to make diagnostic and treatment decisions. Of course, not all recommendations can be based on such studies so it is important to recognize the inherent limitations of less rigorous forms of evidence. One method of accommodating concerns

regarding these limitations is to assign a score defining the strength and quality of the recommendation. The classification scheme proposed for veterinary clinical nutrition [13](#) may be useful for establishing rules of evidence for recommendations regarding veterinary nephrology. These guidelines categorize the quality of evidence into grades I to IV based on applicability to clinical case management (see box below). Grades I and II are evidence with the highest quality for application in the clinical environment, whereas grade IV would be evidence with the lowest quality.

The quality and strength of evidence can be used to make a recommendation about use of a specific therapeutic modality. Although EBM does not always lead to a definitive answer, it does provide a framework for making decisions and understanding the risk-benefit relationship of various therapeutic or preventive plans.

Quality of evidence grading guidelines

Grade I

Evidence obtained from one or more properly designed, randomized controlled clinical trials performed in clinical patients of the target species

Grade II

Evidence obtained from properly designed, randomized controlled studies performed using animals of the target species with spontaneous disease in a laboratory or research animal colony setting

Grade III

Evidence obtained from appropriately controlled studies without randomization, appropriately designed cohort or case control studies, studies using acceptable models of disease or simulations in the target species, case series or dramatic results from uncontrolled studies

Grade IV

Evidence obtained from studies conducted in other species, reports of expert committees, descriptive studies, case reports, pathophysiological justification, and opinions of recognized experts developed on the basis of their clinical experience

Applying evidence-based concepts to therapeutic decisions for CKD

A grade I to IV scoring system [13](#) was used to systematically evaluate the quality of published evidence for the use of 13 categories of interventions in cats with CKD, as discussed below. The strength of evidence was also assessed in making a final recommendation for use of a specific treatment modality.

It is important to note that the use of any specific therapeutic strategy hinges not only on the quality and strength of evidence summarized here but also on the clinical expertise of the individual veterinarian, owner preferences and available resources. In addition, the impact of providing unnecessary or unproven treatments on the pet-owner relationship should be considered. Treatments that are undesirable to the pet or owner may be disruptive to that relationship. As providing high quality health care to pets is based on the pet-family bond, it is important to avoid inadvertently disrupting this relationship when recommending therapy.

Fluid therapy

Progression of CKD is accompanied by decreases in GFR, worsening azotemia and ultimately uremia. Long-term administration of subcutaneous balanced electrolyte solutions has been advocated in patients with CKD to prevent dehydration, thereby maximizing renal blood flow and GFR, increasing urine output and ameliorating clinical manifestations of dehydration and uremia.^{2,14,15} Subcutaneous administration of balanced electrolyte solutions is usually reserved for cats with advanced (stages 3–4) CKD prone to dehydration.

Preliminary studies have been completed in healthy adult cats using intravenous fluids and various diuretics.¹⁶ In healthy cats, the renal effects of mannitol were superior to those of furosemide and dopamine together when used as an adjunct to fluid diuretic therapy.¹⁶ Additional studies with fluid therapy are required to assess physiological and clinical parameters in animals with CKD. Use of diuretics should be limited to patients with uremic crises rather than cats with stable CKD. Anecdotal reports suggest that many cat owners can give subcutaneous fluids successfully in the home environment for a long period of time. Hypertension is a well-recognized complication of CKD in cats and it is unknown if excessive sodium chloride delivered by parenteral fluids will exacerbate clinical problems associated with hypertension or kidney disease. For example, every 100 ml of lactated Ringer's solution provides 13 mEq sodium, which is approximately eight times the 24 h maintenance requirement for sodium in healthy or diseased cats.^{17,18} Although empirical evidence suggests that some individual cats may benefit from long-term subcutaneous fluid therapy, no controlled clinical studies exist to determine whether such therapy prolongs survival and improves quality of life in cats with CKD.

FLUID THERAPY

Summary of the evidence

Weak evidence (grade IV) exists to support long-term subcutaneous fluid therapy in cats with CKD.

Anecdotal evidence supports chronic administration of subcutaneous fluids to cats with advanced (stages 3–4) CKD that are prone to dehydration, or that respond to such therapy with a demonstrable improvement in clinical signs. Randomized controlled clinical trials are desirable to validate these recommendations and better identify those feline patients that will benefit most from this form of therapy.

Calcitriol therapy

The kidneys are responsible for converting 25-hydroxycholecalciferol to its most active metabolite, 1,25-dihydroxycholecalciferol, also known as calcitriol. Calcitriol is one of the major hormones responsible for calcium metabolism along with parathyroid hormone (PTH). Among calcitriol's important functions is modulation of PTH activity at the transcriptional level.¹⁹ As production of 1,25-dihydroxycholecalciferol is typically impaired in CKD, calcitriol deficiency may be one factor promoting renal secondary hyperparathyroidism. Parathyroid hormone has been proposed to act as a 'uremic toxin' and supplementing calcitriol may ameliorate a variety of supposed adverse effects of excess parathyroid hormone in patients with CKD. A study found that 8/26 cats with CKD had renal secondary hyperparathyroidism as detected by elevated serum PTH concentrations.²⁰

Use of calcitriol in patients with CKD is described in numerous textbooks and reviews.^{21–23} Investigators have reported that in cats with CKD receiving calcitriol therapy, patients: (1) were brighter and more alert, (2) had improved appetites, (3) were more physically active, and (4) lived longer.²¹ These findings were based on an uncontrolled survey of veterinarians who used calcitriol in their practice. A study of 10 cats with CKD showed that serum PTH concentrations were not significantly different after 14 days of calcitriol administration, regardless of whether calcitriol was administered daily or intermittently.²⁴ Adverse effects of calcitriol administration (specifically ionized hypercalcemia) were not observed in this short-term study.

However, many of the beneficial effects of calcitriol therapy in patients with CKD may be mediated through activation of vitamin D receptors on many tissues rather than modulation of PTH activity alone.²⁵ A 1-year randomized controlled clinical trial examined the effect of low-dose calcitriol therapy on progression of CKD and clinical signs in cats.²⁶ In this study, calcitriol was ineffective in altering renal mortality or improving appetite, activity or quality of life in cats with stages 2–4 CKD. In contrast, a 1-year randomized controlled clinical trial examining the effect of low-dose calcitriol therapy on progression of CKD and clinical signs in dogs demonstrated that calcitriol therapy was associated with a significant reduction in all-cause mortality compared with placebo.²

CALCITRIOL THERAPY

Summary of the evidence

Weak evidence (grade IV) from pathophysiological rationale, surveys of veterinarians and data from clinical trials in dogs and humans suggests a possible role for calcitriol in the management of CKD in cats. However, a short-term descriptive case series (grade III evidence) and a longer term unpublished, randomized controlled clinical trial (grade I evidence) failed to support the use of calcitriol in cats with CKD. Current evidence fails to support a recommendation either for or against use of calcitriol in cats with CKD.

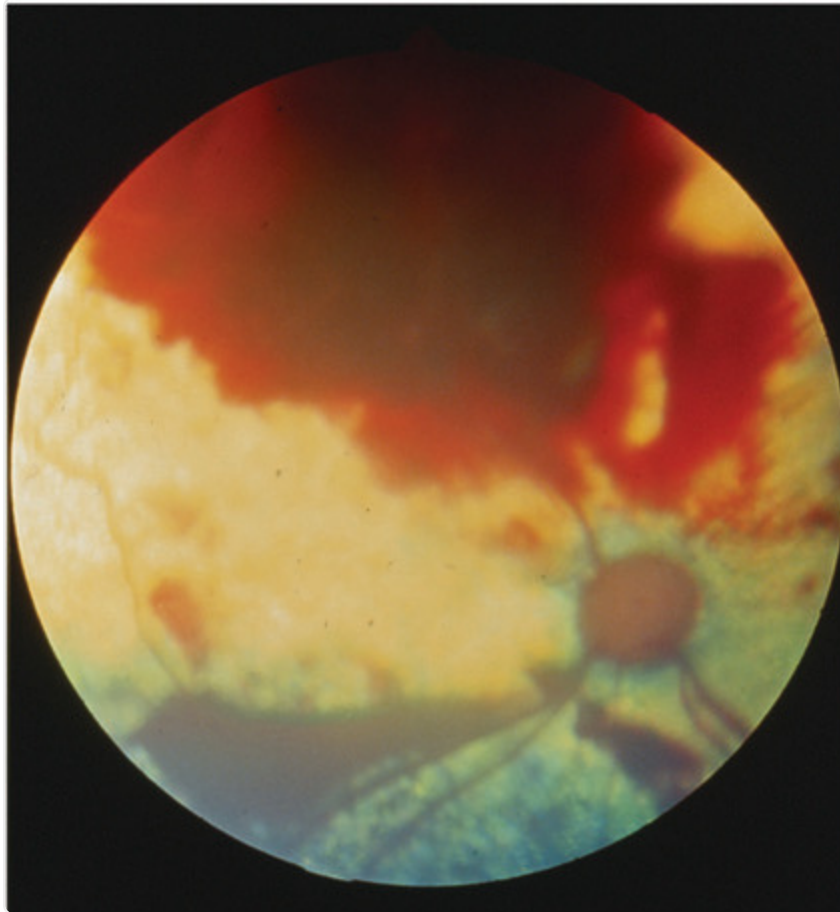
Antihypertensive therapy

Hypertension is a well recognized complication of CKD in cats. Mild to moderate systemic hypertension was originally reported in 17/28 cats that had spontaneous CKD.²⁷ More recent data suggest that the prevalence of hypertension in cats with CKD may be about 20%.²⁸

Sustained systemic hypertension can result in hypertensive retinopathy with retinal detachment, hemorrhage and blindness (Fig 1). Cats with such severe ocular manifestations probably reflect only a small percentage of pets with CKD and hypertension. More subtle ocular lesions such as retinal edema, retinal vessel tortuosity and subtle retinal hemorrhages may be much more common. Hypertension-related central nervous system disorders (eg, seizures, loss of balance, abrupt changes in personality, obtundation, etc) have also been observed.^{29–31} Management of

hypertension reportedly limits postoperative neurological disorders following renal transplantation in cats.^{[32](#)}

Fig 1.



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Retinal Image from a 10-year-old spayed female domestic shorthair cat with acute blindness. Chronic kidney disease and hypertension were diagnosed

Hypertension may also damage other organs including the kidneys, heart and blood vessels. A small study in cats with CKD suggested that glomerulosclerosis is more severe in hypertensive cats compared with normo-tensive cats.^{[33](#)}

Initiation of therapy for hypertension has been advocated for cats with clinical signs consistent with pressure-related end-organ damage (eg, hypertensive retinopathy with retinal detachment, acute-onset neurologic signs) and/or blood pressure values that persistently exceed 180/120 mmHg.^{4,34} However, the touted renoprotective benefit of antihypertensive therapy in cats is largely extrapolated from observations in humans and experimental studies in animals. The potential benefits of intervention in patients without clinical signs might include retarding progression of kidney disease, prolonging survival and reducing the incidence of hypertensive retinopathy and encephalopathy. In one study, proteinuria was the only variable significantly related to survival in hypertensive cats.⁸

Although specific evidence-based guidelines for treatment of systemic hypertension have not been definitively established for cats, studies have provided information on the effectiveness of some antihypertensive treatments. Currently, the calcium channel antagonist amlodipine besylate appears to be the drug of choice for managing hypertension in cats with CKD. Antihypertensive effects have been noted in uncontrolled clinical observations on the use of amlodipine in cats with spontaneous kidney disease.^{8,31,35,36} Blood pressure control with amlodipine appears to influence survival in hypertensive cats in a positive manner and lower blood pressures at the time of diagnosis indicate that treatment is more likely to be effective.⁸ Treatment with amlodipine can result in a significant reduction in proteinuria and, as mentioned, the level of proteinuria appears to be related to survival in hypertensive cats.⁸ Another study showed that amlodipine administration decreased systemic arterial blood pressure and dramatically reduced the prevalence of ocular lesions attributable to hypertension in cats with an induced model of renal failure.³⁷ In this last study, all cats were also fed a therapeutic renal food throughout the study period.

In contrast to amlodipine, angiotensin converting enzyme (ACE) inhibitors and β -blocking agents do not appear to be effective as sole agents for long-term management of systemic hypertension in cats.^{38,39} However, one study showed that the ACE inhibitor benazepril is well tolerated when administered with amlodipine to hypertensive cats and may assist in managing cats with poorly controlled blood pressure.⁴⁰ Amlodipine also appears to have intrinsic antioxidant activity.⁴¹ It is currently unknown whether this in vitro antioxidant activity results in significant improvement in endothelial function in patients with CKD or provides substantive clinical benefits.

ANTIHYPERTENSIVE THERAPY

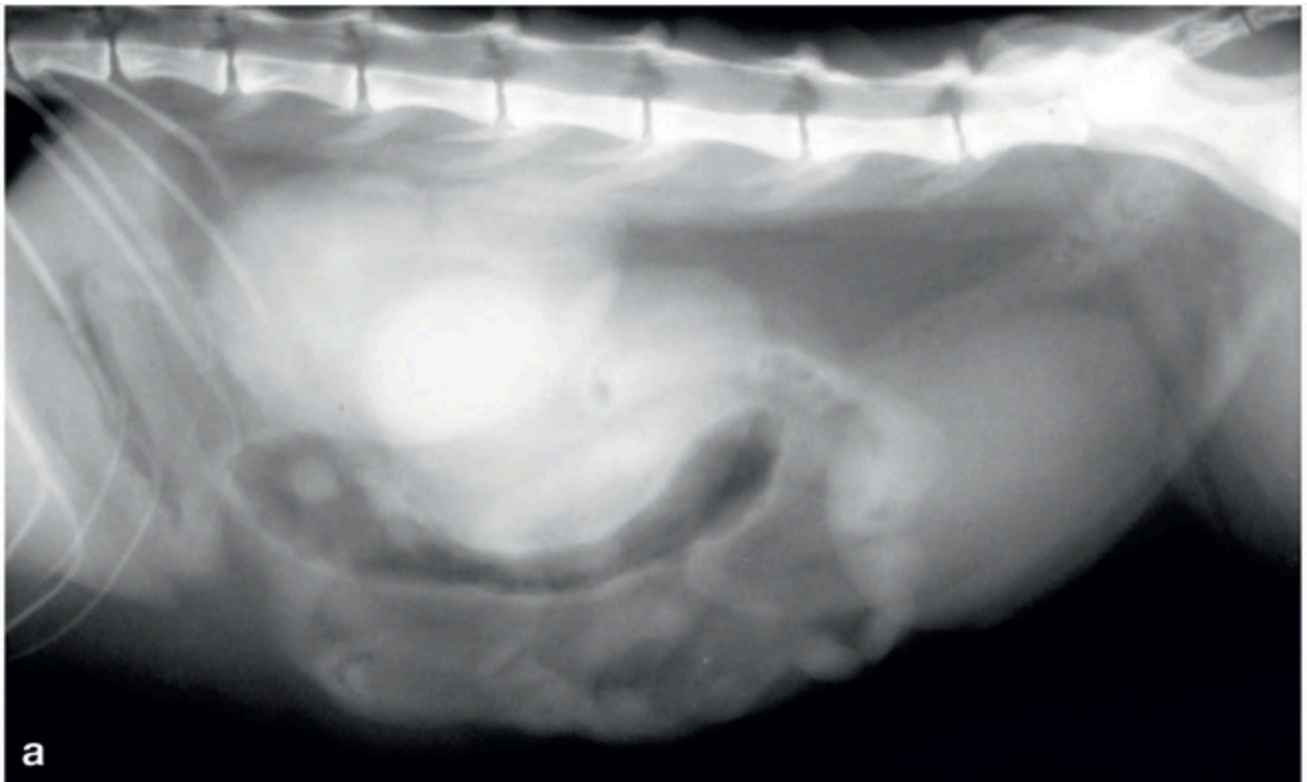
Summary of the evidence

There is moderate evidence (grade III) to recommend the use of amlodipine in the management of systemic hypertension in cats with CKD. Randomized controlled clinical trials are desirable to validate these recommendations and better identify those feline patients that will benefit most from this form of therapy.

ACE inhibitor therapy

Systemic and renal changes that are observed in patients with kidney disease may contribute to the progression of renal injury ([Figs 2](#) and [3](#)). Frequently implicated changes include increased systemic arterial pressure (systemic hypertension), increased intraglomerular pressure (glomerular hypertension), glomerular enlargement (glomerular hypertrophy) and altered glomerular perm-selectivity. Evidence suggests that therapy with ACE inhibitors preserves glomerular structure and function in patients with kidney disease. Administration of ACE inhibitors to rodents with reduced renal mass revealed that renoprotection is associated with lowering of glomerular capillary pressure, blood pressure and glomerular size.[42,43](#) Systematic reviews and meta-analysis of clinical studies of ACE inhibitor use in human patients have revealed beneficial effects in terms of slowing progression of non-diabetic kidney disease.[44,45](#) This beneficial effect seems to be mediated by factors in addition to decreasing blood pressure and urinary protein excretion and is greater in patients with proteinuria.[44,46](#)

Fig 2.



Lateral (a) and ventrodorsal (b) abdominal radiographs of a cat with chronic kidney disease and bilateral renomegaly. Lymphoma, polycystic kidney disease and feline infectious peritonitis were considered as potential causes. Diagnostic evaluation confirmed the presence of lymphoma

Fig 3.



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Kidneys from a cat that died as a result of acute progression of chronic kidney disease. Necropsy confirmed the presence of renal hypoplasia (small kidney) and renal hypertrophy (larger kidney)

The ACE inhibitor benazepril has been licensed in several countries for use in managing cats with CKD. ⁴⁷ Two studies examined the physiological effects of benazepril in cats with induced kidney disease. Systemic arterial and glomerular capillary pressures were shown to be reduced and GFR was increased by such therapy. ^{48,49} However, the reduction in systemic blood pressure was small and a beneficial effect in reducing proteinuria was not evident. These initial studies failed to detect any evidence that administering benazepril resulted in long-term structural or functional renal protection. ⁵⁰

Data have recently been reported from randomized controlled clinical trials of benazepril use in cats with spontaneous CKD. ^{49,51-54} In two controlled studies, benazepril reduced proteinuria in cats with spontaneous CKD; however, it did not significantly prolong survival time. ^{51,54} Cats with a UPC > 1.0 had mean ($\pm$ standard deviation) survival times of 402 ± 202 days with benazepril ($n = 4$) compared with 149 ± 90 days with placebo ($n = 9$); however, this difference was not statistically significant. ⁵¹ Cats with initial UPC > 1.0 treated with benazepril had a significantly better appetite compared with cats treated with placebo. ⁵¹ While administration of ACE inhibitors lessens proteinuria in cats with CKD and proteinuria is a prognostic indicator for cats with CKD, it has yet to be established whether reduction of proteinuria with use of ACE inhibitors improves the long-term prognosis in cats with CKD. ⁵⁰

In a separate study, treatment with enalapril and benazepril did not change plasma renin activity, aldosterone concentration and indirect systolic arterial blood pressure in cats with hypertension associated with CKD. ³⁹ However, another study showed that benazepril is well tolerated when administered with amlodipine in hypertensive cats and may assist in managing cats with poorly controlled blood pressure. ⁴⁰

ACE INHIBITOR THERAPY

Summary of the evidence

Although ACE inhibitors have been shown to reduce the magnitude of proteinuria in cats with spontaneous CKD (grade I evidence), weak evidence exists to support a recommendation for or against use of ACE inhibitors for the purpose of slowing progression and prolonging survival of cats with CKD (limited grade I and grade III evidence). In contrast, results of a randomized controlled clinical study provide moderate support for the recommendation of providing ACE inhibitors for the purpose of increasing appetite in cats with urine UPC > 1.0. Further randomized controlled clinical trials are desirable to validate these recommendations and better identify those feline patients that will benefit most from ACE inhibitor therapy.

Erythropoietic hormone replacement therapy

Low packed cell volume (PCV), erythrocyte count and hemoglobin are characteristic of cats with moderate to advanced CKD.^{[2,55](#)} The anemia is normocytic, normochromic, hypoproliferative and proportional to the severity of the kidney disease. The principal cause is hypoplasia of erythroid elements of the bone marrow secondary to inadequate production of erythropoietin by the kidneys. Shortened erythrocyte life span, erythropoietic inhibitor substances in plasma, chronic gastrointestinal blood loss, nutritional abnormalities (eg, iron deficiency) and bone marrow fibrosis may further exacerbate anemia in some patients with CKD. Recent studies in humans indicate that hypoxia from anemia of kidney disease may contribute to progression of CKD because the anemia reduces oxygen delivery within the kidney, further promoting hypoxia and progressive renal damage from oxidative stress.^{[56](#)}

Hormone replacement therapy using human recombinant erythropoietin (rHuEPO) has been shown to be effective in correcting anemia of CKD in cats. Uncontrolled clinical trials indicated a substantial improvement in appetite and quality of life associated with this treatment.^{[57](#)} Unfortunately, development of antibodies directed against rHuEPO has limited the effectiveness of this therapy in a substantial number of patients. As a consequence, it is best to select carefully those cases most likely to benefit from rHuEPO treatment. Generally, hematocrit or PCV guidelines

can be used to judge the severity of anemia; symptomatic cats with PCV < 20% are the best candidates.⁵⁵ In addition, the degree of azotemia, expected rate of progression of CKD, appetite and willingness to eat therapeutic renal foods, and rate of progression of anemia must all be considered in the risk-benefit analysis of when to start therapy.² Gastrointestinal blood losses may contribute to anemia in patients with CKD and should be investigated before making the decision to use rHuEPO therapy.

Darbepoietin- α is a longer acting form of rHuEPO that has been used in a limited number of cats with anemia of CKD. Compared with regular rHuEPO, darbepoietin has a longer half-life and greater potency enabling the same clinical efficacy with less frequent administration.⁵⁸ Although its use has not been supported by clinical studies in cats with CKD, anecdotal reports indicate that darbepoietin is similar in efficacy and safety to erythropoietin, with the possible benefit of a reduction in the incidence of neutralizing antibodies.

The efficacy and safety of recombinant feline erythropoietin therapy was recently evaluated in cats with anemia of CKD and those with CKD and red cell aplasia induced by rHuEPO therapy. Most cats demonstrated erythroid hyperplasia, reticulocytosis, increased hematocrit and improved quality of life (appetite and energy level) with treatment. Unexpectedly, some cats that initially responded to feline recombinant erythropoietin therapy subsequently developed anemia that was refractory to additional treatments.⁵⁹ Unfortunately, the recombinant feline erythropoietin used in this study is not currently available as a commercial product.

The effect of renal transplantation on erythropoiesis and serum erythropoietin concentrations in cats with CKD has been evaluated.⁶⁰ Although the erythropoietic response was highly variable in cats after renal transplantation, anemia resolved in most cats within 2 months after surgery, suggesting that hormone replacement therapy is not needed in this subset of cats.

ERYTHROPOIETIC HORMONE REPLACEMENT THERAPY

Summary of the evidence

Anemia is a serious complication in many feline patients with moderate to advanced CKD. There is moderate (grade III) evidence to support a recommendation to consider treatment of anemia of CKD using rHuEPO in carefully selected cases. Placebo-controlled clinical trials are needed to confirm the safety and efficacy of darbepoietin for managing anemia of CKD in cats. Future use of specific feline recombinant erythropoietin products may improve the overall success of therapy in cats with anemia of CKD should these products become available.

Potassium supplementation

An association between polyuric renal failure and hypokalemia has been recognized in cats by several investigators.^{[61–63](#)} Hypokalemia occurred in 19% of cases in a clinical series of cats with spontaneous kidney disease and was moderate to severe in more than half of these patients.^{[64](#)}

There is limited evidence that hypokalemia may be a cause of and contributing factor to CKD in cats, rather than simply a consequence of the disease. In an uncontrolled study, evidence of renal lesions and dysfunction developed in 3/9 cats fed a potassium-restricted, acidifying food for several months.^{[65](#)} However, it was not clear whether potassium depletion or hypokalemia preceded the onset of kidney disease. In another study, 4/7 cats with induced CKD fed a food containing 0.3% potassium (dry matter basis) developed hypokalemia but four cats with normal renal function fed the same food did not develop hypokalemia.^{[66](#)} Muscle potassium content has been shown to be decreased in normokalemic cats with spontaneous CKD, indicating that a total body deficit of potassium may develop well before the onset of hypokalemia.^{[67](#)} The latter findings support the concept that reduced renal function precedes the development of hypokalemia.

In many cats with CKD and hypokalemia, renal function may improve after potassium supplementation and restoration of normokalemia, suggesting that hypokalemia may be associated with a reversible, functional decline in GFR. Renal function was shown to be adversely affected in normal cats when an acidified, low-potassium food was fed.^{[68](#)} In that study, potassium depletion

and acidosis appeared to have additive effects in impairing renal function. Routine supplementation with low oral doses of potassium has been recommended for all cats with CKD. [62](#) This recommendation appears to be based on the hypothesis that, in some cats with CKD, hypokalemia and potassium depletion might promote a self-perpetuating cycle of declining renal function, metabolic acidosis and continuing potassium losses. It is proposed that potassium supplementation may stabilize renal function before potassium depletion exacerbates the disease.

The potential benefit of potassium supplementation has been examined; however, the clinical benefits of such therapy remain to be proven. [67,69](#) Although the value of providing supplemental potassium to cats with CKD and normokalemia has not been established, it is clear that muscle potassium and probably total body potassium stores may be reduced in at least some cats with CKD, presumably putting them at risk of hypokalemia. Feeding acidified foods or dietary acidifiers to cats with CKD may exacerbate their tendency to develop potassium depletion. [70](#) However, unpublished studies in cats with induced CKD failed to confirm an adverse effect of mild to moderate metabolic acidosis on hypokalemia or renal lesions. [71](#) There also appears to be an inverse relationship between serum potassium concentrations and blood pressure in cats. [72](#)

POTASSIUM SUPPLEMENTATION

Summary of the evidence

Although supported only by weak (grade III) evidence, potassium supplementation is generally advocated for cats with CKD and hypokalemia, even in the absence of overt clinical signs. There is weak (grade IV) evidence to support a recommendation either for or against routine supplementation of potassium for all cats with CKD for preventing hypertension or progression of CKD. Given that current feline renal foods are generally supplemented with additional potassium, randomized controlled clinical trials are required to establish whether potassium supplementation beyond the amount found in current feline renal foods is necessary or beneficial.

Antioxidant supplementation

The kidney accounts for 10% of the whole body oxygen consumption and is a site of significant aerobic metabolism that produces reactive oxygen species as by-products.⁷³ Increased free radical production and antioxidant depletion have been reported in human patients with CKD.⁷⁴ Increased free radicals appear to cause renal injury and increased oxidative stress may contribute to the progression of CKD.⁷⁵⁻⁷⁷

In a study, cats with spontaneous CKD were fed a dry food supplemented with the anti-oxidants vitamins E and C, and 3-carotene for 4 weeks. When compared with healthy cats, cats with stage 2 CKD had a tendency to develop oxidative stress.⁷⁸ The antioxidant supplements significantly reduced DNA damage in cats with CKD as evidenced by reduced serum 8-hydroxy-2'-deoxyguanosine (8-OHdG) concentrations, and comet assay parameters of DNA damage. Serum 8-OHdG concentrations generally reflect oxidation of DNA and comet assay shows the breakdown of DNA structure in the cells measured.

ANTIOXIDANT SUPPLEMENTATION

Summary of the evidence

Cats with early spontaneous CKD show a tendency towards enhanced oxidative stress, which may worsen as CKD progresses. Dietary supplementation with antioxidant nutrients such as vitamin E, vitamin C and α -carotene significantly reduced oxidative damage to DNA in cats with spontaneous CKD. Based on limited evidence, cats with CKD might benefit from antioxidant fortification of therapeutic renal foods.

Randomized controlled clinical studies are needed to confirm the potential benefits of dietary antioxidant supplementation on progression of CKD and important clinical outcomes (survival, quality of life).

Alkalinization therapy

Metabolic acidosis appears to be a common complication of CKD in cats,^{61,64} and alkalinization therapy designed to correct metabolic acidosis may be an important part of the overall management of such patients. A cross-sectional study involving 59 cats with CKD showed that over half of patients with severe disease had acidemia and low plasma bicarbonate concentrations.^{79,80} These data also suggested that biochemical evidence of severe metabolic

acidosis does not generally occur in cats until late in the course of CKD.⁷⁹ As discussed earlier, there is also a putative relationship between hypokalemia, potassium depletion, chronic metabolic acidosis and CKD in cats.⁸¹

Alkalinization therapy has been recommended for patients with moderate to severe metabolic acidosis associated with CKD based on pathophysiological justification and extrapolation from findings in other species. Potential benefits of alkalinization therapy in patients with CKD include:

- improving clinical signs that may be caused by uremic acidosis;
- preventing the catabolic effects of metabolic acidosis on protein metabolism, thereby
- promoting adaptation to dietary protein restriction;
- limiting skeletal changes resulting from bone buffering; and
- modifying progression of kidney disease.^{2,81,82}

Chronic metabolic acidosis appears to impose an unnecessary metabolic risk that can easily be corrected by administration of potassium citrate or sodium bicarbonate when acidosis is appropriately confirmed.

In an investigation of cats with induced CKD, an acidifying (AD) or non-acidifying (NAD) diet was fed for 39 weeks.⁷¹ Hyperchloremic metabolic acidosis developed in the AD group (mean venous pH = 7.27, compared with 7.33 in the NAD group); the net acid excretions in this group were six-fold higher than in the NAD group. Cats with induced-CKD fed the AD for 6 months did not develop progressive glomerular dysfunction or renal tubulointerstitial injury when compared with cats fed a NAD. This suggests that levels of dietary acidification typically used in manufactured adult maintenance foods may not promote progression of feline CKD. A double-masked placebo-controlled cross-over study examined the effect of supplementing the diet with potassium gluconate (2 mEq twice daily) for 12 weeks.^{69,83} This regimen had no effect on venous blood pH or bicarbonate ion concentrations although it did result in an increase in urine pH. No other controlled clinical trials have been performed to confirm that appropriate alkalinization therapy will significantly alter the progression of disease or clinical signs in cats with spontaneous CKD. Most therapeutic renal foods used in the management of CKD have increased dietary buffering capacity and are already less acidifying than many standard feline maintenance foods.

ALKALINIZATION THERAPY

Summary of the evidence

There is weak (grade IV) evidence to support therapy of acidemia using alkalinizing agents such as sodium bicarbonate or potassium citrate for the purposes of slowing progression of CKD and minimizing clinical signs or uremic acidosis. The benefit of alkali supplementation over and above that provided by therapeutic renal foods in cats with CKD remains to be established. Randomized controlled clinical trials are necessary to validate recommendations for routine use of this therapy, especially in the light of the fact that oral supplementation with alkalinizing agents is a treatment that is often not well accepted by the patient and must be given multiple times per day for the remainder of the pet's life.

Dietary phosphorus restriction and intestinal phosphate binders

Phosphorus is retained in CKD, eventually resulting in hyperphosphatemia, which in turn promotes renal secondary hyperparathyroidism. The role of dietary phosphorus intake and hyperphosphatemia in pets with spontaneous CKD was first discussed nearly 30 years ago.^{[84,85](#)} Today, there appears to be a consensus of opinion that phosphate retention is a major cause of progression of CKD in many species.^{[9](#)} In human beings with CKD receiving hemodialysis therapy, the adjusted relative risk of mortality was stable in patients with serum phosphate concentrations below 6.5 mg/dl, but increased significantly above this level.^{[86](#)} Similar data for mortality risk associated with hyperphosphatemia in cats with spontaneous CKD are lacking. Nonetheless, treatment targets for serum phosphorus concentrations have been established for cats with CKD: serum phosphorus concentration is recommended to be maintained below 4.5 mg/dl (1.45 mmol/l), 5.0 mg/dl (1.6 mmol/l) and 6.0 mg/dl (1.9 mmol/l) in cats with stages 2, 3 and 4 CKD, respectively.^{[9](#)}

Hyperphosphatemia appears to be a reliable clinical indicator of hyperparathyroidism in cats with CKD.^{[87,88](#)} Hyperphosphatemia is detected in approximately 60% of cats with CKD, with the prevalence increasing as renal function declines. In a study of cats with advanced CKD, the prevalence of renal secondary hyperparathyroidism was reported to be 84%.^{[88](#)} In this study, 87%

of cats with clinical signs and 47% of clinically normal cats with only biochemical evidence of CKD were diagnosed with renal secondary hyperparathyroidism. Hyperparathyroidism was even detected in nine cats with CKD having normal serum calcium and phosphorus concentrations.

In a model of induced kidney disease in cats, high dietary phosphorus intake (1.56% dry matter) for up to 11 months was associated with renal mineralization, fibrosis and mononuclear cell infiltration.⁸⁹ Low levels of phosphorus intake (0.42% dry matter) were not associated with these histological abnormalities. Progressive renal dysfunction was not detected in either group during the timeframe of the study. In at least some cats with spontaneous CKD, dietary management with low-phosphorus feline renal foods alone may normalize hyperparathyroidism.⁸⁸

Intestinal phosphate binding agents may be useful in further reducing phosphate retention and hyperparathyroidism in cats with CKD when nutritional management alone (therapeutic renal foods) is insufficient in achieving serum phosphorus concentration targets. However, experimental or controlled clinical studies establishing the effectiveness of adding intestinal phosphate binding agents to dietary phosphate restriction in cats with CKD are limited. Clinical reports and anecdotal evidence suggest that Intestinal phosphate-binding agents are useful in reducing serum phosphate concentrations, especially in uremic, hyperphosphatemic animals.^{9,22} Lanthanum carbonate octahydrate has recently been developed as an intestinal phosphate binding agent and studied in azotemic, normophosphatemic cats with a model of CKD.⁹⁰ Addition of this agent to a moist high-phosphorus (2.2% phosphorus, dry matter basis) feline maintenance food fed to cats with induced CKD resulted in reduced phosphorus digestibility and increased phosphorus excretion.⁹⁰ Controlled clinical studies of this and other phosphate binders to confirm effectiveness in slowing progression of spontaneous CKD in cats have not been reported.

Intestinal phosphate binders are not well tolerated by some cats and may impair food intake. These agents must be given with every meal to be effective, which may decrease owner acceptance and usefulness of this therapy.

DIETARY PHOSPHORUS RESTRICTION AND INTESTINAL PHOSPHATE BINDERS

Summary of the evidence

While there is good evidence for use of dietary phosphorus restriction in therapeutic renal foods for management of CKD in cats (see below), there is only weak evidence (grade IV) supporting the recommended treatment targets for serum phosphorus concentrations. Consequently, only weak evidence exists for adding intestinal phosphate binders to dietary phosphorus restriction for additional control of hyperphosphatemia and hyperparathyroidism. This is relevant in the light of the fact that treatment with phosphate binders is often not well accepted by the patient and must be given multiple times per day for the remainder of the pet's life. The value of strict, target-based recommendations for management of serum phosphorus concentrations should be confirmed by randomized controlled clinical trials.

Therapeutic renal foods

Therapeutic foods have been used for over 50 years in veterinary patients with CKD. Compared with typical commercial and homemade pet foods, therapeutic renal foods usually have reduced protein, phosphorus and sodium, and increased dietary buffering capacity, soluble fiber, B-complex vitamins, antioxidants, potassium and omega-3 fatty acids.⁸¹ Veterinarians are often confronted with the decision of whether to recommend switching to a therapeutic renal food or to continue the current diet with the view that eating any food is better than risking reduced food intake by attempting a potentially unwanted dietary change. This clinical challenge underlines the importance of assessing the quality and strength of evidence underlying the use of therapeutic renal foods.

Several published clinical trials have provided evidence in support of feeding therapeutic renal foods when managing cats with stages 2-A CKD. A retrospective study compared the survival times of cats newly diagnosed with CKD that had been fed a normal diet or one of seven different types of therapeutic feline renal foods.⁹¹ In this study, the median survival time of 175 cats with CKD that received conventional foods was 7 months, whereas the median survival time of 146 cats with CKD fed one of the therapeutic renal foods was 16 months.⁹¹ A prospective study comparing feeding a

renal food with not making a dietary change, showed a striking enhancement of survival time associated with feeding the renal food.⁹² The control group in this study was composed of cats that refused to eat the renal food and continued their usual diet. A principal limitation of this study is that cats electing not to consume the renal food may have had an intrinsically worse prognosis unrelated to the food consumed. However, the magnitude of the difference in outcome (median survival time was increased nearly 2.5 times when the renal food was fed) suggests that the clinical benefit of feeding the renal food was likely to be real.

Of all the treatments recommended for cats with CKD, the only one that has been shown (in a randomized controlled clinical study) to decrease uremic episodes and mortality is therapeutic renal food.

The single most important thing you can do to increase acceptance of therapeutic renal foods is gradually transition, sometimes over a period of at least 3–4 weeks.

The effectiveness of a therapeutic renal food in cats with stages 2 and 3 CKD was evaluated by a randomized controlled clinical trial.⁹³ This study examined whether clinically important benefits accrued when cats with CKD consumed a renal food compared with a standard feline maintenance food. Other than being randomly assigned to either the renal food or the maintenance food, cats were managed in an identical manner with respect to other treatment interventions. Cats fed the maintenance food had a significantly greater number of uremic episodes (26%) compared with cats fed the renal food (0%). In addition, a significant reduction in renal-related mortality was observed in cats fed the renal food (0% versus 22%). Importantly, significant adverse effects of feeding the renal food were not detected in the study and cats fed the renal food maintained body weight and body condition during the 2-year study period.

Seemingly, the greatest problem with advocating renal foods for cats with CKD has been acceptance of the food. In most instances this can be overcome by carefully correcting metabolic complications of CKD and gradually introducing the renal food over several weeks (see box, page 206). In a recent clinical trial, most cats with CKD (90%) readily accepted and continued eating a renal therapeutic food using a gradual transition period and were no more likely to refuse the renal therapeutic food than a more typical adult maintenance food.⁹³

THERAPEUTIC RENAL FOODS

Summary of the evidence

There is strong evidence supporting the recommendation that cats with stages 2, 3 and 4 CKD be fed a therapeutic renal food (grade I evidence for stages 2 and 3 CKD; grade III evidence for stage 4 CKD) for the purpose of minimizing uremic episodes and reducing premature mortality.

Assisted feeding

Malnutrition is usually detected as weight loss, declining serum albumin or total plasma protein concentrations, anemia and evidence of decreased muscle mass. In patients with CKD, malnutrition usually results from inadequate food intake. Commercially available foods designed for patients with CKD contain sufficient protein, calories and other nutrients to sustain adequate nutrient intake when consumed in appropriate quantities. Some cats with stages 3 and 4 CKD fail to eat sufficient food voluntarily, regardless of palatability or nutrient content.^{[94](#)}

When malnutrition is evident or suspected, clinical recommendations often include a stepwise approach designed to facilitate adequate food intake.^{[2,95](#)} The first step is to ensure that metabolic and other medical causes of decreased appetite have been corrected including dehydration, gastrointestinal hemorrhage, metabolic acidosis, hypokalemia, anemia, urinary tract infection and drug-associated anorexia. When metabolic and other medical causes of anorexia have been excluded or corrected, therapy for uremic gastroenteritis should be initiated. A study has shown that hypergastrinemia is common in cats with naturally developing CKD and may contribute to gastric hyperacidity, uremic gastritis, gastrointestinal tract bleeding and associated clinical signs of anorexia and/or vomiting.^{[96](#)} Therapy for gastrointestinal complications of uremia usually includes administration of an H₂ antagonist such as famotidine, which may be combined with an anti-emetic and a gastric mucosal protectant such as sucralfate.^{[97](#)}

If therapy for uremic gastroenteritis fails to restore normal appetite, assisted feeding should be considered. Long-term use of esophagostomy or percutaneous gastrostomy tubes has been successful for delivering food, extra water and medications to patients with CKD. Anecdotal

reports suggest that tube feeding can reverse the progressive weight loss associated with CKD and patients can have extended periods of improved quality of life.[2,55,95](#)

ASSISTED FEEDING

Summary of the evidence

Although there is only weak (grade IV) evidence supporting the use of assisted feeding in the management of CKD in cats, there is no proven effective alternative. Clinical observations suggest that assisted feeding may reverse malnutrition and improve quality of life in many patients. Randomized controlled clinical trials would assist in validating the benefits and risks of assisted feeding in patients with CKD and better identify those cats that would benefit from this form of therapy.

10 tips to encourage acceptance of therapeutic renal foods

1 Provide owner education

Educate pet owners about the role of nutritional management in prolonging survival time and quality of life in cats with kidney disease. In order for treatment to have the best chance of success, owners must commit valuable resources (ie, their time and money), which is more likely to occur if they understand the benefits of their efforts.

2 Implement early

Begin nutritional management sooner rather than later. Current evidence supports feeding a renal food when serum creatinine is ≥ 2 mg/dl ($176\mu\text{mol/l}$). Waiting until creatinine is higher or the patient is showing uremic signs (eg, inappetence, vomiting) is not advised. Uremic patients are less likely to accept a new food and therefore will not receive the optimal benefits of a renal therapeutic food.

3 Transition gradually

The single most important thing you can do to increase patient acceptance of a therapeutic renal food is gradually transition to the new food. The transition period should be a minimum of 7 days; however, many cats may need a transition of 3–4 weeks or longer. It is critical to discuss the need for this transition with pet owners, otherwise they are likely to buy the new food, go home, and switch from the old food to the new food at the next meal. In this scenario, many cats will refuse to eat the new food, resulting in an unhappy owner and a patient that will not receive the benefits of the prescribed nutritional management.

4 Provide old and new

One option for transitioning to a renal food is to mix the old and new food, gradually adding more of the new food over time. Another approach is to

provide both foods (old and new) in food dishes side by side. This latter technique assists with gradual transition and also allows cats to express their preferences. For more information see The Indoor Cat Initiative website (www.vet.osu.edu/indoorcat )

5 Use a suitable food dish

When transitioning cats from dry to moist food, use a flat food dish (eg, saucer) rather than a bowl. This avoids the cat's whiskers rubbing on the food dish, which could affect acceptance of a new food.

6 Avoid stressful periods

Avoid offering therapeutic renal foods in stressful periods (eg, to sick and/or hospitalized cats, or during force-feeding); a food aversion may develop causing decreased acceptance of the food when the patient is feeling better. Put another way — while hospitalized, do not feed your feline patients the food you want them to eat for the rest of their lives. One option would be to feed a maintenance food that avoids excessive protein, phosphorus and sodium until the patient is feeling better and then begin gradual transitioning to a therapeutic renal food.

7 Temperature may be important ...

Use fresh food at room temperature. Some patients may eat refrigerated food that is warmed, but others will only eat food from a newly opened container. Some patients may eat food that has been refrigerated and stored in a plastic container (versus being stored in the original can).

8 ... So may texture and formulation

Offer foods with different textures (eg, minced formulas) or formulations (dry versus moist). Some pets may have shown a preference for either dry or moist food when healthy but, when they develop kidney disease, their preferences may switch (eg, a cat that has eaten only dry food all its life may only eat moist food once kidney disease occurs).

9 Add flavor enhancers

Add flavor enhancers (low-sodium chicken broth, tuna juice, oregano, brewer's yeast, small amount of regular food) to enhance palatability and encourage the patient to eat all the therapeutic food. Excessive use of other food may decrease the beneficial effects of nutritional management.

10 If all else fails, try a different brand

If, having followed all the above steps, there is still reluctance to eat a therapeutic renal food, switch to a different brand. While commercially available renal foods have general features in common, they are not the same foods. In addition, individual pets may express a preference for one brand over another. Avoid giving the owner samples of several different brands of foods at once; this could result in the cat developing a food aversion to all foods, especially if owners offer each sample at successive meals or on consecutive days.

Dialysis

In human medicine, hemodialysis is the most common means of providing replacement or supplemental renal excretory function. Hemodialysis has been used successfully to manage clinical signs of uremia in cats,^{[94,98-100](#)} Although hemodialysis is primarily indicated for managing cats with potentially reversible acute renal failure, there may be patients with CKD for which hemodialysis should be considered as a suitable adjunct to other forms of therapy, specifically in preparing cats for possible renal transplantation. Hemodialysis is most likely to be beneficial when serum urea nitrogen concentrations exceed 90 mg/dl (32 mmol/l urea) and serum creatinine concentrations exceed 8 mg/dl (700 μ mol/l).^{[98,99](#)}

Recent investigations have focused on recruiting the gastrointestinal tract as a means of excreting wastes normally eliminated by the kidney; in essence, enteric dialysis. Polymers designed for oral administration adsorb uremic toxins and excess fluid in the gastrointestinal tract for excretion in the feces.^{[24](#)} These polymers have therapeutic potential as another extra-renal clearance mechanism to be used in conjunction with hemodialysis or other forms of therapy in patients with CKD. Further controlled clinical studies with oral polymer therapy are anticipated. In addition, a

probiotic product purported to enhance gastrointestinal excretion of nitrogenous waste products has recently entered the veterinary market. The manufacturer proposes that enhanced intestinal clearance of these nitrogenous wastes will ameliorate clinical signs of uremia.

Renal transplantation

Renal allograft transplantation has become a viable therapeutic option for cats with CKD. Long-term immunosuppression with a combination of oral corticosteroids and ciclosporin has proven successful in cats, although regular monitoring is required.^{[101,102](#)} However, severe and often fatal complications of renal transplantation may occur in cats.^{[32,103-108](#)}

Recommended criteria for selection of feline transplant recipients include:

- early decompensated CKD for which conventional therapy is no longer effective;
- weight loss of no greater than 20% of healthy body weight;
- no history of recent urinary tract infection;
- no concurrent serious medical conditions (eg, inflammatory bowel disease);
- no evidence of cardiac dysfunction;
- negative tests for chronic systemic viral infections.^{[109](#)}

Other criteria to evaluate when considering renal transplantation include the emotional and financial commitment by the owner to immediate and long-term care of the animal, as well as access to a veterinarian familiar with managing transplant recipients should complications arise.

Table 1.

Summary of evidence grades supporting recommendations for therapy of feline CKD

Evidence grade	Therapeutic recommendations
Grade I	Some therapeutic renal foods ACE inhibitors (reducing magnitude of proteinuria; increasing appetite in cats with UPC > 1.0)
Grade III	Some therapeutic renal foods Antihypertensive therapy (amlodipine in cats with CKD and proven hypertension) Recombinant human or feline erythropoietin (suitably anemic cats with CKD) Potassium supplementation (cats with CKD and hypokalemia) Dietary phosphorus restriction (cats with stages 3 and 4 CKD)
Grade IV	Fluid therapy (long-term administration of subcutaneous fluids) Potassium supplementation (all cats with CKD) Intestinal phosphate binders Alkalinizing therapy (acidemic cats with CKD) Assisted feeding (anorexia and malnutrition in cats with CKD) ACE inhibitors (non-proteinuric CKD) CKD — chronic kidney disease, ACE = angiotensin converting enzyme

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DIALYSIS

Summary of the evidence

In general, it is probably inappropriate to recommend hemodialysis for long-term therapy of cats with CKD until studies have been performed confirming the clinical effectiveness of such therapy, including consideration of the effect of such therapy on overall quality of life for the cat and owner. Preliminary evidence in species other than cats provides some experimental basis for the concept of 'enteric dialysis'. Randomized controlled clinical trials are needed to confirm the effectiveness of the veterinary-licensed probiotic in minimizing clinical signs of uremia in cats with CKD.

RENAL TRANSPLANTATION

Summary of the evidence

Weak evidence currently exists to support a recommendation either for or against renal transplantation as an alternative to medical management of cats with CKD. Studies comparing clinical outcomes in cats with comparable CKD treated with medical management and renal transplantation are desired. High cost, technical complications, inconvenience and limited availability will continue to preclude the routine use of renal transplantation for many cats with CKD.

KEY POINTS

The concepts of evidence-based medicine can be readily applied to management of CKD in cats.

For management of CKD in cats, the highest quality and strength of evidence exists for use of specific therapeutic renal foods ([Table 1](#)). This is the only evidence-supported therapeutic intervention that should be recommended routinely for initial management of CKD in cats.

Most cats will accept a dietary change if it is started early in the course of disease and if a gradual transition period is used.

Moderate strength and quality of evidence exists for use of antihypertensive agents (patients with hypertension associated with CKD), ACE inhibitors (proteinuric kidney disease), erythropoietic hormone replacement (patients with anemia) and renal transplantation.

At present, the lowest strength and quality of evidence exists for use of subcutaneous fluid therapy, calcitriol, alkalinizing agents, intestinal phosphate binders, assisted feeding and dialysis.

Randomized controlled clinical trials are needed to validate the benefits and risks of many treatments recommended and used in patients with CKD to better identify those patients that would benefit most from other forms of management.

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Clinical therapeutics in feline medicine: updates for old and new drugs

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Abstract

Practical relevance:

Over the past 10 years, feline medicine has significantly advanced through the addition of new pharmaceuticals and alternative formulations available for cats, as well as improvements in the knowledge about existing agents. Through continued drug development, the therapeutic options for cats are expanding.

Clinical challenges:

There are many species-specific clinical challenges encountered with the treatment of feline diseases. Additionally, therapeutic options can vary based on geography and change over time.

Evidence base:

This review article discusses the current evidence for some of the newer therapeutic agents that are either presently available for use by feline medicine practitioners or are in development and undergoing clinical trials. Their recent introduction means the evidence for some of these medications is still emerging. The clinical use of these agents, available pharmacokinetic and pharmacodynamic data, and reported adverse effects in cats are presented.

Audience:

The information in this article is relevant to all veterinarians who practice feline medicine.

Introduction

There is an increasing need for medications that can effectively manage both acute and chronic diseases in cats, and recently, there has been an influx of new pharmaceuticals on the market that are available for the treatment of various feline medical conditions. Advancements in clinical pharmacology are also assisting veterinarians with overcoming the unique challenges experienced when diagnosing and treating cats, including patient and client compliance. Consequently, the aim of this article is to review a number of the pharmaceuticals pertinent to feline medicine that are either emerging or have new clinically applicable information available since our previous review.¹ Approved uses for these medications may vary based on geographic location and may also change over time. Therefore, the authors recommend that clinicians are knowledgeable about the drugs available and approved within their geographic area and reference their respective drug licensing authority for additional information.

Buprenorphine

Buprenorphine is a lipophilic opioid derivative that results in mild to moderate sedation or pain control in cats (Table 1), with the potential for fewer systemic adverse effects compared with other opioids.² After systemic absorption of buprenorphine, which is a partial mu receptor agonist and kappa receptor antagonist, it is metabolized in the liver via N-dealkylation by cytochrome P450 3A4 to form norbuprenorphine, an active metabolite. Buprenorphine and norbuprenorphine undergo glucuronidation, forming inactive metabolites, which are excreted mainly in feces via bile and partially in urine.^{2,3} Other feline-specific pharmacologic data are listed in Table 2.

Table 1

Recommended drug dosages, reported adverse effects and clinical indications in cats of the therapeutics reviewed in this article

Drug	Dosage	Adverse effects	Clinical indications* and other comments
Buprenorphine (prolonged duration) ❖ Transdermal 20 mg/ml (Zorbum; Elanco Animal Health) ❖ Subcutaneous high concentration 1.8 mg/ml (Simbadol; Zoetis)	❖ 2.7-6.7 mg/kg applied topically (dorsal cervical area) 1-2 h before surgery ❖ 0.24 mg/kg SC applied ~1 h before surgery	❖ Hyperthermia, mydriasis, euphoria, tachycardia, hyperactivity, dysphoria, sedation, anorexia ❖ Hyperthermia, mydriasis, euphoria, tachycardia, hyperactivity, dysphoria, sedation, anorexia	❖ Postoperative analgesia for 4 days. ^{4,5} Should not be applied to diseased or injured skin ⁴ ❖ Postoperative analgesia for 24 h. Daily application for 3 days ⁵
Pregabalin Oral solution 50 mg/ml (Bonqat; Zoetis)	5 mg/kg PO given ~1.5 h before an anticipated stressful event	Ataxia, lethargy, vomiting, proprioception abnormalities, muscle tremors, anorexia, weight loss	Acute anxiety ¹⁵

Drug	Dosage	Adverse effects	Clinical indications* and other comments
Gabapentin	❖ Sedation: 100 mg/cat or 10-30 mg/kg PO ~2 h before an anticipated stressful event ❖ Analgesia: 10 mg/kg PO q12h¹⁶	Sedation, ataxia, behavioral changes, muscle tremors, elevated third eyelid, mydriasis, anisocoria, vomiting/diarrhea, hypersalivation	❖ Sedation, anxiety, as part of multimodal analgesia ❖ Consider dose reduction in cats with CKD
Frunevetmab (Solensia; Zoetis)	1–2.8 mg/kg SC once every 28 days for cats ≥2.5 kg and older than 7 months of age	Dermatitis, pruritus, alopecia	Osteoarthritic analgesia ^{17–19}
Bexagliflozin Oral tablets 15 mg (Bexacat; Elanco Animal Health)	15 mg/cat PO q24h	Euglycemic diabetic ketoacidosis, diarrhea, vomiting, anorexia, urinary tract infections	DM ^{20,21} See Table 3 for contraindications
Velagliflozin Oral solution 15 mg/ml (Senvelgo; Boehringer Ingelheim Animal Health)	1 mg/kg PO q24h	Euglycemic diabetic ketoacidosis, diarrhea, vomiting, anorexia, urinary tract infections	DM ^{20,21} See Table 3 for contraindications
Remdesivir Intravenous solution (Veklury/compounded formulations)	Recommended q24h IV dosages based on the form of FIP or site of infection[†] ❖ Effusive: 15 mg/kg ❖ Non-effusive: 15 mg/kg ❖ Ocular: 15-20 mg/kg ❖ CNS: 20 mg/kg	Anaphylactic reactions, fluid overload	FIP ^{22–26}
GS-441524 Compounded oral formulation	Dosing once or split twice daily according to response and ease of administration[‡] ❖ Effusive: 7.5 mg/kg (minimum recommended dose; adjust dosing according to response up to 15 mg/kg/day) ❖ Non-effusive: 7.5 mg/kg ❖ Ocular: 7.5-10 mg/kg ❖ CNS: 10 mg/kg q12h	Increased liver enzymes (ALT), diarrhea	FIP ^{22–26}
Molidustat Oral suspension 25 mg/ml (Varenzin-CA1; Elanco Animal Health)	5 mg/kg PO q24h for up to 28 days [‡] After 28 days of treatment a pause of ≥7 days is required	Vomiting, ²⁷ systemic hypertension, mild hyperkalemia, thromboembolism	Non-regenerative anemia associated with CKD ²⁷
Telmisartan Oral solution 10 mg/ml (Semintra; Boehringer Animal Health)	❖ Systemic hypertension: 1.5 mg/kg PO q12h for 14 days, then 2 mg/kg PO q24h with dose reductions of 0.5 mg/kg as indicated vs 2 mg/kg PO q24h ❖ Proteinuria: 1 mg/kg PO q24h	GI upset, lethargy, acute kidney injury, hypotension, increased liver enzymes	Systemic hypertension and proteinuria ^{28,29}
Tamsulosin	0.004–0.006 mg/kg PO q12–24h	Hypotension, vomiting, allergic reaction	Ureterospasms ^{30,31}
Rivaroxaban	0.5–1 mg/kg/day or 2.5–5 mg/cat PO q24h	Vomiting, increased risk of spontaneous hemorrhage	ATE ^{32,33}
Capmorelin Oral solution 20 mg/ml (Elura; Elanco Animal Health)	2 mg/kg PO q24h	Vomiting, hypersalivation, inappetence, behavior changes, lethargy, anemia, dehydration	Weight loss associated with CKD ³⁴
Rapamycin Felycin alpha-CA1, sirolimus delayed-release tablets; TriviumVet)	0.3 mg/kg PO once weekly [§]	DM, immunosuppression, cytopenias, GI upset, polydipsia, arrhythmias, urinary incontinence	Ventricular hypertrophy with subclinical HCM ^{35–37}

ATE = aortic thromboembolism; ALT = alanine transaminase; CNS = central nervous system; DM = diabetes mellitus; CKD = chronic kidney disease; FIP = feline infectious peritonitis; GI = gastrointestinal; HCM = hypertrophic cardiomyopathy

*

Clinical indications listed are approved uses or extra-label if there are no uses that are currently approved. See the text for more detail

†

Remdesivir must be diluted in 100 ml of saline and given slowly as an intravenous constant rate infusion over 4 h, q24h. Compounded formulations of remdesivir available legally in some countries are administered as a diluted solution and given over 20 mins. Remdesivir administration is given during hospitalization over the initial 72 h or until the patient can accept oral medications

‡

Compounded GS-441524 is given PO q24h or divided and given q12h for 84 days, with the dosage adjusted according to response. In cats with effusive FIP, shorter treatment durations (42 days) are reported as effective.³⁸ Oral GS-441524 should be administered on an empty stomach, followed by a small amount of water or wet food, with a full meal only given 30–60 mins later

§

Molidustat and rapamycin are currently conditionally approved for use by the US Food and Drug Administration. Similar to fully approved drugs, their safety has been proven when used according to the label; however, a ‘reasonable expectation of effectiveness’ has been shown opposed to ‘substantial evidence’ required for full approval. These drugs may only be prescribed for labeled use pending a full demonstration of effectiveness for full approval

Table 2

Summary of the clinically relevant available pharmacokinetic data in cats for the therapeutics reviewed in this article³⁹

Drug	Route of administration	Bioavailability	Peak plasma concentrations	Half-life
Buprenorphine	Transdermal	16%	7.3–20 h	65–90 h
Buprenorphine	SC (high concentration)	91%	NA	12.3 h
Pregabalin	PO (solution)	94%	20–60 mins	14.7 h
Gabapentin	PO	90%	65–100 mins	2.8–3.8 h
Frunevetmab	SC	60–73%	3.5 days	10–12 days
Bexagliflozin	PO (tablets)	NA	0.5–2 h	4–6.6 h
Velagliflozin	PO (solution)	NA	0.25 h	3.7–6.4 h
Remdesivir*	IV	NA	1 h	5.1 h
GS-441524	PO (compounded)	NA	1.5 h	3.8–8.6 h
Molidustat	PO (suspension)	NA	1 h	5–6 h
Telmisartan	PO (solution)	NA	30 mins	8 h
Rivaroxaban	PO	NA	2.5 h	7.5 h
Capromorelin	PO (solution)	NA	0.25–1 h	1.1 h

NA = not available

*

Remdesivir pharmacokinetic information is based on GS-441524 concentration due to its rapid conversion to GS-441524⁴⁰

Historically, buprenorphine has been administered either parenterally or via the buccal transmucosal route in cats, requiring repeated dosing within 24 h. Recently approved alternative formulations provide prolonged durations of analgesia in cats via a transdermal solution at 20 mg/ml (Zorbiu; Elanco Animal Health) and a high-concentration formulation for subcutaneous injection at 1.8 mg/ml (Simbadol; Zoetis) (Table 1).^{4,5} These alternative routes of administration can improve compliance compared with daily administration of oral medications to cats.

The transdermal buprenorphine solution, formulated with an absorption enhancer, is applied to the dorsal cervical area. It is sequestered in the stratum corneum, allowing for prolonged systemic delivery. The transdermal solution should not be applied to other locations or diseased or injured skin as absorption characteristics may differ.⁴ In an efficacy study, 89/112 cats receiving transdermal buprenorphine following elective neutering and forelimb onychectomy did not require rescue analgesia or opioid reversal or experience an adverse event during the 96 h postoperative period.^{4,6} This proportion of cats was higher than those that were treated with placebo (42/107 cats), although it is important to note that both groups received a standardized treatment of other pre-emptive analgesia (dexmedetomidine hydrochloride and a lidocaine metacarpal four-point ring block).

The subcutaneous injection takes advantage of buprenorphine's slow dissociation from the mu receptor and its ceiling effect to limit adverse effects, despite its delivery at a higher dose (Table 1).⁵ In an efficacy study, 66/93 cats that received high-concentration buprenorphine subcutaneously did not require rescue analgesia compared with 45/102 cats that received placebo in the 72 h postoperative period after soft tissue surgery (all cats received acepromazine with or without glycopyrrolate as premedication).^{5,7} High-concentration buprenorphine also resulted in similar analgesic effects in cats after dental extractions compared with buprenorphine hydrochloride 0.3 mg/ml given at a dosage of 0.02 mg/kg IM q8h (all cats also received meloxicam and dental nerve blocks with bupivacaine).⁸

The frequency of administration of buprenorphine, which is appropriate for treating mild to moderate pain, should be based on the expected duration of the noxious stimulus.⁹ In the case of surgery, pain can be prolonged postoperatively (2–3 days or longer), and this duration should be considered by clinicians given the importance of preventive analgesia.⁹ Subcutaneous and transdermal delivery of buprenorphine offer ease of administration to prolong the duration of analgesia; however, compared with intravenous and intramuscular administration, these methods might be inferior for postoperative analgesia.¹⁰ The bioavailability of subcutaneous and transdermal buprenorphine is comparatively low and highly variable among individuals.^{11,12} Due to the variable absorption of these newer formulations, as well as anecdotal clinical experience suggesting prolonged adverse effects in cats treated at labeled dosages, more conservative (or lower) dosing may be needed in some cats.

The adverse effects associated with the administration of the new formulations of buprenorphine are summarized in Table 1.^{4,5} Sedation and transient euphoria are the most common behavior changes associated with buprenorphine administration in cats.¹³ Compared with other species, cats also exhibit unique responses to opioids, including mydriasis and hyperthermia.¹³ While in one study cats that received transdermal buprenorphine at the higher end of the dosing range exhibited no increase in adverse effects but a more profound analgesic effect compared with placebo,⁶ in another study evaluating the safety of the transdermal buprenorphine solution, hyperthermia was observed in 66% of the cats on the day after surgery.¹⁴ Clinically, these adverse effects may limit the use of buprenorphine in some cats, and repeated naloxone dosing may be needed to reverse them due to the combination of naloxone's short half-life and the prolonged duration of effect of the newer formulations. Safe use of these longer-duration buprenorphine formulations has not been assessed in cats with renal, hepatic, cardiac or respiratory disease or in cats that are pregnant, lactating, intended for breeding or younger than 4 months of age.^{4,5}

Pregabalin

Pregabalin is primarily indicated for reducing acute anxiety in cats. It is structurally related to the inhibitory neurotransmitter gammaaminobutyric acid (GABA), but it does not interact with or bind to the GABA receptor. Pregabalin is a ligand of the presynaptic alpha-2-delta (α -2- δ) subunit of voltage-gated calcium channels within the central nervous system (Figure 1).

Figure 1

A schematic representation of the mechanism of action of pregabalin and gabapentin. Pregabalin and gabapentin are ligands of the presynaptic alpha-2-delta (α -2- δ) subunit of voltage-gated calcium (Ca^{2+}) channels within the central nervous system. They reduce presynaptic neuronal calcium influx, thus decreasing the subsequent release of many excitatory neurotransmitters, including glutamate, norepinephrine, substance P and calcitonin gene-related peptide. These mechanisms account for pregabalin and gabapentin's anxiolytic, analgesic and antiepileptic properties.^{46,47}

A pregabalin oral solution (Bonqat; Zoetis) is now available and labeled for cats to minimize fear and anxiety associated with veterinary visits (Table 1). Ease of administration is improved through the small volume required and flavoring of the liquid suspension. Extra-label uses of pregabalin include the treatment of neuropathic pain, as an adjunct analgesic therapy for allodynia and hyperalgesia, and as a substitute for gabapentin as an adjunct perioperative sedation agent.^{41,45}

Pregabalin oral solution is labeled as a single oral dose administered approximately 1–2 h before an anticipated stressful event.¹⁵ In both a field study and a randomized, placebo-controlled study, approximately half of the cats that received pregabalin 1.5 h before a veterinary visit had a good to excellent response to the stress of transport and handling, with decreased vocalization, restlessness and panting.^{15,46}

Pregabalin solution has high oral bioavailability (Table 2)^{43,47} and it can be given with a small amount of food. Renal elimination of the active drug and its metabolites in other species suggests caution in cats with kidney disease. It should also be used cautiously in cats with cardiac disease or hypertension as it may cause bradycardia and reflex hyper-tension.¹⁵ The most common adverse effects associated with pregabalin oral solution in cats are listed in Table 1.¹⁵

Gabapentin

Gabapentin is an analog of GABA and is used for its sedative and anxiolytic effects and for the treatment of musculoskeletal pain in cats (Table 1).^{48,54} Despite its frequent and longterm use, the mechanism of action of gabapentin is not completely understood.⁵⁵ Gabapentin does not interact with GABA type-B receptors and, until recently, was not known to influence GABA type-A

receptors.^{56,57} Similarly to pregabalin, it binds to the α -2- δ subunit of voltage-gated calcium channels (Figure 1), and its inhibitory effects on voltage-gated calcium influx are believed to be responsible for many of its therapeutic applications.^{55,58}

Oral gabapentin is highly bioavailable when administered on an empty stomach (Table 2).^{48,59} Repeated dosing (10 mg/kg PO q12h for 14 days) does not alter its pharmacokinetics but is ineffective at maintaining the plasma concentrations correlated with efficacy in other species,⁴⁸ although the optimal plasma concentrations of gabapentin for clinical efficacy in cats are not yet fully understood. In one study, transdermal gabapentin (Lipoderm base) was found to be systemically absorbed in cats but reached lower plasma concentrations than oral administration.⁶⁰ An improvement in pain scores was also observed; however, the fact that the assessment was not masked should be considered. Gabapentin plasma concentrations may increase over several days with transdermal administration, but the time to reach a steady state with this route is unknown due to the short time frames that have been studied.^{48,60} Cats with chronic kidney disease (CKD) have higher dose-normalized serum gabapentin concentrations compared with healthy cats, providing evidence that dose reduction is needed for cats with reduced kidney function; however, the optimal dose reduction strategy is not yet established for these patients.⁶¹

The pharmacodynamic effects of gabapentin have been evaluated in multiple clinical trials in cats. Although there are currently no approved indications for gabapentin in cats, it is frequently prescribed, with its main clinical uses being analgesia, as an anxiolytic to reduce stress and improve compliance during veterinary evaluations, and to decrease the minimum alveolar concentration of gas anesthetics.^{50,52,62,65} Gabapentin may also stimulate appetite; one study showed that at a dose of 5 mg/kg, it improved appetite similarly to mirtazapine in cats immediately post-ovariectomy.⁶²

Studies evaluating gabapentin's effectiveness in anesthesia premedication protocols are limited. When administered alongside buprenorphine, gabapentin provided similar levels of preanesthetic sedation to pregabalin and buprenorphine.⁴⁴ Gabapentin also resulted in similar sedation and stress scores as alprazolam in healthy cats prior to ovariohysterectomy.⁶⁶ In further studies, while oral gabapentin administered to cats prior to general anesthesia was found to decrease the minimum alveolar concentration of isoflurane, intravenous gabapentin was not shown to produce the same effect; the reason for this discrepancy remains unclear.^{67,68}

The evidence for gabapentin's effectiveness as an analgesic is weak,^{50,53,62,65} and stronger evidence is available to support its use as a sedative and anxiolytic. It reduces fear responses in cats in trap–neuter–return programs and accelerates behavior modification.^{49,52} Multiple studies have shown that a single dose of gabapentin, administered approximately 2 h prior to a veterinary visit, reduces stress and improves compliance.^{51,54,69,71} If further management of stress is needed after 2 h, clinicians could consider either an additional dose of gabapentin or other therapies depending on the original dose given (an oral dose of gabapentin of up to 47.6 mg/kg has been reported⁷¹), patient compliance and the availability of time. Sedation does not appear to increase when gabapentin given orally at 10 mg/kg is combined with trazadone given orally at 5 mg/kg compared with trazadone alone.⁷² However, another study found that sedation was increased when these drugs were combined and administered at approximately twice the aforementioned doses.⁷³ Gabapentin can result in a mild decrease in serum cortisol concentration compared with placebo,^{74,75} but another study did not corroborate this finding.⁵¹ Serum glucose concentrations do not appear to decrease with gabapentin.^{51,74} The evidence for administering gabapentin to cats supports its use as a sedative but not for analgesia. If prescribed as an analgesic, it should be used as a part of multimodal therapy and not as a sole agent.

Although gabapentin is useful for improving patient compliance and reducing stress during veterinary evaluations, clinicians should be aware that it may influence some clinical parameters. In a randomized, masked clinical trial, gabapentin (100 mg/cat) was shown to impact neurologic examination findings in healthy cats.⁷⁶ Gabapentin did not alter the level of consciousness, cranial nerve function or spinal reflexes, but did cause proprioceptive ataxia and abnormal postural reactions. It has also been found to not affect horizontal pupil diameter, intraocular pressure, Schirmer tear test and intradermal allergy testing results.^{51,77} Gabapentin did not appear to affect systolic blood pressure in a small study of five healthy cats, but a median decrease of 12 mmHg was observed in both healthy cats and those with CKD in another study.^{78,79} Gabapentin is considered appropriate for sedation prior to cardiac evaluation, since studies investigating its effects on echocardiographic parameters have shown either no changes or changes that were not thought to be clinically meaningful.^{71,80,81} Aside from its intended sedative effects, gabapentin's additional reported adverse effects are summarized in Table 1.^{48,54,69,76,81}

Frunevetmab

Frunevetmab (Solensia; Zoetis) is indicated for the management of osteoarthritic pain in cats (Table 1). It is a feline-specific monoclonal antibody against nerve growth factor (NGF; see 'Role of nerve growth factor in osteoarthritic pain' box) that binds to NGF, blocking NGF-mediated pain transmission (Figure 2). Frunevetmab is given as a subcutaneous injection every 28 days, which is an attractive route and frequency of administration for many patients.

Figure 2

A simplified schematic representation of the role of nerve growth factor (NGF) in osteoarthritic pain. NGF, which functions as a signaling protein and is the target of NGF inhibitors such as frunevetmab, binds the high-affinity NGF-specific tropomyosin receptor kinase A (TrkA) receptor of sensory neurons. The NGF–TrkA complex is endocytosed and transported to the neuronal cell body within the dorsal root ganglion where it increases the expression of cell surface receptors (ie, transient receptor potential vanilloid 1 [TRPV1], acid-sensing ion channels [ASIC], bradykinin receptors [BR]), ion channels (ie, voltage-gated sodium channels [VGSC]) and transcription factors (ie, substance P [SP], calcitonin gene-related peptide [CGRP]) central to nociception.⁸⁷ The NGF inhibitor frunevetmab is a feline-specific IgG monoclonal antibody that binds NGF, making it unavailable to bind the TrkA receptor of sensory neurons, thereby stopping the transmission of noxious stimuli mediated by the overproduction of NGF associated with osteoarthritic pain.

Frunevetmab has a half-life of 10–12 days in cats (Table 2). To reach steady-state trough frunevetmab concentrations, two doses administered 28 days apart are required, although clinical responses have been reported within 3–4 weeks.^{17,18,82} In initial studies evaluating frunevetmab in client-owned cats with osteoarthritis, a single dose of frunevetmab improved activity and pain scores for 6 weeks following treatment compared with placebo, without adverse effects.⁸² In multi-dose studies in cats with osteoarthritis treated with frunevetmab or placebo, owners reported a clinical improvement in activity, there was increased activity measured by accelerometer and the veterinarian-assessed joint pain scores were lower in the frunevetmab-treated cats compared with placebo.^{18,19} Studies evaluating the efficacy of frunevetmab compared with non-steroidal anti-inflammatory drugs (NSAIDs) have not been reported to the authors' knowledge.

In people, concerns have been raised about the development of rapidly progressive osteoarthritis in individuals concurrently treated with NSAIDs and monoclonal antibodies against NGF.⁸³ The safety of frunevetmab combined with NSAIDs has not been evaluated. Recently, a case of rapidly progressive osteoarthritis was reported in a dog treated with bedinvetmab, a monoclonal antibody against NGF approved for canine use;⁸⁴ however, this diagnosis has been challenged.⁸⁵ Furthermore, a recent disproportionality analysis evaluating musculoskeletal adverse events in dogs treated with beninvetmab revealed that ligament/tendon injuries, polyarthritis, fractures, musculoskeletal neoplasia and septic arthritis were reported nine times more frequently in bedinvetmab-treated dogs compared with the dogs treated with six comparator drugs (all NSAIDs with the same indication) combined.⁹¹ Additionally, an expert panel concluded a strong suspicion of an association between bedinvetmab and accelerated joint destruction in 19 cases.⁹¹ These reports in people and dogs should alert feline medicine practitioners to the potential of musculoskeletal adverse events secondary to frunevetmab.

The most common adverse effects reported in cats treated with frunevetmab are dermatologic (Table 1).^{18,19} In a recent case series, five cats developed pruritus not associated with the injection site 3–18 days after receiving frunevetmab;⁹² interestingly, four of the five experienced the skin reaction after the first dose of frunevetmab. The initial pruritus and subsequent self-trauma described in these cases may suggest that some cats treated with frunevetmab experience adverse cutaneous sensations like those reported in people treated with NGF inhibitors.^{92,95}

With limited clinical use and studies reporting treatment durations of ≤ 3 months, the long-term impact of frunevetmab on sensory and sympathetic nerve development is unknown. In non-human primates, prolonged exposure to high doses of NGF inhibitors reduced the size of post-ganglionic neuronal cell bodies, which returned to normal after discontinuation of the NGF inhibitor, suggesting these observed changes were reversible.⁹⁶ Frunevetmab appears well tolerated as an alternative therapy for cats with osteoarthritis, although some cats may experience skin reactions after the first injection. Additional research is needed to provide guidance on the long-term use of frunevetmab in cats, especially those with underlying CKD. Close monitoring during treatment is recommended to evaluate for potential adverse effects and each cat's clinical response, which might be facilitated by incorporating the use of pain scoring and owner questionnaires. No information regarding frunevetmab use in cats with a history of immune-mediated disease is available to guide its use in these patients.

Bexagliflozin/velagliflozin

In people with type 2 diabetes, cardiac or renal disease, sodium-glucose-linked transporter 2 (SGLT2) inhibitors have become widely used, which has sparked an interest in the use of these in veterinary medicine. This interest is particularly high in cats due to the similarities between feline diabetes mellitus and type 2 diabetes in people.⁹⁷ SGLT2 inhibitors are a potential oral option for treating diabetes mellitus in cats as they block glucose reabsorption from the glomerular filtrate and increase urinary glucose excretion, thereby lowering blood glucose concentrations (Figure 3). SGLT2 inhibitors also block sodium reabsorption in the renal proximal tubules and play a beneficial role in treating non-diabetic cardiac and renal diseases. In people, SGLT2 inhibition-induced glucosuria decreases plasma uric acid levels, reducing the risk of cardiovascular complications and congestive heart failure.¹⁰³ In addition, SGLT2 inhibition-induced natriuresis decreases plasma volume and blood pressure, in turn reducing cardiac preload and afterload, respectively.¹⁰³ SGLT2 inhibition-induced natriuresis in people promotes sodium delivery to the macula densa, activating tubuloglomerular feedback, afferent vasoconstriction, reduced glomerular pressure and decreased albuminuria, although healthy cats experience a paradoxical induction of glomerular hyperfiltration with dapagliflozin.^{103,104}

Figure 3

A schematic representation of the functional unit of the kidney, the nephron, is depicted with a focus on the sodium-glucose-linked transporters (SGLTs; SGLT1 and SGLT2) along the nephron and the mechanism of the SGLT2 inhibitors. Most of the filtered glucose is reabsorbed by the kidneys within the proximal renal tubules via two types of SGLTs – SGLT-1 and SGLT-2 – that cotransport glucose and sodium into the tubular epithelial cells.⁹⁸ Glucose is then transported across the basolateral membrane by glucose transporter 2 (GLUT2).⁹⁸ The SGLTs cotransport sodium via a carrier protein down the concentration gradient established by the Na-K ATPase pump and actively transport glucose. SGLT2 is a low-affinity, high-capacity glucose transporter located in the early proximal renal tubule epithelium, responsible for 90% of glucose uptake from the glomerular filtrate, whereas SGLT1 is a high-affinity, low-capacity glucose transporter located later in the renal proximal tubule, reabsorbing 10% of the filtered glucose.^{98,99} SGLT1 plays a more significant role in dietary glucose absorption via its presence on the luminal surface of the small intestinal epithelium. SGLT2 inhibitors decrease renal sodium and glucose reabsorption, thereby inducing glucosuria, natriuresis and osmotic diuresis.^{20,21,100,101} The SGLT2 inhibitors available for cats include bexagliflozin and velagliflozin

Recently, two veterinary-approved SGLT2 inhibitors have become available for use in cats: bexagliflozin (Bexacat; Elanco Animal Health) and velagliflozin (Senvelgo; Boehringer Ingelheim Animal Health) (Tables 1–3). The availability of an SGLT2 inhibitor in both tablet form and as a liquid suspension provide options for oral administration to cats. Both agents are highly selective SGLT2 inhibitors, and their glucosuria-induced effects rapidly decrease blood glucose concentrations independently of endogenous or exogenous insulin, thereby promoting a euglycemic state.^{97,103} Bexagliflozin and velagliflozin may also improve insulin sensitivity and support beta (β)-cell recovery.^{97,105} Although bexagliflozin and velagliflozin are highly selective SGLT2 inhibitors, they may partially inhibit SGLT1, which is responsible for sodium and glucose absorption at the small intestinal lumen and late renal proximal tubular epithelium. This inadvertent SGLT1 inhibition may contribute to gastrointestinal adverse effects.^{100,101}

Although SGLT2 inhibitors decrease blood glucose concentrations, clinical hypoglycemia is uncommon.^{106,107} However, cats remain at risk for diabetic ketoacidosis (DKA), which can manifest as euglycemic DKA, especially when the underlying cause for insulin resistance persists. In such cases, a relative lack of insulin continues to drive lipolysis and ketogenesis. Additionally, SGLT2 is expressed on α-cells of the pancreas, which contributes to the risk of developing DKA as SGLT2 inhibitors promote glucagon release.⁹⁷ Ideally, cats treated with SGLT2 inhibitors must have some β-cell function for endogenous insulin production to reduce the risk of developing DKA.^{97,106,107}

It is important to carefully screen cats for candidacy of treatment with an SGLT2 inhibitor (Table 3). Cats with a new diagnosis of diabetes mellitus that are apparently healthy and without major comorbidities are currently considered the most appropriate candidates. Specific contraindications vary with the drug labels,^{20,21} but in general an SGLT2 inhibitor should not be used in a cat with ketosis, pancreatitis or other disease that can increase the risk of developing DKA. The safety and efficacy of SGLT2 inhibitors with CKD, especially advanced-stage CKD (International Renal Interest Society stages 3 and 4), is unknown due to concerns about dehydration and decreased glucose excretion from nephron loss leading to decreased effectiveness.⁹⁷ During treatment, close clinical monitoring is necessary, as with all diabetic cats, to optimize treatment and to minimize adverse effects, including the development of euglycemic DKA (see ‘Monitoring cats treated with bexagliflozin/velagliflozin’ box).

Table 3

Comparison of bexagliflozin and velagliflozin used for glycemic control in cats with diabetes mellitus that are not on or have not been previously treated with insulin therapy^{20,21,100,102}

SGLT2 inhibitor	Formulation	Pharmacokinetics	Contraindications
Bexagliflozin (Bexacat; Elanco Animal Health)	Flavored tablets	❖ Food increases oral absorption ❖ Metabolism extrapolated from people, including glucuronidation and oxidation ❖ Unchanged drug excreted in feces ❖ Glucuronide metabolite excreted in urine	❖ Cats previously or currently treated with insulin ❖ Insulin-dependent diabetes mellitus ❖ DKA ❖ Anorexia, dehydration or lethargy ❖ Clinical pancreatitis ❖ Hepatic disease (bexagliflozin) or bilirubin >0.5 mg/dl (velagliflozin) ❖ Reduced renal function (bexagliflozin) or creatinine >2 mg/dl (velagliflozin) ❖ Chronic or unresponsive diarrhea (velagliflozin) ❖ Cachexia (velagliflozin)
Velagliflozin (Senvelgo; Boehringer Ingelheim Health)	Oral solution	❖ Food decreases oral absorption ❖ Metabolism extrapolated from people, including glucuronidation and oxidation ❖ Unchanged drug excreted in feces ❖ Glucuronide metabolite excreted in urine	

DKA = diabetic ketoacidosis; SGLT2 = sodium-glucose-linked transporter 2

From available clinical studies in otherwise healthy, newly diagnosed diabetic cats that were not previously treated with insulin, velagliflozin and bexagliflozin lowered blood glucose concentrations in most cats.^{106,107} However, up to 50% of cats experienced at least one drug adverse effect, with 5–7% of cats developing DKA. The most common adverse effects of the SGLT2 inhibitors are listed in Table 1.^{106,107} Other studies have begun investigating the extra-label use of bexagliflozin or velagliflozin in poorly controlled diabetic cats already receiving insulin. These studies have reported some success in lowering blood glucose concentrations, reducing exogenous insulin requirements and improving clinical outcomes in many cats.^{99,108,111} In one, a larger study evaluating velagliflozin in historically poorly controlled diabetic cats receiving insulin, treatment failure or drug adverse effects limited velagliflozin's use in 25% of cats and 7% of cats developed DKA.¹⁰⁸

Remdesivir/GS-441524

Remdesivir is a broad-spectrum antiviral effective against RNA viruses, including feline coronavirus (FCoV). It is a prodrug of the nucleoside analog, GS-441524, which is further phosphorylated into its active nucleoside triphosphate (NTP) metabolite that then interferes with coronavirus replication (Figure 4).^{22,112} Remdesivir and GS-441524 have shown efficacy in treating FCoV associated with feline infectious peritonitis (FIP) based on in vitro studies, animal models and the treatment of FIP in cats.^{23,113,117} Antiviral protocols, including parental remdesivir used in combination with oral compounded GS-441524 or oral GS-441524 alone, vary depending on the cat's clinical signs, availability of reputable drug formulations and ease of drug administration to the patient. Recommended dosages for remdesivir and oral GS-441524 also depend on the cat's clinical signs and are summarized in Table 1.²⁴ The use of remdesivir and oral compounded GS-441524 allows veterinarians to develop more standardized protocols to treat cats with FIP successfully and alleviates the need for owners to rely on unlawful GS-441524.^{24,114}

Figure 4

A schematic representation of the relationship between the antivirals, remdesivir and GS-441524, and their mechanism of action. (a) Remdesivir is a prodrug of the nucleoside analog that aids in attaining intracellular concentrations of GS-441524. GS-441524 is further phosphorylated to its active nucleoside triphosphate (NTP) metabolite that acts as a nucleoside analog and inhibits the RNA-dependent RNA polymerase (RdRp) of coronaviruses.^{22,112} (b) Chemical structure comparison of the active metabolite of remdesivir and GS-441524, NTP metabolite, to adenosine triphosphate (ATP). (c) Schematic representation of the NTP metabolite competing with ATP for incorporation into replicating viral RNA via RNA-dependent RNA polymerase (RdRP), resulting in the termination of viral RNA synthesis. Coronavirus survival requires functional RNA-dependent RNA polymerase for viral replication. It is the NTP metabolite that acts as an inhibitor of RdRP, interfering with coronavirus replication.^{22,112}

Reports indicate that 75–100% of cats treated with antivirals – remdesivir or GS-441524 – show some clinical improvement within 48–72 h.^{23,25} The clinical response to these antivirals is incredible considering FIP was, until a few years ago, considered an almost uniformly fatal disease for cats.¹¹⁸ Both antivirals appear to be well tolerated in cats over the 84 days of treatment in these studies. Adverse effects reported with remdesivir treatment in cats are listed in Table 1;^{23,25} additionally, the formation of urolithiasis has been reported in cats treated with unlawful GS-441524.^{119,120}

Studies are beginning to emerge that support shorter treatment durations in some cat populations with FIP treated with licensed GS-441524. In a small prospective, randomized controlled study in cats diagnosed with effusive FIP, 42 days of treatment with licensed GS-441524 was effective in resolving clinical signs without noted adverse effects.³⁸ In addition, the use of acute-phase protein – alpha-1 acid glycoprotein or serum amyloid A – monitoring may offer an objective marker of clinical response in cats with FIP. For example, alpha-1 acid glycoprotein offers a non-specific monitoring tool to help clinicians assess FIP recovery and tailor antiviral treatment durations.¹²¹ The use of GS-441524 therapeutic drug monitoring has also begun to be evaluated in cats but, in the absence of feline population pharmacokinetic data, it is not routinely available for clinical decision-making.⁴⁰ Cats with more severe FIP or younger cats may require GS-441524 dosage increases with increasing body weight.

With new antivirals becoming available, the opportunity to successfully treat cats with FIP is promising and continues to evolve. Overall, remdesivir and GS-441524 are well tolerated in most cats. Effective antiviral protocols, including more oral options, will likely continue to develop. Therapeutic options for cats may vary based on clinical presentation, severity of illness and the antivirals available to veterinarians and owners.

Molidustat

Molidustat is indicated for use in cats with non-regenerative anemia associated with CKD and acts by increasing the levels of endogenous erythropoietin, an important hematopoietic growth factor. Erythropoietin is primarily produced by renal peritubular interstitial cells, and its production is stimulated by hypoxia-inducible factor (HIF). During normoxia, HIF is degraded by hypoxia-inducible factor prolyl hydroxylase (HIF-PH) enzymes; when there is renal hypoxia, however, this inhibits HIF-PH enzymes, stimulating erythropoiesis (Figure 5). With CKD comes a decrease in the number of erythropoietin-producing renal peritubular interstitial cells; this leads to the decrease in erythropoietin levels well recognized in cats with this condition. However, a relatively higher oxygen tension in the kidneys remains due to the reduced metabolic activity secondary to CKD. Experimentally, a hypoxic stimulus that inhibits HIF-PH enzymes still results in erythropoiesis,^{122,123} and it is this mechanism that therefore provides the basis for the use of molidustat, an HIF-PH inhibitor, to treat anemia in CKD.

Figure 5

A simplified schematic diagram of the mechanism of action of molidustat. During renal normoxia, hypoxia-inducible factor (HIF) is degraded by hypoxia-inducible factor prolyl hydroxylase (HIF-PH) enzymes that are oxygen sensing. During renal hypoxia, or in the presence of molidustat, these enzymes are inhibited, leading to stabilization of HIF and erythropoiesis. The HIF complex binds to hypoxic response elements of target genes, ultimately leading to endogenous erythropoietin production¹²⁴

Molidustat oral suspension (Varenzin-CA1; Elanco Animal Health) can be administered daily for up to 28 consecutive days based on its conditional approval status (Table 1).²⁷ Its formulation as a liquid suspension improves ease of administration for many cats. Molidustat effectively increased erythropoietin concentrations, as detected by a human enzyme-linked immunosorbent assay, as well as hematocrit in healthy cats.¹²⁵ Minimal drug accumulation occurs, and the reported elimination half-life is 4–6 h, supporting once daily dosing for cats with decreased renal function. In a randomized, masked, placebo-controlled clinical trial, on day 28 of treatment, approximately half of the cats with CKD and concurrent anemia had an increase in hematocrit by either >25% from baseline or an absolute increase of at least 4% when treated with molidustat.¹²⁶

Concurrent iron supplementation with the administration of erythrocyte-stimulating agents is typically recommended in cats. However, most human CKD patients experience a decrease in hepcidin when treated with HIF-PH inhibitors, which may increase the bioavailability of iron.¹²⁴ Young, healthy cats did not develop apparent iron deficiency with molidustat when assessed by cell hemoglobin in reticulocytes.¹²⁶ Iron status has not been assessed in a molidustat-treated feline CKD population, however, and consequently it is currently unknown if cats with anemia associated with CKD treated with HIF-PH inhibitors would benefit from concurrent iron supplementation. Molidustat appears to be relatively well tolerated in both healthy cats and cats with CKD, aside from vomiting.^{125,126} Reported adverse effects of molidustat in cats are provided in Table 1.^{27,126}

It is unknown if HIF-PH inhibitors are more effective than erythrocyte-stimulating agents, such as darbepoetin, for the treatment of anemia in CKD in cats; however, no other therapies are currently approved for treatment in this species. Clinicians should consider multiple factors when deciding on treatment options for anemia in CKD (Table 4). Additionally, recommendations on when to initiate treatment are evolving. Therapy was traditionally recommended when anemia was severe (packed cell volume [PCV] <20%) or if it was affecting the patient's quality of life.^{128,129} However, anemia is a negative prognostic indicator in cats with CKD,^{130,131} and a PCV <27% was recently found to be associated with poorer quality of life scores.¹³² Consequently, clinicians should consider starting treatment of anemia in CKD earlier, with a PCV <28% suggested as a new threshold for cats.¹³³

Table 4

Clinical differences between molidustat and darbepoetin used in cats for treatment of non-regenerative anemia associated with chronic kidney disease

	Molidustat ^{27,125,126}	Darbepoetin ¹²⁷
Mechanism of action	Hypoxia-inducible factor prolyl hydroxylase inhibitor	Erythrocyte-stimulating agent
Approval	Conditionally approved	Extra-label
Route of administration	Oral solution	Subcutaneous injection
Frequency of administration	Daily	Weekly
Adverse effects	❖ Vomiting ❖ Systemic hypertension ❖ Mild hyperkalemia ❖ Thromboembolism	❖ Vomiting ❖ Systemic hypertension ❖ Seizures ❖ Pure red cell aplasia
Cost*	\$	\$\$\$
Iron supplementation	Potentially not required	Recommended

★

Each \$ represents 100 US dollars and is approximated from the cost of a 27 ml bottle of Varenzin-CA1 (Elanco Animal Health; molidustat) and a 1 ml vial of Aranesp (Amgen; darbepoetin alfa) 25 µg/ml from the authors' institutions

Telmisartan

Telmisartan (Semintra; Boehringer Ingelheim Animal Health) is an angiotensin receptor blocker (ARB) used for the treatment of systemic hypertension in cats, as well as for the treatment of proteinuria associated with CKD. It is formulated for cats as a liquid suspension that is rapidly absorbed and can be given fasted or with food.^{28,134} It is a selective ARB due to its affinity for binding to the angiotensin II type 1 receptor and not the angiotensin II type 2 receptor, allowing it to mediate some of the pathologic effects of angiotensin II while sparing certain protective effects (Figure 6).^{135,136} Telmisartan undergoes effective glucuronidation in cats, as indicated by an in vitro study.¹³⁷

Figure 6

A simplified flow diagram of the renin–angiotensin–aldosterone system with effects of stimulation of the classic angiotensin receptor types 1 and 2 by angiotensin II. The angiotensin receptor blocker telmisartan is a selective antagonist of the angiotensin II type 1 receptor. ACE = angiotensin converting enzyme; ARB = angiotensin receptor blocker; AT = angiotensin II type

Telmisartan dosage recommendations for treating systemic hypertension in cats vary between studies (Table 1).^{29,138,140} A dose-finding study in healthy cats and a prospective, randomized, double-masked, placebo-controlled clinical trial evaluating telmisartan therapy for cats with systemic hypertension used an escalating dosage strategy, initially 1.5 mg/kg q24h for 14 days and then 2 mg/kg q24h, with further dose reductions of 0.5 mg/kg for individual patients.^{138,139} A similar study was conducted concurrently that investigated telmisartan at a dosage of 2 mg/kg PO q24h.¹⁴⁰ The cohorts in both studies included cats with systolic arterial blood pressures between 160 mmHg and 200 mmHg. Although dosing strategies differed slightly between these two studies, there was an overall reduction in blood pressure by approximately 21 mmHg by day 14 and approximately 24 mmHg by day 28 of treatment. Additionally, approximately 52% of these cats achieved a reduction in blood pressure to <150 mmHg or a decrease of at least 15% from baseline by day 28. Although evidence is limited for telmisartan's efficacy in cats with a blood pressure of >200 mmHg, it may still be effective.¹⁴¹ For refractory cases, combination therapy with amlodipine may be considered with careful monitoring for hypotension.

Telmisartan was determined to be noninferior to benazepril for the treatment of proteinuria associated with CKD.¹⁴² It significantly reduced the urine protein:creatinine ratio at all timepoints, whereas benazepril did not. Furthermore, a higher percentage of telmisartan-treated cats were reclassified from proteinuric to non-proteinuric or borderline proteinuric (57.2%) compared with the benazepril-treated cats (30.8%) by the end of the study. Telmisartan may be superior to angiotensin-converting enzyme inhibitors for the treatment of proteinuria associated with CKD in cats; however, further studies are needed to verify this and to evaluate its impact on survival.

Reported adverse effects of telmisartan are summarized in Table 1.^{28,138,140,142} In cats receiving telmisartan, if a cat experiences a decreased appetite or anorexia, telmisartan should be discontinued to prevent acute kidney injury.¹³⁹ Although telmisartan appears to be well tolerated by most cats, close monitoring within 1–2 weeks of starting telmisartan or dose escalation for deterioration in renal function and hyperkalemia is recommended, particularly in cats with advanced stages of CKD.

Tamsulosin

Tamsulosin is a selective α_1 -adrenergic antagonist that is used for relaxation of smooth muscle of the urinary tract. It is considered uroselective due to its specificity for α_{1A} - and α_{1D} -adrenergic receptors, which predominate in the urinary tract smooth muscle, compared with α_{1B} -adrenergic receptors, which are found in the vasculature.¹⁴³ As a result, its use should relax smooth muscle in the urinary tract without causing significant systemic hypotension. Currently, there is limited to no data on its pharmacokinetics or pharmacodynamics in cats. Due to the available strength (0.4 mg capsules), compounding is required to achieve the recommended dosing in cats (Table 1). Consequently, it is not uncommon for cats to receive a higher dose in clinical practice.

Smooth muscle relaxation is often used as part of the medical management for feline urethral and ureteral obstructions. Nevertheless, prazosin, another non-selective α_1 -adrenergic antagonist, is currently not recommended¹⁴⁴ based on several recent studies showing no benefit of its use.^{145, 147} There is a paucity of data on tamsulosin in cats. Tamsulosin dosed at 0.1 mg/cat (~0.017 mg/kg) PO q24h for 10 days did not significantly alter urethral pressure profile parameters in a small population of healthy male cats.¹⁴⁸ Its effectiveness in treating urethral obstructions also remains uncertain, especially as obstructions are more common distal to the smooth muscle, in the skeletal muscle of the prostatic urethra.^{149, 151} In contrast to the urethra, the feline ureter contains smooth muscle that surrounds the mucosa throughout its length.¹⁵¹ In a retrospective study of cats with obstructive ureterolithiasis, which can cause acute kidney injury, stone passage was documented in 22/70 (31%) cats treated with tamsulosin as part of their regimen.³⁰ Due to the lack of a comparator in this study, however, the benefit of tamsulosin for medical management of obstructive ureterolithiasis in cats cannot be ascertained, and further research is required. It is also difficult to compare these findings with other studies evaluating the success of medical management for benign ureteral obstructions. In one study, serial imaging was only reported in 11/52 cats with obstructive ureteral calculi that were treated medically;¹⁵² passage of stones was reported in nine of those cases, but an improvement in creatinine was only seen in 4/9 cats. In a more recent study, successful medical management was reported in 17/75 (23%) ureters with obstructive calculi.¹⁵³ No adverse effects of tamsulosin, such as systemic hypertension, vomiting or allergic reaction, were reported in the abovementioned study.³⁰

Rivaroxaban

Rivaroxaban is a direct Xa inhibitor used in cats at risk for thrombosis. Clotting factor X is essential for secondary hemostasis and, when activated (Xa), it converts prothrombin to thrombin (Figure 7). Rivaroxaban directly inhibits Xa, reducing thrombin formation and indirectly inhibiting thrombin-induced platelet aggregation.¹⁵⁴ Having gained a better understanding of the role of thrombin and platelets in arterial and venous thrombosis, rivaroxaban is now used in people for the prevention of both conditions.¹⁵⁴ Rivaroxaban offers advantages such as high oral bioavailability, rapid onset of action and predictable anticoagulant activity, reducing the need for frequent clinical monitoring in people.¹⁵⁵ Similar efficacy and safety of rivaroxaban are anticipated in other species, leading to its extra-label use as an antithrombotic agent for cats (Table 1).^{32, 156, 157}

Figure 7

A limited schematic representation of the coagulation cascade that highlights the key coagulation factors to illustrate the mechanism of action of rivaroxaban. The intrinsic and extrinsic coagulation pathways activate factor X, which is essential for thrombin formation (factor IIa) and clot development through the common coagulation pathway. The critical activated factors of the intrinsic pathway triggered by endothelial damage include XIIa, XIa, IXa and VIIIa. Factor VII of the extrinsic pathway is activated by tissue factor (III). Prothrombin is activated by both factors Xa and Va within the common pathway to produce the thrombin needed to activate fibrinogen into fibrin and, with the help of factor XIIIa, the development of a cross-linked fibrin clot. Rivaroxaban is a direct factor Xa inhibitor that does not require a cofactor to bind Xa to produce its anticoagulant effects,³² and it binds both circulating and bound Xa. By limiting thrombin (IIa) formation, rivaroxaban prevents fibrin formation and, ultimately, clot formation. Indirectly through its effects on thrombin, rivaroxaban inhibits thrombin-induced platelet aggregation.³²

Antiplatelet and anticoagulant drugs are used to prevent thrombus propagation or recurrence in cats with aortic thromboembolism (ATE) associated with underlying hypertrophic cardiomyopathy (HCM). The preferred antithrombotic therapy for cats at risk of ATE is unclear, but options include clopidogrel (antiplatelet), low molecular weight heparin (anticoagulant) or rivaroxaban (anticoagulant), with dual therapy (antiplatelet and anticoagulant) sometimes administered.¹⁵⁶ Emerging studies support the use of rivaroxaban with or without clopidogrel to prevent thrombus formation in cats at risk of ATE and report few adverse effects.^{157, 159} In healthy cats treated with clopidogrel and rivaroxaban, platelet responsiveness to agonists, platelet activation and thrombin generation were reduced compared with treatment with either drug alone.¹⁵⁸ In a small prospective study comparing rivaroxaban with clopidogrel combined with enoxaparin in cats with HCM-associated ATE, rivaroxaban-treated cats showed no bleeding or thrombus recurrence.¹⁵⁷ In a multicenter prospective study evaluating recurrence of ATE with rivaroxaban or clopidogrel, the ATE recurrence rate of rivaroxaban (39%) was similar to clopidogrel (37%).¹⁵⁹ A small retrospective study in cats presenting for an ATE event or diagnosed with a cardiac thrombus reported a low rate of ATE recurrence (16.7%) when treated with both clopidogrel and rivaroxaban.¹⁶⁰

Although the formulation as a tablet may pose an administration challenge for cats, the available pharmacokinetics and pharmacodynamics of rivaroxaban in healthy cats indicate that it is well tolerated.^{32,157} Like in people, rivaroxaban peak plasma concentrations in cats occur approximately 2.5 h after oral dosing, with a dose-dependent increase in anti-Xa activity (Table 2).³² In people, rivaroxaban undergoes hepatic metabolism and is eliminated via feces and urine with both the active drug and its metabolites renally excreted.¹⁶¹ It is recommended to avoid prescribing rivaroxaban in cats with severe liver disease, and to use it with caution in cats with underlying kidney disease.³² Rivaroxaban is a p-glyco-protein substrate; coadministration with a p-glycoprotein inhibitor (eg, ciclosporin, azole antifungals, erythromycin) may increase plasma rivaroxaban concentrations and prolong its anticoagulant effects.^{32,33,162} In people, abrupt discontinuation of rivaroxaban can precipitate thrombotic events, so tapering is recommended before discontinuation.^{33,162}

Adverse effects reported in cats receiving rivaroxaban are summarized in Table 1.^{33,162} Spontaneous bleeding has not been reported in published studies involving healthy cats or cats with ATEs receiving rivaroxaban alone.^{32,157,158} However, when rivaroxaban was combined with clopidogrel in cats at risk for ATEs, clinical bleeding was observed in 15%, and included cases of epistaxis, hematuria and gastrointestinal bleeding.¹⁵⁸

Despite limited studies, rivaroxaban shows promise as an anticoagulant option for cats, either as a monotherapy or in combination with clopidogrel. The interest in rivaroxaban in cats at risk for ATE continues to grow, as it provides an oral anticoagulant option that avoids the need for daily injections of low molecular weight heparin and potentially reduces the need for therapeutic drug monitoring.

Capromorelin

Capromorelin is a growth hormone secretagogue (GHS) that stimulates appetite by acting as a selective agonist of the ghrelin receptor (or GHS-receptor 1a) and promoting growth hormone secretion via the hypothalamus and pituitary (Figure 8).¹⁶³ Capromorelin (Elura; Elanco Animal Health) is a veterinary formulated ghrelin receptor agonist for managing weight loss in cats with CKD and is used off-label for appetite stimulation and weight loss management in cats without CKD (Table 2).^{34,164,167} Its formulation as a liquid suspension and small volume given allow for easy administration. Capromorelin-treated healthy laboratory cats gained weight, increased food intake and had elevated concentrations of insulin-like growth factor-1 (IGF-1) compared with placebo-treated cats.^{165,166}

Figure 8

A simplified schematic representation of the role of ghrelin (a growth hormone secretagogue [GHS]) in appetite regulation and the mechanism of action of growth hormone secretagogues (eg, capromorelin). During fasting, the stomach produces ghrelin, which acts via the hypothalamus and peripherally to stimulate appetite. Ghrelin binds the ghrelin receptor (GR or GHS-receptor) on the hypothalamus and pituitary to produce growth hormone.¹⁶³ Indirectly, the hypothalamus triggers growth hormone production through growth hormone-releasing hormone (GHRH) binding the GHRH receptor in the pituitary. Growth hormone prompts the liver to secrete insulin-like growth factor-1 (IGF-1), leading to an increase in muscle mass.¹⁶³ Ghrelin, at least transiently, decreases glucose-stimulated insulin secretion via the GR on the delta cells of the pancreas, resulting in variable increases in blood glucose concentrations.¹⁶⁸ Capromorelin is a GR agonist, selectively targeting the GHS-receptor 1a to augment growth hormone secretion and increase IGF-1 concentrations, collectively improving appetite and promoting weight gain.^{164,165}

Capromorelin is absorbed in the proximal gastrointestinal tract, metabolized by hepatic enzymes and excreted primarily in the feces within 72 h, with minor urinary excretion in dogs.¹⁶⁴ It has not been studied in cats with hepatic disease. In fasted, healthy cats, peak concentrations occur 0.25–1 h after administration, with a half-life of 1.1 h (Table 2).³⁴ When given with food, the time to peak concentrations is delayed and reduced by approximately 50%.³⁴ Reported capromorelin adverse effects are listed in Table 1.^{34,166,168}

Capromorelin offers an effective oral solution for short-term appetite stimulation in anorexic cats. However, capromorelin affects pancreatic delta cells in cats, reducing glucose-stimulated insulin secretion and leading to variable increases in glucose concentrations;¹⁶⁸ in healthy cats, capromorelin causes transient increases in serum glucose and IGF-1 concentrations.^{34,167,168} Capromorelin is therefore contraindicated in cats with acromegaly and should be used cautiously in cats with current or historical diabetes mellitus, due to its impact on glucose metabolism and IGF-1 concentrations.^{34,167,168} Capromorelin should also be avoided in cats at risk for hypotension due to the potential effects low molecular ghrelin receptor agonists have on vascular receptors.¹⁶⁹

Rapamycin

Rapamycin (sirolimus) is a macrolide that appears to have beneficial effects on the progression of feline HCM, and may also offer potential benefits for other chronic conditions such as CKD and neoplasia. Rapamycin inhibits the mechanistic target of rapamycin (mTOR), a serine/threonine protein kinase present in two multiprotein complexes: mTOR complex 1 (mTORC1) and mTOR

complex 2 (mTORC2) (Figure 9). It is generally accepted that the benefits of rapamycin are primarily attributed to its inhibition of mTORC1, while many adverse effects arise from mTORC2 inhibition.^{170,171}

Figure 9

Highly simplified diagram representing the differences between the mechanistic target of rapamycin complex 1 (mTORC1) and mTORC2. Understanding of the intricate physiology associated with mTOR continues to evolve and is described in detail elsewhere.^{170,176} mTORC1 is largely a positive regulator of several anabolic processes, including protein and lipid synthesis, but it is also a negative regulator of autophagy. Less is known about the activity of mTORC2, but it is involved in lipid and glucose metabolism as well as cell survival, proliferation and apoptosis. Along with differences in the functions of these two complexes, there are differences in their sensitivities to rapamycin. Rapamycin acutely inhibits mTORC1, whereas mTORC2 is less sensitive, with inhibition occurring after chronic exposure. The goal of intermittent dosing of rapamycin is to primarily inhibit mTORC1 while minimizing inhibition of mTORC2

The mTOR signaling pathway plays a key role in the multifactorial process of aging. Rapamycin's antiaging effects are believed to result from its modulation of protein homeostasis, endoplasmic reticulum stress, mitochondrial respiration and autophagy, which may offer potential benefits for various chronic diseases (Table 1).¹⁷² A clinical trial is currently underway to investigate its effects on CKD in cats.

Few studies have evaluated rapamycin's effects as an immunosuppressive or anticancer agent in feline medicine. Rapamycin inhibits feline lymphocyte proliferation in vitro; however, its immunosuppressive effects have not yet been studied in vivo.¹⁷³ mTOR and its activated form, phosphor-mTOR, are expressed more frequently in feline triple negative (ie, absence of estrogen and progesterone receptor expression and human epidermal growth factor 2 [HER2] overexpression) as well as HER2-negative feline mammary carcinomas.¹⁷⁴ Additionally, the mTOR pathway is frequently activated with feline squamous cell carcinomas.¹⁷⁵ Given that mTORC1 activity promotes protein synthesis, cell growth and proliferation, this pathway may represent a therapeutic target for a subset of feline mammary carcinomas, squamous cell carcinomas and potentially other neoplasms.

Rapamycin appears to have beneficial effects on the progression of feline HCM. Sirolimus (TriviumVet; rapamycin) delayed-release tablets (felycin-CA1) have recently been granted conditional approval for the management of ventricular hypertrophy in cats with subclinical HCM.³⁵ The diagnosis of subclinical HCM can be made in a cat with a left ventricular wall thickness at end diastole of ≥ 6 mm that does not have systemic hypertension, other causes of compensatory myocardial hypertrophy, current or past clinical signs of congestive heart failure, ATE or severe left ventricle outflow tract obstruction.³⁵ A multiomics study in a colony of research cats with HCM showed that delayed-release rapamycin, given orally once weekly at a low dose (0.15 mg/kg) or high dose (0.3 mg/kg) for 60 days, resulted in dose-dependent suppression of myocardial hypertrophy and increased autophagic activity, and had anti-inflammatory and potential anticoagulant effects.³⁶ The exact mechanism of rapamycin's effects on cardiac remodeling is unknown, but it has an inhibitory effect on left ventricular hypertrophy in cats. In a prospective, randomized, double-masked, placebo-controlled study, delayed-release rapamycin was administered orally once weekly at a low dose (0.25–0.38 mg/kg) or high dose (0.52–0.73 mg/kg) over 180 days to cats with subclinical, non-obstructive HCM.³⁷ Cats receiving the low-dose rapamycin had a significantly lower maximum left ventricular wall thickness compared with those receiving placebo by the end of the study, with no differences in adverse events noted among treatment groups. Further research is needed to determine if rapamycin delays or prevents progression to congestive heart failure and improves survival.

Table 1 summarizes the adverse effects associated with rapamycin.^{170,171} The dosing strategies used in two feline HCM studies seem to minimize adverse effects;^{36,37} however, of the 30 cats treated with rapamycin in one of these studies, one developed hyperglycemia and one developed DKA, and they died after 1 month and 5 months of therapy, respectively.³⁷ The onset of diabetes mellitus in cats treated with rapamycin is of concern. Additionally, while its formulation as a tablet could pose administration challenges for cats, it only needs to be administered once a week. As the potential uses of rapamycin in cats become better understood, its profile and frequency of adverse effects should also become clearer.

Summary

As the landscape of feline medicine continues to evolve, the development of new drugs must be accompanied by proper implementation in clinical practice. It is essential for feline practitioners to stay informed about emerging therapies. Continued use of these pharmaceutical agents will provide a better understanding of their potential long-term benefits and adverse effects. As demonstrated in this review, new insights into established therapies are often being reported. For convenience, the 12 feline therapeutics discussed in this manuscript are summarized in Table 1, including the recommended dosages along with clinical indications and adverse effects. In addition, Table 2 summarizes the clinically relevant available pharmacokinetic data for these feline therapeutics.

Key Points

- ❖ Recent advancements in feline medicine are transforming disease management, making it essential for veterinarians to understand both the clinical application and regulatory considerations of novel therapies.

- ❖ Maintaining current knowledge of veterinary pharmaceuticals is critical for ensuring safe and effective treatment, as ongoing research continues to refine drug indications, dosages and safety profiles.
- ❖ Close clinical monitoring is vital when implementing new therapies, allowing for early detection of adverse effects and ensuring optimal therapeutic outcomes.

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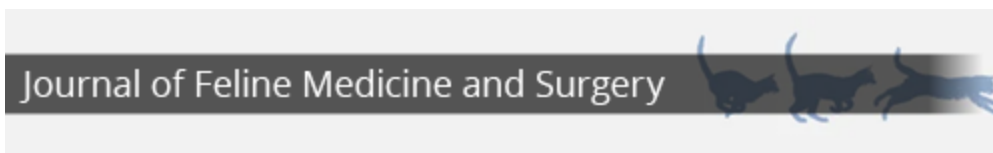
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Anemia of Renal Disease

What it is, what to do and what's new

[Serge Chalhoub](#)¹, [Cathy Langston](#)^{2,✉}, [Adam Eatroff](#)³

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Abstract

Patient group It is estimated that 15–30% of geriatric cats will develop chronic kidney disease (CKD), and that 30–65% of these cats will develop anemia as their renal disease worsens. Anemia of renal disease is multifactorial in its pathogenesis, but the main cause is reduced production of erythropoietin, a renal hormone that controls the bone marrow's production of red blood cells, as kidney disease progresses.

Practical relevance It is important to recognize the presence of anemia of renal disease so that adequate treatment may be instituted to improve quality of life and metabolic function. Erythrocyte-stimulating agents (ESAs), such as epoetin alfa, epoetin beta and darbepoetin alfa, have been developed to counteract the effects of decreased erythropoietin production by the kidneys. These treatments, which are the focus of this review, have 83% similarity in amino acid sequence to the feline hormone. On average, the target packed cell volume (>25%) is reached within 3–4 weeks of ESA therapy.

Clinical challenges The use of ESAs has been associated with a number of complications, such as iron deficiency, hypertension, arthralgia, fever, seizures, polycythemia and pure red cell aplasia (PRCA). Darbepoetin has a prolonged half-life compared with epoetin and thus can be given only once a week, instead of three times a week. The incidence of PRCA appears to be decreased with darbepoetin use when compared with epoetin use in cats.

Evidence base There is limited published evidence to date to underpin the use of ESAs in cats. This review draws on the relevant publications that currently exist, and the authors' personal experience of using these therapies for over 5 years.

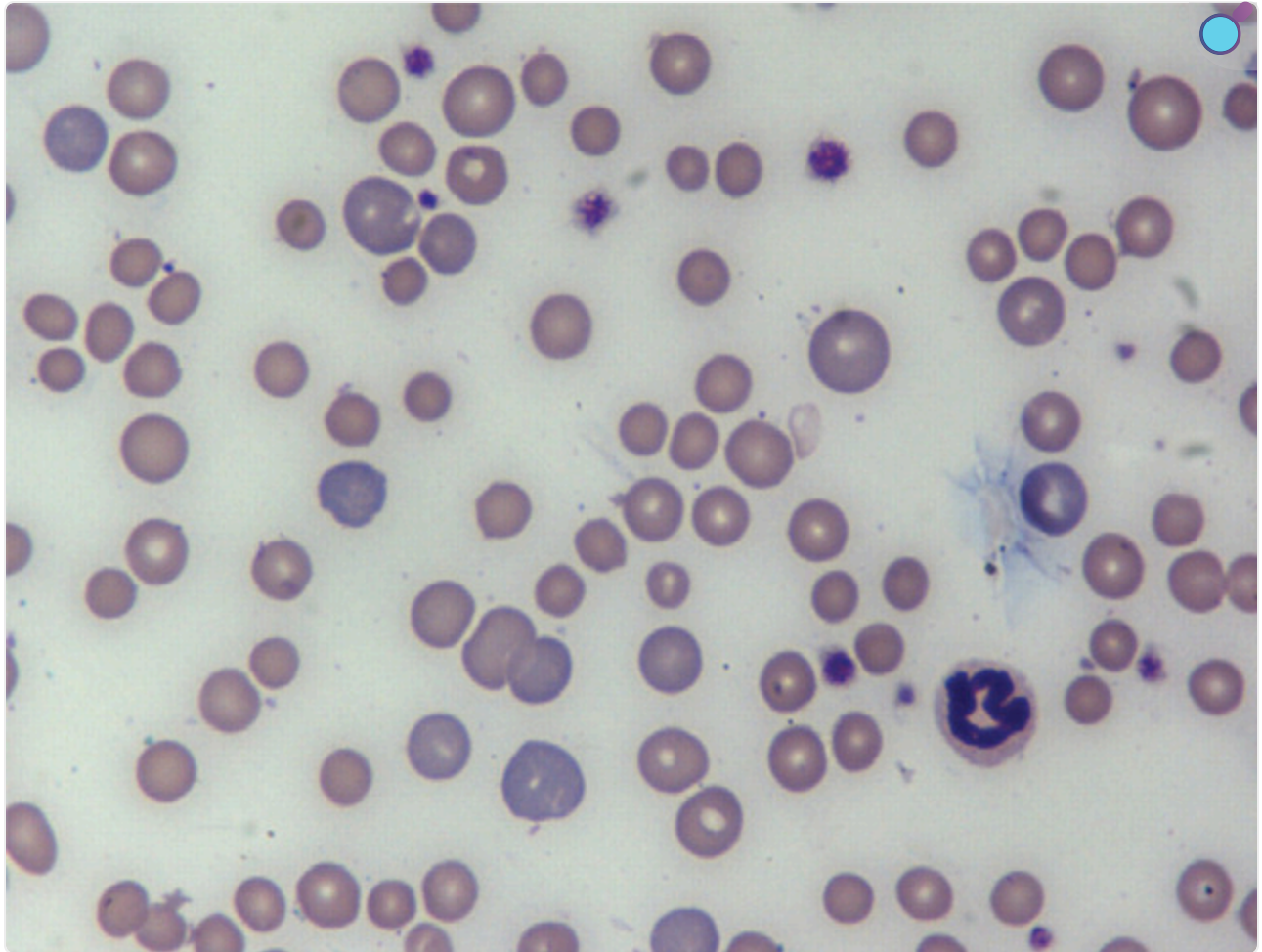


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Anemia — a quality of life issue

Chronic kidney disease (CKD) is a common disorder of domestic cats. Previous studies have indicated that 15–30% of geriatric cats may develop some degree of renal insufficiency or overt azotemia.^{1,2} The kidneys perform multiple important metabolic and endocrine functions in the body, including contributing to erythropoiesis. It is estimated that 30–65% of cats with CKD develop anemia as their renal disease worsens ([Fig 1](#)).^{1,3–5} In addition to lack of appropriate erythropoiesis, uremia from progressing renal disease can decrease red blood cell (RBC) survival.

FIG 1.



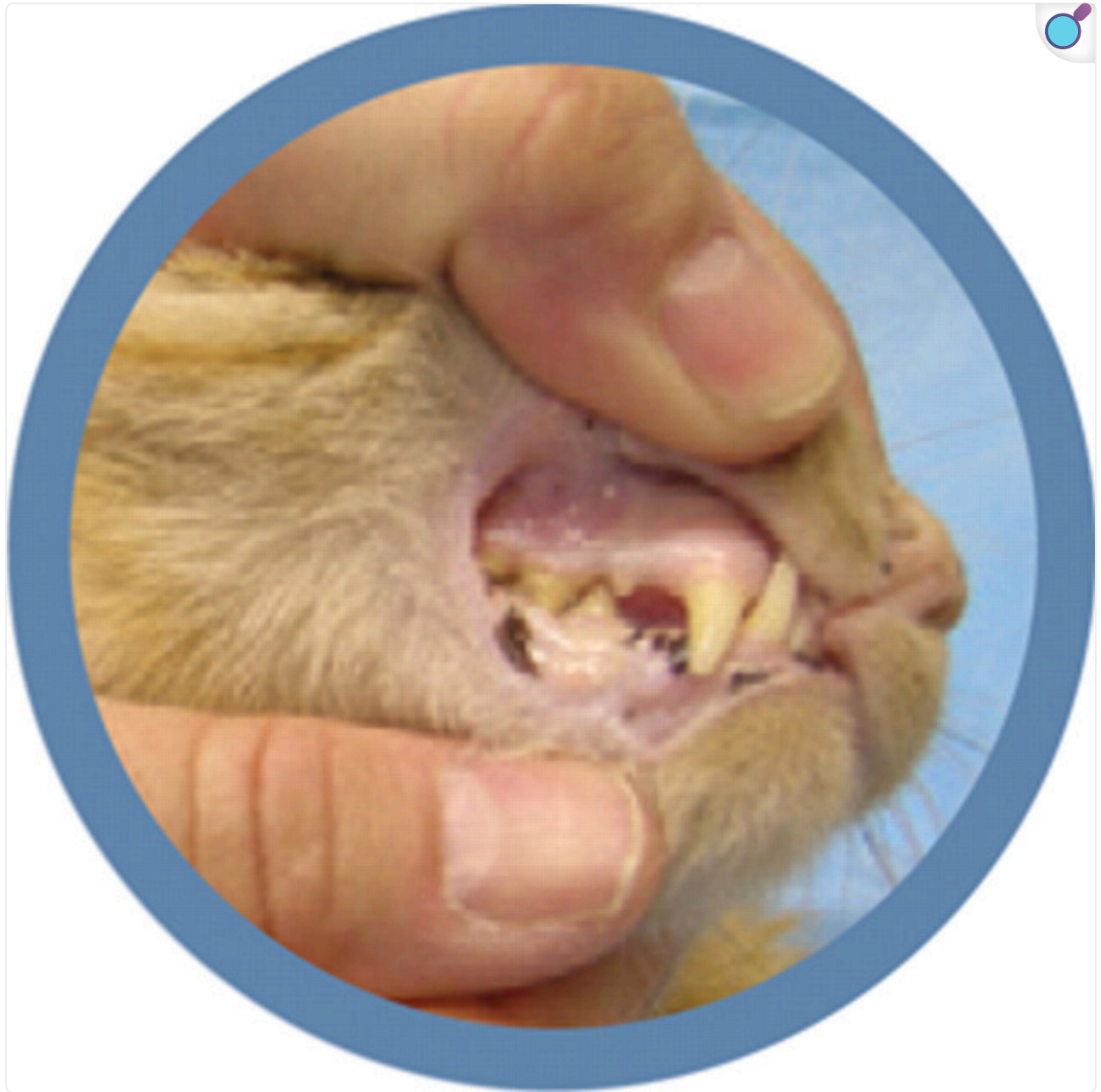
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The anemia of CKD is typically normocytic, normochromic and non-regenerative. In this blood smear from a cat with longstanding anemia of CKD, acute gastrointestinal hemorrhage and iron deficiency, the RBCs demonstrate anisocytosis (variable sizes), polychromasia (younger RBCs have more grayish-blue color) and central pallor. Hypochromia is not commonly seen in cats compared with dogs with iron deficiency anemia. Wright-Giemsa, $\times 100$. *Courtesy of Andrea Siegel, ALX Laboratory*

There is contradictory evidence as to whether anemia represents a predictor of survival in cats with CKD, but certainly chronic anemia affects the body negatively in a number of ways, leading to an overall decrease in quality of life.^{2,6,7} It is, therefore, important to recognize and diagnose

anemia of renal disease and to eliminate other causes of anemia so that appropriate therapy may be implemented.

An estimated 30–65% of cats with CKD develop anemia as their renal disease worsens.



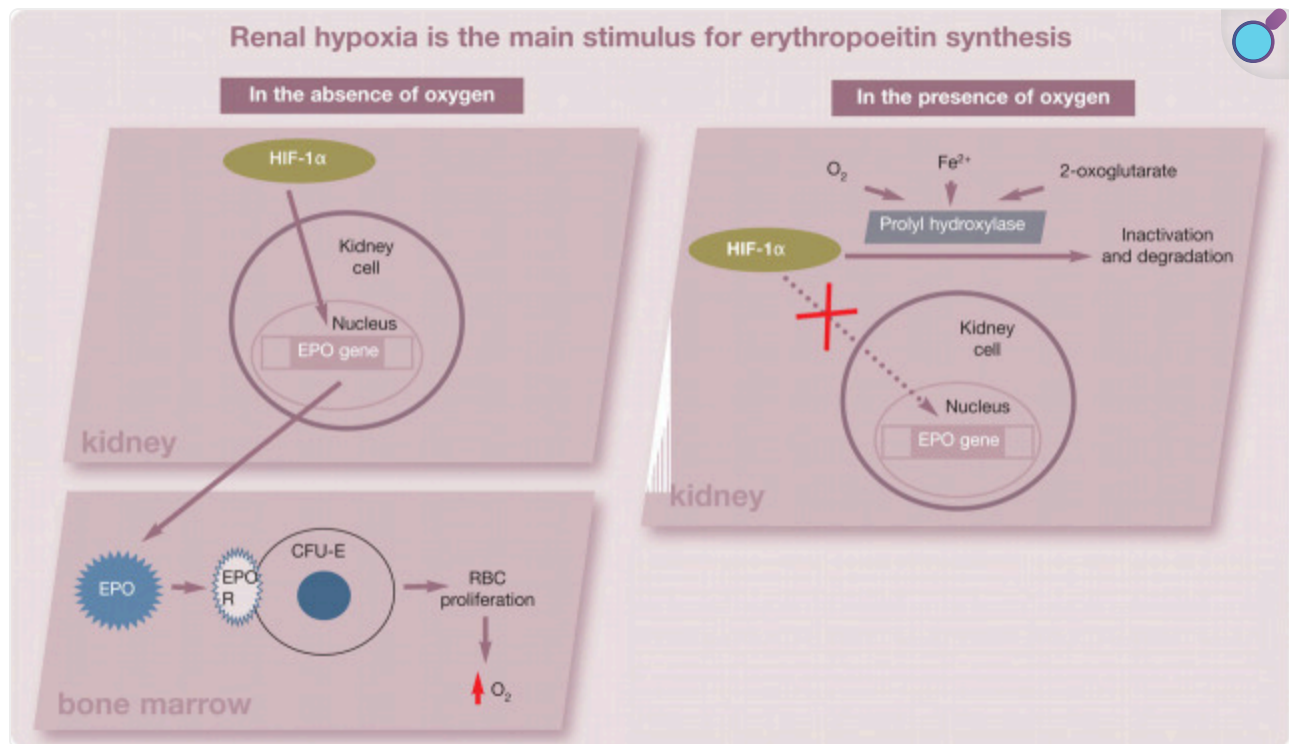
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Erythropoietin's role in erythropoiesis

Erythropoiesis is controlled by the production of the hematopoietic growth factor erythropoietin in response to anemia.^{8,9} Erythropoietin is mainly produced in the peritubular interstitial cells of the inner renal cortex and outer medulla in the kidney. The liver, brain, uterus, peripheral endothelial cells, muscles and insulin-producing cells also produce erythropoietin, but to a much lesser degree.⁸⁻¹¹ As kidney disease progresses, there are fewer erythropoietin-producing cells within the kidneys.^{1,9,11}

The main stimulus for erythropoietin synthesis is renal hypoxia. When hypoxia is present, degradation of hypoxia-inducible factor 1 (HIF-1 α) is inhibited, and HIF-1 α is free to bind to hypoxia-response elements of oxygen-regulated genes ([Fig 1](#)). These response elements control the erythropoietin gene within the kidneys, and binding of HIF-1 α stimulates an increase in production of erythropoietin.¹²⁻¹⁴ The rate of production of erythropoietin is inversely proportional to the oxygen-carrying capacity of blood.⁹ The main site of erythropoietin action is the bone marrow, where it binds to its receptor expressed on the surface of erythroid progenitor cells and leads to increased erythropoiesis ([Fig 2](#)).¹¹ Erythropoietin has the greatest influence on colony-forming unit-erythron (CFU-E) cells; these cells contain the largest number of erythropoietin receptors.¹⁵

FIG 1.

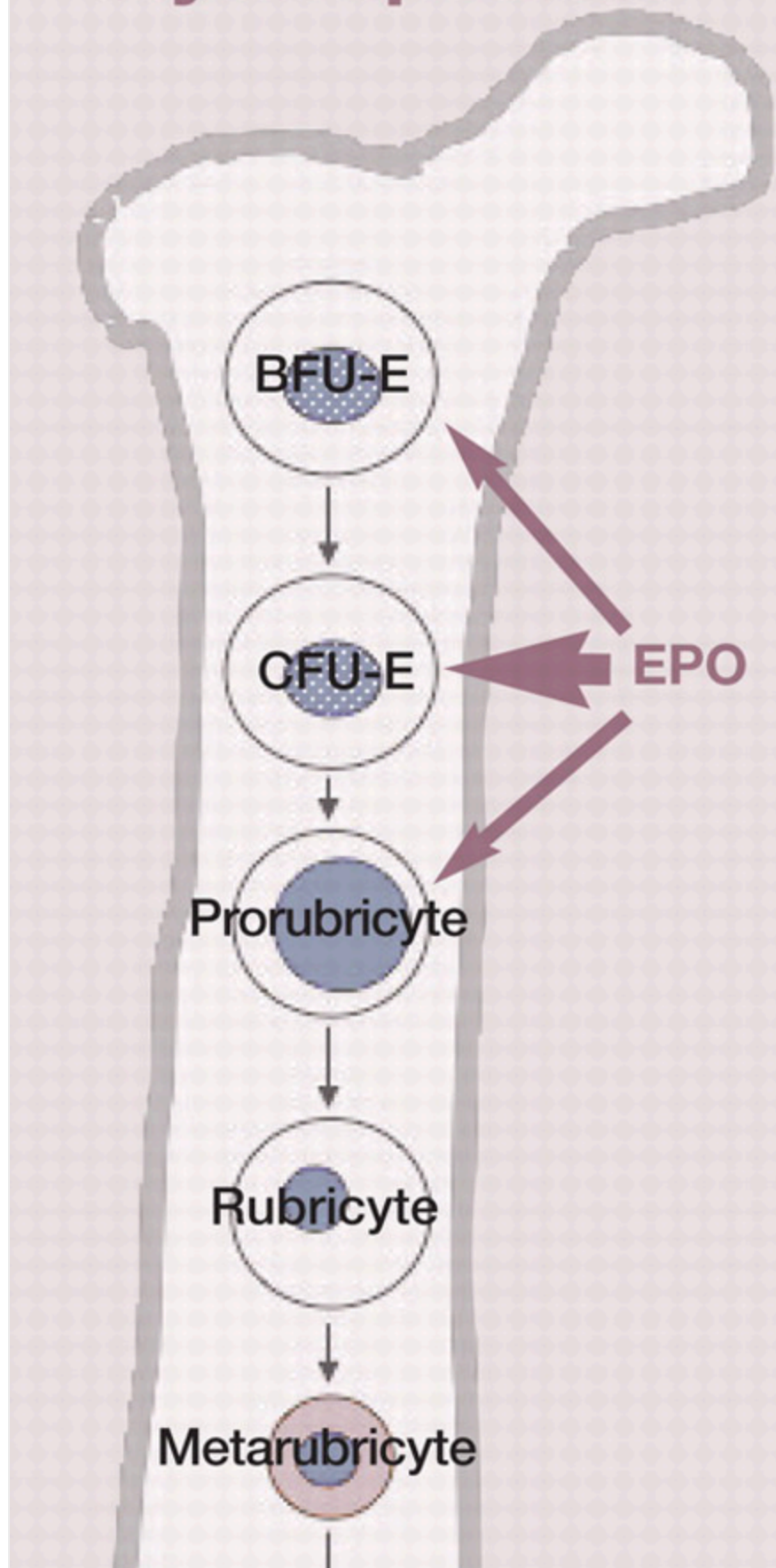


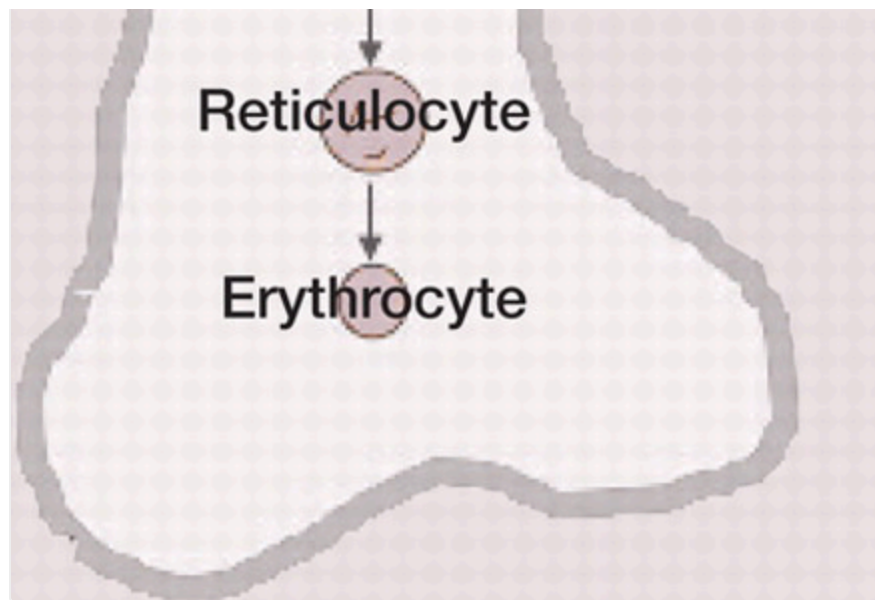
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In the absence of oxygen, HIF-1 α increases transcription of the erythropoietin (EPO) gene, creating more EPO. EPO binds to its receptor on colony-forming unit-erythron (CFU-E) cells in the bone marrow and prevents apoptosis, enhancing proliferation and maturation to RBCs, which increase oxygen-carrying capacity. When oxygen is present, prolyl hydroxylase is stabilized and increases degradation of HIF-1 α , preventing EPO production. HIF-1 = hypoxia-inducible factor 1, EPO R = erythropoietin receptor

FIG 2.

Stages of erythropoiesis





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Colony-forming unit-erythron (CFU-E) cells have the highest number of erythropoietin (EPO) receptors. Blast-forming unit-erythron (BFU-E) cells and prorubricytes are also capable of responding to EPO

Anemia and its consequences

Anemia is defined as a state of deficient mass of circulating RBCs and hemoglobin, which results in reduced oxygen delivery to all organs and a subsequent decline in cell metabolism. ^{16,17} Anemia triggers numerous adaptive response mechanisms due to tissue hypoxia, some of which may be detrimental in the long term. Adaptive mechanisms include increased release of plasma norepinephrine, renin, angiotensin II and aldosterone, all of which can lead to increased sympathetic nervous system response, increased heart contractility and rate, and increased blood pressure. ¹⁸

As anemia becomes more severe, cats may also develop left heart enlargement secondarily to hemodynamic compensation, and become more prone to congestive heart failure, especially if receiving intravenous fluid therapy. ¹⁹

Pathogenesis of anemia of renal disease

Anemia of renal disease is multifactorial in its pathogenesis (see top box, page 631). The lack of erythropoietin is the main cause of anemia in CKD patients. ²⁰ Additional factors contributing to

anemia of renal disease include anemia of inflammation, other causes of suppressed erythropoiesis, the influence of uremic toxins on RBC survival, and gastrointestinal hemorrhage or other sources of blood loss.

Causes of anemia in CKD.

Decreased erythropoiesis

- Lack of erythropoietin
- Inflammatory cytokines
- Absolute iron deficiency
- Functional iron deficiency
- Uremic toxins
- ACE inhibitors, angiotensin receptor antagonists
- Hyperparathyroidism
- Marrow fibrosis/infiltration

Shortened RBC survival

- Uremic toxins
- Hemolysis
- Premature removal by reticuloendothelial cells

Increased RBC loss

- Thrombocytopathy
- Gastrointestinal ulcers
- Blood sampling

Decreased erythropoiesis

Anemia of inflammation (both acute and chronic) contributes to anemia of renal disease, and CKD itself is known to be associated with a chronic pro-inflammatory state.^{[21](#)} Cytokines such as

interferon- α , - β and - γ , as well as tumor necrosis factor- α , interleukin-1 and interleukin-6, are produced in inflammatory states and create a relative iron deficiency by induction of hepcidin, an acute phase protein. [21,22](#)

Control of hepcidin production.

Production is stimulated by:

Iron presence

Inflammation

Chronic kidney disease

Production is suppressed by:

Iron deficiency

Erythropoietic activity

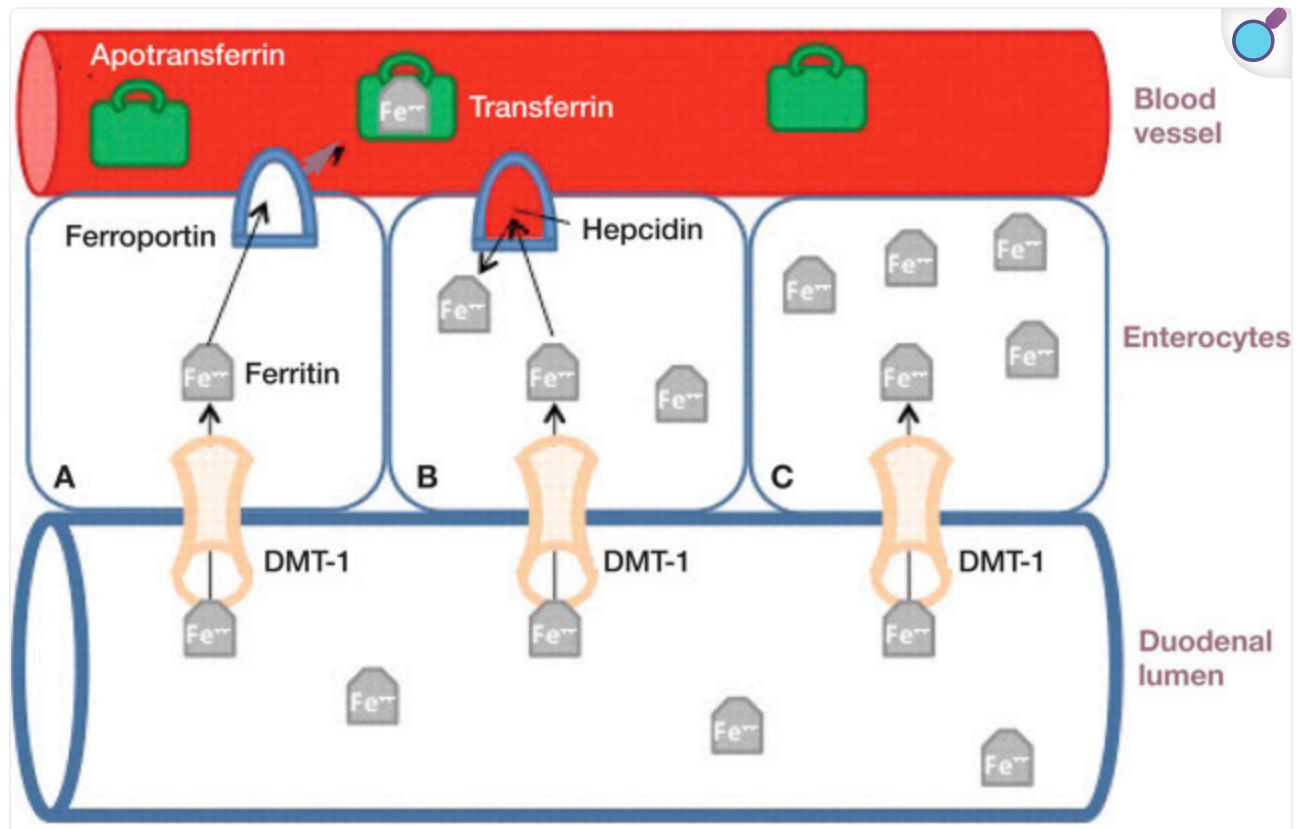
Hepcidin is considered the central regulator of systemic iron homeostasis.

Hepcidin is produced by the liver in response to inflammation and iron loading, and is suppressed by normal erythropoietic activity and iron deficiency. Hepcidin is considered to be the central regulator of systemic iron homeostasis and the primary inflammatory mediator causing its increased production is interleukin-6. [22,23](#)

Hepcidin's biological actions are mediated by its binding to and internalization of ferroportin, which is the principal cellular efflux channel for iron. Orally ingested iron is absorbed into the duodenal enterocytes via the apical protein DMT-1, and enters the circulation via ferroportin on the basolateral side ([Fig 3](#)). Hepcidin traps iron in the enterocytes by enhancing degradation of ferroportin. Hepcidin prevents release of iron stored in macrophages and hepatocytes as well. [23,24](#) The decrease in enteral absorption and the sequestration of iron in macrophages eventually leads

to anemia by decreasing the availability of iron for hemoglobin production. Hepcidin production is suppressed in the face of active erythropoiesis, which enhances availability of iron for hemoglobin production, but the mechanism by which this happens is poorly understood.²⁴ Hepcidin concentrations are increased in human CKD patients, even in those without significant inflammation, in part due to decreased renal clearance.^{22,24}

FIG 3.



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Ingested iron is absorbed from the duodenal lumen into the enterocyte through the divalent metal transporter (DMT-1) on the luminal side. In the absence of hepcidin, iron leaves the enterocyte on the basolateral side via ferroportin (cell A). Apotransferrin bound to iron is called transferrin, and is the main method of transporting iron in the circulation. Hepcidin binds to ferroportin, causing rapid internalization and degradation of ferroportin (cell B). Without ferroportin, iron efflux from the cell is prevented (cell C)

In addition to induction of hepcidin production, the aforementioned inflammatory cytokines can negatively affect RBC survival and directly decrease responsiveness to erythropoietin, all contributing to worsening anemia. [21,25](#)

Several uremic inhibitors of erythropoiesis have also been identified. However, their relative contribution to anemia is controversial. It is thought that each of these substances affects erythropoiesis at various stages within the bone marrow. [26,27](#) Secondary hyperparathyroidism as a consequence of CKD can cause erythropoietin hyporesponsiveness. Parathyroid hormone in excess may have a direct toxic effect on erythroid progenitor cells, and also an indirect effect by inducing bone marrow fibrosis. [28](#) Deficiency of B vitamins may also impair erythropoiesis.

Common therapies used for CKD may impair RBC production. Angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers are suspected to decrease angiotensin II-induced release of erythropoietin, as well as prevent pluripotent hemopoietic stem cell recruitment. [28,29](#) Aluminum toxicity, from aluminum containing phosphate binders, can interfere with iron metabolism, leading to a microcytic anemia that is not responsive to iron administration. [30-33](#)

Lack of erythropoietin is the main cause of anemia in CKD patients.

Shortened RBC survival

Uremia has been associated with decreased RBC survival, but the pathophysiology is unclear and most likely multifactorial. [34](#) There is suspicion of a uremic toxin circulating and affecting RBC lifespan. In part, the decreased survival is due to increased low-grade hemolysis of RBCs as uremia progresses. [35,36](#) In addition, there is evidence that lipid peroxidation of red cell membranes may occur as uremia progresses, resulting in premature clearance by the reticuloendothelial system, which is activated in the inflammatory state. [37](#)

Increased RBC loss

Blood loss and blood sampling contribute to the anemia of CKD. Uremia affects platelets and can lead to uremic thrombocytopathy. This may be due to retention of small substances normally eliminated by functioning kidneys that affect platelet function, vascular damage caused by uremic toxins, changes in platelet responsiveness and function, changes in platelet-endothelial interactions, increased levels of nitric oxide (a potent platelet antagonist) in uremic states, and dysregulation of coagulation factors responsible for normal platelet function. [38-40](#) Manifestations

of platelet dysfunction include gastrointestinal bleeding, bleeding from surgical sites and mucocutaneous bleeding.⁴¹

Significant blood loss can occur through the gastrointestinal tract in people with CKD in the absence of identifiable discrete bleeding ulcers.^{42,43} Studies of the prevalence of upper gastrointestinal ulceration in humans with CKD yield conflicting results, although recent evidence suggests that CKD patients do not have a higher prevalence of ulceration compared with patients with normal kidney function.⁴³⁻⁴⁶ Although gastrin, a hormone responsible for stimulating hydrochloric acid secretion in the stomach, is cleared by the kidneys, and gastrin concentration in the blood is elevated in people and cats with CKD, hyperacidity and gastric ulceration are variably present.⁴⁷⁻⁴⁹ Gastric pathology is common in dogs with kidney disease, but ulceration is uncommon;⁵⁰ similar studies in cats are lacking. Acute upper gastrointestinal bleeding may manifest as an acute onset of anemia, hypoproteinemia, hypovolemia or hypotension.

Therapy with erythrocyte-stimulating agents (ESAs)

Human erythropoietin is a 30,400 Dalton glycosylated protein that contains a 165 amino acid residue backbone and has a half-life of approximately 6–10 h.⁵¹ The gene sequence for erythropoietin was discovered in 1983, which allowed for the production of recombinant erythropoietin in 1985.⁵² Recombinant human erythropoietin (rHuEPO) products have an identical amino acid sequence to the natural hormone.^{15,53-55} Multiple recombinant erythropoietin products are available, including epoetin alfa (Epogen, Amgen; Procrit, Centocor Ortho Biotech Products; Eprex, Janssen), epoetin beta (Neo-Recormon, Roche), darbepoetin alfa (Aranesp, Amgen) and continuous erythropoietin receptor activators (Mircera, Roche). These vary in their degree of glycosylation, but there is no difference in their clinical efficacy.⁵⁶ The amount of glycosylation affects renal clearance, thus influencing the frequency of administration.⁵⁷

Canine erythropoietin shares an 81.3% homology with the amino acid sequence of human erythropoietin, while homology in cats is 83.3%.^{58,59} The relative conservation of amino acid sequence allows for rHuEPO products to bind to and interact with erythropoietin receptors of dogs and cats.

Epoetin

Epoetin was the first ESA used in client-owned pets with naturally occurring CKD. The usual starting dose of 100 IU/kg administered subcutaneously three times a week is recommended until

the packed cell volume (PCV) reaches the low end of the target range (see guidelines, page 633).⁶⁰ The authors recommend a target PCV of $\geq 25\%$ in cats. A response is usually seen within 3–4 weeks. Once the target range is attained, maintenance doses of 50–100 IU/kg twice weekly are usually sufficient. Supplementation of iron at the initiation of and during epoetin therapy is also recommended to ensure an adequate response to the ESA.

Darbepoetin

Darbepoetin, a hyperglycosylated recombinant human erythropoietin analogue, was developed in the early 1990s, and differs from epoetin in that it contains five N-linked carbohydrate chains.^{61,62} The development of the molecule was based on the hypothesis that adding an N-linked carbohydrate chain — thus increasing the amount of sialic acid residues — would result in a molecule with a longer circulating half-life.^{60,63} By increasing the half-life, darbepoetin could be administered less often. Although data is not available in cats, pre-clinical studies of darbepoetin in dogs demonstrated a threefold increase in half-life when compared with epoetin (25.0 vs 7.2 h), and a correspondingly reduced mean clearance rate (2.4 ml/kg/h vs 8.4 ml/kg/h).⁶² Extensive human studies evaluating the use of darbepoetin to correct anemia of renal disease in CKD patients have demonstrated clinical efficacy. The proportion of patients achieving a satisfactory hemoglobin concentration has been similar to those patients receiving epoetin.^{62,64–67}

The authors have been using darbepoetin alfa as the sole ESA therapy for anemia of renal disease for many years with good success.

A darbepoetin dose of 0.45 $\mu\text{g}/\text{kg}$ once weekly administered subcutaneously has been demonstrated to provide optimal responses in 60–70% of human patients over a 4-week period;^{62,68} an optimal response in people is defined as a hemoglobin increase of 1–3 g/dl over the first 4 weeks. There are no published effective dosages for companion animals. Based on the authors' database, 12% of cats responded adequately to a starting dose of 0.45 $\mu\text{g}/\text{kg}$ once weekly, whereas 62% responded at a dose of 1.0 $\mu\text{g}/\text{kg}$ once weekly. Thus, we recommend a starting dose of 1.0 $\mu\text{g}/\text{kg}$ once weekly until the target PCV is attained, thereafter decreasing the frequency of administration to every 2–3 weeks. On average, a response is expected within 2–3 weeks. The dose is then adjusted to maintain the PCV within the target range (25–35%).

The authors have been using darbepoetin as the sole ESA therapy for anemia of renal disease for many years with a good success rate. Feline recombinant erythropoietin has also been developed, but studies have demonstrated that the incidence of pure red cell aplasia (PRCA), a complication

associated with the production of neutralizing antibodies, was not significantly reduced when compared with the incidence in cats administered epoetin. [69](#)

We recommend monitoring the PCV weekly ([Fig 4](#)) until both the PCV and dosing regimen are stable. In addition, a reticulocyte count should be evaluated weekly to ensure adequate bone marrow stimulation until the maintenance phase of therapy is reached, and thereafter monthly. However, it has been our experience that reticulocyte counts of cats receiving darbepoetin often do not reflect the effectiveness of therapy (ie, the PCV reaches target, but the reticulocyte count may not increase). We speculate that as darbepoetin is administered once weekly, the rise in reticulocyte count occurring in the few days after darbepoetin administration may be missed when evaluated 7 days after administration. Blood pressure should also be assessed at each examination ([Fig 5](#)) as hypertension is one of the most common side effects of ESA therapy.

FIG 4.



The PCV should be monitored weekly at the start of ESA therapy. Frequency of monitoring can be decreased once the PCV is stable and ESA dose adjustments are no longer needed

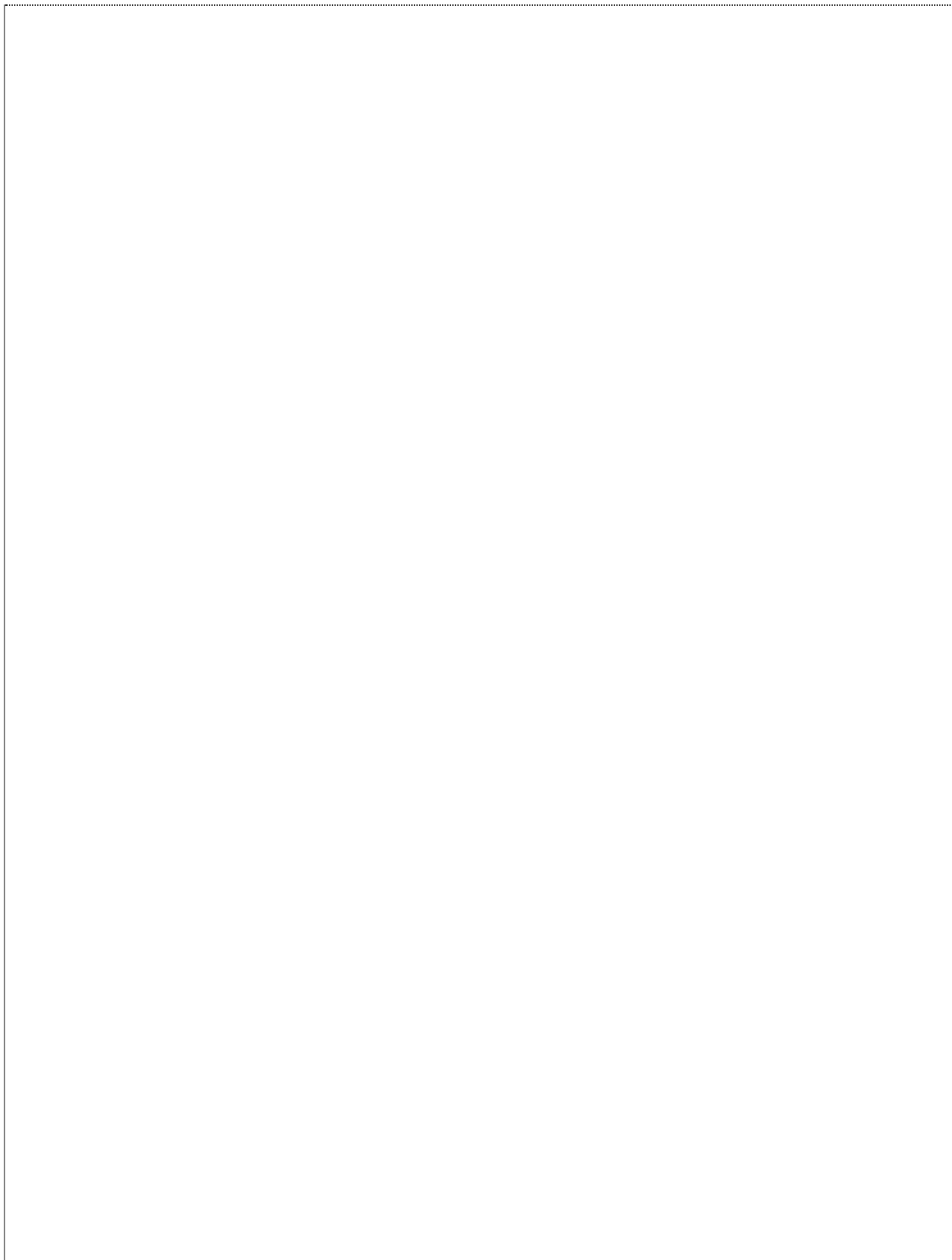
FIG 5.



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Because hypertension is a reported side effect of ESA therapy, blood pressure should be monitored at each visit, especially during the initial phases of treatment when the PCV is rising

Guidelines for using ESAs.



Should you use an ESA?

Is the anemia unlikely to resolve on its own?

and

Is the cat symptomatic for anemia?

Which ESA?

Darbepoetin
(less risk of PRCA)

or

Epoetin
(slightly less expensive)

Starting dose

Darbepoetin
1 µg/kg SC once a week

or

Epoetin
100 IU/kg SC three times a week

Iron supplementation

Iron dextran (preferred)
50 mg/cat IM monthly

or

Oral iron
10–20 mg/cat elemental iron q24h

Initial monitoring

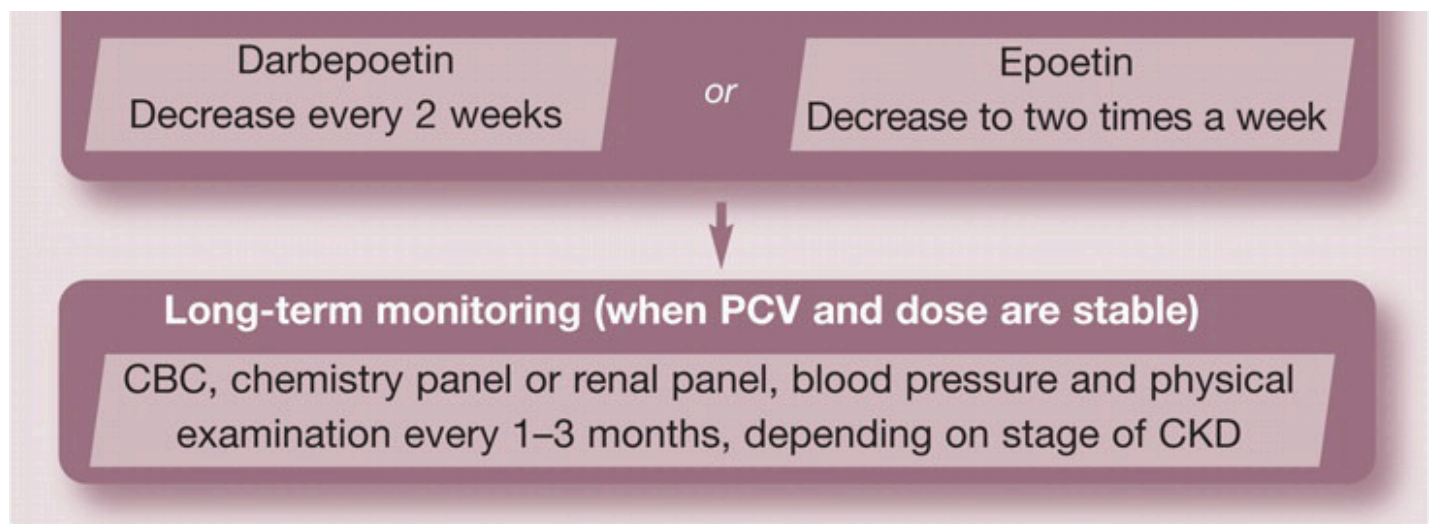
Weekly PCV, reticulocyte count, blood pressure and physical examination until PCV and dose are stable

Target

Ideal PCV range: 25–35%, with 1–3% increase per week.
A faster rise in PCV may increase risk of hypertension

Dose adjustment

Decrease dose by 20–25% when target PCV is reached



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Iron supplementation

Lack of available iron for erythropoiesis is recognized as the leading cause of treatment failure in humans on ESA therapy.^{70–72} Iron is necessary for hemoglobin formation and RBC function, and should be administered to all cats receiving ESAs. Cats with CKD that responded to treatment with feline erythropoietin had a decrease in mean cell volume serum iron concentration and percentage transferrin saturation despite receiving oral iron supplementation.⁶⁹ The decrease in circulating iron parameters during ESA therapy may represent utilization of iron for RBC production in excess of intake or in excess of available iron stores, inadequate intestinal iron absorption despite oral iron supplementation, or functional iron deficiency due to sequestration of iron in storage sites such as the bone marrow with an inability to utilize that iron due to alterations in hepcidin metabolism.^{20,69}

There is a state of relative iron deficiency in many cats with CKD during ESA therapy. Evaluation of serum iron parameters is recommended at the start of ESA therapy and monthly thereafter. Cats with true iron deficiency would be expected to have low serum iron, ferritin and transferrin saturation. In cases of chronic inflammation (including CKD patients), ferritin and total iron binding capacity may be normal or above the reference interval, whereas serum iron and transferrin saturation should be below the reference interval (see below).

True iron deficiency or chronic inflammation?

Iron deficiency

- Low serum iron
- Low ferritin
- Low total iron binding capacity
- Low % saturation

Chronic inflammation

- Low serum iron
- High ferritin
- Normal-high total iron binding capacity
- Low % saturation

Iron should be administered to all cats receiving ESA therapy.

Oral iron supplements, such as ferrous sulfate compounds, tend to be poorly absorbed in the gastrointestinal system, and the bitter taste may cause some cats to reject them. The recommended dosage is 50–100 mg/cat per day administered orally (10–20 mg of elemental iron). A popular oral liquid multivitamin (Pet-Tinic; Pfizer) contains 12.5 mg of elemental iron per 5 ml. Tablet formulations of iron typically contain anywhere from 35–100 mg of elemental iron. Iron dextran given by intramuscular injection every 3–4 weeks along with an ESA may be a better alternative to oral iron supplementation. The dose of iron dextran is typically 50 mg/cat. Intramuscular injection can be painful, but iron dextran should not be administered intravenously as there is a risk of anaphylaxis.

Iron therapy in patients with chronic inflammation is considered controversial. Iron is an essential nutrient for microorganisms and its sequestration may be a defense mechanism against infection, and iron promotes formation of toxic hydroxyl radicals. [25,73](#)

Other treatments

Anabolic steroids

The use of nandrolone and stanzolol has been reported in the human literature for erythrocyte stimulation, but their efficacy is doubtful and hepatic side effects are common.^{74–78} These products are *not* recommended for treating anemia of CKD in cats.

Transfusion therapy

Transfusion of whole blood or packed RBC preparations is indicated when there is acute blood loss or when a patient has severe clinical signs of anemia ([Fig 6](#)). Disadvantages of blood transfusions include the possibility of immune reactions and donor—recipient incompatibility, the limited availability of blood products, the reduced lifespan of infused blood products in a uremic patient, the associated costs of blood product transfusion, and the lack of long-term effectiveness of these products.⁵

FIG 6.



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Transfusion of whole blood or packed RBCs is the most rapid way to correct anemia of CKD and may be indicated if clinical signs of anemia need urgent treatment or if aggressive intravenous fluid therapy is anticipated and congestive heart failure is a concern

Oxyglobin

Oxyglobin is a hemoglobin-based oxygen carrier that is a sterile solution of purified, polymerized bovine hemoglobin and a modified lactated Ringer's buffer. It has a lower viscosity than blood and has enhanced oxygen offloading characteristics.⁷⁹ Its main indications are for temporary oxygen-carrying capacity in anemia caused by hemorrhage or inadequate erythropoiesis. Effects are short-lived, cost is a limiting factor, and circulatory overload is a concern in cats with concurrent cardiac disease.⁸⁰

Complications of ESA therapy.

The use of ESAs can create a number of complications, such as iron deficiency, hypertension, arthralgia, fever, seizures, polycythemia and PRCA. In humans, it is known that darbepoetin has a similar adverse event profile to that of epoetin.⁶² Hypertension is reported in 23% of those administered darbepoetin, with seizures in 2%, and cerebrovascular disorders and PRCA in less than 1% each.⁸⁵

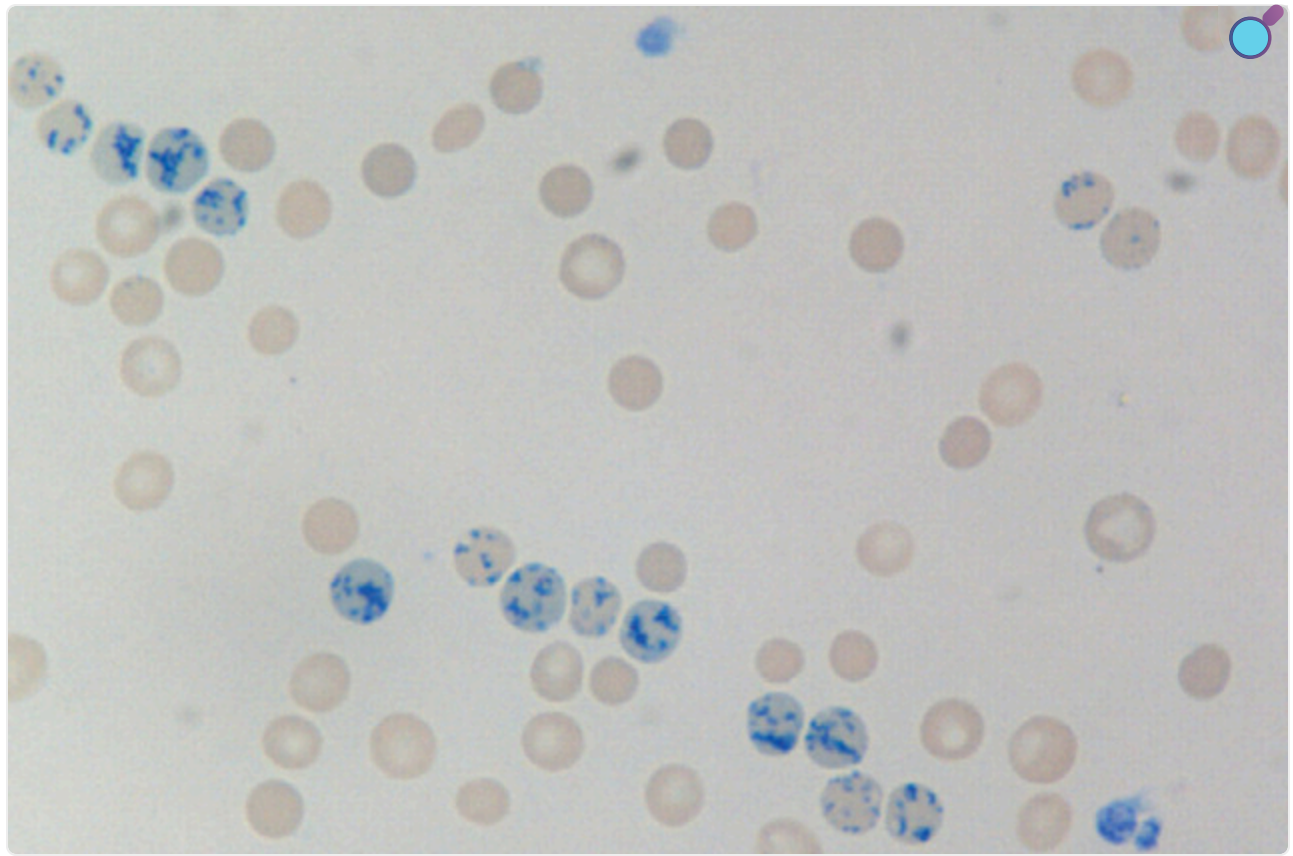
Hypertension

Hypertension is a side effect of epoetin administration in 40–50% of dogs and cats.^{57,86} Based on the authors' experience, similar percentages are seen with darbepoetin therapy. The pathophysiology of hypertension with ESA therapy includes an increase in blood viscosity, improved cardiac output, reduced vasodilation as compensation for anemia, and imbalance in vasoactive hormones with increased vascular responsiveness.⁷³ Uncontrolled hypertension may contribute to the development of seizures in cats treated with darbepoetin.

Pure red cell aplasia

PRCA is caused by the production of neutralizing anti-erythropoietin antibodies that cross-react with all ESAs, as well as with endogenous erythropoietin.^{87–89} It is characterized by a severe, non-regenerative anemia with an almost complete lack of bone marrow RBC precursors. The antibodies can persist for over 8 months, and patients with PRCA become transfusion dependent.⁵ The cytological definition of PRCA varies significantly in the veterinary literature.

The use of ESAs can create a number of complications, such as iron deficiency, hypertension, arthralgia, fever, seizures, polycythemia and PRCA.



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New methylene blue stained blood smear from a cat receiving darbepoetin showing multiple reticulocytes (RBCs with speckled blue stain uptake). Detection of reticulocytosis may be affected by sample timing. A decrease in reticulocyte count in conjunction with a decrease in PCV may portend development of PRCA. $\times 100$. Courtesy of Andrea Siegel, ALX Laboratory

Myeloid to erythroid (M:E) ratios of 4:1 to 299:1 have been reported in dogs and cats with PRCA. [20,57,86](#) Antibody-mediated PRCA is an extremely rare occurrence in humans, with fewer than 300 cases reported worldwide, the majority of which have been associated with use of epoetin. [89](#) In contrast, PRCA is an important concern in companion animals using human-based ESAs. As mentioned, the amino acid sequence of canine and feline erythropoietin shares approximately 83% homology with rHuEPO. Anti-erythropoietin antibodies are directed against the protein backbone, and this lack of complete homology is thought to be the cause of the increased rate of PRCA in animals compared with humans. [90](#)

PRCA seems to occur at an incidence of 25–30% in dogs and cats receiving epoetin. [86,91](#) In the experience of the authors, the occurrence of PRCA seems to be much lower with darbepoetin compared with epoetin, with a frequency of less than 10%.

PRCA is difficult to treat and requires immediate discontinuation of ESA administration, as well as possible institution of immunosuppressive therapy. There is no established best immunosuppressive protocol in humans or animals. [92,93](#) With repeated transfusions, identifying cross-match compatible blood becomes more difficult and, even with transfusion of cross-matched blood, the rate of RBC destruction seems to accelerate, and many cats need weekly transfusions. Because of quality of life concerns, as well as financial implications, the development of PRCA leads many owners to consider euthanasia.

This product is no longer available in the United States.

Vitamins and L-carnitine

Commercially prepared kidney diets contain adequate vitamin supplementation, making vitamin deficiency unlikely in a patient consuming an amount adequate to maintain body weight. However, partial anorexia is common in cats with CKD. Several B vitamins are necessary for erythropoiesis, including cobalamin (B₁₂), folic acid (B₉), niacin (B₃) and pyridoxine (B₆). Deficiencies of these vitamins may exacerbate anemia and increase erythropoietin resistance. Although there is a paucity of data on whether CKD patients commonly develop deficiencies of these vitamins, supplementation of these water soluble vitamins is unlikely to cause harm, but they are not effective as sole therapies for anemia. [81,82](#)

L-carnitine is believed to increase CFU-E cells (see Fig 2) and stimulate erythropoiesis. Although L-carnitine deficiency is common in people on long-term hemodialysis, and there is a modest benefit of L-carnitine supplementation in these patients if they are erythropoietin-resistant, there is little evidence that pre-dialysis patients are L-carnitine deficient. [83,84](#) There exists no data describing L-carnitine status in cats with CKD.

Treatment failure with ESAs

Failure to reach the target PCV during ESA therapy can be due to a number of factors (Table 1). Over 90% of people receiving ESA therapy respond adequately. [62](#) However, based on the authors'

experience of treating over 70 cats with epoetin or darbepoetin, only about 60–65% had an adequate response. In humans, a decreased response to ESA therapy is defined as a significant decrease in hemoglobin level at a constant ESA dose, a significant increase in ESA dose to preserve a certain hemoglobin level, or a failure to raise the hemoglobin level to greater than 11 g/dl (110 g/l) despite an ESA dose equivalent to greater than 500 IU/kg a week of epoetin or 1.5 µg/kg a week of darbepoetin.²⁹

The approach to a cat with an inadequate response to ESA therapy involves evaluation of multiple parameters (see box on the right). The order of evaluation may vary based on the clinical picture, and several parameters may be evaluated simultaneously. Iron deficiency is present in 25–38% of anemic people with CKD and is the leading cause of failure to achieve target hematocrit in people on ESA therapy^{28,29} Changes in the complete blood count (low mean cell volume, low mean cell hemoglobin content) are not sensitive indicators of iron deficiency.⁹⁴ Serum iron parameters (total iron, total iron binding capacity, ferritin) can be difficult to interpret in the face of concurrent infection, inflammation or functional iron deficiency.

Infection and inflammation are common disorders in people with CKD and can have a dramatic impact on erythropoiesis and erythropoietin resistance. As discussed earlier, inflammatory cytokines directly suppress erythropoiesis and decrease iron availability by upregulation of hepcidin production. A thorough search for infectious complications (ie, urinary tract infection, dental infection, feeding tube exit site infection) and aggressive treatment may restore responsiveness to ESA.

TABLE 1.

Causes of inadequate ESA response described in people

Iron deficiency (absolute or functional)

Infection/inflammation

Chronic blood loss

Osteitis fibrosis

Aluminum toxicity

Hemoglobinopathy (sickle cell anemia, β -thalassemia)

Folate or vitamin B₁₂ deficiency

Multiple myeloma

Malnutrition

Hemolysis

Dialysis-related carnitine deficiency

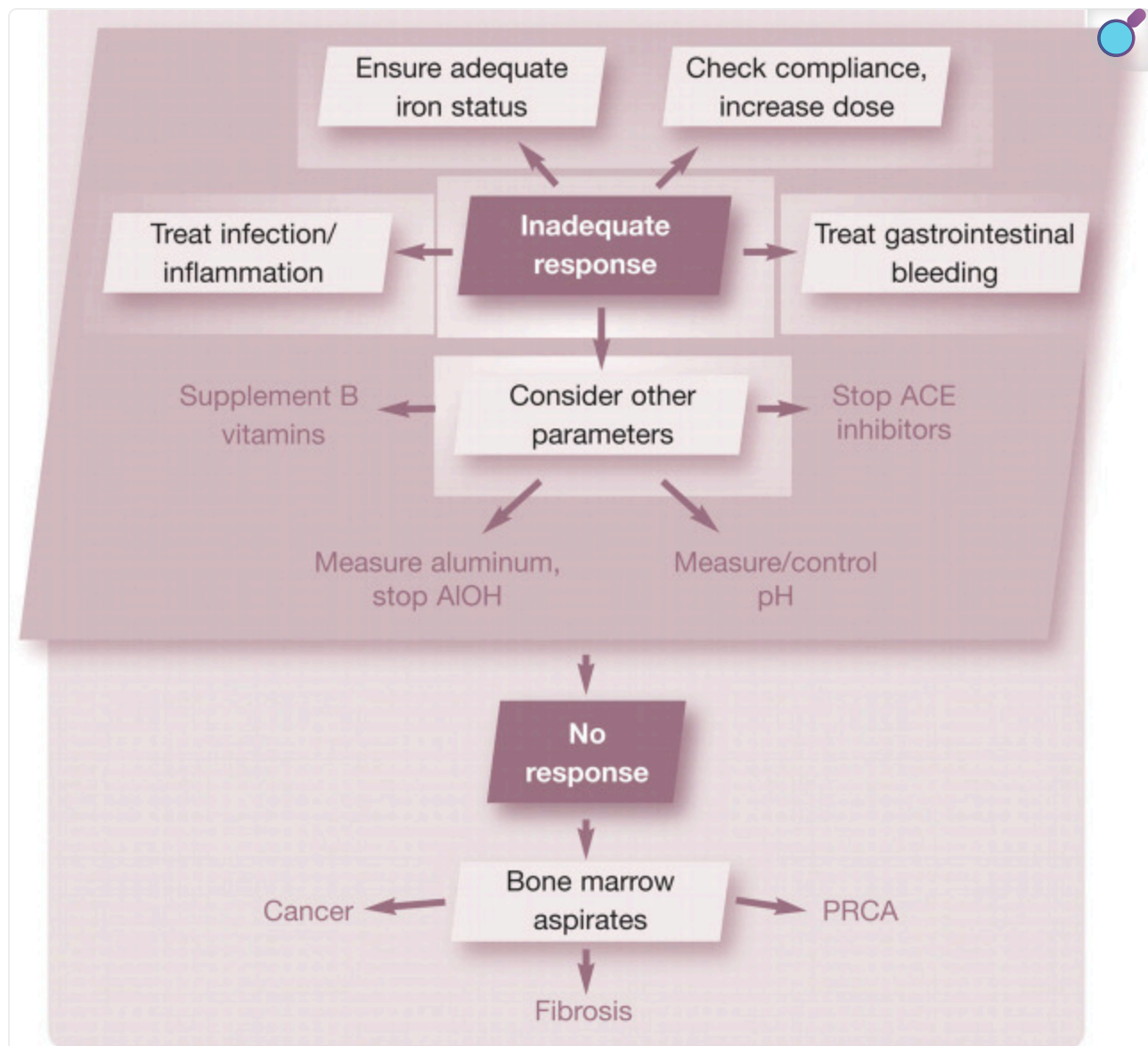
PRCA

ACE inhibitors/angiotensin receptor antagonists

Inadequate dialysis dose/frequency

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Inadequate response to ESA therapy?



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Based on the authors' experience of treating over 70 cats with epoetin or darbepoetin, only about 60–65% of cats had an adequate response.

In some cases, the ESA dose may be too low for the patient or be incorrectly administered. A higher ESA dose may partially overcome ESA resistance, although doses higher than 500 IU/kg a week for epoetin or 1.5–2.0 µg/kg a week of darbepoetin are rarely used by the authors.

If the PCV does not improve after addressing these issues, other reasons for hypo-responsiveness (deficiency of B vitamins, hyperparathyroidism, aluminum toxicity, inhibition of erythropoiesis by ACE inhibitors) should be investigated and addressed. If the PCV still remains low, a bone marrow aspirate or core biopsy should be considered to rule out occult neoplasia, bone marrow fibrosis or PRCA.

Future directions

Darbepoetin has become the mainstay for treatment of anemia of renal disease in cats. However, use of this drug does not come without risks and complications. Newer drugs in development may carry less risk and be more effective. An oral prolyl hydroxylase inhibitor is under development for humans and companion animals. This drug prevents degradation of HIF-1 α (see Fig 1), thus enhancing native erythropoietin production.^{[95,96](#)} There is also a need for diagnostic assays that allow for better characterization of anemia of renal disease, so that treatment may be tailored to the individual needs of each patient. The development of a hepcidin assay that is consistent and reliable may prove useful in determining iron status and aid in iron management.^{[97](#)}

Case notes.

Bobby is a 12-year-old castrated male domestic shorthair cat who was diagnosed with International Renal Interest Society stage III CKD when he was presented for polyuria and polydipsia. He had a blood urea nitrogen (BUN) of 71 mg/dl (25.3 mmol/l), creatinine of 4.5 mg/dl (398 μ mol/l), hematocrit of 29.4% and a systolic blood pressure of 220 mmHg. Treatments started at that time included switching to a renal diet, every other day subcutaneous fluid therapy and oral amlodipine. His blood pressure normalized at 140 mmHg. On a routine recheck 8 months after his initial presentation, the owners were happy with his clinical status at home. However, he had lost 0.5 kg, his BUN was 60 mg/dl (21.4 mmol/l), his creatinine was 3.3 mg/dl (292 μ mol/l), his hematocrit was 21% and his reticulocyte count was 18,200/ μ l.

Ongoing symptomatic treatment Given the lack of clinical signs of anemia, Bobby's treatments were not changed at that time, but the owners were instructed to return him for a recheck examination in a month. At the time of this recheck, he had lost an additional 0.2 kg, had a BUN of 68 mg/dl (24 mmol/l), a creatinine of 3.5 mg/dl (309 μ mol/l), a hematocrit of 17.9% and a reticulocyte count of 4180/ μ l. The following week, his BUN was 95 mg/dl (34 mmol/l), his creatinine was 3.7 mg/dl (327 μ mol/l), his hematocrit was 15.1% and his reticulocyte count was 4660/ μ l.

As the progressive increase in the BUN to creatinine ratio may suggest dehydration or gastrointestinal bleeding, treatment with gastroprotectants (ie, sucralfate and famotidine) was initiated. Although treatment to stop blood loss from an ulcer may allow the anemia to correct itself, the rate of improvement can be very slow in patients with CKD. The following day, Bobby was presented for weakness, with a hematocrit of 13.6%, a reticulocyte count of 3200/ μ l and a systolic blood pressure of 105 mmHg.

Given the rapid decrease in PCV and the appearance of clinical signs associated with anemia, a transfusion of one unit of packed RBCs was administered, increasing the PCV to 19%.

ESA therapy Treatment with darbepoetin was initiated at a dose of 1 μ g/kg once a week subcutaneously. The PCV the following week was 24%, the reticulocyte count was 36,790/

μl and all other clinical parameters remained stable. Two weeks after starting darbepoetin, the PCV decreased to 21%. On week 3 of therapy, the PCV was 18% and his reticulocyte count was 4250/ μl .



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Bobby's pale mucous membranes are a classic sign of anemia

Chronic gastrointestinal blood loss can cause iron deficiency. A serum iron panel was submitted and 50 mg of iron dextran was administered intramuscularly. The dose of darbepoetin was also increased to 2 $\mu\text{g}/\text{kg}$ once weekly. The dose increase was made to overcome erythropoietin resistance, despite concerns that the decrease in PCV may have been due to PRCA. The following week, the PCV had increased to 22%, and the reticulocyte count increased to 16,560/ μl ; at this time another 2 $\mu\text{g}/\text{kg}$ dose of darbepoetin was administered. The results of the serum iron panel showed hypoferremia (18 $\mu\text{g}/\text{dl}$, reference interval [RI] 33–134 $\mu\text{g}/\text{dl}$; 3.2 $\mu\text{mol}/\text{l}$, RI 6–24 $\mu\text{mol}/\text{l}$), low % saturation (6%, RI

20–33%), and a high ferritin (399 ng/ml, RI 31–144 ng/ml; 896.6 pmol/l, RI 70–324 pmol/l), suggesting functional iron deficiency associated with inflammation.

Bobby's PCV the following week was 29%, and his darbepoetin dose was decreased to 1 µg/kg once weekly. A further week on, the PCV was 33%, so the darbepoetin dose was decreased again, to 1 µg/kg every other week. The PCV stabilized between 27 and 32% on a dose of 1 µg/kg administered every other week.

WHAT THIS CASE DEMONSTRATES

Bobby's case provides a typical example of renal anemia developing in a cat with CKD, and illustrates the active follow-up necessary to reach and maintain target PCV.

KEY POINTS.

- Anemia related to kidney disease is multifactorial in its pathogenesis; however, the main cause is decreased erythropoietin production by the failing kidneys.
- A large percentage of cats with kidney disease will develop anemia.
- Acute and chronic inflammation contributes to anemia of renal disease by the production of inflammatory cytokines and substances such as hepcidin that will decrease erythropoietin function, red cell survival and available iron.
- Multiple therapies for treating anemia of renal disease exist. However, the use of erythrocyte-stimulating agents (ESAs), such as darbepoetin or epoetin, and elimination of other potential causes of anemia, seem to be most efficient in the long term.
- While both darbepoetin and epoetin can be effective in reversing anemia of renal disease, darbepoetin has a longer half-life and is less likely to cause pure red cell aplasia when compared with epoetin in cats.
- Iron is essential for hemoglobin formation and red cell function, and should be administered to all cats receiving ESA therapy.

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Use of molidustat, a hypoxia-inducible factor prolyl hydroxylase inhibitor, in chronic kidney disease-associated anemia in cats

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Abstract

Abstract

Background

Erythropoietic effects of molidustat, a hypoxia-inducible factor prolyl hydroxylase (HIF-PH) inhibitor, were previously demonstrated in healthy cats.

Objective

To evaluate the safety and erythropoietic effects of daily PO administration of molidustat in anemic cats with chronic kidney disease (CKD).

Animals

Twenty-one client-owned CKD cats (4-17 years old) with anemia.

Methods

Multicenter field study; randomized, masked, and placebo-controlled. Cats were treated PO once daily for 28 days with suspensions of control product (CP; n = 6) or 5 mg/kg of molidustat (n = 15). Hematocrit (HCT) was evaluated at weekly intervals. Individual cat treatment success was defined as a $\geq 4\%$ point increase in HCT compared to baseline.

Results

Control group mean HCT remained low throughout the study (20.1%-23.4%). Mean HCT of molidustat-treated cats increased weekly, and a significant increase compared to baseline (23.6%) was first observed on Day 21 (27.3%; $P < .001$; 95% confidence interval [CI], 1.69-5.67). Compared to CP group, mean HCT was significantly higher on Day 21 (27.3% vs 20.1%; $P < .001$; 95% CI, 2.91-10.75) but not significantly higher on Day 28 (27.8% vs 23.4%; $P = .06$; 95% CI, -0.23 to 9.88). The number of individual treatment successes on Day 28 was higher among remaining molidustat-treated cats (7/14) compared to remaining control cats (1/5), but there was no significant difference between groups.

Conclusions and Clinical Importance

Daily PO molidustat administration may stimulate a clinically relevant erythropoietic response in anemic cats with CKD. This HIF-PH inhibitor may be an alternative for managing anemia in cats compared to recombinant EPO treatment.

Abbreviations

adverse event

autoregressive

confidence interval

chronic renal disease

control product

erythropoietin

erythropoietin stimulating agent

Europe

hematocrit

mercury

hypoxia-inducible factor

hypoxia-inducible factor prolyl hydroxylase

renal erythropoietin-producing

repeated measures analysis of covariates

visit

1

INTRODUCTION

Anemia associated with chronic kidney disease (CKD) develops in 30%-65% of CKD cats.¹ Current erythropoietin (EPO) replacement treatment can be associated with life-threatening complications in some cats.^{1 2 3}

Production of EPO is controlled by hypoxia-inducible factor (HIF), found in specialized renal EPO-producing (REP) cells, and degraded under normal oxygen conditions by HIF prolyl hydroxylase (HIF-PH) enzymes, giving mammals the ability to adapt to environmental changes in oxygen levels.^{4 5 6 7} In CKD cats, decreased metabolic activity of failing kidneys leads to relative hyperoxia, decreasing EPO production by the degradation of HIF by HIF-PH enzymes, despite a preserved EPO control mechanism.^{4 7 8} It was therefore postulated that administration of a HIF-PH inhibitor, such as molidustat sodium, may stimulate production of erythrocytes in cats. The erythropoietic effect of daily PO administrations of 5 mg/kg molidustat on hematological parameters was first investigated in healthy adult cats showing an increase in EPO concentration and subsequent increase in hematocrit (HCT) despite normoxic conditions (internal communication). Molidustat administration was well tolerated in study cats.

Considering these results, our aim was to evaluate the ability of molidustat to stimulate production of erythrocytes in client-owned anemic cats with CKD using quantitative assessments of HCT responses over time.

2

MATERIALS AND METHODS

2.1

Study design

Ours was a multicenter, randomized, double-masked, parallel, placebo-controlled field study. It was conducted in accordance with guidelines of the Animal Care Review Committee of Bayer Animal Health (Protocol #205375) and with informed owner consent. It was designed to investigate the effects and safety of molidustat as a PO suspension in client-owned cats in the United States and Europe with anemia associated with CKD when administered daily for 28 days.

2.2

Inclusion and exclusion criteria

To be eligible for inclusion, cats had to be ≥ 1 year of age with body weight ≥ 2.0 kg, diagnosed with anemia (PCV $\leq 27\%$ at screening and on Day 0) associated with CKD (based on historical patient data and baseline serum creatinine concentration > 1.8 mg/dL without staging), negative urine culture and a systemic blood pressure (SBP) ≤ 165 mm Hg pretreatment. Cats could be receiving crystalloid fluids SC and comorbidities had to be stable for a minimum of 2 weeks or for at least 6 months in the presence of hyperthyroidism.

Cats were excluded if they were intended for breeding, pregnant or lactating, had an untreated urinary tract infection, or were difficult to handle. Cats also were excluded if they had received previous treatment with recombinant EPO or a blood transfusion or both, severe clinical signs of inflammation not attributed to CKD, other uncontrolled clinically relevant medical conditions, had been diagnosed with feline leukemia or feline immunodeficiency virus, or had clinical signs of gastrointestinal bleeding.

2.3

Randomization and masking

All study personnel involved in clinical evaluations, observations, or collection of samples were masked to treatment and study group. Dispensers and any study personnel witnessing or administering treatments were not involved in any other roles. To ensure masking, 2 alphabetical treatment codes were randomly assigned to the molidustat or control product (CP), and cats were randomly assigned to 1 of 2 treatment groups at a ratio of 2:1 (molidustat: CP) based on order of entry into the study. A global randomization form was created using SAS Statistical Software (version 9.3, Cary, North Carolina) and exported to the MEDRIO system (MEDRIO, San Francisco, California). The assignment of cats to study groups was performed in MEDRIO by the dispenser at the study site.

2.4

Treatment phase and primary hematological parameters

Primary clinical endpoints were HCT and PCV responses over time. Each cat qualifying for enrollment underwent a screening evaluation on Day -7 (± 2 days) which included physical examination, SBP and body weight measurements, urinalysis and bacterial culture, routine serum biochemistry and hematology, and PCVs. Starting on Day 0, cats were treated PO once daily by the owner before feeding for at least 28 consecutive days with a suspension of either CP

(vehicle only) or molidustat (25 mg/mL). Molidustat was administered at a dosage of 5 mg/kg, equivalent to 0.2 mL/kg. The same volume of vehicle was administered to CP cats.

Blood was collected to assess hematological parameters with body weights recorded during each visit (V1-V5), which occurred on Days 0 (V1; pretreatment), 7 (V2), 14 (V3), 21 (V4), and 28 (V5) $\pm$ 2. Blood also was collected to evaluate serum biochemistry on Days 0 and 28. Three to 5 SBP measurements (within 20% variation) were recorded and then averaged before blood sampling on Days 0 (pretreatment), 14, and 28 after the cat had a chance to acclimate to a quiet room. Each study site used either an ultrasonic doppler or oscillometric SBP device. A central laboratory was used for evaluation of clinical pathology specimens in the United States (Antech GLP, Morrisville, North Carolina) and Europe (IDEXX GmbH, Ludwigsburg, Germany).

Treatment success was evaluated up to and including Day 28. Any clinical findings were recorded as adverse events (AEs) or serious AEs (causing death or life-threatening condition) and immediately reported to the Investigator whether associated with treatment or not.

2.5

Hematological effect evaluation

Individual treatment success for each cat was defined as the primary endpoints reaching a clinically relevant increase. A clinically relevant increase in hematological parameters was defined as a >25% increase or an absolute 4% point increase or both in PCV or HCT from baseline up to and including Day 28.

2.6

Continuation phase

After the 28-day treatment phase, owners could voluntarily include their cats in a nonplacebo-controlled continuation phase for up to 8 additional weeks (File ^{S1}).

2.7

Statistical analysis

Hematocrit and PCV, measured multiple times during the study including baseline, were analyzed using a repeated measures analysis of covariance (RMANCOVA), with treatment, time (week), and treatment by time terms in the model as fixed effects, and animal identified as subject in the repeated statement (MIXED procedure in SAS). Pretreatment (baseline) values were used as a covariate, and remained in the model, regardless of significance. Random effects were included in the model consisting of the clinic (SITE) and region (Europe or United States). The best covariance structure in the RMANCOVA was investigated and determined to be the autoregressive (1) AR (1). If treatment by time interaction was significant at the .05 level, then treatment group comparisons for each day were performed. If the 2-way interaction was not significant, the main effect of treatment group was evaluated for any significant treatment group effect. All terms in the model were evaluated at an unadjusted alpha = .05. Mean HCT % ($\pm$ SD) over a 28-day treatment phase for cats treated with either CP or molidustat was graphed.

Various treatment success definitions (binary in structure, reported in percentages) were analyzed using 2 methods: a nonparametric Fisher's exact test and a generalized linear mixed model (PROC GLIMMIX, SAS). A binomial distribution was assumed, and a logit link was used, adjusting the degrees of freedom using Kenward Rogers

adjustment and including baseline values as a covariate. The number of treatment successes in each group at each visit from Day 7 up to and including Day 28 were counted and compared to baseline.

Separate analyses of both HCT and PCV values also were performed using a model testing for differences in study day for each treatment group (ie, baseline control analyses). An analysis of covariance with fixed term of Day (0, 7, 14, 21, and 28) and pretreatment baseline (Day 0) as a covariate was used to test for Day differences. All pairwise comparisons of mean HCT and PCV values compared to baseline for each visit were tested at the .05 level of significance.

3

RESULTS

3.1

Study population

From 65 cats screened at 11 American and 9 European sites, 23 client-owned CKD cats with anemia were enrolled in the 28-day study (Figure ¹).

FIGURE 1

Enrollment flowchart of the global study population of the randomized pilot field study evaluating the erythropoietic effects of daily oral administration of molidustat in client-owned anemic cats with chronic kidney disease (CKD). CP, control product; EU, Europe; US, United States.

Baseline characteristics of the study population showed that groups of mostly domestic short-haired cats (n = 20/23) were evenly matched in age and body weight, ranging from 4 to 17 years and 2 to 6 kg, respectively, and nearly evenly distributed by sex (Table ¹).

TABLE 1

Baseline mean characteristics (±SD) of the efficacy study population on Day 0, Visit 1 (n = 21).

Variables	CP (n = 6)	Molidustat (n = 15)
Demographics		
Age (years)	10.6 (5.4)	12.2 (3.8)
Sex (M/F) ^a	1M/5F	8M/7F
Breed (DSH)	6	13
Region (% from US)	33	60
Physical examination		
Body weight (kg)	3.9 (1.26)	3.5 (1.07)
SBP (mm Hg)	147.8 (17.97)	148.0 (24.43)
Laboratory results		

Variables	CP (n = 6)	Molidustat (n = 15)
Blood urea nitrogen (mg/dL) reference interval: 34.2-81	170.4 (57.12)	177.6 (71.52)
Creatinine (mg/dL) reference interval: <1.8	5.0 (2.8)	5.0 (2.8)
HCT%	23.2 (2.50)	23.6 (3.23)
PCV%	21.3 (2.98)	22.5 (2.91)
Hb (g/dL)	6.9 (1.03)	7.49 (1.06) ^b

Abbreviations: CP, control product; DSH, domestic short haired; F, female; Hb, hemoglobin; HCT, hematocrit; M, male; NA, not applicable; SBP, systemic blood pressure; US, United States.

a

All cats were neutered or spayed except for 1 intact female in the CP group.

b

n = 14.

There were no comorbidities unrelated to CKD in the CP group, whereas there were 3 cats with hyperthyroidism, 1 with coccidiosis, 1 with cystitis, and 1 with suspected osteoarthritis in the molidustat-treated group (Table ^A, Supporting Information). Several concomitant medications were used during the study, administered to treat clinical signs of CKD or comorbidities at the discretion of the attending veterinarian (Table ^A, Supporting Information).

3.2

Hematological parameters

Mean ($\pm$ SD) baseline HCT of the CP group (n = 6) was 23.2% ($\pm$ 2.50) and weekly responses remained low (20.1%-23.4%) at all assessed post-treatment timepoints (Figure ²).

FIGURE 2

Mean hematocrit (HCT % $\pm$ SD) values over a 28-day treatment phase with either molidustat (n = 15 to 14 due to missing data on Day 28) or CP (n = 6 to 3 due to missing data on Day 28) of anemic cats with CKD. *Significant difference at $P < .05$ between groups on Day 21 is evident; the 95% CI for the difference is 2.91-10.75. CI, confidence interval; CKD, chronic kidney disease; CP, control product.

Mean HCT of the molidustat-treated group increased weekly and a significant increase compared to baseline (23.6%) was first observed on Day 21 (27.3%; $P < .001$; 95% confidence interval [CI] for the difference, 1.69-5.67; Table ²). Mean PCV significantly increased as early as Day 14 in the molidustat-treated group compared to baseline (24.7% vs 22.5%; $P = .02$). In contrast, no significant increases relative to baseline were observed in the CP group at any timepoint.

TABLE 2

Pairwise comparisons with mean baseline HCT and PCV values ($\pm$ SD) in the molidustat-treated group (n = 15).

Hematological parameter	Visit 1 (Day 0)	Visit 2 (Day 7)	Visit 3 (Day 14)	Visit 4 (Day 21)	Visit 5 ^a (Day 28)
HCT (%)	23.6 ± 3.23	25.2 ± 4.34	25.5 ± 5.94	27.3 ± 4.82	27.8 ± 5.25
Difference from baseline	1.6	1.9	3.7	4.2	
<i>P</i> value	.1	.06	<.001	<.001	
PCV (%)	22.5 ± 2.91	24.2 ± 4.16	24.7 ± 4.95	26.4 ± 4.51	26.2 ± 5.24
Difference from baseline	1.7	2.2	3.9	3.7	
<i>P</i> value	.07	.02	<.001	<.001	

Note: Bold indicates statistically significant values.

Abbreviation: HCT, hematocrit.

^a

n = 14.

Compared to the CP group, mean HCT of the molidustat-treated group was significantly higher on Day 21 (27.3% vs 20.1%; $P < .001$; 95% CI, 2.91-10.75) but was not significant on Day 28 (27.8% vs 23.4%; $P = .06$; 95% CI, -0.23 to 9.88; Figure ²).

The number of individual treatment successes on Day 28 was higher among molidustat-treated cats (7/14) compared to CP cats (1/5; Table ³). However, no significant difference was found between groups. Other hematological parameters such as red blood cell and reticulocyte counts and Hb also increased over time, but statistical analysis was not performed (Table ^B, Supporting Information).

TABLE 3

Percent treatment success defined as a >25% increase and/or a 4% point increase in HCT from baseline (Day 0) up to and including Day 28 in anemic cats with CKD given either molidustat or the control product (CP).

Group	Day 7	Day 14	Day 21	Day 28
CP	16.7% (1/6)	20% (1/5 ^a)	0% (0/5)	20% (1/5)
Molidustat	20% (3/15)	30% (5/15)	46.7% (7/15)	50% (7/14 ^b)
<i>P</i> value				.37

Note: Bold indicates statistically significant values.

Abbreviations: CKD, chronic kidney disease; HCT, hematocrit.

^a

One cat died shortly after Day 7.

^b

One cat was excluded because of potential impact on HCT from dehydration.

3.3

In-use safety

Any clinical findings evident on physical examinations before and after the 28-day treatment period were reported as AEs (Table ⁴). It was possible for a cat to have multiple reported AEs. Biochemistry variables were within reference intervals (Table ^C, Supporting Information) except for 2 molidustat-treated cats (Table ⁴). A slight increase in liver enzyme activity was noted for 1 cat on Day 28 and mild leukocytosis, with absolute and relative neutrophilia, absolute and relative lymphocytopenia, relative monocytopenia, and absolute eosinophilia were noted in another cat on Day 7 without further investigation or treatment (Table ⁴). Body weight measurements did not change over time, and SBP means varied between 139.4-150.8 mm Hg for the CP group and 143.9-157.3 mm Hg for the molidustat group (Figure ³).

TABLE 4

Summary of number of cats with adverse events (AEs) reported during a 28-day treatment period with either molidustat or the control product (CP), listed in descending order of frequency.

Adverse event	CP (n = 6)	Molidustat (n = 15)
Vomiting	0	6
Decreased appetite	0	2
Death	1	0
Respiratory distress	1	0
Thrombosis	0	1
Increased liver activity	0	1
Lateral recumbency	0	1
Renal lymphoma	0	1
Renal failure	0	1
Dehydration	0	1
Polyuria/polydipsia	0	1
Stomatitis	0	1
Increased CK activity ^a	0	1
Leukocytosis/neutrophilia/lymphopenia/monocytopenia/eosinophilia ^b	0	1
Lethargy	0	1
Cystitis	0	1
Mild agitation	0	1
Ear discharge	0	1

Adverse event	CP (n = 6)	Molidustat (n = 15)
Skin scab	0	1
Abnormal test result (coccidiosis)	0	1

Abbreviations: ALP, alkaline phosphatase; ALT, alanine transaminase; AST, aspartate transaminase; CK, creatine kinase; EU, Europe; RI, reference interval; WBC, white blood cells.

a

One cat (Case ID #1) with ALP 83 U/L (RI: 0-73); ALT 1058 U/L (RI: 0-175); AST 558 U/L (RI: 0-71); CK 3923 U/L (RI: 0-542) on Day 28.

b

One cat (Case ID #19) with WBC $20.6 \times 10^3/\mu\text{L}$ (RI: 3.5-16.0); absolute neutrophils $17.9 \times 10^3/\mu\text{L}$ (RI: 2.5-8.5) and relative neutrophils 87.1% (RI: 35-75); absolute lymphocytes $1.07 \times 10^3/\mu\text{L}$ (RI: 1.2-8) and relative lymphocytes 5.2% (RI: 20-45); relative monocytes 0.7% (RI: 1-4); absolute eosinophils $1.29 \times 10^3/\mu\text{L}$ (RI: 0-1) on Day 7.

FIGURE 3

Mean body weight measurements ($\text{kg} \pm \text{SD}$) and mean SBP measurements ($\text{mm Hg} \pm \text{SD}$) of anemic cats with CKD over a 28-day treatment phase with either molidustat (dotted lines) or CP (full lines) and for an additional 8-week molidustat treatment during the continuation phase. CKD, chronic kidney disease; CP, control product; SBP, systemic blood pressure.

Vomiting was the most frequently reported AE, occurring in 6 of 15 (40%) cats in the molidustat group. The number of vomiting events was 17, 9 of which came from 1 cat with decreased appetite during the first week of study. Of the other 5 cats, 3 vomited once, 1 vomited twice, and 1 vomited once 3 consecutive days during the second week of the study. None of the CP cats vomited (Table ⁴).

Three serious AEs occurred during the study. One CP cat showed respiratory distress and died on the way to the hospital (Day 14), 1 molidustat-treated cat was euthanized because of end-stage renal failure (Day 27), and another molidustat-treated cat was presented in lateral recumbency with a cold front leg and was euthanized without necropsy (Day 28; Table ⁴).

3.4

Continuation phase

Continuation phase results are presented in File ^{S1} and in Figures ^{2, 3, 4}.

FIGURE 4

Mean ($\pm \text{SD}$) PCV of 8 anemic cats with CKD treated with molidustat sodium during a 28-day treatment phase and for an additional 8 weeks during the continuation phase. CKD, chronic kidney disease.

4

DISCUSSION

We investigated the ability of molidustat, a novel HIF-PH inhibitor, to stimulate production of erythrocytes in client-owned CKD cats with anemia. Results showed the potential of molidustat to improve HCT and PCV in affected cats when compared to baseline values as early as 14-21 days after the start of daily PO administration of 5 mg/kg molidustat sodium. Although significant differences in reaching clinically relevant increases in HCT or PCV compared to CP cats were not consistently achieved, the number of individual treatment successes on Day 28 was higher among molidustat-treated cats (7/14) compared to CP cats (1/5). This lack of significant difference may be attributed to the limited sample size of the study. The number of cats achieving treatment success increased during the continuation phase. Although lack of EPO is considered to be a major factor in the anemia of CKD, other possible contributing factors potentially present in these client-owned cats may have included comorbid conditions causing inflammation, undiagnosed hyperthyroidism, coadministration of medications, or insufficient iron stores (functional or absolute). These factors may have slowed or blunted the erythropoietic response. In addition, because CKD staging was not performed pretreatment, its impact on study results is unknown. Molidustat's in-use safety profile was demonstrated in these CKD cats after up to 12 weeks of use when administered alone or in combination with other medications needed to alleviate clinical signs of CKD.

Production of EPO progressively decreases in failing kidneys because of the continued loss of functioning REP cells to fibrosis, uremic substances causing oxidative stress, chronic presence of inflammatory cytokines, and impaired iron mobilization.^{1, 8, 9, 10, 11} Overexpression of HIF inconsistently has been associated with renal fibrosis, leading to concern that HIF-PH inhibitors may promote renal fibrosis. However, studies in mice have shown that HIF-PH inhibitors do not increase renal fibrosis despite HIF stabilization and increased EPO production in pericytes.¹² Erythrocyte-stimulating agents (ESAs) such as recombinant human EPO are used as EPO replacements for anemic cats.^{1, 13, 14} This parenteral treatment has been an important tool in the management of anemia in cats. However, numerous complications such as iron deficiency, hypertension, arthralgia, fever, seizures, polycythemia, and life-threatening pure red cell aplasia have been reported in treated cats.^{1, 2, 3} Our results confirm that the EPO control mechanism is preserved in the remaining functioning cells of failing kidneys of CKD cats, and that HCT and PCV are adequate quantitative surrogate parameters to measure the erythropoietic response to daily administration of molidustat in a clinical setting. Molidustat also was well tolerated without affecting SBP in cats during our study. The use of different SBP devices at different sites and antihypertensive medications in several cats may have impacted this latter finding. A larger and longer study would provide more information regarding SBP.

Molidustat, a novel HIF-PH inhibitor, administered once daily for 28 days at 5 mg/kg significantly increased HCT and PCV as early as 14-21 days after the start of treatment compared to baseline in anemic cats with CKD.

Limitations of our study include low power (small sample size) and short duration, lack of CKD staging at baseline and unknown impact of CKD stage on study results, use of different laboratories (Europe and United States) for clinical pathology assessment, data missing for several cats at critical timepoints and use of concomitant medications in the presence of comorbidities. Despite these limitations, our objectives of investigating the ability of molidustat to safely stimulate erythropoiesis in failing kidneys of anemic cats were fulfilled. Future studies evaluating the use of molidustat in a larger number of anemic CKD cats, using HCT (or PCV) as surrogate parameters to conveniently monitor erythropoietic response, are necessary because molidustat potentially could be an innovative therapeutic option to address an unmet clinical need while safely improving the quality of life of anemic cats with CKD.

CONFLICT OF INTEREST DECLARATION

Dr S. Charles, Dr R. Süssenberger, and T. Settje are or were employed by Elanco. Drs C. Langston and C. Lainesse received payments as independent consultants.

OFF-LABEL ANTIMICROBIAL DECLARATION

Authors declare no off-label use of antimicrobials.

INSTITUTIONAL ANIMAL CARE AND USE COMMITTEE (IACUC) OR OTHER APPROVAL DECLARATION

This study was conducted in accordance with guidelines of the Animal Care Review Committee of Bayer Animal Health (Protocol #205375) and with informed owner consent.

HUMAN ETHICS APPROVAL DECLARATION

Authors declare human ethics approval was not needed for this study.

Supporting information

File S1. Continuation phase—molidustat in CKD cats.

Table A: Comorbidities and concomitant medications administered to study cats.

Table B: Summary statistics for RBC parameters of interest.

Table C: Summary statistics for biochemistry parameters.

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Hypoxia-inducible factor prolyl hydroxylase inhibitor as a new therapeutic option for treating anemia in feline chronic kidney disease

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Abstract

Anemia is a frequent complication in cats with chronic kidney disease (CKD), mainly due to decreased erythropoietin (EPO) production resulting from a loss of renal erythropoietin-producing (REP) cells. Anemia affects both survival and quality of life by inducing clinical signs such as lethargy and anorexia. Early detection of anemia by monitoring hematocrit/packed cell volume (Htc/PCV) trends is essential to initiate treatment before progressive renal damage limits therapeutic effectiveness. Molidustat, a hypoxia-inducible factor prolyl hydroxylase inhibitor (HIF-PHI), is a novel oral drug stimulating endogenous erythropoietin synthesis. However, treatment has to be started when REP cells are still functional, with PCV target ranges between 30–40%. Regular monitoring following the International Renal Interest Society (IRIS) guidelines is essential to keep PCV within the target range and to determine when treatment should be started, stopped, or restarted. Moreover, monitoring the iron status is important since both functional and absolute iron deficiency might be seen in cats with CKD and can impair response to treatment.

Keywords: Cat, HIF-Inhibitor, Molidustat, hematocrit, treatment, PCV

Introduction

Chronic kidney disease (CKD) is commonly observed in elderly cats [1](#). Decreasing renal function results in many systemic complications, among which anemia is particularly important as it significantly contributes to the reduction in quality of life and survival [1](#). The pathogenesis of anemia in CKD is multifactorial [1](#). It is primarily caused by reduced production of erythropoietin (EPO) due to loss of functional renal erythropoietin-producing (REP) cells [2](#). Chronic inflammation, uremic toxins, nutritional deficiencies, and gastrointestinal blood loss might contribute as well [1](#). Without early diagnosis and management, anemia progresses and leads to clinical signs such as lethargy, anorexia, and weight loss that significantly reduce the cat's quality of life [3](#).

With the introduction of newer therapeutic agents such as molidustat, a hypoxia-inducible factor prolyl hydroxylase (HIF-PH) inhibitor, there is a new option for managing anemia in cats with CKD. Molidustat acts by increasing endogenous EPO production, but its efficacy depends on the presence of functional REP cells, making early intervention critical [4](#). In this context, the importance of regular monitoring – especially tracking packed cell volume (PCV) or hematocrit value (HCT) trends – becomes apparent. Subsequently, the term PCV will be used for both PCV or HCT irrespective of the method of measurement.

Early detection of decreasing PCV allows for timely treatment initiation, thus optimizing outcome [4](#).

This paper explores the pathophysiology of anemia in feline CKD, its clinical impact, and the critical need for early detection and intervention. It also provides a detailed review of molidustat's mechanism, the IRIS treatment guidelines, and considerations such as side effects and iron supplementation.

Pathophysiology of anemia in feline chronic kidney disease

Anemia is a frequent and clinically significant complication of feline chronic kidney disease (CKD), particularly in advanced stages (IRIS stages 3–4). Approximately 30–65% of cats with CKD develop anemia during progression of renal disease [5](#) [6](#) [7](#) [8](#). It is typically normocytic, normochromic and non-regenerative, mainly caused by the reduced synthesis of erythropoietin

(EPO), a hormone primarily produced by specialized renal interstitial fibroblasts, the renal erythropoietin-producing (REP) cells ⁹. REP cells sense oxygen tension in renal tissue and respond to hypoxia by increasing EPO production, thereby stimulating erythropoiesis in the bone marrow ⁹. In CKD, there is a progressive loss and myofibroblastic transformation of REP cells, finally resulting in inadequate red blood cell (RBC) production and scar tissue formation ⁹.

Moreover, in CKD, REP cells also contribute to development of anemia by active production of proinflammatory cytokines ⁹. Proinflammatory cytokines such as interleukin-6 (IL-6) and tumor necrosis factor-alpha (TNF- α) can suppress erythropoiesis and decrease the bone marrow's response to EPO ¹⁰. These cytokines also induce hepatic production of hepcidin, a hormone inhibiting iron mobilization from macrophages, decreased iron transport and intestinal iron absorption, finally resulting in functional iron deficiency / iron limited erythropoiesis despite adequate total body iron ¹¹.

Uremic toxins, including indoxyl sulfate and p-cresyl sulfate, have been shown to exert direct toxic effects on erythroid precursor cells and may also interfere with EPO receptor signaling ¹². Furthermore, CKD often results in decreased RBC lifespan due to oxidative stress, altered membrane permeability, and increased erythrocyte osmotic fragility. Additional contributors to anemia in feline CKD include gastrointestinal (GI) blood loss, which may occur secondary to uremia-induced gastric mucosal injury or associated with medications ¹. Chronic GI hemorrhage may result in both iron deficiency and progressive anemia ¹.

The effect of anemia on longevity and quality of life in cats

While anemia is prevalent across all species with CKD, cats respond even to subtle changes in RBC concentration so that even a PCV at the lower limit of the reference interval might result in renal ischemia and progression of CKD ⁴. Although anemia tends to develop gradually in cats with progressive CKD, it often goes under-recognized until it becomes moderate to severe ⁴.

The most commonly reported clinical signs of anemia in cats include lethargy, reduced appetite, decreased grooming behavior, pale mucous membranes, tachycardia, and reduced activity. Owners frequently describe their cats as being “less interactive,” “quieter,” or “slowing down,” which are subjective but clinically important quality-of-life indicators. These symptoms often overlap with those of CKD itself. However, there is also an interrelationship between the development of anemia and the progression of CKD ⁴. Anemia reduces the oxygen-carrying capacity of the blood, resulting in tissue hypoxia. Along with already compromised renal perfusion, reduced oxygen delivery

further exacerbates tubular dysfunction and may contribute to progression of kidney damage by hypoxia, inflammation, and fibrosis [4](#).

Several studies have shown a strong association between lower PCV values and shorter survival times in cats with CKD [13](#) [14](#) [15](#) [16](#). It has been reported that cats with a hematocrit value below 25% [14](#) and 26% [16](#), respectively, consistent with a mild anemia (lower reference limit 30% in this study) had significantly reduced median survival times compared to those with values above this threshold [16](#). Anemia (i.e., a PCV below the lower reference limit of 27%) [18](#) or even a median PCV<31% as defined by Cox multivariate regression model [17](#) is associated with progression of CKD. Conversely, a minimal increase of PCV of just 1% is associated with a reduced risk of progression of CKD of 10% [18](#). While PCV cut off values used in the studies vary between 25% and 31% due to different reference intervals and statistical methods, they all have in common that they are close to or consistent with the lower limit of the reference interval. Thus, it has been recommended that the PCV in cats with CKD should not drop below the lower limit of the reference interval [4](#).

Quality of life is equally important as survival in feline CKD as clinical signs associated with untreated anemia contribute to the owners' distress and impact decisions regarding euthanasia [3](#), successful treatment resulted in clinical improvement in many cats [4](#).

Importance of monitoring PCV trends for early detection of anemia

In feline CKD, PCV monitoring plays a pivotal role in the early identification of anemia. While a single PCV measurement merely represents a snapshot of red cell status, longitudinal tracking of PCV trends offers superior clinical insight, enabling veterinarians to detect subtle declines in red cell mass before anemia becomes overt and irreversible renal damage and loss of erythropoietin-producing cells develops [4](#). Monitoring trends rather than isolated values helps identifying a decline in PCV well before critical thresholds are reached.

PCV measurements are influenced by biological and analytical variation. Intraindividual biological variation of PCV might be up to 5.5% in healthy cats, so that true clinically relevant changes in PCV (i.e., the reference change value) become only apparent when a change of 15.5% is exceeded [19](#). For example, in a cat with a PCV of 0.30 l/l, a decrease of 15.5% (0.045 l/l) to 0.25 l/l has to be considered clinically relevant exceeding the impact that can be solely explained by biological and analytical variation.

In cats with CKD, other factors might contribute to PCV variability. Dehydration commonly seen in CKD cats may transiently increase PCV, masking early anemia. Conversely, overhydration or fluid therapy may dilute RBC mass, leading to falsely low PCV results. Nevertheless, in clinically stable cats with CKD, the reference change value might be at least a guideline, by which changes of PCV can be considered as clinically relevant.

According to IRIS (International Renal Interest Society guidelines), regular PCV monitoring in cats diagnosed with chronic kidney disease (CKD) is extremely important and assessments at diagnosis as well as every 3–6 months thereafter depending on disease stage and stability are recommended [20](#). More frequent monitoring may be warranted when PCV declines or clinical signs of anemia become apparent. Recording PCV trends also facilitates evaluation of treatment efficacy once treatment has been initiated. In addition to PCV, complete blood count (CBC) analysis offers valuable complementary information, including red blood cell indices, reticulocyte counts and reticulocyte indices [21](#).

Early intervention and REP cell function

The efficacy of anemia treatment in CKD is closely linked to the functional status of renal erythropoietin-producing (REP) cells. These specialized interstitial fibroblasts are responsible for sensing tissue hypoxia and EPO production, a glycoprotein hormone that stimulates erythropoiesis in the bone marrow [9](#). As CKD progresses, renal fibrosis and loss of functional nephrons lead to depletion of REP cells, resulting in a diminished endogenous capacity to produce EPO. Consequently, anemia worsens and becomes less responsive to therapy.

Molidustat, a hypoxia-inducible factor prolyl hydroxylase inhibitor (HIF-PHI), has emerged as a promising therapeutic agent to address anemia by stimulating endogenous EPO production in people [22](#) and cats [23](#) [24](#). It acts by inhibiting prolyl hydroxylase enzymes that target hypoxia-inducible factors (HIFs) for degradation under normal oxygen conditions. Stabilized HIFs translocate to the nucleus, promoting the transcription of genes including the EPO gene. By mimicking a hypoxic state, molidustat enhances endogenous EPO synthesis even in diseased kidneys with relative hyperoxia due to decreased oxygen utilization [25](#). Treatment with HIF-PH inhibitors is considered a more physiological treatment than the replacement therapy with human recombinant EPO [4](#).

However, the drug's effectiveness is contingent upon the presence of viable REP cells capable of responding to HIF stabilization. If treatment is delayed until late-stage CKD when REP cells have been extensively lost or fibrotic replacement is profound, molidustat's capacity to induce adequate

EPO production is compromised. Therefore, early initiation of therapy, ideally when anemia is mild and REP cell numbers remain substantial, optimizes outcomes [4](#) [17](#) [18](#) . From a practical standpoint, early detection via regular PCV trend monitoring, as outlined in previous sections, is critical to identify patients at a stage when molidustat therapy will be most beneficial. Veterinarians should monitor cats with CKD closely and initiate treatment promptly when PCV begins to decline, ideally before values fall below 30% where oxygen delivery to the kidney starts to decrease rapidly as it has been demonstrated for dogs [26](#) . This concept is even more important for cats, where normal PCVs are often below 35% [4](#) . The threshold of 30% aligns with the IRIS molidustat treatment algorithm, which recommend maintaining PCV within a target range of 30–40% for optimal clinical outcomes [20](#) .

Additionally, preserving REP cell function may have broader benefits beyond anemia correction. By improving renal oxygenation and reducing hypoxia-related fibrosis, therapies targeting the HIF pathway might contribute to slowing CKD progression, although this hypothesis requires further investigation.

Molidustat monitoring protocol and target PCV

The introduction of molidustat as a treatment option for anemia in feline CKD necessitates a structured monitoring protocol to ensure safety, efficacy, and optimal clinical outcomes. The International Renal Interest Society (IRIS) provides comprehensive guidelines for the use of molidustat in cats, which emphasize regular hematologic controls to maintain packed cell volume (PCV) within a target therapeutic range of 30–40% [20](#) .

Maintaining PCV within this range is critical. A PCV below 30% is generally considered anemic and associated with clinical signs that compromise quality of life, whereas PCV values exceeding 40% raise concerns for erythrocytosis, which can increase blood viscosity and predispose to thromboembolic complications. Although erythrocytosis has been no concern in the first study reporting the use of molidustat in a group of anemic CKD cats (n=15), the aim is to correct anemia adequately while avoiding excessive erythrocytosis [20](#) .

As per conditional license by the US Food and Drug Administration (FDA) [27](#) , the IRIS molidustat treatment algorithm recommends initiating therapy at a dose of 5 mg/kg p.o. daily for a cycle of 28 days in cats with a persistent PCV<25% or cats demonstrating a downward trend in PCV, particularly when values approach or drop below 25–28%. Subsequent PCV measurements should initially be conducted at 2-week intervals, to monitor the response and guide treatment pause. If

PCV rises rapidly or exceeds 40%, temporary discontinuation of molidustat treatment may be necessary to mitigate risks [20](#) . Treatment should be started again if the PCV declines below 30%.

In addition to PCV, monitoring of hemoglobin concentration and reticulocyte counts with reticulocyte indices (reticulocyte hemoglobin content) can provide additional insight into the erythropoietic response. A regenerative response characterized by increased reticulocyte counts typically confirms treatment efficacy. In the authors' opinion, a suboptimal reticulocyte response, hypochromasia or microcytosis may signal iron deficiency or other complicating factors, warranting further diagnostic evaluation.

Regular assessment of renal function, blood pressure, and hydration status is also recommended throughout therapy [20](#) , as changes in these parameters can affect hematologic results and overall patient stability. Since molidustat acts via modulation of hypoxia-inducible factors, its effects on systemic vascular dynamics warrant vigilance for hypertension or related adverse events as demonstrated in people [28](#) .

Potential side effects and safety monitoring

While molidustat represents a promising alternative to traditional erythropoiesis-stimulating agents (ESA) for the management of anemia in feline CKD, it is essential, as with any therapeutic intervention, to carefully consider potential adverse effects.

One of the primary concerns with molidustat therapy is the risk of erythrocytosis, or excessive increase of packed cell volume (PCV). Increased PCV raises blood viscosity, which can predispose cats to thromboembolic events, hypertension, and impaired microcirculation. Thus, regular PCV monitoring and adherence to target therapeutic ranges (30–40%) as recommended by IRIS is essential [20](#) .

Gastrointestinal signs, including vomiting and anorexia, have been reported in some cats treated with ESA and may occur with molidustat, albeit infrequently [23](#) [24](#) . If such symptoms occur, symptomatic treatment should be initiated.

Another potential side effect is hypertension possibly due to chronic hypoxia due to HIF signaling in the carotid artery [28](#) . Data in people suggest that the use of HIF-PH inhibitors was associated with an increased risk of hypertension when compared to placebo, though lower than during treatment with ESAs such as erythropoietin or darbepoetin [28](#) . Although data specific to feline

patients remain limited, blood pressure should be monitored regularly during treatment to detect and manage emerging hypertension promptly.

Unlike recombinant erythropoietin or darbepoetin alfa, molidustat's oral administration is not associated with a risk of antibody formation against EPO as endogenous EPO production is stimulated. In contrast, there are even few case reports in people with anti-EPO antibodies due to previous treatment with recombinant human erythropoietin successfully treated with an HIF-PH inhibitor [29](#).

Nevertheless, awareness for signs of treatment resistance or unexpected anemia progression remains important due to the multifactorial etiology of anemia in CKD patients.

Renal function should be monitored regularly, as the long-term effects of HIF-PH inhibitors on kidney disease progression in cats are not fully elucidated. Although nephroprotective effects via improved tissue oxygenation and iron metabolism [30](#) can be hypothesized and myofibroblast-transformation of REP cells is reversible to a certain degree [9](#), close follow-up is warranted to detect any deleterious changes.

Consideration of iron supplementation

Many cats with CKD have iron limited erythropoiesis, although recent studies suggest that iron deficiency is rather functional due to chronic inflammation than absolute [31](#) [32](#). Nevertheless, stimulation of erythropoiesis by EPO, darbepoetin or HIF-PH inhibitors results in an increased iron demand potentially resulting in absolute iron deficiency if iron stores are not sufficient.

In practice, assays for ferritin reflecting the iron stores are not easily available. Thus, clinicians often start iron supplementation empirically when reticulocyte response is insufficient 2–3 weeks after initiating an ESA, the PCV remains low or if there are signs of iron restriction such as a low mean cellular volume of erythrocytes (MCV) indicative of chronic iron limited erythropoiesis. Irrespective of the etiology, modern hematology analyzers also provide the hemoglobin content of aggregate reticulocytes (CHr, Ret-He, Retic-HGB) reflecting iron status of the last 2–4 days [21](#) [33](#) [34](#), thus allowing a much earlier detection of iron limited erythropoiesis than with the MCV.

Potential iron supplementation in addition to molidustat treatment is part of the treatment algorithm given by the IRIS guidelines [20](#). However, there are also some risks associated with iron supplementation. Oral iron supplementation (ferrous gluconate, ferrous sulfate) is preferred in

stable patients due to lower costs and higher safety, but might cause gastrointestinal side effects [35](#) and absorption is limited in cats [1](#). Iron dextran has been commonly administered intramuscularly, however, injection-site or anaphylactic reactions might occur [35](#). Over-supplementation of iron is also a risk and might lead to oxidative stress or intestinal dysbiosis [36](#). Further studies are needed to better characterize the indication for iron supplementation.

In people, there is evidence that HIF-PH inhibitors optimize iron availability by reduction of hepatic hepcidin production resulting in a higher iron mobilization from macrophages and intestinal absorption [37](#). HIF-PH inhibitors indirectly reduce hepcidin by stimulating erythropoiesis resulting in increased concentrations of erythroferrone, a hormone secreted by erythroid precursor cells and suppressor of hepatic hepcidin expression [38](#). In people, HIF-PH inhibitors tend to reduce the requirements of intravenous iron supplementation compared to injectable EPO, and thus, the need for hospitalization [39](#). Thus, there is hope that the requirements for iron supplementation might decrease by use of HIF-PHI in cats as well. Interestingly, concurrent intake of oral iron supplements and molidustat in people resulted in a significantly reduced bioavailability of molidustat and markedly decreased endogenous EPO production. This effect markedly reduced if oral iron supplementation was administered 4 h prior molidustat [40](#). Although comparable data for cats are not available, concurrent administration of molidustat and oral iron supplements should be avoided until more data in cats are available.

Footnotes

Interessenkonflikt Die Autoren erklären, dass sie wiederholt Referate gehalten, als Berater tätig waren und Forschungsk Kooperationen mit verschiedenen pharmazeutischen Unternehmen (im Veterinärbereich) durchgeführt haben.

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Conclusion for practice.

Anemia is a common and clinically significant complication of chronic kidney disease (CKD) in cats, arising primarily from diminished erythropoietin (EPO) production due to progressive renal damage and loss of renal erythropoietin-producing (REP) cells. This anemia adversely affects both the longevity and quality of life of affected cats, manifesting as lethargy, decreased appetite, and reduced activity. Early detection through regular monitoring of (PCV) trends is important, as it allows timely intervention before anemia becomes severe and less responsive to treatment. The HIF-PHI molidustat is a promising new compound that can be administered safely and easily and can help to maintain quality of life in feline CKD patients.

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Hypoxie-induzierbarer Faktor Prolylhydroxylase-Inhibitor als neue Therapieoption zur Behandlung der Anämie bei feline chronischer Nierenerkrankung

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Zusammenfassung

Anämie ist eine häufige Komplikation bei Katzen mit chronischer Nierenerkrankung (chronic kidney disease, CKD), hauptsächlich aufgrund einer verminderten Produktion von Erythropoetin (EPO), die aus einem Verlust von renalen Erythropoetin-produzierenden (REP) Zellen resultiert. Anämie beeinflusst sowohl das Überleben als auch die Lebensqualität, indem sie klinische Symptome wie Lethargie und Appetitlosigkeit hervorruft. Die frühzeitige und regelmäßige Hämatokrit-Bestimmung (Htk/PCV) ist für eine Behandlungsentscheidung wichtig, bevor progressive Nierenschäden die therapeutische Wirksamkeit einschränken. Molidustat, ein Hypoxie-induzierbarer Faktor (HIF)-Prolylhydroxylase-Inhibitor (PHI), ist ein neues orales Medikament, das die endogene EPO-Synthese stimuliert. Die Behandlung sollte jedoch begonnen werden, wenn die REP-Zellen noch funktionsfähig sind, mit PCV-Zielwerten zwischen 30–40%. Eine regelmäßige Überwachung gemäß den Richtlinien der International Renal Interest Society (IRIS) ist entscheidend, um den PCV-Wert innerhalb des Zielbereichs zu halten und zu bestimmen, wann die Behandlung begonnen, beendet oder wiederaufgenommen werden sollte. Darüber hinaus ist die Überwachung des Eisenstatus wichtig, da sowohl funktioneller als auch absoluter Eisenmangel bei Katzen mit CKD auftreten können und die Antwort auf die Behandlung beeinträchtigen können.

Schlüsselwörter: Katze, HIF-Inhibitor, Molidustat, Hämatokrit, Therapie, PCV

Einleitung

Die chronische Nierenerkrankung (chronic kidney disease, CKD) tritt bei älteren Katzen häufig auf [1](#). Der fortschreitende Nierenfunktionsverlust führt zu vielen systemischen Komplikationen, wobei die Anämie besonders wichtig ist, da sie maßgeblich zur Verschlechterung der Lebensqualität und der Überlebensrate beiträgt [1](#). Die Pathogenese der Anämie bei CKD ist multifaktoriell [1](#). Sie wird hauptsächlich durch eine verminderte Produktion von Erythropoetin (EPO) verursacht, die auf den Verlust funktioneller renaler EPO-produzierender Zellen (REP-Zellen) zurückzuführen ist [2](#). Chronische Entzündung, urämische Toxine, Nährstoffmängel und gastrointestinale Blutverluste können ebenfalls zur Anämie beitragen [1](#). Ohne frühzeitige Diagnose und Behandlung schreitet die Anämie voran und führt zu klinischen Symptomen wie Lethargie, Anorexie und Gewichtsverlust, die die Lebensqualität der Katze erheblich verringern [3](#).

Mit der Einführung neuer Therapieoptionen wie Molidustat, einem Hypoxie-induzierbaren Faktor-Prolylhydroxylase (HIF-PH)-Inhibitor, eröffnet sich eine neue Möglichkeit zur Behandlung der Anämie bei Katzen mit CKD. Molidustat wirkt, indem es die endogene EPO-Produktion steigert;

seine Wirksamkeit hängt jedoch von der Anwesenheit funktioneller REP-Zellen ab, wodurch eine frühzeitige Intervention entscheidend wird [4](#). In diesem Kontext wird die Bedeutung einer regelmäßigen Überwachung betroffener Tiere besonders deutlich – insbesondere das Tracking des Zentrifugenhämatokrits (PCV, packed cell volume) oder Hämatokritwert (HCT) im Zeitverlauf. Im Folgenden wird der Begriff PCV sowohl für PCV als auch für HCT unabhängig von der Messmethode verwendet. Eine frühzeitige Erkennung eines sinkenden PCV ermöglicht die rechtzeitige Therapieeinleitung und optimiert somit die Behandlungsergebnisse [4](#).

Diese Arbeit beschreibt die Pathophysiologie der Anämie bei feline CKD, ihre klinische Bedeutung und die Notwendigkeit einer frühzeitigen Erkennung und Intervention. Sie liefert zudem eine detaillierte Übersicht über das Wirkprinzip von Molidustat, die IRIS-Behandlungsempfehlungen und wichtige Hinweise zu Nebenwirkungen und Eisenpräparaten.

Pathophysiologie der Anämie bei chronischer Nierenerkrankung

Anämie ist eine häufige und klinisch bedeutende Komplikation der CKD bei Katzen besonders in fortgeschrittenen Stadien (IRIS-Stufen 3–4). Etwa 30–65% der Katzen mit CKD entwickeln im Verlauf der Nierenerkrankung eine Anämie [5](#) [6](#) [7](#) [8](#). Sie ist typischerweise normozytär, normochrom und nicht regenerativ und wird hauptsächlich durch die verringerte Synthese von Erythropoetin (EPO) verursacht, einem Hormon, das vorwiegend von spezialisierten interstitiellen Nierenfibroblasten, den Nieren-REP-Zellen (Regenerations- und Progenitorzellen), produziert wird [9](#). REP-Zellen detektieren die Sauerstoffspannung im Nierengewebe und reagieren auf Hypoxie, indem sie die EPO-Produktion steigern, wodurch die Erythropoese im Knochenmark angeregt wird [9](#). Bei CKD gibt es einen progressiven Verlust und eine myofibroblastische Transformation der REP-Zellen, was schließlich zu einer unzureichenden Produktion von Erythrozyten (RBC) und zur Bildung von Narbengewebe führt [9](#).

Darüber hinaus tragen REP-Zellen bei CKD zur Entstehung der Anämie bei, indem sie aktiv proinflammatorische Zytokine produzieren [9](#). Proinflammatorische Zytokine wie Interleukin-6 (IL-6) und Tumornekrosefaktor-Alpha (TNF- α) können die Erythropoese hemmen und die Reaktion des Knochenmarks auf EPO verringern [10](#). Diese Zytokine induzieren ebenfalls die hepatische Produktion von Hepcidin, einem Hormon, das die Eisenmobilisierung aus Makrophagen hemmt, den Eisentransport sowie die intestinale Eisenaufnahme reduziert und letztlich zu einem funktionellen Eisenmangel bzw. eisenlimitierter Erythropoese trotz adäquatem Körpereisen führt [11](#).

Urämische Toxine, darunter Indoxyl-Sulfat und p-Cresyl-Sulfat, haben nachweislich direkte toxische Effekte auf erythroide Vorläuferzellen und könnten auch die Signaltransduktion des EPO-Rezeptors stören [12](#) . Ferner führt CKD oft zu einer verkürzten Lebensdauer der RBC durch oxidativen Stress, veränderter Membranpermeabilität und erhöhter osmotischer Fragilität der Erythrozyten. Weitere Faktoren, die bei feline CKD eine Anämie begünstigen, sind gastrointestinale Blutungen, die sekundär durch urämische Toxine, Magenschleimhautschäden oder durch Medikamente entstehen können [1](#) . Chronische gastrointestinale Blutungen können sowohl einen Eisenmangel als auch die Verschlechterung der Anämie verursachen [1](#) .

Auswirkungen von Anämie auf Langlebigkeit und Lebensqualität bei Katzen

Während eine Anämie bei allen Tierarten mit einer CKD vorkommt, reagieren Katzen selbst auf geringe Veränderungen der RBC-Konzentration, sodass ein PCV am unteren Limit des Referenzintervalls bereits zu einer Nierenischämie und zum Fortschreiten der CKD führen kann [4](#) . Obwohl die Anämie bei Katzen mit fortschreitender CKD tendenziell langsam auftritt, bleibt sie häufig unbemerkt, bis sie mittel- bis hochgradig ist [4](#) .

Die häufig berichteten klinischen Anzeichen einer Anämie bei Katzen umfassen Lethargie, verminderter Appetit, reduziertes Pflegeverhalten, blasse Schleimhäute, Tachykardie und verminderte Aktivität. Besitzer beschreiben ihre Katzen häufig als „weniger interaktiv“, „ruhiger“ oder „langsamer“, was subjektive, aber klinisch wichtige Indikatoren für Lebensqualität darstellen. Diese Symptome überschneiden sich oft mit denen der CKD selbst. Dennoch besteht eine Wechselwirkung zwischen der Entwicklung einer Anämie und dem Fortschreiten der CKD [4](#) . Anämie reduziert die Sauerstofftransportkapazität des Blutes, was zu Gewebehypoxie führt. In Verbindung mit bereits beeinträchtigter Nierenperfusion verstärkt der verminderte Sauerstofftransport die tubuläre Dysfunktion und kann zur progressiven Nierenschädigung durch Hypoxie, Entzündung und Fibrose beitragen [4](#) .

Mehrere Studien haben eine starke Assoziation zwischen niedrigem PCV und kürzeren Überlebenszeiten bei Katzen mit CKD gezeigt [13](#) [14](#) [15](#) [16](#) . So wurde berichtet, dass Katzen mit einem PCV unter 25% [14](#) bzw. 26% [16](#) (entsprechend einer milden Anämie; unterer Referenzwert 30% in der Studie) signifikant reduzierte mediane Überlebenszeiten im Vergleich zu solchen mit Werten über diesem Schwellenwert [16](#) aufwiesen. Anämie (d. h. ein PCV unter dem unteren Referenzwert von 27% in dieser Studie) [18](#) oder sogar ein medianer Hämatokritwert < 31%, wie im Cox-Multivariaten-Regressionsmodell definiert [17](#) , ist mit einer Progression der CKD assoziiert. Im Gegensatz dazu ist ein minimaler Anstieg des PCV um nur 1% mit einem reduzierten Risiko der Verschlechterung der CKD um 10% verbunden [18](#) . Obwohl die PCV-Grenzwerte in den Studien

zwischen 25% und 31% variieren (aufgrund unterschiedlicher Referenzintervalle und statistischer Methoden), weisen sie alle darauf hin, dass der PCV nahe am unteren Limit des Referenzintervalls liegt. Daher wird empfohlen, dass der PCV bei Katzen mit CKD nicht unter das untere Referenzintervall fallen sollte ⁴.

Die Lebensqualität ist gleichermaßen wichtig wie die Überlebensrate bei feline CKD, da klinische Zeichen einer unbehandelten Anämie das Leid der Besitzer verstärken und Entscheidungen hinsichtlich der Euthanasie beeinflussen ³. Es ist empirisch belegt, dass die Behandlung der Anämie bei vielen Katzen zur klinischen Besserung führt, sobald die RBC-Konzentration auf normale Werte zurückkehrt ⁴.

Bedeutung der regelmäßigen PCV-Messung für die frühzeitige Erkennung einer Anämie

Bei feline CKD spielt die PCV-Überwachung eine entscheidende Rolle für die frühzeitige Erkennung einer Anämie. Während ein einzelner PCV-Messwert lediglich eine Momentaufnahme des Erythrozytenstatus darstellt, liefert die Langzeitverfolgung des PCV-Trends wesentlich bessere Erkenntnisse. Tierärzt*innen können damit früher eine geringe Reduktion der RBC erkennen, bevor die Anämie klinisch sichtbar wird und irreversible Nierenschädigungen oder der Verlust von EPO-produzierenden Zellen entstehen ⁴. Das Monitoring von Trends statt einzelner Werte ermöglicht die frühzeitige Erkennung eines Abfalls des PCV, noch bevor kritische Schwellenwerte unterschritten werden.

PCV-Messungen werden durch biologische und analytische Variation beeinflusst. Die intraindividuelle biologische Variation eines PCV kann bei gesunden Katzen bis zu 5,5% betragen; erst bei Überschreitung einer Veränderung von 15,5% wird eine Änderung klinisch relevant (d.h. dem Referenzänderungswert; engl: Reference change value) ¹⁹. Beispiel: Bei einer Katze mit einem PCV von 0,30 l/l stellt eine Abnahme um 15% (0,045 l/l) auf 0,25 l/l einen klinisch relevanten Abfall dar, der nicht allein durch biologische und analytische Schwankungen erklärbar ist.

Bei CKD-Katzen können weitere Faktoren die PCV-Variabilität beeinflussen: Dehydration, die bei CKD häufig vorkommt, kann den PCV kurzfristig erhöhen und eine geringgradige Anämie verschleiern. Umgekehrt kann eine Hyperhydratation oder Flüssigkeitstherapie die RBC-Masse verdünnen und zu falsch niedrigen PCV-Ergebnissen führen. Trotzdem bleibt bei klinisch stabilen CKD-Katzen der Referenzänderungswert zumindest ein Leitfaden, nach dem PCV-Änderungen als klinisch relevant gelten können.

Laut den Richtlinien der International Renal Interest Society (IRIS) ist eine regelmäßige PCV-Überwachung bei Katzen mit diagnostizierter CKD von größter Bedeutung. Eine Bewertung zum Zeitpunkt der Diagnose sowie je nach Krankheitsstufe und Stabilität wird eine Messung alle 3–6 Monate empfohlen [20](#). Bei PCV-Abnahme oder dem Auftreten klinischer Anämiezeichen kann eine häufigere Kontrolle sinnvoll sein. Die Aufzeichnung von PCV-Trends erleichtert zudem die Bewertung der Therapieeffektivität, sobald die Behandlung begonnen wurde.

Neben dem PCV liefert das komplette Blutbild (CBC) wertvolle ergänzende Informationen, darunter RBC-Indizes, Retikulozytenzahlen und Retikulozytenindizes [21](#).

Frühe Intervention und Funktion der REP-Zellen

Die Effektivität der Therapie der Anämie bei CKD ist eng mit dem funktionellen Zustand der renalen EPO-produzierenden Zellen (REP-Zellen) verknüpft. Diese spezialisierten interstitiellen Fibroblasten sind für die Erkennung von Gewebehypoxie und die Produktion von EPO verantwortlich – einem Glycoproteinhormon, das die Erythropoese im Knochenmark stimuliert [9](#). Mit fortschreitender CKD führen die Nierenfibrose und der Verlust funktioneller Nephrone zur Depletion der REP-Zellen, was die endogene EPO-Produktion reduziert. Folglich verschlechtert sich die Anämie und die Effektivität der Therapie nimmt ab.

Molidustat, ein HIF-Prolylhydroxylase-Inhibitor (HIF-PHI), hat sich als vielversprechendes therapeutisches Mittel zur Behandlung der Anämie erwiesen, indem es die endogene EPO-Produktion bei Menschen [22](#) und Katzen [23](#) [24](#) stimuliert. Es hemmt Prolylhydroxylase-Enzyme, die Hypoxie induzierbare Faktoren (HIF) unter normalen Sauerstoffbedingungen abbauen. Stabilisierte HIF gelangen in den Zellkern und fördern die Transkription von Genen, einschließlich des EPO-Gens. Durch die Simulation eines hypoxischen Zustands erhöht Molidustat die endogene EPO-Synthese, auch im Nierengewebe mit relativ hohem Sauerstoffgehalt aufgrund reduzierter Sauerstoffverwertung [25](#). Die Behandlung mit HIF-PHI gilt als physiologischere Alternative zur Ersatztherapie mit rekombinantem humanem EPO [4](#).

Wie gesagt hängt die Effektivität des Medikaments jedoch vom Vorhandensein vitaler REP-Zellen ab, die auf die HIF-Stabilisation reagieren können. Beginnt die Behandlung erst spät in der CKD, wenn REP-Zellen stark reduziert oder durch Fibrose ersetzt sind, wird die Fähigkeit von Molidustat, ausreichendes EPO zu erzeugen, stark eingeschränkt. Damit ist eine frühzeitige Therapieeinleitung – idealerweise, wenn die Anämie noch mild ist und die REP-Zellzahl noch beträchtlich ausgeprägt ist – entscheidend für optimale Behandlungsergebnisse [4](#) [17](#) [18](#).

Aus praktischer Sicht ist die frühzeitige Erkennung über die regelmäßige PCV-Trendüberwachung, wie oben beschrieben, entscheidend. Tierärzte*innen sollten CKD-Katzen engmaschig beobachten und die Therapie umgehend einleiten, sobald der PCV beginnt zu sinken – idealerweise vor dem Abfallen unter 30% (0,30 l/l), denn ab diesem Wert beginnt die Sauerstoffversorgung der Niere rasch zu sinken, was bei Hunden gezeigt [26](#) wurde. Dieser Schwellenwert von 30% (0,30 l/l) ist mit den IRIS-Richtlinien und dem Molidustat-Behandlungsalgorithmus abgestimmt, die den PCV-Zielwert von 30–40% (0,30–0,40 l/l) für optimale klinische Resultate empfehlen [20](#).

Die Aufrechterhaltung der REP-Zellfunktion hat über die reine Anämiekorrektur hinaus wahrscheinlich noch weitere Vorteile. Durch verbesserte Nierenoxygenierung und die Reduktion der Hypoxie bedingten Fibrose könnten Therapien, die den HIF-Weg modulieren, das Fortschreiten der CKD verlangsamen – jedoch bedarf diese Hypothese weiterer Untersuchungen.

Molidustat-Monitoringprotokoll und Ziel-PCV

Die Einführung von Molidustat als Behandlungsoption bei Anämie bei CKD-Katzen erfordert ein strukturiertes Monitoringprotokoll, um Sicherheit, Wirksamkeit und optimale klinische Ergebnisse zu gewährleisten. Die International Renal Interest Society (IRIS) stellt umfassende Richtlinien für die Anwendung von Molidustat bei Katzen bereit. Regelmäßige hämatologische Kontrollen sind angeraten, um den PCV im therapeutischen Zielbereich von 30–40% (0,30–0,40 l/l) zu halten [20](#).

Die Aufrechterhaltung des PCV innerhalb dieses Bereichs ist entscheidend. Ein PCV unter 30% (0,30 l/l) gilt allgemein als anämisch und ist mit klinischen Symptomen verbunden, die die Lebensqualität beeinträchtigen. PCV-Werte über 40% (0,40 l/l) geben Anlass zur Sorge hinsichtlich einer Erythrozytose, die die Blutviskosität erhöhen und das Risiko thromboembolischer Komplikationen erhöhen kann. Obwohl in der ersten Studie zur Anwendung von Molidustat bei einer Gruppe CKD-Katzen (n=15) keine Erythrozytose auftrat, ist es das Ziel, die Anämie adäquat zu korrigieren und gleichzeitig eine übermäßige Erythrozytose zu vermeiden [20](#).

Gemäß der vorläufigen Zulassung durch die US-amerikanische Arzneimittelbehörde (FDA) [27](#) empfiehlt der IRIS-Behandlungsalgorithmus für Molidustat einen Therapiebeginn mit einer Dosis von 5 mg/kg p.o. Bei Katzen mit einem persistierenden PCV<25% (0,25 l/l) oder einem sinkenden PCV-Trend, insbesondere bei Werten nahe oder unter 25–28% (0,25–0,28 l/l), wird Molidustat über einen 28-tägigen Zyklus täglich verabreicht. Die nachfolgenden PCV-Messungen sollten anfänglich alle 2 Wochen erfolgen, um das Ansprechen auf die Therapie zu überwachen und eine Behandlungspause einzuleiten. Steigt der PCV rasch an oder übersteigt 40% (0,40 l/l), kann eine

vorübergehende Unterbrechung der Molidustat-Behandlung erforderlich sein, um Risiken zu minimieren [20](#). Die Behandlung sollte wieder aufgenommen werden, sobald der PCV unter 30% (0,30 l/l) sinkt.

Zusätzlich zum PCV kann die Überwachung der Hämoglobinkonzentration und der Retikulozytenzahl mit Retikulozytenindizes (Hämoglobingehalt der Retikulozyten) weitere Einblicke in die erythropoetische Reaktion ermöglichen. Eine regenerative Reaktion, die durch eine erhöhte Retikulozytenzahl gekennzeichnet ist, bestätigt in der Regel die Wirksamkeit der Behandlung. Nach Ansicht der Autoren können eine suboptimale Retikulozytenreaktion, Hypochromasie oder Mikrozytose auf einen Eisenmangel oder andere Komplikationen hinweisen und erfordern eine weiterführende Diagnostik. Die regelmäßige Kontrolle der Nierenfunktion, des Blutdrucks und des Hydratationsstatus wird während der gesamten Therapie empfohlen [20](#), da Veränderungen dieser Parameter die hämatologischen Ergebnisse und die allgemeine Stabilität des Patienten beeinflussen können. Da Molidustat über die Modulation von Hypoxie induzierbaren Faktoren wirkt, müssen seine Effekte auf die systemische Gefäßdynamik hinsichtlich Bluthochdruck oder damit zusammenhängenden Nebenwirkungen, wie sie beim Menschen gezeigt wurden, aufmerksam überwacht werden [28](#).

Mögliche Nebenwirkungen und Sicherheitsüberwachung

Während Molidustat eine vielversprechende Alternative zu herkömmlichen Erythropoese-stimulierenden Agentien (ESA) für die Anämie-Behandlung feline CKD darstellt, ist es wie bei jeder Therapie notwendig, auf potenzielle Nebenwirkungen zu achten.

Eines der Hauptbedenken bei der Molidustattherapie ist das Risiko einer Erythrozytose (Polyzythämie), d. h. einer übermäßigen Erhöhung des PCV. Ein erhöhter PCV erhöht die Blutviskosität, was bei den Katzen zu thromboembolischen Ereignissen, Bluthochdruck und beeinträchtigter Mikrozirkulation führen kann. Regelmäßiges PCV-Monitoring und die Einhaltung der von IRIS empfohlenen therapeutischen Zielbereiche (30–40%, 0,30–0,40 l/l) sind daher unerlässlich [20](#).

Gastrointestinale Symptome, einschließlich Erbrechen und Appetitlosigkeit, wurden bei einigen mit ESA behandelten Katzen berichtet und können auch mit Molidustat in seltenen Fällen auftreten [23](#) [24](#). Tritt ein solches Symptom auf, sollte eine symptomatische Behandlung eingeleitet werden. Eine weitere mögliche Nebenwirkung ist Bluthochdruck, möglicherweise aufgrund chronischer Hypoxie durch HIF-Signalgebung in der Halsschlagader [28](#). Daten beim Menschen legen nahe, dass der Einsatz von HIF-PHI mit einem erhöhten Risiko von Bluthochdruck im Vergleich zu Placebo

verbunden ist, jedoch weniger als bei der Behandlung mit ESAs wie Erythropoetin oder Darbepoetin [28](#). Obwohl die Daten zu Katzen begrenzt sind, sollte der Blutdruck regelmäßig während der Behandlung überwacht werden, um eine Hypertonie frühzeitig zu erkennen und zu behandeln.

Im Gegensatz zu der Verabreichung von rekombinantem Erythropoetin oder von Darbepoetin ist die orale Einnahme von Molidustat nicht mit einem Risiko der Antikörperbildung gegen EPO verbunden, da die körpereigene EPO-Produktion stimuliert wird. Im Gegensatz dazu gibt es sogar wenige Fallberichte bei Menschen mit Anti-EPO-Antikörpern aufgrund früherer Behandlung mit rekombinantem menschlichem Erythropoetin, die erfolgreich mit einem HIF-PHI behandelt wurden [29](#). Es ist wichtig auf Anzeichen eines ausbleibenden Therapieerfolgs oder einer unerwarteten Verschlimmerung der Anämie zu achten, weil die Ätiologie der Anämie bei CKD-Patienten multifaktoriell ist. Die Nierenfunktion sollte auch regelmäßig überwacht werden, weil Langzeitdaten in der Behandlung von Katzen mit HIF-PHI noch nicht verfügbar sind. Obwohl nephroprotektive Effekte durch verbesserte Gewebeoxygenierung und Eisenstoffwechsel [30](#) vorstellbar sind und die Myofibroblasten-Transformation von REP-Zellen bis zu einem gewissen Grad reversibel ist [9](#), ist eine enge Nachverfolgung erforderlich, um schädliche Veränderungen zu erkennen.

Überlegung zur Eisensupplementierung

Viele Katzen mit CKD weisen eine eisenlimitierte Erythropoese auf. Neuere Studien legen nahe, dass der Eisenmangel durch eine chronische Entzündung eher funktionell als absolut bedingt ist [31](#), [32](#). Dennoch führt die Stimulation der Erythropoese durch EPO, Darbepoetin oder HIF-PHI zu einem erhöhten Eisenbedarf, der zu einem absoluten Eisenmangel führen kann, wenn die Eisenspeicher nicht ausreichend sind. In der Praxis stehen Laborparameter zur Überprüfung des Eisenstatus nicht leicht zur Verfügung (z. B. Ferritin). Daher beginnen Kliniker häufig empirisch mit einer Eisensupplementierung, wenn die Retikulozytenantwort 2–3 Wochen nach Beginn eines ESA nicht ausreichend ist, der PCV niedrig bleibt oder wenn Zeichen eines Eisenmangels, z. B. etwa ein niedriges mittleres Zellvolumen (MCV) der Erythrozyten, auftreten, was auf eisenlimitierte Erythropoese hinweisen kann. Moderne Hämatologiegeräte reportieren den Hämoglobingehalt der Retikulozyten (CHr, Ret-H e, Retic-HGB). Dieser Parameter zeigt die Eisenverfügbarkeit für die Erythropoese in den letzten 2–4 Tagen [21](#) [33](#) [34](#) und ermöglicht – verglichen mit dem MCV – eine deutlich frühere Erkennung des Eisenmangels.

Eine potenzielle Eisensupplementation zusätzlich zur Molidustat Therapie ist Teil des Behandlungsalgorithmus der IRIS-Richtlinien [20](#). Dennoch gibt es auch Risiken in diesem

Zusammenhang. Die orale Verabreichung von Eisen (Eisen(II)-sulfat, Eisenglukonat) wird bei stabilen Patienten wegen geringerer Kosten und höherer Sicherheit bevorzugt, kann jedoch gastrointestinale Nebenwirkungen verursachen [35](#) und die Absorption bei Katzen ist begrenzt [1](#). Außerdem kommt oral verabreichtes Eisen bei einem funktionellen Eisenmangel nicht bzw. weniger in der Erythropoese an. Eisendextran wird häufig intramuskulär verabreicht, jedoch können lokale Reaktionen am Injektionsort auftreten [35](#). Eine Überdosierung von Eisen birgt ebenfalls die Gefahr von oxidativem Stress oder intestinaler Dysbiose [36](#). Weitere Studien sind daher erforderlich, um die Indikation für eine Eisensupplementierung besser zu charakterisieren.

Bei Menschen gibt es Hinweise, dass HIF-PHI die Eisenverfügbarkeit optimieren, indem sie die hepatische Hepcidin-Produktion senken, was zu einer höheren Eisenmobilisierung aus Makrophagen und intestinaler Absorption führt [37](#). HIF-PHI reduzieren indirekt Hepcidin durch Stimulation der Erythropoese. Dies führt zu erhöhten Konzentrationen von Erythroferron, einem Hormon aus erythroiden Vorläuferzellen, das die hepatische Hepcidin Expression hemmt [38](#). Bei Menschen führt die Verwendung von HIF-PHI im Vergleich zu injizierbarem EPO zu einer Reduktion des Bedarfs an intravenöser Eisensupplementierung und damit zu weniger Krankenhausaufenthalten [39](#). Es besteht also Hoffnung, dass auch bei Katzen die Notwendigkeit einer Eisensupplementierung durch den Einsatz von HIF-PHI sinkt. Interessanterweise führte die simultane Einnahme von oralem Eisen und Molidustat beim Menschen zu einer signifikant reduzierten Bioverfügbarkeit von Molidustat und stark nachlassender körpereigener EPO-Produktion. Dieser Effekt konnte deutlich reduziert werden, wenn oral Eisen 4 h vor Molidustat eingenommen wurde [40](#). Obwohl vergleichbare Daten für Katzen nicht vorliegen, sollte die gleichzeitige Verabreichung von Molidustat und oralem Eisen vermieden werden, bis weitere Daten vorliegen.

Fazit für die Praxis.

Anämie ist eine häufige und klinisch signifikante Komplikation der chronischen Nierenerkrankung (CKD) bei Katzen, die hauptsächlich aus einer verminderten EPO-Produktion (EPO: Erythropoetin) aufgrund progressiver Nierenschäden und Verlust von renalen EPO-produzierenden (REP) Zellen resultiert. Diese Anämie beeinträchtigt die Lebensdauer und die Lebensqualität der betroffenen Katzen negativ, manifestiert sich als Lethargie, verringerter Appetit und reduzierte Aktivität. Frühzeitige Erkennung durch regelmäßige Überwachung des PCV-Trends ist wichtig, da sie eine rechtzeitige Intervention ermöglicht, bevor die Anämie schwerwiegend wird und schlechter auf die Therapie anspricht. Der HIF-PHI Molidustat ist ein vielversprechendes neues Medikament, das sicher und einfach verabreicht werden kann und dazu beitragen kann, die Lebensqualität von Katzen mit chronischer Nierenerkrankung zu erhalten.

Changes in Systolic Blood Pressure over Time in Healthy Cats and Cats with Chronic Kidney Disease

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Abstract

Abstract

Background

Hypertension is a common problem in older cats, most often associated with chronic kidney disease (CKD). Cross-sectional studies have suggested that blood pressure in cats increases with age.

Hypothesis/Objectives

To determine whether blood pressure in cats increases with age and whether this occurs independently of the presence of CKD. To investigate risk factors for developing hypertension.

Animals/Subjects

Two hundred and sixty-five cats with CKD and 133 healthy cats ≥ 9 years were retrospectively identified.

Methods

Four groups were created according to status at initial evaluation (CKD or healthy) and blood pressure at the last included visit (normotensive [NT] or developed hypertension [DH]): Healthy-NT, Healthy-DH, CKD-NT and CKD-DH. Systolic blood pressure (SBP) over time slopes were compared with 0 and between groups. Risk factors for the development of hypertension were investigated, and associations of biochemical and clinical variables with SBP were examined.

Results

Cats that were hypertensive at CKD diagnosis ($n = 105$) were not included in further analyses. Twenty-seven cats with CKD and 9 healthy cats developed hypertension ≥ 3 months after diagnosis of CKD or their first visit. Systolic blood pressure significantly increased with age in all cats ($P < .001$). Healthy cats were at less risk than cats with CKD to become hypertensive (hazard ratio 0.2, $P < .001$), with creatinine being an independent risk factor for the development of hypertension.

Conclusions and Clinical Importance

The high prevalence of hypertension in azotemic cats in this study shows the importance of monitoring of SBP in elderly cats, and in particular in cats with CKD.

Keywords: Feline, Hypertension, Renal disease, Risk factors

Abbreviations

beats per minute

chronic kidney disease

developed hypertension

interquartile range

remained normotensive

packed cell volume

renin-angiotensin-aldosterone system

systolic blood pressure

urine specific gravity

Hypertension is a common problem in older cats but, in contrast to humans, where 95–99% cases of hypertension are considered to be primary or essential hypertension, hypertension in cats is most often associated with underlying diseases such as chronic kidney disease (CKD).¹ Increased serum creatinine concentrations occur in up to 74% of hypertensive cats and 19–65% of cats with CKD are hypertensive.^{2, 3, 4, 5} In humans, the prevalence of CKD increases with increasing age, and the same is true for cats.^{6, 7} Mean values of systolic blood pressure (SBP) also increase with age in most human populations, as does the prevalence of clinical hypertension.^{8, 9} In cats, it has been suggested that age is a predisposing factor for the development of hypertension. A cross-sectional study grouping healthy cats into 3 age groups (<5, 5–10 and older than 10 years) showed higher SBP in the older age groups, although renal function was not assessed in all cats that were classified as healthy.¹⁰ There is a positive correlation between age and mean arterial pressure, but the age distribution of the cats included in this study is not known, which precludes interpretation of the results.¹¹

Chronic kidney disease and hypertension are both common in older cats and current consensus is that hypertension is secondary to CKD, although the possibility still exists that the hypertension occurs independently of the presence of CKD or actually causes the renal injury.¹² The question can be raised whether this increase in prevalence of hypertension and CKD in older cats is purely age related, or whether the development of hypertension in most cats is secondary to an inability of the kidney to regulate blood pressure.¹² Investigations into the risk factors for blood pressure to increase in both healthy older cats and cats with CKD may aid in answering this question. The hypothesis of this study is that the blood pressure increases with age in senior cats and that the rate of this increase, and therefore the occurrence of clinically significant hypertension, is greater in cats with CKD.

Materials and Methods

Case Selection

Cats that were 9 years of age or older and examined for the first time at 2 first opinion practices in central London (People's Dispensary for Sick Animals in Bow and Beaumont Sainsbury's Animal Hospital in Camden) between August 2000 and August 2012 were retrospectively identified. All cats had a follow-up period of ≥ 3 months and had visited 1 of the 2 veterinary practices a minimum of 3 times. On every visit, a full history was obtained and a physical

examination was performed. Systolic blood pressure was measured at all visits using a noninvasive Doppler technique¹ after a period of acclimatization and the SBP was calculated as the average of 5 consecutive readings. Indirect fundoscopy after applying 1 drop of tropicamide 1% to both eyes was performed at the end of the consultation if average SBP was ≥ 160 mmHg. At recruitment into the study, the owner consented to collection of blood samples via jugular venipuncture and urine samples by cystocentesis. Blood samples were collected into lithium heparin and held on ice (4°C) for a maximum of 6 hours before centrifugation² and separation. Plasma biochemistry was performed at an external laboratory³. Total thyroxine concentration was assessed in all cats that showed clinical signs of hyperthyroidism (eg, polyphagia, weight loss, tachycardia, and palpable goiter) or had plasma biochemical findings that raised concern (increased alanine aminotransferase or alkaline phosphatase activity). If the bladder was palpable and a urine sample could be collected, a dipstick evaluation was performed and a urine specific gravity (USG) was measured in all cats. In addition to this, the sediment of samples was evaluated microscopically and, if bacteria or an active sediment was found, urine was sent for bacterial culture and sensitivity testing. All cats diagnosed with hyperthyroidism (total T4 > 55 nmol/L) were excluded from the study, unless a thyroidectomy had been performed ≥ 90 days before inclusion, to allow stabilization of glomerular filtration rate.¹³ Visits from cats that were being treated with glucocorticoids, nonsteroidal anti-inflammatory drugs, antihypertensive medications, erythropoietin, or intravenous fluid therapy were excluded.

For the purposes of this study, systemic hypertension was diagnosed when cats had a SBP ≥ 170 mmHg on 2 consecutive visits (reliably placing them in the category that is at moderate risk to develop target organ damage)¹² or SBP ≥ 160 mmHg with concurrent evidence of hypertensive retinopathy or choroidopathy on indirect fundoscopy. All hypertensive cats were started on amlodipine at a dose of 0.625 mg once daily at the time of diagnosis. CKD was diagnosed in cats with renal azotemia (plasma creatinine concentration ≥ 2.0 mg/dL either on 2 consecutive visits or in conjunction with USG <1.035). All owners of cats diagnosed with CKD were advised to feed their cat a commercially available renal diet. Cats were considered healthy if no significant clinical signs were reported by the owner, no abnormalities were found on clinical exam, and the blood results showed no evidence of CKD or other pathologies. Cats were classified according to the diagnosis of CKD and their SBP—the CKD-normotensive group (CKD-NT-group) was composed of cats diagnosed with CKD that did not meet the hypertensive criteria during follow-up, cats in the CKD-developed-hypertension group (CKD-DH-group) developed hypertension ≥ 3 months after diagnosis of CKD, cats that met the criteria for being healthy were included in the healthy-normotensive group (Healthy-NT-group), and cats were included in the healthy-developed-hypertension group (Healthy-DH-group) if the criteria for being healthy were met, but the cat was diagnosed with idiopathic hypertension during follow-up of ≥ 3 months.

Re-examination of cats diagnosed with CKD was offered at a maximum of 8 week intervals and blood- and urine samples were obtained at a maximum of 16 week intervals. Re-examination of healthy cats was offered at a maximum of 6 month intervals, at which point blood- and urine samples were obtained. If a healthy cat was diagnosed with CKD, its penultimate visit before developing azotemia was excluded from the analysis as it is likely that nonazotemic CKD was already present at that stage.

Statistical Analyses

All statistical analyses were performed using commercially available software⁴. *P*-values ≤ 0.05 were considered significant. Descriptive statistics are presented to define the population, SBP, biochemical and clinical variables at the

first and last visit, and are presented as median and interquartile range (IQR). Comparisons between groups were performed using a nonparametric Kruskal-Wallis or parametric ANOVA with *post-hoc* comparisons where appropriate. Continuous variables were graphically assessed for normality and values were log transformed if normality criteria were not met to allow use in the linear mixed effects model and the time-dependent Cox proportional hazards model.

Systolic blood pressure at the first visit between all 4 groups was compared using a Kruskal–Wallis test with a Bonferroni adjusted *post-hoc* comparison. The rates of change in SBP over time and associations between SBP and biochemical (sodium, chloride, phosphate, total calcium, potassium, creatinine, packed cell volume (PCV), albumin, cholesterol) and clinical (weight, heart rate) variables over time were compared between all groups using a linear mixed effects model with subjects as random effects, and time, group and sex as fixed effects. In the CKD-DH-group and Healthy-DH-group, the actual SBP measurement that resulted in the diagnosis of hypertension was not included in the analysis. In the CKD-NT-group and Healthy-NT-group all available visits were included. Not all cats had all information available at all visits as, in general, blood samples were taken every other visit, but having a SBP measurement was a requirement for the visit to be included in the models. Change in SBP over time is expressed as $\Delta\text{mmHg}/100$ days. A time-dependent Cox proportional hazards model and Kaplan–Meier curves were used to assess the association of the biochemical and clinical variables with the risk of becoming hypertensive. The assumption of proportional hazards was checked for all variables included in the model and cases were censored if they died, were lost to follow-up (defined as not seen for >6 months and not contactable by telephone), or if the study end point was reached. To investigate the independent association of the significant variables with SBP (linear mixed model) and the risk of becoming hypertensive (time dependent Cox proportional hazards model), all significant variables at the $P < .10$ level were included in a multivariable analysis.

Results

A total of 265 cats were diagnosed with azotemic CKD between August 2000 and August 2012 and were seen for 3 or more visits, over the course of more than 3 months. Of these cats, 105 were hypertensive at diagnosis of CKD or were diagnosed with systemic hypertension within 3 months of CKD diagnosis. These cats were excluded from further analyses. Of the remaining 160 cats with CKD, 133 (83%) remained normotensive throughout follow-up (CKD-NT group), and 27 cats of the 160 cats that were initially normotensive (17%) developed hypertension ≥ 3 months after CKD diagnosis (CKD-DH group). A total of 133 cats were included as healthy cats in this study of which 9 (7%) developed hypertension (Healthy-DH group), and 124 healthy cats remained normotensive throughout follow-up (Healthy-NT group).

Demographic Data

An overview of the clinical and biochemical variables at baseline and at the last visit (NT-groups: last included visit; DH-groups: hypertensive visit) is presented in Tables ¹ and ². The majority of cats included in the 4 groups were domestic short hair ($n = 229$), followed by domestic long hair ($n = 34$), and 147 cats were female (of which 4 not neutered) and 146 cats male (of which 4 not neutered). Of the 105 cats that were diagnosed as hypertensive at or within 3 months of CKD diagnosis, 73 (70%) had visible retinal lesions, whereas of the CKD-DH group 52% (14 cats) and of the Healthy-DH group 3 cats (33%) had evidence of hypertensive retinopathy.

Table 1

Clinicopathologic variables for all groups at initial visit

Variable	CKD-NT (n = 133)	CKD-DH (n = 27)	Healthy-NT (n = 124)	Healthy-DH (n = 9)
Age (years)	14 (12, 16) ^a	14 (13, 15) ^a	12 (10, 13) ^b	11 (10, 14) ^{ab}
Weight (kg)	3.9 (3.3, 4.7) ^a	3.7 (3.4, 4.7) ^{ab}	4.6 (3.6, 5.3) ^b	4.7 (3.9, 5.8) ^{ab}
SBP (mmHg)	133.2 (121.2, 144.6) ^a	147.2 (140.4, 156.1) ^b	131.6 (115.0, 143.7) ^c	145.6 (139.5, 154) ^{ab}
Heart rate (bpm)	186 (165, 204)	184 (164, 204)	180 (168, 197)	180 (176, 190)
PCV (%)	34 (32, 38) ^a	36 (31, 39) ^a	39 (36, 41) ^b	36 [34, 38) ^{ab}
Albumin (g/dL)	3.1 (3.0, 3.3) ^a	3.1 (2.9, 3.2) ^{ab}	3.2 (3.1, 3.4) ^b	3.3 (2.9, 3.5) ^{ab}
Creatinine (mg/dL)	2.4 (2.2, 2.9) ^a	2.5 (2.3, 2.8) ^a	1.5 (1.4, 1.7) ^b	1.5 (1.5, 1.6) ^b
Urea (mmol/L)	16.9 (14.4, 20.2) ^a	17.0 (13.6, 20.0) ^a	9.9 (8.9, 11.2) ^b	9.5 (8.4, 10.5) ^b
Phosphate (mg/dL)	4.22 (3.63, 5.12) ^a	3.97 (3.43, 4.73) ^{ab}	3.84 (3.3, 4.73) ^b	4.12 (3.47, 4.46) ^{ab}
Total calcium (mg/dL)	10.2 (9.9, 10.4) ^a	9.9 (9.7, 10.6) ^{ab}	9.8 (9.5, 10.3) ^b	10.1 (9.8, 10.6) ^{ab}
Sodium (mEq/L)	153.0 (151.3, 154.8) ^a	151.7 (150.5, 152.5) ^a	152.4 (151.3, 153.7) ^a	151.0 (150.0, 152.5) ^b
Potassium (mEq/L)	4.10 (3.80, 4.40)	4.10 (3.90, 4.38)	3.90 (3.70, 4.20)	3.80 (3.60, 4.25)
Chloride (mEq/L)	117.8 (116.3, 120.2) ^{ab}	116.0 (114.8, 118.9) ^a	119.3 (117.2, 120.6) ^b	120.4 (116.5, 121.2) ^{ab}
Cholesterol (mg/dL)	200 (154, 239)	201 (160, 234)	178 (151, 212)	166 (154, 205)
USG	1.020 (1.017, 1.024) ^a	1.017 (1.016, 1.022) ^a	1.050 (1.038, 1.060) ^b	1.042 (1.039, 1.050) ^b

CKD-DH, CKD-developed-hypertension; CKD-NT, CKD-remained normotensive; SBP, systolic blood pressure; PCV, packed cell volume; USG, urine specific gravity.

Parameters shown are at first visit for all cats. Superscript letters identify groups, which differed significantly.

Table 2

Clinicopathologic variables for all groups at last visit

Variable	CKD-NT (n = 133)	CKD-DH (n = 27)	Healthy-NT (n = 124)	Healthy-DH (n = 9)
Follow-up (days)	316 (225, 512)	379 (281, 771)	666 (495, 1065)	939 (685, 1217)
Age (years)	15 (13, 17) ^a	15 (14, 18) ^a	14 (12, 16) ^b	15 (14, 15) ^{ab}
Weight (kg)	3.6 (3.0, 4.5) ^a	3.7 (3.2, 4.3) ^{ab}	4.3 (3.4, 5.1) ^b	4.7 (3.4, 5.7) ^b
SBP (mmHg)	138.4 (120.5, 148) ^a	181.6 (175.0, 188.8) ^b	133.4 (121.1, 146.1) ^a	181.2 (174.8, 190) ^b
Heart rate (bpm)	186 (168, 210)	180 (160, 204)	184 (176, 204)	192 (186, 208)
PCV (%)	34 (30, 38) ^a	36 (29, 39) ^{ab}	37 (34, 40) ^b	37 (36, 40) ^{ab}

Variable	CKD-NT (n = 133)	CKD-DH (n = 27)	Healthy-NT (n = 124)	Healthy-DH (n = 9)
Albumin (g/dL)	3.1 (2.9, 3.3)	3.1 (2.9, 3.3)	3.1 (3.0, 3.4)	3.2 (3.1, 3.4)
Creatinine (mg/dL)	2.5 (2.1, 3.3) ^a	2.7 (2.4, 3.0) ^a	1.4 (1.3, 1.7) ^b	1.6 (1.4, 1.9) ^b
Urea (mmol/L)	16.9 [13.6, 22.0) ^a	17.1 (15.2, 21.1) ^a	10.5 (8.8, 12) ^b	10.9 (9, 11.8) ^b
Phosphate (mg/dL)	4.15 (3.55, 5.04) ^a	4.28 (3.72, 5.16) ^a	3.81 (3.50, 4.31) ^b	4.06 (3.53, 4.56) ^{ab}
Total calcium (mg/dL)	10.4 (10.0, 10.8) ^a	10.6 (10.1, 11.0) ^a	9.9 (9.5, 10.2) ^b	9.9 (9.8, 10.2) ^{ab}
Sodium (mEq/L)	154.3 (152.2, 156.4) ^a	153.8 (151.7, 154.9) ^{ab}	153.6 (152.0, 154.9) ^b	152.7 (151.0, 153.6) ^{ab}
Potassium (mEq/L)	4.20 (3.95, 4.50) ^a	4.10 (3.95, 4.55) ^{ab}	4.05 (3.80, 4.30) ^b	4.20 (3.79, 4.40) ^{ab}
Chloride (mEq/L)	119.1 (117.4, 120.8)	117.7 (115.8, 119.6)	118.9 (117.2, 120.8)	118.7 (117.3, 119.4)
Cholesterol (mg/dL)	200 (157, 247) ^a	212 (177, 256) ^{ab}	177 (147, 219) ^b	201 (151, 236) ^{ab}
USG	1.018 (1.016, 1.022) ^a	1.018 (1.015, 1.019) ^a	1.047 (1.032, 1.054) ^b	1.034 (1.034, 1.038) ^b

CKD-DH, CKD-developed-hypertension; CKD-NT, CKD-remained normotensive; SBP, systolic blood pressure; PCV, packed cell volume; USG, urine specific gravity.

Parameters shown are at last included visit for the-NT groups and first hypertensive visit for the-DH groups. Superscript letters identify groups, which differed significantly.

Initial Visit

Chronic kidney disease-cats that remained normotensive during follow-up had significantly lower mean SBP at their first visit than CKD-cats that developed hypertension, and Healthy-NT cats had a significantly lower SBP than Healthy-DH cats ($P < .05$) (Table ¹). The SBP at baseline between the NT-groups did not significantly differ, and neither did the DH-groups (Table ¹). Cats included in the Healthy-NT group were significantly younger than both groups of cats with CKD ($P < .001$) but the other groups did not significantly differ in age (Table ¹).

Rate of Change in Systolic Blood Pressure over Time

The CKD-DH, CKD-NT and Healthy-NT cats all had a significant increase in SBP over time, although the rates of change were not significantly different from each other ($P = .09$; CKD-NT: 0.5 ± 0.1 , Healthy-NT: 0.4 ± 0.1 mmHg, CKD-DH: 1.1 ± 0.3 , Healthy-DH: 0.3 ± 0.5 mmHg/100 days). All rates of change, aside from the Healthy-DH group, were significantly different from 0 ($P < .001$).

Association Between Systolic Blood Pressure and Biochemical Variables over Time

Using measurements from all time points available for each cat, no significant association between creatinine, sodium, phosphate, total calcium, potassium, and weight and SBP over time could be found. Cholesterol, PCV, albumin and heart rate were significantly and positively associated with SBP and chloride was negatively associated with SBP in the univariable analysis (Table ³). In the multivariable model heart rate (0.07 ± 0.02 mmHg/bpm, $P < .001$) and PCV (0.2 ± 0.1 mmHg/L/L, $P < .05$) remained significantly and positively associated with SBP.

Table 3

Linear mixed model investigating the association between biochemical and clinical variables and systolic blood pressure over time

	Estimate $\pm$ SE	<i>P</i> -value
Creatinine (mg/dL)	-3.0 ± 2.2	.17
Sodium (mEq/L)	-0.03 ± 0.1	.75
Chloride (mEq/L)	-0.2 ± 0.08	.04
Phosphate (mg/dL)	0.4 ± 0.4	.34
Total calcium (mg/dL)	0.6 ± 0.6	.30
Potassium (mEq/L)	0.6 ± 1.1	.56
Cholesterol (mg/dL)	0.02 ± 0.009	.04
PCV (%)	0.30 ± 0.10	.001
Albumin (g/dL)	4.5 ± 1.8	.013
Weight (kg)	0.08 ± 0.07	.27
Heart rate (bpm)	0.07 ± 0.01	<.001

PCV, packed cell volume; SE, standard error.

Values depicted in bold were significantly associated with SBP in the univariable analysis and were included in the multivariable linear mixed model. PCV and heart rate remained significantly associated with SBP.

Risk of Becoming Hypertensive

The Cox proportional hazards model showed that at all time points, cats that were healthy were at less risk than cats with CKD of becoming hypertensive (Hazard ratio = 0.2 [95% confidence interval 0.08–0.36], $P < .001$) (Fig ¹). Sodium, chloride, total calcium, potassium, cholesterol, PCV, albumin, weight and heart rate were not significant risk factors for the development of hypertension (Table ⁴). Increased phosphate and creatinine concentrations were found to be significant risk factors for the development of hypertension in the univariable analysis ($P < .05$). Creatinine concentration was the only independent risk factor for a cat becoming hypertensive (Hazard ratio = 7.3 [95% confidence interval 1.2–45.4], $P = .03$).

Figure 1

Kaplan–Meier curve of probability to become hypertensive. CKD; chronic kidney disease, time is in days from first visit. Cats with CKD have a greater probability to be hypertensive at each time point than healthy cats ($P < .001$). Censored cases are represented by ticks.

Table 4

Cox proportional hazards model investigating the risk factors for the development of hypertension

	HR	95% Confidence interval	P-value
Creatinine (mg/dL)	9.39	2.06–42.82	.004
Sodium (mEq/L)	1.07	0.90–1.27	.47
Chloride (mEq/L)	1.00	0.92–1.08	.93
Phosphate (mg/dL)	1.34	1.03–1.74	.03
Total calcium (mg/dL)	0.88	0.51–1.53	.65
Potassium (mEq/L)	0.91	0.28–2.98	.87
Cholesterol (mg/dL)	1.00	1.00–1.01	.38
PCV (%)	0.95	0.86–1.05	.3
Albumin (g/dL)	0.72	0.17–3.11	.66
Weight (kg)	0.81	0.58–1.14	.22
Heart rate (bpm)	1.00	0.98–1.00	.16

PCV, packed cell volume.

Values depicted in bold were a significant risk factor for the development of hypertension in the univariable analysis and included in the multivariable Cox proportional hazards model. Creatinine remained the only independent risk factor for the development of hypertension.

Discussion

Of the initially normotensive cats with CKD, 17% developed hypertension, whereas of the initially normotensive healthy cats included in this study, 7% did. In addition to this, 105/264 azotemic cats were hypertensive at or within 3 months of CKD diagnosis. In cross-sectional studies, 23% of hypertensive cats are nonazotemic and do not have an identifiable cause for their hypertension,¹⁴ and 12% of nonazotemic, nonhyperthyroid cats are hypertensive at baseline.⁷ This study only included healthy cats that developed hypertension at a later time-point and excluded cats that were presented as an idiopathic hypertensive cat at their first visit to the clinic, which probably explains the lower prevalence. The criteria for defining hypertension and the composition of the study population greatly influence reported prevalence, and the prevalence of hypertension in cats with CKD in the current study is lower than the 65% reported in a study performed in a referral setting,⁴ and greater than the 19.4% previously reported in a cross-sectional study using a higher SBP cut-off.³ In addition to this, all cats included in this study had a minimum follow-up of 3 visits over the course of 3 months. This could have artificially increased the size of the group of cats that were hypertensive at diagnosis of CKD, as these cats came back for regular blood pressure checks on antihypertensive medication.

Both healthy cats and cats with CKD included in this study showed a significant increase in SBP with increasing age. No significant increase in SBP over time could be found for the healthy-DH group, although this was most likely because of the small group size. The significant increase in blood pressure with age is comparable to the human situation.¹⁵ Studies investigating the age-effect on blood pressure in cats are conflicting, but all studies published to

date commenting on this phenomenon were cross-sectional in design.^{5, 10, 16, 17, 18} A moderate correlation between all blood pressure variables and age has been reported in cats,¹⁸ although a study including only healthy cats did not find a significant correlation between SBP and age.¹⁷ Sansom and co-workers reported a higher blood pressure in older cats if the cats were grouped according to age (younger than 5, 5–10 years and older than 10). The cats that were included in this study did, however, not all have serum creatinine concentration measured, and the possibility exists that cats with subclinical CKD have been included, especially in the oldest age group.¹⁰ Another study performed in a population of healthy cats, however, could not demonstrate a significant difference between the SBP of cats <10 years of age and cats that were 10 years or older.⁵ A further study including healthy and ill cats did report an age cut-off, with cats over the age of 11 years having significantly higher blood pressure than cats younger than 11, although renal function was not assessed in the apparently healthy population of cats.¹⁸ Hypertensive cats have been reported to be significantly older than their normotensive counterparts, although the study population consisted of a mixture of cats including cats with renal disease and hyperthyroidism.¹⁶ Both diseases are associated with hypertension,¹² and the prevalence of CKD and hyperthyroidism also increases with age,^{6, 7, 19} which means that the age effect found could be due to the inclusion criteria of this study. To our knowledge, this study is the first investigating changes in blood pressures in individual healthy cats and cats with CKD over time and is therefore the first to definitely show by longitudinal analysis that pressure rises with age in cats, as it does in humans.

The positive association between heart rate and SBP has commonly been reported in human literature. Mean arterial pressure is determined by cardiac output and peripheral resistance, and the association between heart rate and SBP and can be attributed to the fact that cardiac output is increased with an increased heart rate.²⁰ A previous cross-sectional study found no significant correlation between heart rate and SBP in healthy cats.²¹ Heart rate has been reported to not be significantly different in cats with higher blood pressures and CKD when compared to healthy cats with lower blood pressures,¹¹ although a previous study reported 17% of cats with hypertension to be tachycardic.¹⁴ To our knowledge, this study is the first to investigate the association between heart rate and SBP over time in individual cats. A positive independent association between PCV and SBP was found in this study. Hematocrit is the most important determinant of blood viscosity, which in turn determines peripheral resistance together with vascular diameter. Human studies have also described this independent positive association between PCV and SBP, and these studies also report an increased PCV in hypertensive subjects compared to normotensive subjects.²²

Cats that develop hypertension have a higher baseline blood pressure than their normotensive counterparts. In the human literature the term ‘prehypertension’ is often used to describe a nonoptimal baseline blood pressure. Blood pressure and the incidence of hypertension increases with age, and human subjects with higher baseline blood pressure are at higher risk of becoming hypertensive.^{23, 24} The CKD-DH group had a significantly higher blood pressure than both the CKD-NT group and the healthy-NT group. The small size of the healthy-DH group makes interpretation of the result more challenging, although the SBP at time of inclusion in the study seems comparable to the SBP of the CKD-DH group. The World Health Organisation-International Society of Hypertension has guidelines to classify nonhypertensive individuals into blood pressure categories (optimum, normal or high-normal). Based on these categories, screening intervals are recommended for human subjects, although these intervals differ between organizations.²⁴ No optimum recommendations exist yet for screening intervals for individual elderly normotensive cats, but based on the results obtained in this study it could be argued that more frequent blood pressure measurements need to be encouraged if an owner presents a cat with a mean SBP ≥ 140 mmHg. Cats with CKD have a significantly

greater probability of becoming hypertensive, and creatinine is the only independent predictor for becoming hypertensive, which could call for an even shorter interval between screening visits for cats with CKD. Of the cats that were diagnosed as hypertensive, a greater percentage showed signs of target organ damage if they were diagnosed with CKD and hypertension around the same time point (70%), when compared with the cats that already came to the clinic for regular blood pressure checks after their CKD diagnosis (52%). The overall relatively high incidence of target organ damage provides clinical validity to the diagnosis of hypertension in most cats included in this study. A shorter screening interval can aid in early diagnosis and treatment of hypertension and therefore decrease the development of target organ damage.

Having CKD increases the probability a cat develops systemic hypertension, but their blood pressure does not increase at a significantly greater rate than cats that are healthy and become hypertensive, although this result may have been affected by group size. Cats with CKD included in this study were significantly older than healthy cats that remained normotensive, and even though the difference was not significant for the healthy-DH group, the age of these cats seems comparable to the healthy-NT group. This is a confounding factor of this study, and difficult to control for as CKD prevalence increases with age.⁶ There was no correlation between creatinine concentration and SBP when the data from all time points for each cat was used, similar to the result of a cross-sectional study in cats with CKD.⁵ A study investigating differences between hypertensive and normotensive cats also reported no significant difference in creatinine concentration between groups.³

Human patients with end stage renal disease almost invariably suffer from hypertension, but in human CKD patients there is no correlation between serum creatinine concentration and blood pressure,^{25, 26} although a decline in creatinine clearance is mildly correlated with mean blood pressure in healthy subjects.¹⁵ It has been demonstrated that baseline creatinine is an independent risk factor for the development of azotemic CKD in cats.⁷ This study shows that creatinine is also an independent risk factor for the development of hypertension and that CKD cats are more likely to be diagnosed with clinically significant hypertension. Historically it has been suggested that hypertension in most cats is secondary to CKD, and several mechanisms, such as activation of the renin-angiotensin-aldosterone system (RAAS) and fluid retention have been described.^{11, 27, 28} A recent study showed that plasma renin activity is suppressed in hypertensive cats, whereas plasma aldosterone concentration is increased, although there was a substantial overlap among groups.²⁸ However, it could be hypothesized that CKD and hypertension share a pathophysiological basis that is not yet understood. The possibility exists that feline hypertension has a genetic component, like essential hypertension in humans,²⁹ but to date no studies into the genetics of feline hypertension have been published. This is an area that warrants further investigation.

Conclusion

In conclusion, blood pressure increases with age in all cats. Cats that develop clinically significant hypertension have a higher blood pressure at initial evaluation than their normotensive counterparts and cats with CKD are more likely to develop hypertension. The high prevalence of hypertension in azotemic cats in this study shows the importance of monitoring of SBP in elderly cats and, in particular, in cats with CKD. It could, therefore, be suggested that cats with a higher baseline blood pressure at diagnosis of CKD should be more closely monitored than cats with a lower baseline blood pressure. Creatinine concentration is an independent risk factor for the development of hypertension, and early diagnosis of CKD is essential so that appropriate management can be offered.

Acknowledgments

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Off-label Antimicrobial Declaration: Authors declare no off-label use of antimicrobials.

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Footnotes

1

Parks Electronic Doppler Model 811B, Perimed UK, Bury St Edmunds, UK

2

Mistral 3000, Sanyo-Gallenkamp, Leicestershire, UK

3

IDEXX Laboratories, Wetherby, Yorkshire, UK

4

R i386 3.0.1; R Foundation for Statistical Computing, Vienna, Austria and GraphPad Prism 6; GraphPad Software, La Jolla, CA, USA.

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Comparison of Efficacy of Long-term Oral Treatment with Telmisartan and Benazepril in Cats with Chronic Kidney Disease

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Abstract

Abstract

Background

The efficacy and benefits of telmisartan in cats with chronic kidney disease (CKD) have not previously been reported.

Hypothesis

Long-term treatment of cats with CKD using telmisartan decreases urine protein-to-creatinine ratio (UP/C) similar to benazepril.

Animals

Two-hundred and twenty-four client-owned adult cats with CKD.

Methods

Prospective, multicenter, controlled, randomized, parallel group, blinded clinical trial with noninferiority design. Cats were allocated in a 1 : 1 ratio to either telmisartan (1 mg/kg; n = 112) or benazepril (0.5–1.0 mg/kg; n = 112) PO q24 h. The primary endpoint was prospectively defined as the change in proteinuria (benazepril:telmisartan) based on a log transformed weighted average of UP/C change from baseline (AUC 0→t/t) as a percentage compared using a confidence interval (CI) approach. Changes of UP/C from baseline were assessed on all study days and corrected for multiple comparisons.

Results

Telmisartan proved noninferior to benazepril in controlling proteinuria (CI, −0.035 to 0.268). At Day 180, UP/C compared to baseline in the telmisartan group was significantly lower (-0.05 ± 0.31 ; $P = .016$), whereas in the benazepril group the change (-0.02 ± 0.48) was not statistically significant ($P = .136$). Similar results were obtained at all assessment points with significant decrease in UP/C occurring with telmisartan but not benazepril.

Conclusion and Clinical Importance

Both telmisartan and benazepril were well tolerated and safe. Telmisartan proved to be noninferior to benazepril and significantly decreased proteinuria relative to baseline at all assessment points whereas benazepril did not.

Keywords: ACE inhibitor, Angiotensin II receptor blocker, Benazepril, Proteinuria

Abbreviations

angiotensin converting enzyme

adverse event

angiotensin-II

angiotensin receptor blocker

1

angiotensin II type 1 receptor

2

angiotensin II type 2 receptor

area under curve

colony forming units

confidence interval

chronic kidney disease

control product (benazepril)

good clinical practice

glomerular filtration rate

intention-to-treat population

investigational product (telmisartan)

International Renal Interest Society

New York Heart Association

urine protein-to-creatinine ratio

peroxisome proliferator-activated receptor gamma

per-protocol-population dataset

renin angiotensin aldosterone system

systolic blood pressure

Chronic kidney disease (CKD) is irreversible, progressive, and one of the most common medical reasons for medical evaluation of older cats. The prevalence increases with age and up to 35% of the geriatric cat population is affected.^{1, 2} Clinical signs such as polyuria and polydipsia, lethargy, decreased appetite and weight loss often are described, and a considerable effect of CKD on the quality of life can be assumed.³ Furthermore, CKD negatively impacts survival, with an average life expectancy of 1–3 years once clinical signs become apparent.^{4, 5} Early management can improve quality of life and prognosis.^{1, 6} Compensatory chronic activation of the renin–angiotensin–aldosterone system (RAAS) in CKD to maintain glomerular filtration rate (GFR) increases angiotensin-II (AT-II) production with renal concentrations exceeding plasma concentrations.^{7, 8} Angiotensin-II is a central mediator of renal injury because of its

ability to produce glomerular hypertension that results in glomerular damage, proteinuria, and activation of pro-inflammatory and profibrotic pathways.^{9, 10} Furthermore, even mild proteinuria is strongly associated with and predictive of onset of azotemia,¹¹ progression of CKD,¹² and survival in both humans and cats.^{4, 13} Several AT-II receptors have been identified.¹⁴ The major detrimental renal effects of AT-II described above are mediated by AT₁ receptors. The AT₂ receptors modulate actions of AT-II that are renoprotective, namely vasodilation, natriuresis, inhibition of renin secretion, and anti-inflammatory, anti-ischemic and antifibrotic effects. The expression of the AT₂ receptor is increased in pathologic circumstances.^{15, 16}

Angiotensin converting enzyme (ACE)-inhibitors prevent enzymatic conversion of angiotensin-I, decreasing AT-II concentrations. However, in mammals, including the cat, alternative pathways (ACE-escape) exist for AT-II generation.^{17, 18, 19} Although increased plasma renin activity has been reported during treatment with angiotensin receptor blockers (ARBs) in humans, a phenomenon that also could partially overcome competitive receptor blockade, this phenomenon seems to be without clinical relevance.²⁰

The ARBs, such as telmisartan,¹ selectively block the AT₁ receptor with high affinity and displace AT-II, while leaving beneficial effects of the AT₂ receptor activation unaffected. Furthermore, the efficacy of ARBs is unaffected by ACE-escape mechanisms. The renoprotective properties of telmisartan have been demonstrated in ex vivo models and in vivo studies in human patients with diabetic and nondiabetic nephropathies.^{21, 22, 23, 24} In contrast to other ARBs, telmisartan also is a partial agonist of the peroxisome proliferator-activated receptor gamma (PPAR-γ) receptor and its renoprotective properties are increased by this dual action.^{22, 23, 25}

There are no published studies on telmisartan or other ARBs in cats with naturally occurring CKD, whereas benazepril has been shown to decrease the time-averaged urine protein-to-creatinine ratio (UP/C) relative to that of placebo-treated cats.²⁸ The primary objective of this study was to compare the antiproteinuric effect of telmisartan with that of benazepril in cats with naturally occurring CKD. Additional variables were evaluated to assess effect on quality of life.

Materials and Methods

Animals

Client-owned adult cats suspected of having CKD were screened for CKD at 48 centers across Europe including: Germany (15), France (12), United Kingdom (11), Netherlands (4) Belgium (4), and Italy (2).

Inclusion Criteria

Cats of either sex were eligible for inclusion only if owner informed consent was given and they were >2.0 kg body weight and diagnosed with clinically stable CKD International Renal Interest Society (IRIS)²⁷ stage 2–3. Diagnosis of CKD was based on history, physical examination and laboratory findings. The following criteria had to be met: plasma or serum creatinine concentration ≥1.6 and <5.0 mg/dL, urine specific gravity <1.035, UP/C ≥ 0.2 and <2.0, plasma or serum T4 concentrations ≤3.1 μg/dL, and systolic blood pressure (SBP) ≤170 mmHg. Cats with SBP > 170–<180 mmHg were eligible if they had been stable on amlodipine treatment for ≥4 weeks.

Cats receiving a renal diet were eligible if they had received the diet for ≥4 weeks.

Exclusion Criteria

Cats were ineligible for inclusion if they had received ACEinhibitors, ARBs or other vasodilating agents, or diuretics (eg, furosemide) <14 days before screening or had their diet changed <4 weeks before screening. Cats with urinary tract infection identified by urine cytology and culture (>1,000 colony forming units [CFU]/mL), palpably enlarged kidneys, renal neoplasia, pre- or postrenal azotemia, heart failure (New York Heart Association [NYHA] class II, III, IV), acute kidney injury, pregnancy, or lactation were excluded.

Study Design

A multicenter prospective, randomized, investigator-blinded, positive-controlled, parallel group design was adopted. At each study site, the investigator was blinded by use of a dispenser, who was responsible for allocation and dispensing of medication as well as its return only. The outer package of both medications was visually identical, and the weight of each package was adjusted to prevent differentiation between medications to ensure that all persons involved in the evaluation were not aware of the medication given.

Client-owned cats were allocated to receive telmisartan¹ or benazepril² (1 : 1 ratio). The protocol was prepared in consultation with independent experts in CKD of cats and approved by European regulatory agencies and by an ethical review committee at all sites where required. The study was powered to assess noninferiority regarding the primary endpoint UP/C change from baseline of telmisartan against benazepril, and was conducted according to good clinical practice (GCP)³ guidelines (VICH GL9).

The randomization sequence was generated as a single list, which was held by a single center, not located at a study site.

Procedures were available to allow unblinding in the event of medical emergency.

An solution of telmisartan¹ was administered PO by the cat owner at a dosage of 1 mg/kg (0.25 mL/kg) q24 h. Benazepril² was administered at 0.5–1.0 mg/kg PO q24 h as a flavored 2.5 mg tablet in accordance with the product label. Cats >2 and <5 kg in weight received 1 tablet, and cats >5 and <10 kg body weight received 2 tablets.

Routine treatments with no impact on CKD (eg, vaccinations, antiparasitic drugs) were allowed. Any concomitant treatment was documented. Amlodipine (0.625–1.25 mg PO q24 h) was allowed if cats had been stabilized for ≥4 weeks before inclusion, and treatment was continuous. Antimicrobial treatment was not permitted in the 7 or 21 days before inclusion for short-acting and long-acting agents, respectively. If sedation of a cat was required for a diagnostic procedure during the study, a standard sedation protocol was used. Anti-inflammatory or anti-infective treatment was only permitted as acute treatment.

Schedule of Events

Cats with a history typical of CKD or those >7 years and suspected of having CKD at a routine screening underwent further examination to diagnose and stage CKD based on IRIS guidelines.²⁷ Fasted serum creatinine concentrations were determined in well-hydrated cats at least twice (screening and Day 0). Cystocentesis was performed for urine collection whenever possible. Urinalysis was performed on site. Laboratory examinations were conducted at IDEXX Laboratory Germany, except for those from the United Kingdom, which were performed at the IDEXX Laboratory United Kingdom. A cross-validation of the 2 IDEXX Laboratories confirmed their comparability.

Systolic blood pressure was measured, if indicated, by an indirect method for safety purposes only, using a consistent method for an individual cat (usually Doppler ultrasound examination).

The inclusion date (Day 0) was the day on which the cat first received telmisartan or benazepril. Detailed follow-up physical examinations and urinalyses were scheduled on Days 7, 30, 60, 90, 120, and 180. Routine hematology and blood biochemistry were repeated on Days 30, 90, and 180. Additional diagnostic tests could be undertaken whenever considered indicated by the investigator. All adverse events (AEs) were reported in accordance with local regulations.

Outcome Measures

The primary efficacy outcome of the study was prospectively defined as the decrease in proteinuria, measured by UP/C, compared to baseline. Telmisartan was hypothesized to be similarly efficacious to benazepril. Representing secondary efficacy variables, the changes in general demeanor and appetite in comparison to baseline were recorded as surrogates for quality of life variables. These variables were assessed by the owners who scored appetite as normal, increased, decreased, or no eating and demeanor as normal, alert, listless, or lethargic.

The number of treatment failures, defined as death or euthanasia of any cause, owner noncompliance or deterioration of clinical signs requiring hospitalization, was recorded. Furthermore, according to GCP any AE, defined as “any observation in animals that is unfavorable and unintended and occurs after the use of a veterinary product or a investigational veterinary product, whether or not considered to be product related”, was recorded.

Statistical Methods

Cats included in the efficacy analysis were required to be followed to Day 90 (per-protocol-population dataset [PPS], Fig ¹) whereas all treated cats (intention-to-treat [ITT] population) were included in the safety analysis. Baseline characteristics of the groups were compared for homogeneity.

Figure 1

Participant flow. *More than 1 reason for exclusion possible. **Benazepril group: anemia; telmisartan group: increase in renal parameters (2 cats), uremic crisis, anorexia, cerebellar mass on MRI, intra-abdominal neoplasia, hind limb ataxia (1 cat for each). ***Exclusion UPC “>2 of 5” Rule: For the first 5-study days at most 2 missing or excluded because of bacteria contamination (CFU >1,000/mL) UPC values are considered to be acceptable to yield reliable AUC values.

The same approach was followed as used in the reference study,²⁸ in which benazepril was shown to be superior to placebo in a randomized controlled trial assessing the effect of treatment on time-averaged proteinuria (area under the curve [AUC] UP/C). Because internal pilot and published data⁴ showed a skewed distribution of the calculated AUC UP/C values, sample size determination and statistical analysis of the primary efficacy variable were based on the log transformed AUC UP/C values. For the primary efficacy analysis, a summary value based on a log transformed weighted average of UP/C change from baseline (AUC UP/C 0→t/t) in percentage of each cat over time was calculated (Day 0 = 100%). The AUC principle was used to account for progressive loss of cats to follow-up as used in the reference study.²⁸ Noninferiority to benazepril was based on a confidence interval (CI) approach using the normal approximation.²⁹ The null hypothesis “inferiority of treatment to positive control” was rejected if the 95% 2-sided CI of the log (AUC UP/C 0→t/t) difference lay entirely above the noninferiority margin Δ (−0.108 to 0.116). A noninferiority margin of $\Delta = 0.108$ on the log scale, corresponding to a ratio $AUC_{(IVP)}/AUC_{(CP)}$ of approximately

1.11, was assumed and determined from an internal pilot study with 37 animals over 8–12 weeks in which an average AUC 0→t/t value of 106.83% with a SD of 28.28% was observed. Based on these data, an absolute margin of 119% corresponding to a relative margin of approximately 1.11 was defined a priori and was accepted as clinically meaningful by the licensing authorities.

In addition, absolute change in UP/C from baseline was analyzed within each group using a 1-sample signed-rank test. *P*-values were adjusted for multiplicity by the Bonferroni–Holm procedure.²⁶

Other secondary efficacy variables, including behavioral changes and number of treatment failures, were summarized statistically for the PPS population. Secondary efficacy variables were not compared between groups statistically.

In accordance with the recommendations of CONSORT,³⁰ a valid assessment of this noninferiority study is provided. Statistical analyses were performed using the SAS System Version 9.2.⁴ *P* < .05 was considered to indicate significance.

Results

Cats were recruited between March 2008 and August 2010, with case follow-up completed in January 2011. More than 2000 cats were screened and 224 cats (ITT population) were recruited with 112 cats randomized to each group (Fig ¹). Clinical signs suggestive of CKD were present for >3 months in most cats before inclusion. The baseline demographics, and clinical and laboratory characteristics were distributed homogeneously between groups (Table ¹). Cats >11 years predominated in both groups, which had similar mean and median ages (Table ¹). Most cats (approximately 80%) were classified as IRIS CKD stage 2 (84/112 and 91/112 for telmisartan and benazepril, respectively). Four cats in IRIS CKD stage 4 were included in the study as a protocol deviation (Table ²), the remainder being IRIS CKD stage 3. Most cats enrolled were borderline proteinuric (UP/C 0.2–0.4), with similar numbers in each treatment group; 3 cats with UP/C < 0.2 were included as a protocol deviation because all other criteria for CKD were fulfilled (Table ²). Fifty-one percent of the cats had SBP results in the mild to moderate hypertension class (IRIS classification; 150–179 mmHg; Table ²). A small number of cats with SBP >170 but <180 mmHg that were stabilized on amlodipine treatment before Day 0 was included in both groups (Table ¹). Approximately, 30% of cats in both groups were being fed a renal diet at inclusion (Table ¹). Deviations from planned time schedule were unavoidable under field conditions. They did not exceed 12 days and were considered to be of no influence on the general study outcome.

Table 1

Baseline characteristics at inclusion (intention-to-treat population)

Variable	Benazepril (n = 112)	Telmisartan (n = 112)		
Mean (min–max)	No (%)	Mean (min–max)	No (%)	
Age (years)	12.8 (2–22)		13.6 (3–21)	
Weight (kg)	4.3 (2–11)		4.4 (2–9)	
Gender				
Female		58 (51,8)		55 (49.1)

Variable	Benazepril (n = 112)	Telmisartan (n = 112)		
Mean (min–max)	No (%)	Mean (min–max)	No (%)	
Male		54 (48,2)		57 (50.9)
Neutered		104 (92,9)		106 (94.6)
Breed				
DLH		6 (5.4)		9 (8.0)
DSH		72 (64.3)		75 (67.0)
Mixed		9 (8.0)		4 (3.6)
Other		20 (17.9)		16 (14.3)
Persian		5 (4.5)		8 (7.1)
Serum Biochemistry		% >ULN		% >ULN
TT4 (µg/dL)	1.9 (0.3–5.4)	0.9	1.8 (0.3–4.3)	0.0
Creatinine (mg/dL) ^a	2.4 (1.6–6.6)	58	2.5 (1.1–6.6)	64.3
Urea (BUN) (mg/dL)	48.7 (21.3–139.8)	84.8	52.1 (23.3–299.4)	79.5
Sodium (mEq/L)	154.9 (147.0–167.0)	0.9	155.3 (147.6–174.0)	1.8
Potassium (mEq/L)	4.5 (3.0–6.4)	7.1	4.5 (3.0–8.1) ^b	3.6
Calcium (mg/dL)	10.4 (7.6–13.2)	6.3	10.4 (8.0–13.2)	2.7
Phosphate (mg/dL)	5.0 (2.2–10.2)	7.1	5.0 (2.2–15.8)	6.3
Albumin (mg/dL)	3.4 (2.2–4.6)	0.0	3.4 (2.6–4.5)	0.0
ALT (U/L)	65.2 (22.0–210.8)	1.8	65.1 (21.4–279.4)	1.8
ALP (U/L)	32.8 (10.0–111.8)	0.9	30.4 (12.0–129.0)	0.9
CBC		% <LLN, >ULN		%<LLN, >ULN
RBC ^c 10 × 12/L	7.5 (3.4–10.9)	<3.6, >6.3	8.0 (3.9–14.6)	<2.7, >10.7
HGB g/L	108.2 (55–170)	<17.3, >1.2	113.4 (67–170)	<11.5, >2.6
HCT %	34.9 (19.0–54.0)	<10.8, >1.8	37.3 (21.0–69.8)	<5.4, >3.6
WBC ^c g/L	10.7 (1.9–28.9)	<17.1, >18.9	9.6 (2.8–49.2)	<17.0, >8.0
Urine Variables				
Specific gravity ^c	1.022		1.022	
UPC	0.41 (0.1–2.2)		0.41 (0.1–3.0)	
Systolic BP (mmHg)	147.7 (95–180)		146.2 (98–176)	

Variable	Benazepril (n = 112)	Telmisartan (n = 112)		
Mean (min–max)	No (%)	Mean (min–max)	No (%)	
Amlodipine treatment ^c		6 (5.4)		10 (8.9)
Kidney diet		36 (32.1)		39 (34.8)
Disease History	Mean Duration (years)		Mean Duration (years)	
PU/PD	0.66	63 (56.3)	0.56	71 (63.4)
Decreased appetite	0.41	28 (25.0)	0.36	33 (29.5)
Weight loss	0.53	48 ^c (42.9)	0.52	56 (50.0)
Palpably small kidneys		21 (18.8)		18 (16.1)

a

Upper reference limit (ULN) laboratory 2.0 mg/dL.

b

Laboratory error suspected at D0, at recheck examinations potassium levels of this patient were well within reference ranges.

c

One missing value, LLN/ULN = lower/upper reference limit.

Table 2

Telmisartan and Benazepril Group (intention-to-treat population, n = 112 each), IRIS stage at baseline including substaging

Telmisartan (n = 112) ^a	IRIS Stage 2a (n = 36)	IRIS Stage 2b (n = 48)	IRIS Stage 3 (n = 21)	IRIS Stage 4 (n = 3) ^b
Substaging by SBP				
<150 mmHg	17	30	6	1
150–159 mmHg	11	5	5	1
160–179 mmHg	8	13	10	0
≥180 mmHg	0	0	0	0
Substaging by UP/C ^a				
<0.2	1	1	0	0
0.2–0.4	26	40	13	0
>0.4	9		8	2

Benazepril (n = 112)	IRIS Stage 2a (n = 47)	IRIS Stage 2b (n = 44)	IRIS Stage 3 (n = 19)	IRIS Stage 4 (n = 2)
Substaging by SBP				
<150 mmHg	24	18	8	1
150–159 mmHg	7	9	5	1
160–179 mmHg	16	16	6	0
≥180 mmHg	0	1	0	0
Substaging by UP/C				
<0.2	0	1	0	0
0.2–0.4	42	31	15	1
>0.4	5	12	4	1

a

IRIS stage 1 (n = 4) not shown.

b

One missing value for IRIS 4.

The primary variable log (AUC UP/C 0→t/t) based on the CI approach showed noninferiority of telmisartan compared to benazepril in the PPS population (n = 176) and in the ITT population (n = 224). Although the study was designed as a noninferiority study, telmisartan was close to being superior to benazepril (Fig ²).

Figure 2

Results of the noninferiority and superiority analysis for the primary variable log AUC 0→t/t in the PPS-population. The delta (−Δ) indicates the noninferiority margin. The observed 95% confidence interval limits (CL) for the treatment difference are shown as the shaded area.

The changes from baseline over time are presented in Table ³. On Day 180, overall mean UP/C values were numerically lower in both groups compared to baseline. In the telmisartan group, the mean decrease (-0.05 ± 0.31) was significantly different from baseline ($P = .016$). In the benazepril group, the mean decrease (-0.02 ± 0.48) was not significantly different from baseline ($P = .136$). Furthermore, in the telmisartan group statistically significant decreases in UP/C were found at all assessment points, whereas in the benazepril group no statistical change in UP/C compared to baseline was found at any assessment point. These findings also are reflected in the number of cats that changed from being classified as proteinuric at baseline to borderline proteinuric or nonproteinuric at Day 180, which was 30.8% (4/13) in the benazepril and 57.2% (8/14) in the telmisartan group. In cats that were classified as borderline proteinuric at baseline the number of cats that either shifted to nonproteinuric (approximately 33%) or remained borderline proteinuric (approximately 50%) at Day 180 was comparable in both groups with 85.7% (54/63) in benazepril and 87.9% (51/58) in telmisartan-treated cats.

Table 3

UP/C changes from baseline by treatment and study day for the PPS population

Treatment	Day	N	Mean	SD	Median	Range	IQ Range	P-value
Benazepril	7	87	−0.06	0.25	0.0	−1.4 to 1.0	0.10	.040
30	88	−0.06	0.28	0.0	−1.5 to 0.5	0.15	.023	
60	87	−0.07	0.38	0.0	−1.6 to 1.1	0.10	.046	
90	86	+0.06	0.61	0.0	−1.5 to 3.5	0.20	.072	
120	85	+0.06	0.77	−0.1	−1.6 to 5.2	0.20	.143	
180	76	−0.02	0.48	0.0	−1.5 to 2.3	0.20	.136	
Telmisartan	7	80	−0.07	0.28	−0.1	−1.7 to 1.0	0.10	<.0001 ^a
30	83	−0.08	0.26	−0.1	−1.6 to 0.5	0.10	<.0001 ^a	
60	82	−0.09	0.33	−0.1	−1.7 to 0.8	0.10	.0002 ^a	
90	83	−0.08	0.31	−0.1	−1.4 to 1.3	0.20	<.0001 ^a	
120	75	−0.02	0.51	−0.1	−1.6 to 2.4	0.20	.002 ^a	
180	72	−0.05	0.31	0.0	−1.1 to 1.1	0.10	.016 ^a	

The *P*-values are adjusted for multiplicity by the Bonferroni–Holm procedure (*P*-values sorted from lowest to highest and compared to the adjusted *P*-values (ie, .0083; .01; .0125; .016; .025; .05).

a

Significantly different from baseline.

The majority of cats completed the study, and the proportion of cats removed from the study was comparable in both treatment groups (Fig ¹). The percentage of treatment failures in the PPS population was 15.3% (13/85) for telmisartan and 19.8% (18/91) for benazepril-treated cats. The number of deaths in telmisartan-treated cats (ITT population) was 13/112 and 22/112 for benazepril-treated cats. The changes in general behavior indicative of quality of life in treated cats were assessed relative to baseline to describe clinical benefit. Cat owners reported the most visible change in behavior during the first 60 days with a small numerical increase in active cats being reported in both groups. Most cats (approximately 80%) had normal appetite and little change in appetite was seen throughout the study.

Telmisartan and benazepril were well tolerated and most of the AEs reported were typical of the signs of CKD, and the majority was unlikely to be related to treatment. Adverse events that were reported in both groups (benazepril: 65 cats and telmisartan: 61 cats with ≥1 AE) included signs of deterioration of CKD, gastrointestinal disorders, and very few cats with urinary tract infections, cardiac, oral cavity, skin, or respiratory disorders, neurological signs, or neoplasia. Figure ¹ gives details of the AEs that resulted in withdrawal of cases from the study before Day 90.

At the end of the study, the mean values for red blood cell count (RBC), hemoglobin (HGB) and hematocrit (HCT) were comparable between treatment groups. Clinical evidence of anemia was reported in 2 cats (1 in each group) with CKD; both also were receiving amlodipine. No clinically relevant changes in serum calcium, potassium, sodium, or phosphate concentrations were recorded in either treatment group.

Discussion

The aim of our study was to compare the efficacy of telmisartan against benazepril with respect to the primary endpoint, change in UP/C from baseline. Although the study was designed as a noninferiority study, telmisartan was close to meeting the a priori agreed upon criterion for superiority to benazepril.

The UP/C was significantly decreased compared to baseline at all assessment points in telmisartan-treated cats, whereas statistical significance was not reached for benazepril-treated cats.

Telmisartan¹, a selective AT₁ receptor antagonist, does not bind to AT₂ receptors or other key physiologically important receptors or enzymes at therapeutic concentrations.³¹ Selective blockade of AT₁ receptors offers an alternative approach to the modulation of the RAAS. Telmisartan is licensed for 2 indications in human adults: the treatment of essential hypertension, as well as the reduction in cardiovascular morbidity (cardiovascular prevention) in patients with atherothrombotic cardiovascular disease or type 2 diabetes mellitus with target organ damage. Renoprotective properties of telmisartan in humans are thought to be related to selective blockade of AT₁ and partial agonism of PPAR- γ receptors.^{23, 32} The apparently more pronounced decrease in proteinuria in cats in this study might be explained by this dual mode of action of telmisartan or circumvention of potential ACE-escape mechanisms.

A large number of cats had to be screened to identify suitable cats with CKD for inclusion in this study. The study population (predominantly IRIS CKD stage 2) was considered to be appropriate for the intended target population because early treatment is logical if the therapeutic rationale is that of slowing progression. Cats in this study were older and had a slightly higher body weight compared to the reference study,²⁸ findings that probably are indicative of recent demographic changes in the cat population.

According to the requirements of CONSORT,³⁰ efficacy of the reference treatment had been established in a randomized placebo-controlled study.²⁸ For a valid assessment of noninferiority, it is necessary that participants, study conduct, and outcome measures are similar to the reference study.²⁸ This study closely matched the reference study with respect to study design, target population, and outcome measures. The primary outcome, decrease in proteinuria, calculated by UP/C change from baseline (as assessed by the AUC UP/C 0→t/t), was chosen because it was used in the reference study. Furthermore, proteinuria has been previously demonstrated to be an independent predictor of progression and survival of cats with CKD.^{4, 11, 12, 13}

Limitations reported for the reference study²⁸ were addressed to allow an unbiased assessment of outcome in this study. Thus, before inclusion, urine culture and screening for hyperthyroidism were undertaken. Hyperthyroidism is a known comorbidity in cats with CKD and may influence proteinuria and liver enzyme activity.³³ In order not to bias efficacy and safety conclusions, cats with a T4 concentration >3.1 $\mu\text{g/dL}$, which is within the upper third of the reference range, were excluded, with the aim of excluding cats with concomitant hyperthyroidism and CKD in which suppressed T4 concentrations may complicate diagnosis of hyperthyroidism.³⁴

More than 90% of urine samples collected in this study was obtained by cystocentesis, which compares favorably with 60% collected by this method in the reference study.²⁸ Abnormal urine sediment triggered urine culture in this study. In 14 cats (5 in the benazepril and 9 in the telmisartan group), bacterial urinary tract infection was confirmed leading to exclusion from the efficacy analysis.

Benazepril previously has been reported to decrease proteinuria in cats with CKD in comparison to placebo when assessed over a variable follow-up period, but not to significantly improve survival time,²⁸ or slow CKD progression.³⁵ The reference study showed that time-averaged UP/C increased above baseline in the placebo group, and benazepril seemed to prevent this increase. Similar to the reference study,²⁸ in this study, benazepril prevented an increase in proteinuria while a decrease in proteinuria relative to baseline after 180 days was present only in telmisartan-treated cats. This study was conducted for 180 days, thus the effect of telmisartan on survival of cats with CKD could not be evaluated. Additional studies are warranted to assess telmisartan's effect on survival time in cats with CKD.

It might be questioned whether the decrease in UP/C relative to baseline seen in this study is clinically relevant. A previous study demonstrated that high UP/C (median value of 0.26 versus 0.15 in progressive versus nonprogressive cases, respectively) and high plasma phosphate concentration at diagnosis are significant independent risk factors for progression of azotemia in cats with CKD and that small differences in UP/C at diagnosis are clinically relevant.¹² In a postmortem study, proteinuria was associated with interstitial fibrosis and glomerular hypertrophy.³⁶ These associations between proteinuria and negative CKD outcomes do not prove causality but provide the rationale for investigating telmisartan's effect on progression of CKD or survival time in cats because this study demonstrated telmisartan's efficacy in decreasing proteinuria. These associations also support the contention that, if a decrease in proteinuria is renoprotective and prolongs survival in the cats with CKD (which remains to be proven), relatively small decreases in UP/C (eg, from proteinuric to borderline proteinuric or from borderline proteinuric to nonproteinuric) or prevention of an increases in UP/C over time could be clinically relevant.

Unlike the reference study,²⁸ in our study SBP was monitored at inclusion and at study conclusion, because hypertension is known to be associated with CKD and proteinuria⁴ in cats. The SBP of the PPS population at the end of the study was comparable to baseline in both groups. Our study was not designed to assess antihypertensive efficacy, and additional work is necessary to assess effects on blood pressure. In the ITT population, 16 cats (Table 1), and in the PPS population 11 cats were pretreated with amlodipine (3 and 8, in the benazepril and telmisartan group, respectively). To exclude a potential effect of amlodipine on the primary efficacy variable in the both groups, the same analysis was performed without the amlodipine-treated cats, with almost the same results: observed treatment difference for the primary variable $\log \text{AUC} \rightarrow t/t$: 0.112, and significant change in overall mean UP/C value ($P = .013$) for the telmisartan but not ($P = .16$) for the benazepril group. Furthermore, the percentages of cats classified as proteinuric, borderline proteinuric or nonproteinuric at Day 180 in comparison to baseline were very similar when amlodipine-treated cats were excluded.

Hematology variables were monitored to assess changes in RBC counts, which might be caused either by progression of CKD or by drug interaction with the RAAS. At baseline, approximately 3% of cats had RBC counts below the reference range ($5.0 \times 10^{12}/\text{L}$). In the reference study,²⁸ approximately 10% of the cats had RBC counts below the reference range at baseline. The mean RBC counts were comparable between groups and within the reference range at the end of the study. The proportion of cats with RBC counts below the reference range slightly increased in both groups by study end. Clinical evidence of anemia was reported in 2 cats (1 per group). Both cats received concomitant amlodipine treatment, and thus the role of treatment could not be assessed. Medications that inhibit the RAAS may decrease red cell mass by blocking the effects of AT-II on erythropoiesis.^{37, 38} The results of this study suggest this

potential decrease in RBC count is not clinically relevant. Hypokalemia occurs in 20–25% of cats with CKD.^{39, 40} In this study, neither hypo- nor hyperkalemia was found in any cat over the study period.

Because a primary requirement for the administration of any medication is safety, our study also collected data on AEs. The number of deaths and withdrawals from the study was small in both groups. Adverse events reported in both groups were those expected for cats with CKD and in most cases were judged as unlikely to be treatment related.

In conclusion, telmisartan effectively decreased proteinuria and was safe for treatment of cats with CKD. For the primary variable, telmisartan was at least as effective as benazepril. Indeed, telmisartan's antiproteinuric effect appeared to be more pronounced because significant decreases in UP/C compared to baseline were identified for telmisartan but not benazepril. The results of our study indicated that telmisartan was a safe and effective treatment to decrease proteinuria associated with CKD in cats.

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Conflict of Interest Declaration

This project was funded by Boehringer Ingelheim Vetmedica GmbH, a company representative read and approved the final draft. Ulrike Sent, Rüdiger Gössl and Tanja Zimmering are employees of Boehringer Ingelheim Pharma GmbH&Co.KG and Vetmedica GmbH, respectively. Prof. Elliott has acted as a paid independent consultant to Boehringer Ingelheim Vetmedica GmbH providing advice on study design and critically evaluating the results of the study for registration purposes.

Off-label Antimicrobial Declaration

Authors declare no off-label use of antimicrobials.

Footnotes

1

Semintra oral solution for cats (telmisartan 4 mg/mL), Boehringer Ingelheim Vetmedica GmbH, Ingelheim, Germany

2

Fortekor 2.5, flavored benazepril, Novartis Santé Animale S.A.S Huningue Cedex, France

3

VICH GL9 (GCP), London, UK, July 1, 2001

4

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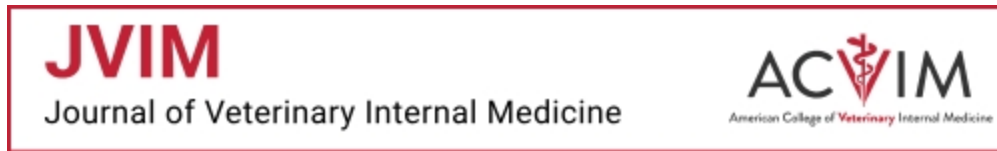
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Circulating renin–angiotensin–aldosterone system markers in cats with non-hypertensive chronic kidney disease or systemic arterial hypertension

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Abstract

Background

Information on the classical and alternative circulating renin–angiotensin–aldosterone systems (RAAS) in cats with non-hypertensive chronic kidney disease (NHT-CKD) or systemic arterial hypertension (SAH) is limited. Age and diet affect the RAAS.

Hypothesis/Objectives

To compare serum equilibrium concentrations of angiotensin peptides and aldosterone in healthy cats and cats with NHT-CKD or untreated SAH, and to evaluate changes in these concentrations in hypertensive cats administered amlodipine.

Animals

Client-owned cats with NHT-CKD (serum creatinine ≥ 1.6 mg/dL, systolic blood pressure [SBP] < 160 mmHg; $n = 17$), SAH (SBP ≥ 160 mmHg or ≥ 150 mmHg with hypertensive ocular lesions; $n = 6$), or normal SBP (< 140 mmHg) and kidney function ($n = 17$).

Methods

Prospective, single-center, observational study. Cats underwent indirect SBP measurement and clinicopathologic analyses. Serum was obtained contemporaneously in all cats, and after 2-4 weeks of amlodipine therapy in hypertensive cats, for evaluation of angiotensin I, II, III, IV, 1-5, and 1-7, and aldosterone.

Results

Adjusting for diet, age, or both modified the conclusions of statistical comparisons for several variables. Mean serum angiotensin I, II, and III concentrations were lower in cats with untreated SAH vs healthy controls across unadjusted and diet-, age-, or both diet- and age-adjusted linear mixed models (geometric mean [95% CI] angiotensin I, 9.02 [3.05-26.62] vs 29.78 [21.15-41.94] pmol/L; angiotensin II, 33.63 [12.25-92.26] vs 124.24 [90.52-170.51] pmol/L; angiotensin III, 1.56 [0.88-2.79] vs 7.96 [5.35-11.86] pmol/L; all $P \leq .038$). Controlling for diet but not age, angiotensin II ($P = .043$) and III ($P = .019$) were also lower in cats with NHT-CKD vs controls. All angiotensin peptides, except for angiotensin IV, significantly increased in hypertensive cats during amlodipine therapy. Serum aldosterone did not differ between groups, or before and after amlodipine in hypertensive cats.

Conclusions and clinical importance

Evidence of circulating RAAS activation in cats with NHT-CKD or untreated SAH was not found.

Keywords: alternative RAAS, amlodipine, classical RAAS, healthy cats

Introduction

The renin–angiotensin–aldosterone system (RAAS) plays a pivotal role in the regulation of kidney function and the pathophysiology of kidney diseases.¹ Although the RAAS is commonly said to be activated in chronic kidney disease (CKD),² this is inconsistently supported by published studies of its circulating markers in cats with kidney dysfunction.³ Plasma renin activity (PRA) and angiotensin II and aldosterone concentrations were higher in cats with experimentally induced CKD relative to preinjury values in 1 study,⁴ as well as in cats with spontaneous CKD and unknown blood pressure (BP) status relative to healthy cats in another.⁵ In other studies, cats with naturally occurring hypertensive CKD had unchanged or lower PRA and higher plasma aldosterone concentrations relative to cats without systemic arterial hypertension (SAH).⁶⁻⁸ Overall, patterns of circulating RAAS (cRAAS) activity appear dependent on certain disease characteristics, such as whether azotemia or SAH is present,^{8,9} and concurrent treatment with amlodipine, an L-type calcium channel blocker known to activate the RAAS.⁸ Age, sex, and dietary sodium and potassium intake also influence RAAS activity.¹⁰⁻¹²

The RAAS is composed of 2 pathways that mediate opposing effects. The first is the so-called *classical* pathway, which culminates in the binding of angiotensin II to its type 1 receptor and mediates vasoconstriction and renal sodium and water retention, among other physiologically relevant effects.¹³ The second is the counterbalancing *alternative* pathway, which culminates in the interaction of angiotensin II with its type 2 receptor and the interaction of angiotensin 1-7 peptide (primarily generated by angiotensin-converting enzyme 2 [ACE2]) with the Mas receptor, both of which mediate vasodilation, natriuresis, and diuresis.¹⁴ To date, most studies of the RAAS in cats with CKD have focused on a limited number of cRAAS activity markers, such as PRA, angiotensin II, and aldosterone.^{2,5-9,15} Two studies describe the activities of both the classical and alternative pathways in cats with SAH, some of which had CKD.^{16,17} However, extensive characterization of the cRAAS in cats with non-hypertensive, naturally occurring CKD is lacking.

The objectives of the present study were (1) to describe equilibrium concentrations of circulating angiotensin peptides and aldosterone in cats with naturally occurring International Renal Interest Society (IRIS) stages 2-4, non-hypertensive chronic kidney disease (NHT-CKD), in cats with naturally occurring SAH, and in apparently healthy cats; and (2) to characterize changes in the equilibrium concentrations of circulating angiotensin peptides and aldosterone in hypertensive cats started on amlodipine, relative to pretreatment values. We hypothesized that (1) compared with age-matched, healthy control cats, cats with NHT-CKD or SAH would have angiotensin peptide and aldosterone concentrations and estimated renin activity levels that indicate activation (NHT-CKD and SAH groups) or a lack of appropriate suppression (SAH group) of the cRAAS; and that (2) antihypertensive treatment with amlodipine besylate would significantly increase angiotensin peptide and aldosterone concentrations.

Materials and methods

Study design

This was a prospective, cross-sectional, single-center, observational study conducted between March 2021 and September 2023. In all cats, the cRAAS was evaluated by equilibrium analysis at a single time point. For hypertensive cats prescribed amlodipine by the attending clinician, cRAAS markers were also reevaluated 2-4 weeks after initiating therapy. All study activities were approved by the Clinical Research Committee of the University of Georgia College of Veterinary Medicine (approval number CR-594). Informed owner consent was obtained from all cat owners prior to enrollment.

Animals

Eligible participants were companion cats of any age, sex, or breed with IRIS stages 2-4 NHT-CKD or SAH of any etiology, presented routinely to the University of Georgia Veterinary Teaching Hospital or recruited from staff- and student-owned cats. In addition, healthy control cats that were age- and sex-matched to NHT-CKD cats were recruited from staff- and student-owned cats during the same period.

Screening visit

For all cats, a screening visit included a complete medical history; a physical examination including indirect fundic evaluation; systolic blood pressure (SBP) measurement using Doppler sphygmomanometry and guidelines set forth by the American College of Veterinary Internal Medicine¹⁸; a CBC, a serum biochemistry profile including symmetric dimethyl arginine (SDMA) concentration, and a total thyroxine (TT4) concentration; urinalysis and a urinary protein-to-creatinine ratio (UPC); and transthoracic echocardiographic evaluation (for all but one NHT-CKD cat). The one cat not undergoing echocardiography was the first enrolled in the study and had no historical or physical examination abnormalities suggestive of cardiovascular disease. Anxiolytics or sedatives were not administered to any cat before SBP determination.

Enrollment criteria

Cats were excluded if they had an unsuitable temperament; had concurrent disease(s) or condition(s) with a known or likely effect on the RAAS (eg, heart disease with any degree of left

atrial enlargement, congestive heart failure, volume depletion, diabetes mellitus, and uncontrolled hyperthyroidism); or had received a RAAS inhibitor (eg, angiotensin-converting enzyme inhibitor, angiotensin receptor blocker, and mineralocorticoid receptor blocker) in the 30 days preceding enrollment, had received a corticosteroid by any administration route in the 30 days preceding enrollment or in the 6 months prior to enrollment for depot formulations; had received other medications known to modify the RAAS directly or indirectly (eg, amlodipine, hydralazine, β - or α -adrenergic blockers, and diuretics) in the 7 days preceding enrollment; or if the owner did not consent to participation.

Control group

Cats were included in the healthy control group based on normal findings on physical examination; indirect SBP measurement (normal, < 140 mmHg); CBC; serum biochemistry profile, SDMA, and TT4; urinalysis; UPC; and echocardiography. Cats were deemed to have normal urine concentrating ability if the urine specific gravity (USG) was ≥ 1.035 . Cats with a spot USG of 1.030-1.034 during screening were also eligible for enrollment provided physical examination and clinicopathologic variables reflective of water balance suggested optimal hydration status, owners reported normal drinking and urination, and if subsequent renal ultrasonographic examination (performed for other unrelated research studies in some cats) was normal, USG at follow-up testing was ≥ 1.035 , or both. To be screened for inclusion as a healthy control, cats had to be age-, sex-, and neuter-status-matched as a matched-pair with a cat enrolled in the NHT-CKD group. Control cats were considered age-matched if the age difference was $\leq 20\%$ (using the NHT-CKD cat as a reference).

NHT-CKD group

Cats were included in the NHT-CKD group if diagnosed with CKD based on evidence of chronic changes in renal function and assigned an IRIS CKD stage of 2 or higher (ie, serum creatinine concentration ≥ 1.6 mg/dL with contemporaneous USG < 1.035 and the absence of an apparent extra-renal cause) and substaged as normotensive (ie, SBP < 140 mmHg) or prehypertensive (ie, SBP, 140-159 mmHg). Cats of any proteinuria substage were included. Systemically non-hypertensive cats with concurrent hyperthyroidism (controlled or uncontrolled) were excluded from the NHT-CKD group.

Hypertensive group

Cats were included in the SAH group if an average session SBP ≥ 160 mmHg was documented during 2 separate measurement sessions at least 1 h apart, or if a single average session SBP ≥ 150 mmHg was noted in a cat with ocular target organ damage (eg, retinal hemorrhage, bullous retinal detachment). Cats could be diagnosed with idiopathic or secondary hypertension. Historically hyperthyroid cats definitively treated with I-131 and whose hyperthyroidism was controlled (ie, serum TT4 concentration $\leq$ upper limit of the laboratory reference interval) for at least 4 weeks prior to screening could be included. All treatment decisions were at the discretion of the attending clinician. Hypertensive cats prescribed amlodipine alone at any dosage were invited back 2-4 weeks after initiation of therapy for repeat evaluation.

Follow-up visit for amlodipine-treated cats

For hypertensive cats prescribed amlodipine at the screening visit, a follow-up visit included a physical examination with indirect fundic evaluation; indirect SBP measurement; and blood and urine sampling for a renal biochemistry profile, urinalysis, UPC measurement, and cRAAS evaluation.

Sample collection and clinicopathologic analysis

At all visits, after SBP determination and physical examination, blood and urine samples were collected via venipuncture and cystocentesis, respectively. Samples were most often collected in unsedated cats. For 8 cats ($n = 5$ and 3 in the NHT-CKH and healthy control groups, respectively), butorphanol (0.2-0.3 mg/kg IM or IV) was administered to facilitate sample collection. Blood was collected from the jugular, saphenous, or cephalic vein and immediately placed into tubes containing no additive, where it was allowed to clot, or EDTA. Urine was collected into glass tubes. Hematology, serum biochemistry, serum TT4 concentration, urinalysis, and UPC determination were performed in-house by the University of Georgia Veterinary Clinical Pathology Laboratory (Athens, Georgia). Serum was shipped on ice to a commercial laboratory (IDEXX Laboratories, Westbrook, Maine) for measurement of SDMA concentration. Within 30 min of collection, serum was separated by centrifugation and stored at -80°C until batch shipment for measurement of cRAAS analytes as described below.

Cats were fasted overnight and food was withheld on the morning of evaluation. Although time of day of sample collection was not standardized, most visits occurred between 9 AM and 12 PM; when possible, reevaluation of hypertensive cats happened at the same time of day as the corresponding screening visit.

Analysis of cRAAS components

The cRAAS was assessed by measurement of serum equilibrium concentrations of angiotensin I, II, III, IV, 1-5, and 1-7 and aldosterone using a previously described LC–MS/MS-based technique (RAS-Fingerprint; Attoquant Diagnostics GmbH, Vienna, Austria).¹⁹ In addition, surrogate markers of enzyme activity and adrenal responsiveness to angiotensin II signaling were calculated from these concentrations, as previously described.²⁰⁻²² These included surrogate measures of ACE activity ($\text{ACE-S} = [\text{angiotensin II}] / [\text{angiotensin I}]$; ACE2 activity ($\text{ACE2-S} = [\text{angiotensin 1-5}] / [\text{angiotensin II}]$; PRA ($\text{PRA-S} = [\text{angiotensin I}] + [\text{angiotensin II}]$); and adrenal responsiveness to angiotensin II (aldosterone-to-angiotensin II ratio, AA2 ratio) = $[\text{aldosterone}] / [\text{angiotensin II}]$.

Echocardiography

All echocardiographic examinations were performed by a board-certified veterinary cardiologist or cardiologist-in-training, using an ultrasound unit (EPIQ CVx, Philips Healthcare) outfitted with phased-array transducers. Complete transthoracic echocardiographic studies were recorded from standard right and left parasternal views on manually restrained animals.²³ Findings were used to guide enrollment decisions as described above. Diagnosis and classification of cardiomyopathies conformed to guidelines set forth by the American College of Veterinary Internal Medicine.²⁴ The hypertrophic cardiomyopathy (HCM) phenotype was assigned to cats with left ventricular concentric hypertrophy (ie, end-diastolic wall thickness ≥ 6 mm) and a non-dilated left ventricle.²⁴ Left atrial enlargement was diagnosed in cats with a left-atrial-to-aortic root diameter ratio (LA:Ao, see below) > 1.5 .^{25,26}

For the purposes of statistical analyses, the following data were recorded for each enrolled cat: (1) LA:Ao calculated from 2-dimensional (2D) short-axis measurements,²⁶ (2) maximum 2D- or M-mode-derived end-diastolic left ventricular wall thickness at any segment (maxLVWTd, mm; a measure of left ventricular hypertrophy severity), and (3) left ventricular fractional shortening (FS, %; a measure of ventricular systolic performance) calculated from M-mode measurements. For each variable, the recorded value reflected the average of values taken from 3 to 5 cardiac cycles.

Statistical analyses

Statistical analyses were performed using commercially available (Graphpad Prism 10 for MacOS, version 10.4.2) and open-source (R project, version 4.4.1²⁷) software packages. For all analyses, P -values $< .05$ were considered statistically significant.

Continuous data were evaluated for normality of distribution by visualization of histograms and the Shapiro–Wilk test. Baseline demographic and clinical data are presented as mean $\pm$ SD or median (first quartile; third quartile) if normally or non-normally distributed, respectively. For RAAS data compared with linear mixed models, which were log-transformed as described below, geometric means and their 95% CIs were also calculated. For the purposes of statistical analysis, USG values > 1.060 were assigned a value of 1.061. Angiotensin peptide concentrations less than the assay lower limit of quantification were reported as that limit divided by 2.

For normally and non-normally distributed variables, intergroup comparisons were performed using a one-way analysis of variance (ANOVA) test or the Kruskal–Wallis test, respectively, with pairwise comparisons adjusted for using the Benjamini–Hochberg method. Categorical data were compared using Fisher’s exact test.

Linear mixed models were built using the R package lme4²⁸ (version 1.1-37), and marginal means and group comparisons were calculated with the package emmeans²⁹ (version 1.11.0). Default package settings were used unless otherwise specified. Angiotensin peptide and aldosterone concentrations, and RAAS surrogate indices calculated from these concentrations, were log-transformed to improve the approximation of a normal distribution. Linear mixed models were then used to compare RAAS markers between the 3 groups. The group (ie, healthy control, NHT-CKD, or SAH) was set as the fixed effect with the healthy control group set as the reference category. Pairing between healthy control and NHT-CKD cats was set as a random effect and the RAAS marker was set as the dependent variable. Multiple comparisons between the 3 groups were adjusted for using the Tukey method. Additional models were built with the feeding of a therapeutic renal diet, age, or both, also set as a fixed effect to adjust for possible effects of diet or age on these group comparisons. In SAH cats only, the Wilcoxon signed rank test was used to evaluate differences in RAAS markers before vs after amlodipine treatment.

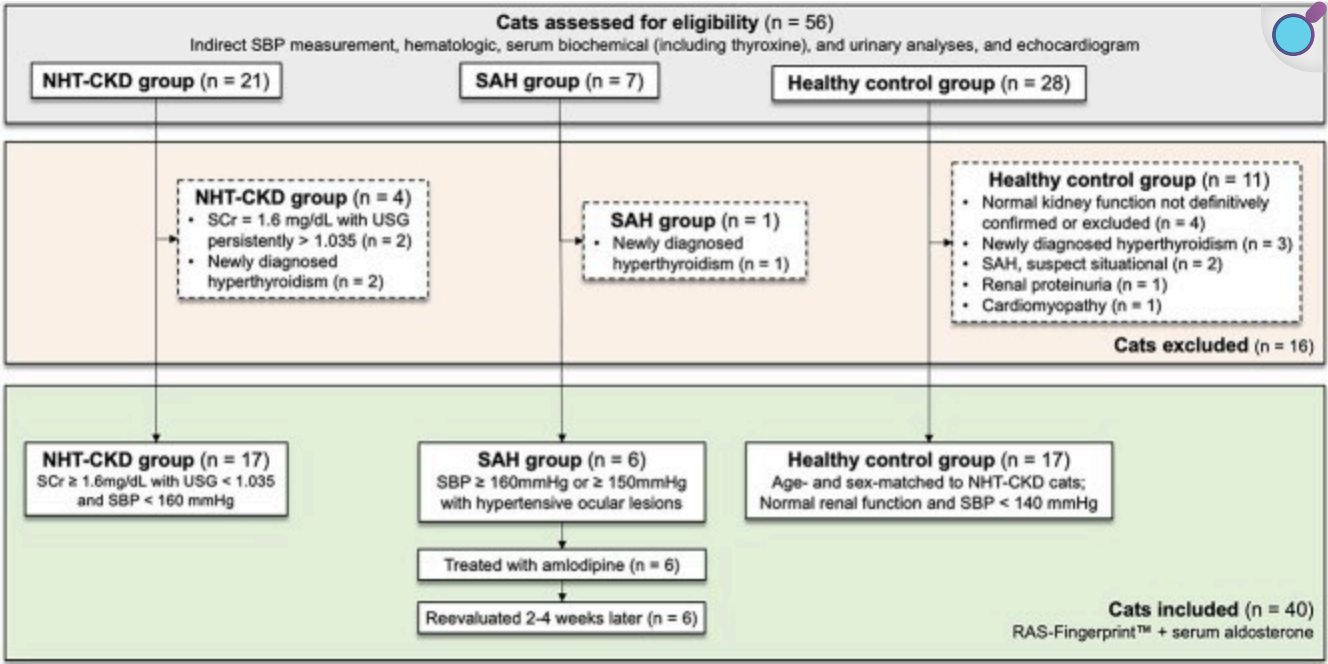
Because of the known interaction between the RAAS and electrolyte balance, and kidney and cardiovascular function,³⁰ exploratory Spearman correlation analysis was conducted between the cRAAS markers evaluated and selected clinical, clinicopathologic, and echocardiographic variables. Specifically, correlations between RAAS markers and SBP; hematocrit; serum sodium, potassium, creatinine, and SDMA concentrations; UPC; and 3 echocardiographic variables (ie, LA:Ao, maxLVWTD, and FS) were explored. Data from all enrolled cats were used in correlation analysis with only pre-amlodipine values used for SAH cats.

Results

Animals

Fifty-six cats were screened for eligibility, of which 40 were included ($n = 17, 17$, and 6 in the NHT-CKH, healthy control, and SAH groups, respectively; [Figure 1](#)). The most common reasons for exclusion included newly diagnosed hyperthyroidism ($n = 6$ cats) and inability to definitively establish normal kidney function in cats screened for the healthy control group ($n = 4$ cats).

Figure 1.



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Flow diagram illustrating the progress of cats through the present study, which evaluates the circulating renin–angiotensin–aldosterone system in cats with NHT-CKD in International Renal Interest Society stages 2 or higher, cats with SAH (all of which had CKD), and healthy controls. Abbreviations: NHT-CKD = non-hypertensive chronic kidney disease; SAH, systemic arterial hypertension.

For cats included in the SAH group, SBP ranged from 157 to 182 mmHg. Evidence of hypertensive ocular target organ damage was noted in 4/6 (67%) cats, including the cat with the lowest SBP. Both treatment-naïve hypertensive cats without ocular lesions had a preexisting diagnosis of CKD.

In one of these cats without ocular lesions (with an average SBP at enrollment of 179 mmHg measured twice, 3 h and 15 min apart), mild aortic root dilatation and insufficiency were observed echocardiographically. In the other cat (with an average SBP at enrollment of 182 and 176 mmHg, measured 2 h and 20 min apart), SAH (SBP > 160 mmHg) was confirmed prior to enrollment using Doppler sphygmomanometry performed by the owner (a trained veterinary technician) in the home environment.

On average, cats in the NHT-CKD and SAH groups were moderately azotemic ([Table 1](#)). All 17 NHT-CKD cats were IRIS stage 2 ($n = 13$) or 3 ($n = 4$), with the majority ($n = 13$ [76%]) classified as normotensive and the remainder ($n = 4$ [24%]) classified as prehypertensive. Of these NHT-CKD cats, 11 (65%) were non-proteinuric, 3 (18%) were borderline proteinuric, and 3 (18%) were overtly proteinuric. All 6 cats in the SAH group had secondary hypertension and CKD of IRIS stage 2 ($n = 5$) or 3 ($n = 1$), with 4 (67%) and 2 (33%) being borderline or non-proteinuric, respectively. One hypertensive cat had a history of hyperthyroidism that was definitively treated with I-131 63 days prior to enrollment.

Table 1.

Demographic, clinical, and clinicopathologic information for enrolled cats with NHT-CKD (International Renal Interest Society stages 2 or higher), treatment-naïve SAH (all of which had CKD), or normal kidney function and blood pressure (healthy control group). Cats in the healthy control group were age- and sex-matched to those in the NHT-CKD group. Significant *P*-values are indicated in bold.

Variable	Healthy control group (<i>n</i> = 17)	NHT-CKD group (<i>n</i> = 17)	SAH group (<i>n</i> = 6)	<i>P</i> -value
Age (years)	11.6 (9.8; 14.1)	13.5 (12.2; 14.6)	15.9 (13.9; 17.3)	N/A ^{a,b}
Sex/neuter status (<i>n</i>)				N/A ^{a,b}
Female spayed	10	10	4	
Male neutered	7	7	2	
Body weight (kg)	4.0 (3.5; 5.2)	3.9 (3.6; 5.0)	4.2 (3.8; 4.8)	.99
Body condition score	5 (5; 7)	6 (5; 6)	6 (5; 7)	.92
Breed				
DSH	10	12	5	.57
DMH	3	0	0	
DLH	2	4	1	
Bengal	1	0	0	
Ragdoll	1	0	0	
Siamese	0	1	0	
Average SBP ^c (mmHg)	119 ± 14	128 ± 15	174 ± 10	N/A ^a
Hematocrit (%) (RI, 33.6-49.0)	40 ± 5 ^d	34 ± 4 ^d	33 ± 4 ^d	.002
Total protein (g/dL) (RI, 6.5-8.2)	7.7 ± 0.7	7.6 ± 0.6	7.7 ± 0.4	.94

Variable	Healthy control group (<i>n</i> = 17)	NHT-CKD group (<i>n</i> = 17)	SAH group (<i>n</i> = 6)	<i>P</i> -value
Serum creatinine (mg/dL) (RI, 0.7-1.8)	1.1 (1.0; 1.3)	2.1 (1.9; 2.8)	2.4 (1.9; 2.7)	N/A _{a,b}
Serum urea nitrogen (mg/dL) (RI, 18-35)	23 ± 3	44 ± 12	39 ± 7	N/A _{a,b}
Serum SDMA (µg/dL) (RI, 0-14)	10 ± 2	19 ± 6	20 ± 3	N/A _{a,b}
Serum sodium (mmol/L) (RI, 148-156)	151 ± 2	152 ± 2	152 ± 2	.32
Serum potassium (mmol/L) (RI, 3.7-5.3)	4.1 (4.0; 4.4)	4.3 (4.2; 4.5)	4.3 (4.1; 4.5)	.62
Serum total calcium (mg/dL) (RI, 9.5-11.0)	9.6 ± 0.5 _d	10.4 ± 0.6 _d	10.2 ± 0.2 _d	<.001
Serum phosphorus (mg/dL) (RI, 2.8-5.7)	3.5 (3.2; 4.3)	4.1 (3.9; 5.2)	4.1 (3.6; 4.5)	.08
Serum total thyroxine (µg/dL) (RI, 1.3-3.5)	2.5 ± 0.5	2.2 ± 0.6	2.0 ± 0.8	.1
Urine specific gravity	1.050 (1.047; 1.057)	1.019 (1.018; 1.024)	1.016 (1.015; 1.023)	N/A _{a,b}
Urinary protein-to-creatinine ratio	0.13 (0.11; 0.16)	0.18 (0.13; 0.28)	0.24 (0.11; 0.32)	N/A _{a,b}
LA:Ao	1.15 (1.10; 1.25)	1.25 (1.12; 1.31)	1.10 (1.02; 1.16)	.22
maxLVWTD (mm)	3.9 (3.3; 4.5) _d	4.5 (4.1; 5.3) _d	5.0 (4.7; 5.0) _d	.026
LVFS (%)	56 (54; 59)	57 (52; 59)	59 (57; 62)	.45

Data are shown as mean $\pm$ SD and compared with the one-way ANOVA test or median (first quartile; third quartile) compared with the Kruskal–Wallis test, as appropriate.

^aData not statistically compared between the 3 groups because it was part of the defining criteria for each group (eg, markers of renal function, systolic blood pressure) or it was used for case-matching (ie, age and sex for the NHT-CKD and SAH groups).

^bParameters did not differ significantly between the NHT-CKD and SAH groups using the Fisher's exact or Mann–Whitney *U* test, as appropriate.

^cFor each cat, 5 individual measurements were averaged to determine SBP.

^dGroups sharing a superscript letter are *not* significantly different at a 5% significance level. Pairwise comparisons were adjusted using the Benjamini–Hochberg method.

Abbreviations: ANOVA, analysis of variance; DLH, domestic longhaired; DMH, domestic medium hair; DSH, domestic shorthair; LA:Ao, left-atrial-to-aortic dimension ratio; LVFS, left ventricular fractional shortening; maxLVWTd, maximum end-diastolic left ventricular wall thickness; NHT-CKD, non-hypertensive chronic kidney disease; RI, reference interval; SAH, systemic arterial hypertension.

Echocardiographic findings were normal in 69%, 67%, and 100% of cats in the NHT-CKD, SAH, and healthy control groups, respectively. An HCM phenotype was diagnosed in 4 (25%) of 16 cats in the NHT-CKD group (the first enrolled cat did not have an echocardiogram), 3 of which also had systolic anterior motion of the mitral valve. Although no cat in the SAH or healthy groups had an HCM phenotype, the median maxLVWTd was significantly higher in SAH cats relative to healthy controls ([Table 1](#)). Discrete upper septal thickening³¹ was present in 1 SAH cat. In keeping with the study's enrollment criteria, left atrial enlargement was not present in any cat.

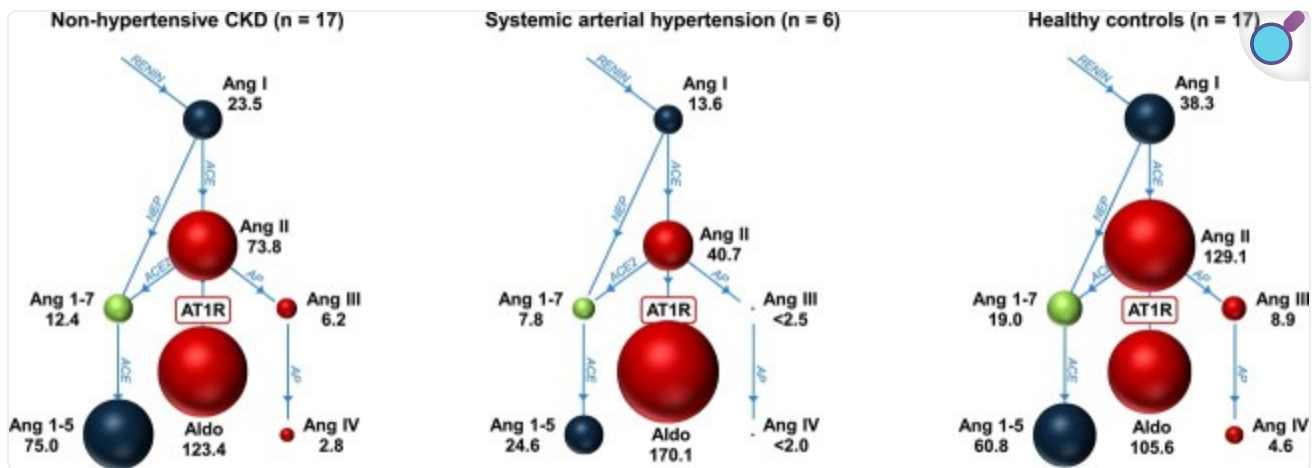
Cats were eating a variety of commercial diets, with a therapeutic renal diet being exclusively fed to 8 (47%) of 17 and 4 (67%) of 6 cats in the NHT-CKD and SAH groups, respectively. A complete list of fed diets is included in [Table S1](#).

Approximately one-half of the 40 included cats ($n = 9$ NHT-CKD, $n = 1$ SAH, and $n = 9$ healthy cats) were receiving no medications or supplements. Ectoparasite or heartworm preventives were being regularly administered to 35% of cats ($n = 4$ NHT-CKD, $n = 2$ SAH, and $n = 8$ healthy). Other medications being regularly administered included mirtazapine ($n = 1$ NHT-CKD cat), maropitant ($n = 1$ NHT-CKD cat), and SQ immunotherapy for atopy ($n = 1$ SAH cat). A list of administered supplements is included in [Table S2](#).

Comparison of cRAAS markers between groups

Median serum equilibrium concentrations of angiotensin peptides and aldosterone by group are depicted in [Figure 2](#). Geometric means and corresponding 95% CIs for these concentrations and calculated surrogates of RAAS activity are listed in [Table S3](#). Circulating RAAS markers did not differ significantly between NHT-CKD and healthy control cats when compared using unadjusted linear mixed models ([Figure 3](#); [Table 2](#)). Cats with untreated SAH had significantly lower mean concentrations of angiotensin I (geometric mean [95% CI], 9.02 [3.05-26.62] vs 29.78 [21.15-41.94] pmol/L; $P = .008$), angiotensin II (geometric mean [95% CI], 33.63 [12.25-92.26] vs 124.24 [90.52-170.51] pmol/L; $P = .002$), angiotensin III (geometric mean [95% CI], 1.56 [0.88-2.79] vs 7.96 [5.35-11.86] pmol/L; $P = .002$), angiotensin IV (geometric mean [95% CI], 1.43 [0.77-2.66] vs 3.75 [2.59-5.45] pmol/L; $P = .026$), and angiotensin 1-5 (geometric mean [95% CI], 22.59 [7.64-66.8] vs 68.44 [48.45-96.68] pmol/L; $P = .026$) than healthy control cats, and of angiotensin I (geometric mean [95% CI], 9.02 [3.05-26.62] vs 22.89 [15.05-34.81]; $P = .045$) and angiotensin III (geometric mean [95% CI], 1.56 [0.88-2.79] vs 4.69 [2.68-8.20]; $P = .038$) when compared with the NHT-CKD group. Mean angiotensin II concentration was numerically, but not statistically ($P = .05$), higher in NHT-CKD cats when compared to SAH cats. The mean aldosterone concentration was not significantly different between groups. For the surrogate markers of RAAS activity, mean PRA-S was significantly lower and mean AA2 ratio was significantly higher in the SAH group relative to both the NHT-CKD and healthy control groups. There were no significant differences in mean ACE-S or ACE2-S between groups.

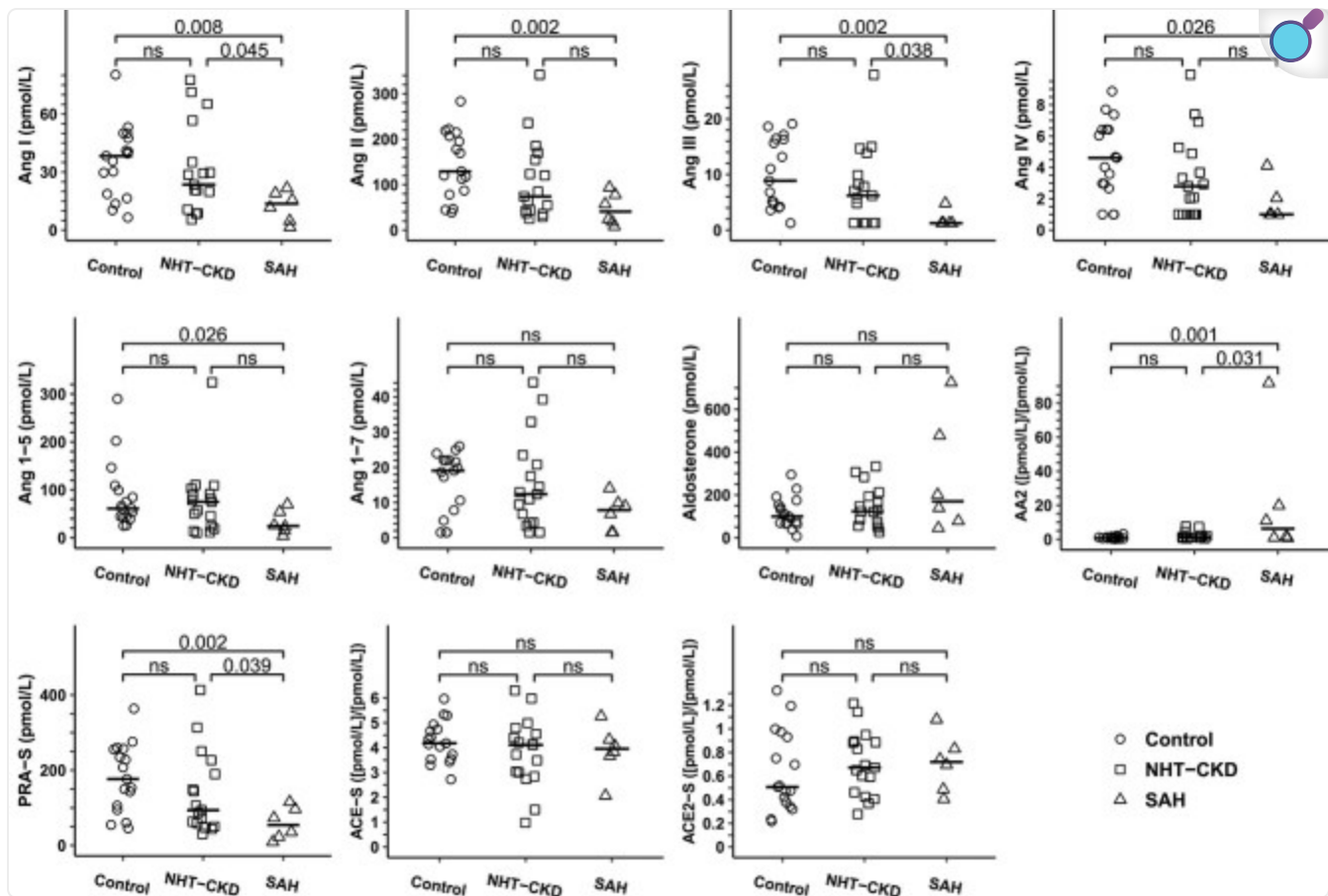
Figure 2.



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Serum equilibrium concentrations of angiotensin peptides and aldosterone in cats with non-hypertensive-CKD (International Renal Interest Society stages 2 or higher), cats with systemic arterial hypertension naïve to antihypertensive therapy (all of which had CKD), and healthy controls that were age- and sex-matched to those in the non-hypertensive-CKD group. The size of each sphere is proportional to median concentrations of each analyte, with median concentrations (pmol/L) shown next to the spheres. Abbreviations: ACE, angiotensin-converting enzyme; ACE2, angiotensin-converting enzyme 2; Aldo, aldosterone; Ang, angiotensin; AP, aminopeptidase; AT1R, angiotensin type-1 receptor; CKD, chronic kidney disease; DAP, aspartyl aminopeptidase; NEP, neutral endopeptidase (neprilysin).

Figure 3.



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Dot plots of circulating renin-angiotensin-aldosterone system markers in cats with NHT-CKD of International Renal Interest Society stages 2 or higher (open squares; $n = 17$), cats with SAH naïve to antihypertensive therapy (open triangles; $n = 6$, all of which had CKD), and healthy controls that were age- and sex-matched to those in the NHT-CKD group ($n = 17$; open circles). Horizontal bars depict the median value for each group. Statistical comparisons between groups were conducted using linear mixed-effect models to account for paired matching between NHT-CKD and healthy control cats. No adjustment for the feeding of a therapeutic renal diet or age was included in this analysis. Significant P -values are noted above relevant brackets. Abbreviations: AA2, angiotensin-to-angiotensin II ratio, a surrogate measure of adrenal responsiveness to angiotensin II; ACE-S, surrogate measure of angiotensin-converting enzyme (ACE) activity; ACE2-S, surrogate measure of ACE2 activity; Ang, angiotensin; ns, not statistically significant; NHT-CKD, non-hypertensive

chronic kidney disease; PRA-S, surrogate measure of plasma renin activity; SAH, systemic arterial hypertension.

Table 2.

Results of intergroup comparisons of circulating renin–angiotensin–aldosterone system markers in cats with NHT-CKD (International Renal Interest Society stages 2 or higher; $n = 17$), cats with untreated SAH ($n = 6$, all of which had CKD), and healthy controls ($n = 17$) using linear mixed models with or without including feeding of a renal diet, age, or both as covariates.

Variable	Group comparison	Unadjusted models			Diet-adjusted models			Age-adju
		Estimate	SE	P-value	Estimate	SE	P-value	Estimate
Ang I	Control vs NHT-CKD	0.30	0.312	.603	0.50	0.364	.366	0.26
	Control vs SAH	1.38	0.432	.008	1.66	0.507	.006	1.27
	NHT-CKD vs SAH	1.07	0.432	.045	1.16	0.437	.032	1.01
Ang II	Control vs NHT-CKD	0.52	0.282	.188	0.83	0.319	.043	0.40
	Control vs SAH	1.53	0.417	.002	1.97	0.476	.001	1.22
	NHT-CKD vs SAH	1.02	0.417	.05	1.15	0.412	.023	0.82
Ang III	Control vs NHT-CKD	0.51	0.245	.123	0.85	0.286	.019	0.33
	Control vs SAH	1.57	0.411	.002	2.05	0.457	.000	1.09
	NHT-CKD vs SAH	1.06	0.411	.038	1.20	0.398	.013	0.76
Ang IV	Control vs NHT-CKD	0.51	0.318	.270	0.61	0.376	.259	0.39

Variable	Group comparison	Unadjusted models			Diet-adjusted models			Age-adju
		Estimate	SE	P-value	Estimate	SE	P-value	Estimate
Ang 1-5	Control vs SAH	1.21	0.443	.026	1.35	0.527	.039	0.89
	NHT-CKD vs SAH	0.70	0.443	.270	0.74	0.455	.251	0.50
	Control vs NHT-CKD	0.30	0.290	.566	0.64	0.330	.154	0.12
	Control vs SAH	1.22	0.445	.026	1.70	0.507	.005	0.73
Ang 1-7	NHT-CKD vs SAH	0.92	0.445	.113	1.06	0.439	.055	0.61
	Control vs NHT-CKD	0.10	0.341	.956	0.34	0.394	.674	−0.02
	Control vs SAH	0.73	0.472	.278	1.07	0.553	.142	0.44
	NHT-CKD vs SAH	0.64	0.472	.378	0.74	0.477	.282	0.45
Aldo	Control vs NHT-CKD	−0.22	0.339	.793	−0.16	0.400	.916	−0.27
	Control vs SAH	−0.79	0.470	.227	−0.70	0.561	.432	−0.91
	NHT-CKD vs SAH	−0.57	0.470	.457	−0.54	0.484	.509	−0.64
	Control vs NHT-CKD	−0.51	0.267	.167	−0.67	0.329	.128	−0.46
AA2	Control vs SAH	−1.60	0.412	.001	−1.83	0.483	.002	−1.48
	NHT-CKD vs SAH	−1.09	0.412	.031	−1.16	0.418	.023	−1.02

Variable	Group comparison	Unadjusted models			Diet-adjusted models			Age-adju
		Estimate	SE	P-value	Estimate	SE	P-value	Estimate
PRA-S	Control vs NHT-CKD	0.46	0.287	.267	0.75	0.327	.081	0.36
	Control vs SAH	1.53	0.418	.002	1.93	0.481	.001	1.26
	NHT-CKD vs SAH	1.07	0.418	.039	1.19	0.416	.019	0.90
ACE-S	Control vs NHT-CKD	0.50	0.342	.34	0.72	0.399	.189	0.31
	Control vs SAH	0.31	0.474	.788	0.64	0.556	.493	−0.19
	NHT-CKD vs SAH	−0.18	0.474	.922	−0.09	0.480	.982	−0.49
ACE2-S	Control vs NHT-CKD	−0.36	0.346	.569	−0.27	0.410	.788	−0.49
	Control vs SAH	−0.42	0.479	.653	−0.30	0.571	.857	−0.79
	NHT-CKD vs SAH	−0.07	0.479	.989	−0.03	0.492	.998	−0.29

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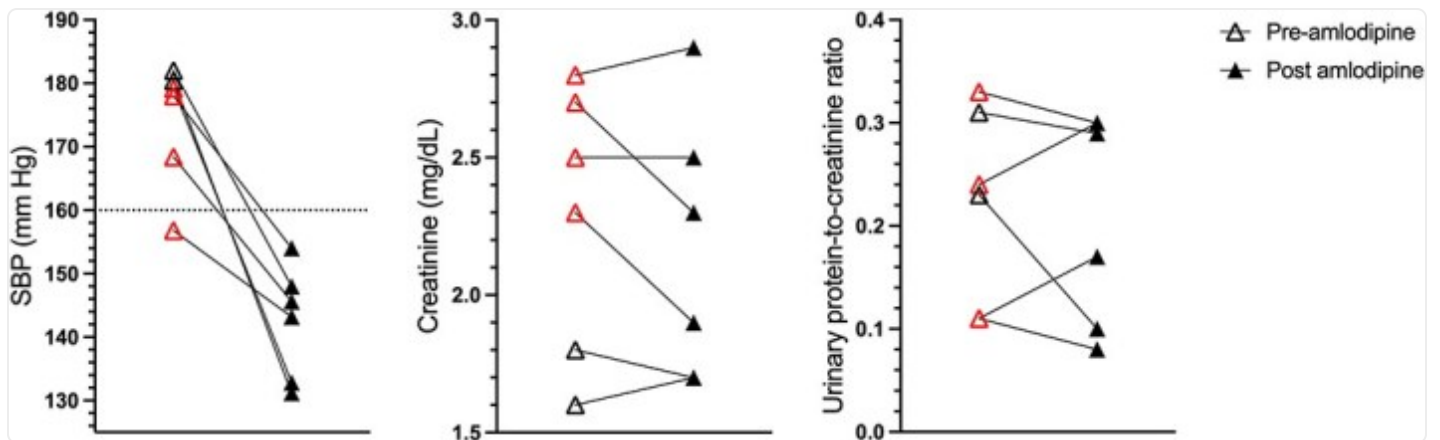
Abbreviations: AA2=angiotensin-to-angiotensin II ratio, a surrogate measure of adrenal responsiveness to angiotensin II; ACE-S, surrogate measure of angiotensin-converting enzyme (ACE) activity; ACE2-S, surrogate measure of ACE2 activity; Aldo=aldosterone; Ang, angiotensin; NHT-CKD=non-hypertensive chronic kidney disease; PRA-S=surrogate measure of plasma renin activity; SAH=systemic arterial hypertension.

Results of intergroup comparisons using linear mixed models adjusted for diet, age, or both are displayed in [Table 2](#). Conclusions for several comparisons differed depending on whether diet- or age-adjustment was conducted. When adjusted for being fed a therapeutic renal diet, without age adjustment, mean concentrations of angiotensin II and III differed significantly between all 3 groups (for both peptides, SAH < NHT-CKD < control). Angiotensins II and III were significantly lower in SAH cats compared with NHT-CKD cats ($P = .023$ and $.013$, respectively) and control cats ($P = .001$ and $P < .001$), and in NHT-CKD cats compared with healthy controls ($P = .043$ and $.019$, respectively). Conclusions for comparisons of all other markers between groups remained unchanged after this adjustment. However, when adjusting for age, or both age and diet, differences between the NHT-CKD and SAH groups in angiotensin I, angiotensin III, AA2, and PRA-S were not statistically significant. In addition, differences between the control and SAH groups for angiotensin IV and 1-5 were also not significant.

Comparison of clinical variables and cRAAS markers in hypertensive cats before and after therapy with amlodipine

All 6 treatment-naïve hypertensive cats were prescribed amlodipine besylate (0.625 mg PO q24h) and reevaluated after 2-4 weeks of therapy. For each cat, follow-up SBP was both lower compared to the pretreatment value and < 160 mmHg, while changes in serum creatinine concentration and UPC were small ([Figure 4](#)). Except for angiotensin IV, median serum equilibrium concentrations of all angiotensin peptides were significantly greater with amlodipine therapy relative to pretreatment ([Figures 5 and 6](#); [Table S4](#)). Concentrations of aldosterone did not differ before and after amlodipine therapy.

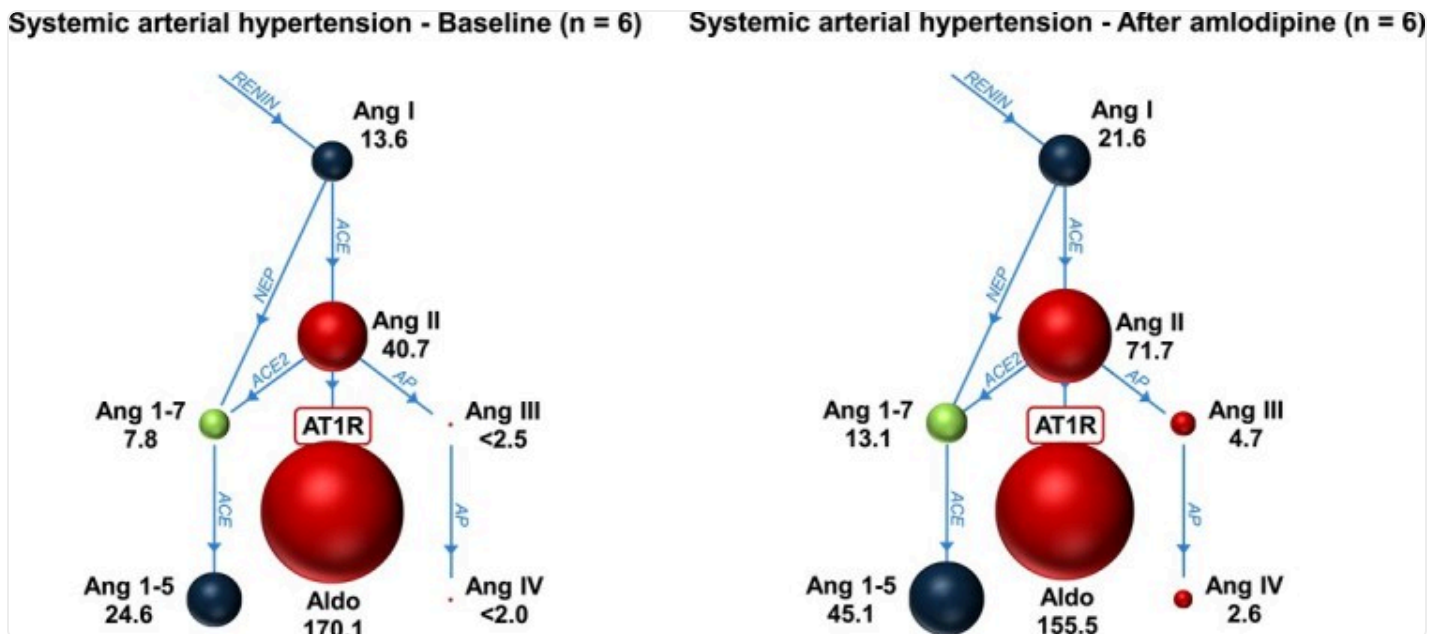
Figure 4.



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Dot plots for systolic blood pressure (SBP), serum creatinine concentration, and urinary protein-to-creatinine ratio in systemically hypertensive cats ($n = 6$, all of which had CKD) before (open triangles) and 2-4 weeks after the start of treatment with amlodipine besylate (0.625 mg PO q24h; solid triangles). Data points from cats with hypertensive ocular damage are denoted in red. Lines connect data from the same cat.

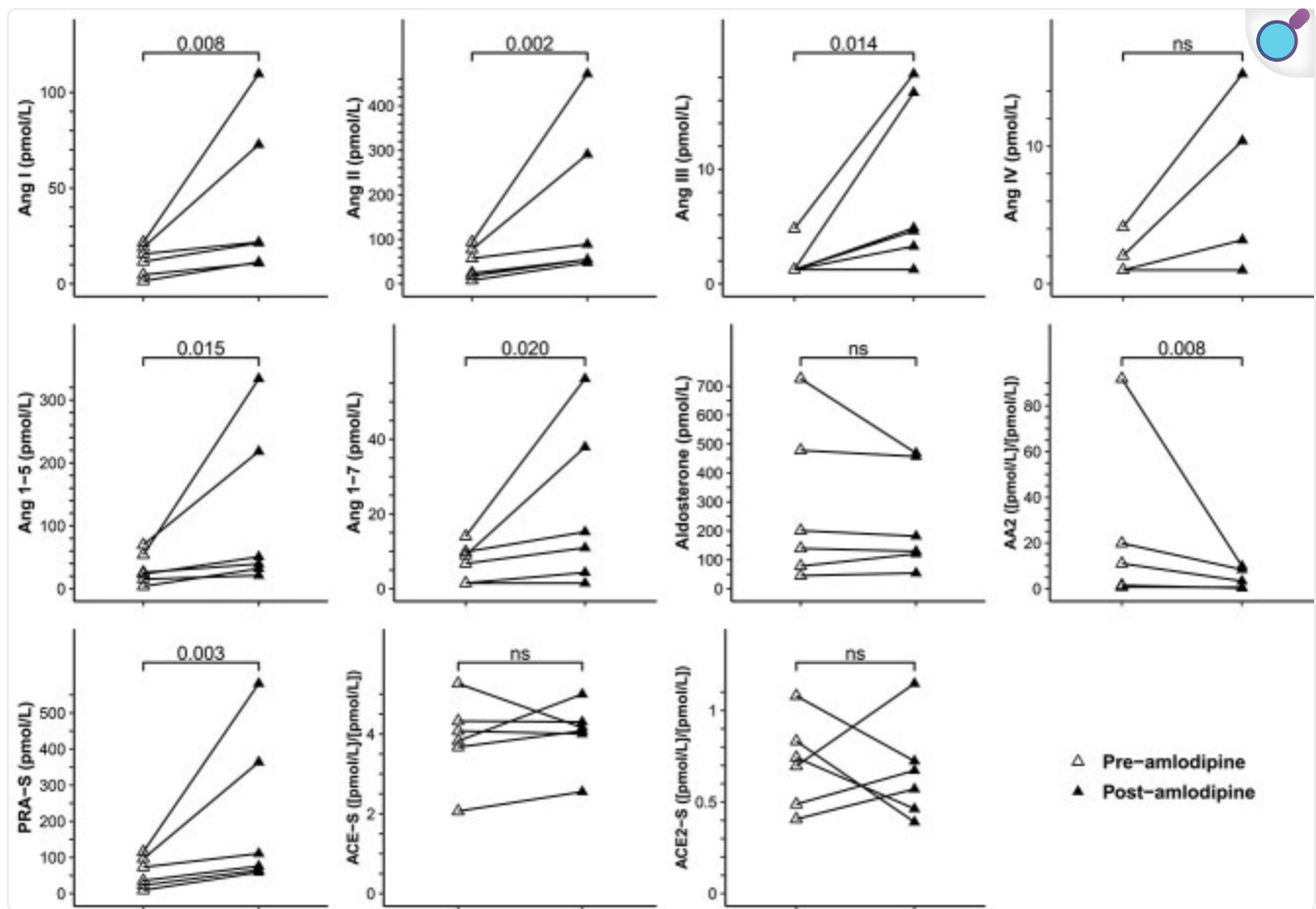
Figure 5.



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Serum equilibrium concentrations of angiotensin peptides and aldosterone of cats with systemic arterial hypertension before (baseline) and 2-4 weeks after the start of treatment with amlodipine besylate (0.625 mg PO q24h). See [Figure 2](#) legend for details and abbreviations.

Figure 6.



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Dot plots for serum equilibrium concentrations of circulating renin-angiotensin-aldosterone system markers in cats with systemic arterial hypertension (all of which had CKD) before (open triangles) and 2-4 weeks after the start of treatment with amlodipine besylate (0.625 mg PO q24h; solid triangles). Statistical comparisons between time points were conducted with the Wilcoxon signed-rank test. Significant *P*-values are noted above relevant brackets. Lines connect data from the same cat. See [Figure 3](#) legend for abbreviations.

Correlation between cRAAS markers and selected clinical, clinicopathologic, and echocardiographic variables

There was significant, though mostly mild to moderate, correlation between several cRAAS markers and SBP, hematocrit, serum sodium concentration, UPC, LA: Ao, and left ventricular FS ([Table 3](#)). Significant negative correlations between SBP and serum equilibrium concentrations of angiotensin I, II, III, IV, and 1-5, and PRA-S were noted, as were positive correlations between SBP and the AA2 ratio. Conversely, hematocrit was positively correlated with serum concentrations of angiotensin I, II, and III, and PRA-S. Serum sodium concentration was positively correlated with serum angiotensin II concentration, whereas UPC was negatively correlated with serum angiotensin 1-5 concentration. Finally, LA: Ao was negatively and LVFS was positively correlated with serum aldosterone concentrations.

Table 3.

Results of Spearman's correlation analysis of circulating renin–angiotensin–aldosterone system markers and select clinical, clinicopathologic, and echocardiographic variables in cats with non-hypertensive chronic kidney disease (International Renal Interest Society stages 2 or higher; $n = 17$), cats with untreated systemic arterial hypertension ($n = 6$; all of which had CKD), and healthy controls ($n = 17$). Significant P -values and corresponding correlation coefficients (ρ) are indicated in bold.

Variable		SBP	Hct	sSodium	sPotassium	sCreatinine	sSDMA	UPC
Ang I	P -value	.009	.038	.085	.78	.24	.07	.24
	ρ	−0.42	0.33	0.28	0.05	−0.19	−0.29	−0.19
Ang II	P -value	.005	.038	.33	.66	.15	.12	.15
	ρ	−0.44	0.33	0.16	0.07	−0.23	−0.25	−0.23
Ang III	P -value	.006	.012	.51	.89	.14	.10	.18
	ρ	−0.44	0.40	0.11	0.02	−0.24	−0.26	−0.22
Ang IV	P -value	.012	.055	.22	.46	.10	.11	.30
	ρ	−0.40	0.31	0.20	0.12	−0.26	−0.26	−0.17
Ang 1-5	P -value	.011	.25	.54	.85	.67	.39	.039
	ρ	−0.40	0.19	0.10	0.03	−0.07	−0.14	−0.33
Ang 1-7	P -value	.051	.066	.039	.89	.72	.19	.36
	ρ	−0.31	0.29	0.33	0.02	−0.06	−0.21	−0.15
Aldo	P -value	.61	.58	.14	.73	.87	.99	.80
	ρ	0.09	0.09	0.24	−0.06	0.03	0.00	0.04
AA2	P -value	.035	.31	.56	.73	.19	.27	.47
	ρ	0.34	−0.16	0.10	−0.06	0.21	0.18	0.12
PRA-S	P -value	.006	.04	.23	.76	.16	.11	.15
	ρ	−0.43	0.33	0.19	0.05	−0.23	−0.26	−0.23
ACE-S	P -value	.41	.51	.25	.92	.59	.96	.63
	ρ	−0.14	0.11	−0.19	−0.02	−0.09	−0.01	−0.08

Variable		SBP	Hct	sSodium	sPotassium	sCreatinine	sSDMA	UPC
ACE2-S	<i>P</i> -value	.52	.36	.45	.51	.27	.84	.11
	ρ	-0.11	-0.15	-0.12	-0.11	0.18	0.03	-0.25

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Abbreviations: AA2, angiotensin-to-angiotensin II ratio, a surrogate measure of adrenal responsiveness to angiotensin II; ACE-S, surrogate measure of angiotensin-converting enzyme (ACE) activity; ACE2-S, surrogate measure of ACE2 activity; Aldo, aldosterone; Ang, angiotensin; Hct, hematocrit; LA:Ao, left-ventricular-to-aorta ratio; LVFS, left ventricular fractional shortening; maxLVWTd, maximum left ventricular wall thickness in diastole; PRA-S, surrogate measure of plasma renin activity; sCreatinine, serum creatinine concentration; sPotassium, serum potassium concentration; sSDMA, serum symmetric dimethylarginine concentration; UPC, urinary protein-to-creatinine ratio.

Discussion

In the present study, serum equilibrium concentrations of angiotensin peptides and aldosterone were compared between cats with NHT-CKD, untreated SAH (in all cases with concurrent CKD as the presumed cause), and apparently healthy cats of similar age and sex relative to those of the NHT-CKD group. Evidence of cRAAS activation in cats with hypertensive or non-hypertensive CKD was lacking. Regardless of whether unadjusted or diet-, age-, or both diet- and age-adjusted models were used for intergroup comparisons, mean concentrations of angiotensin I, II, and III were consistently lower in untreated hypertensive cats relative to healthy cats, as was the value for the surrogate marker of PRA (PRA-S) used. Although the overall pattern indicated that cRAAS activity was lower in SAH cats, intermediate in NHT-CKD cats, and higher in healthy cats, controlling for diet, age, or both altered whether differences in various cRAAS markers between groups were statistically significant. For example, when adjusted for the feeding of a therapeutic renal diet but not age, mean angiotensin II and III concentrations were significantly different between cats with treatment-naïve hypertensive CKD, cats with NHT-CKD, and healthy cats (listed from lowest to highest mean concentrations). Although serum aldosterone concentrations did not differ between groups, the AA2 ratio was higher in the SAH group relative to healthy control groups across all statistical models, suggesting a circulating angiotensin II-independent release of aldosterone in those cats. Hypertensive cats treated with amlodipine experienced activation of the classical and

alternative cRAAS characterized by a significant increase in circulating concentrations of almost all measured angiotensin peptides (ie, angiotensin I, II, III, 1-7, and 1-5) and PRA-S.

The findings of our study are similar to those of a previous large study of 196 cats aged 9 years or older that were presented to a first-opinion practice in the United Kingdom. In that study, PRA and plasma aldosterone concentrations were compared between cats grouped according to the presence or absence of azotemia (ie, plasma creatinine > 2.0 mg/dL) and systemic BP status.⁸ Directly measured PRA was significantly lower in azotemic and non-azotemic cats with untreated systemic hypertension compared with non-azotemic, normotensive cats. In addition, plasma aldosterone concentrations were higher in azotemic cats with untreated systemic hypertension compared to azotemic or non-azotemic normotensive cats. Although our assay techniques and cutoffs for defining renal functional impairment and systemic hypertension differ from those of the previous study, we similarly documented lower PRA-S in hypertensive cats than in normotensive cats. While no significant differences in serum equilibrium concentrations of aldosterone were observed in the present study, only 6 hypertensive cats were evaluated, and a type II error cannot be excluded.

In the previous, larger study,⁸ plasma aldosterone-to-renin ratio was higher in azotemic hypertensive cats than in azotemic normotensive cats, suggesting that the higher aldosterone observed in hypertensive cats is not driven by an increase in PRA and downstream increase in circulating angiotensin II. Somewhat similarly, the present study found a significantly higher mean AA2 ratio in hypertensive cats when compared with the control group across all statistical models and to the NHT-CKD in models without age adjustment, though this appeared to be largely driven by data from 2 individuals. Others who have evaluated smaller samples of cats ($n \leq 16$) have also documented either the same or lower PRA and higher plasma aldosterone concentrations in cats with hypertensive CKD compared to healthy controls,^{6,7} again suggesting that aldosterone release appears independent of angiotensin II production in these cats, as is true for people with low-renin/high aldosterone essential hypertension.³²

Lower-than-normal PRA in the setting of systemic hypertension is an expected physiologic response and suggests normal regulation of the RAAS; that is, elevations in systemic BP should be sensed by the juxtaglomerular apparatus and suppress renin release.³⁰ As expected from normal physiology, we found that the concentrations of almost all angiotensin peptides and PRA-S were negatively correlated with SBP when all cats were considered. This seemingly contradicts the hypothesis that increased activation of the cRAAS is partially responsible for the development of systemic hypertension in patients with CKD. Interestingly, despite our and others' observation of lower PRA (or its surrogate) and circulating angiotensin II concentrations in untreated

hypertensive cats, selective blockade of the angiotensin II type 1 receptor using telmisartan significantly reduces systemic BP in both hypertensive^{33,34} and normotensive³⁵ cats. In recent decades, the role of the intrarenal renin–angiotensin system (RAS) in the regulation of kidney function and BP and in the pathophysiology of SAH has been increasingly highlighted.^{36,37} Notably, regulation of the intrarenal RAS appears to be independent of that of the circulating system,³⁸ and markers of cRAAS activity are poor predictors of renal angiotensin peptide concentrations in cats.³⁹ Intrarenal RAS activity remains uncharacterized in cats with hypertensive CKD, as reliable information on SBP is lacking in prior studies.⁴⁰⁻⁴² Therefore, it is possible (though unproven) that activation of the intrarenal RAS, uncoupled from and undetectable through measurement of the cRAAS, could contribute to SAH in cats with CKD.

In contrast with the present and other studies of cats with spontaneous CKD,⁶⁻⁸ previous work in surgically induced CKD revealed that cats with systemic hypertension after remnant-wrap have higher PRA and plasma aldosterone concentrations than control cats,⁹ and that cats undergoing 7/8 nephrectomy have significantly increased PRA and plasma concentrations of angiotensin I, angiotensin II, and aldosterone approximately 2 months after surgery, relative to baseline.⁴ It would be reasonable to conclude that these findings are specific to surgical models created by reductions in functional renal mass. However, a recent study that used a remnant-kidney CKD model found no difference in serum equilibrium concentrations of angiotensin peptides and aldosterone using RAS-Fingerprint in young adult, purpose-bred cats with induced CKD ($n = 15$, of which 6 were systemically hypertensive and treated with amlodipine) compared to life-stage- and sex-matched, community-owned healthy control cats ($n = 15$).⁴³ In that study, serum aldosterone concentrations and AA2 ratio were higher in cats with induced CKD, even after adjustment for SBP and amlodipine treatment. The time between kidney injury and evaluation of cRAAS markers could factor importantly into these studies' conclusions, as the 2 demonstrating greater-than-normal PRA in cats with hypertensive CKD^{4,9} evaluated markers within 3 months of injury, whereas the most recent, which did not demonstrate differences in angiotensin peptides, evaluated markers at least 27 months after injury.⁴³ In addition, whereas diet was standardized across groups in the earlier studies,^{4,9} in the most recent,⁴³ cats with induced CKD were fed a therapeutic renal diet while control cats were fed maintenance diets. Dietary intakes of sodium and potassium have a known influence on RAAS activity.^{11,12} Compared to maintenance diets, therapeutic renal diets for cats have lower sodium and higher potassium content,⁴⁴ which would be expected to stimulate the RAAS and adrenal release of aldosterone. This would amplify or mask group differences in circulating concentrations of RAAS markers, depending on the group receiving the diet higher in sodium and potassium. In the present study, only 8/17 (47%) cats with NHT-CKD and 4/6 (67%) cats with SAH were fed a therapeutic renal diet, suggesting that dietary influences of lesser magnitude would be expected here compared to the earlier study. Indeed, statistical conclusions for intergroup comparisons of only 2 RAAS components, serum angiotensin II and III,

were altered when these comparisons were adjusted for diet. However, after this adjustment, but not before, angiotensin II and III concentrations were significantly *lower* in NHT-CKD cats relative to healthy controls, and angiotensin II was significantly lower in SAH cats relative to NHT-CKD cats (though this effect was lost if age was concurrently adjusted for). Because a therapeutic renal diet would be expected to increase RAAS activity, the fact that 2 groups (NHT-CKD and SAH) in which these diets were fed with the highest frequency had lower RAAS activity than the group (healthy controls) for which they were not used (after adjustment for diet) suggests that diet might have contributed somewhat by counteracting or masking the tendency of other factors to suppress RAAS activity.

In the present study and the above-cited study,⁸ which evaluated client-owned cats, PRA or its surrogate significantly increased in animals receiving amlodipine, relative to pretreatment values. Since the design of our study, 3 other published studies reported serum equilibrium concentrations of angiotensin peptides in hypertensive cats treated with amlodipine, showing increased concentrations of components of both the classical and alternative RAAS pathways relative to non-hypertensive cats.^{16,17,43}

Correlation analyses were performed between RAAS markers and various physiologically relevant clinical and clinicopathologic variables. Due to the exploratory nature of this analysis, *P*-values reported were not adjusted for multiple comparisons, and findings require further validation. Several variables were significantly, though weakly ($\rho \leq 0.4$), correlated with cRAAS markers in our study. Significant correlations with $\rho > 0.4$ were only found for various cRAAS markers and each of SBP and LA:Ao. In a previous study,¹⁷ cats with cardiomyopathy had significantly higher serum equilibrium concentrations of angiotensin I, angiotensin 1-7, and aldosterone, and PRA-S values than healthy cats. In our study, which excluded cats with substantial heart disease, serum aldosterone concentrations were significantly negatively correlated with LA:Ao, which could be linked to the effects of volume status on the RAAS. For example, blood volume contracture, which would be accompanied by a smaller left atrial size (and therefore, lower LA:Ao), would be expected to be associated with RAAS activation, although angiotensin II was not significantly correlated with this variable.

Strengths of the present study include the comprehensive screening of cats, which included echocardiography to characterize co-existing cardiac pathologies; the exclusion of cats with concurrent conditions and cats receiving medications known to affect the RAAS; the evaluation of both classical and alternative RAAS pathways; and the recruitment of control cats that were age- and sex-matched to the NHT-CKD group.

This study also has important limitations. First, the sample size was small, particularly in the SAH group, which limits statistical power to detect mild or moderate differences between groups, and nonsignificant findings should be interpreted cautiously. Also, as cats were recruited from those regularly presented to a veterinary teaching hospital, time of day of sample collection was not standardized. Because RAAS activity is subject to circadian periodicity,⁴⁵ this omission could have confounded our results. In addition, diet was not standardized, although all cats were fasted prior to sampling, which should decrease the influence of feeding on the cRAAS.⁴⁶ In addition, neither imaging of the urinary tract nor measurement of blood or urinary markers of active kidney injury was performed during screening, which limited our characterization of the cats' kidney diseases; however, all cats were clinically stable at the time of enrollment. Finally, our assessment of the RAAS was limited to circulating markers of this system, and the intrarenal RAS was not explored.

In conclusion, we did not detect activation of the cRAAS in cats with NHT-CKD or SAH, relative to age- and sex-matched controls. The intrarenal RAS should be explored in cats with these conditions for a more complete understanding of the role of this system in their pathophysiology.

Supplementary Material

aalag079_cRAAS_in_cats_with_CKD_or_SAH_supplement_R1

[aalag079_craas_in_cats_with_ckd_or_sah_supplement_r1.docx](#) (26.8KB, docx)

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Abbreviations

2D

2-dimensional

BP

blood pressure

CKD

chronic kidney disease

cRAAS

circulating renin–angiotensin–aldosterone system

IRIS

International Renal Interest Society

LA:Ao

left-atrial-to-aortic dimension ratio in short axis

FS

left ventricular fractional shortening

maxLVWTd

maximum end-diastolic left ventricular wall thickness

NHT-CKD

non-hypertensive chronic kidney disease

RAAS

renin–angiotensin–aldosterone system

RAS

renin–angiotensin system

SAH

systemic arterial hypertension

SBP

systolic blood pressure

SDMA

symmetric dimethylarginine

TT4

total thyroxine

UPC

urinary protein-to-creatinine ratio

USG

Urine specific gravity

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Conflicts of interest

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Data availability

All data are included in the manuscript or supplemental material.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

aalag079_cRAAS_in_cats_with_CKD_or_SAH_supplement_R1

[aalag079 craas in cats with ckd or sah supplement r1.docx](#) (26.8KB, docx)

Data Availability Statement

All data are included in the manuscript or supplemental material.

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Severity, etiology and outcomes of ionized hypocalcemia in azotemic cats and dogs

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Objective: To describe the severity, timing, etiology, and outcome of ionized hypocalcemia in a population of dogs and cats with all-cause azotemia.

Methods: Medical records of dogs and cats with concurrent ionized hypocalcemia and azotemia, in which ionized calcium (iCa) was measured within 24 h of serum biochemistry testing were retrospectively reviewed from a single institution over a 10-year period (January 1st, 2014, to December 31st, 2023). Ionized hypocalcemia was defined as iCa < 1.21 mmol/L in cats and < 1.30 mmol/L in dogs. Moderate hypocalcemia was defined as iCa < 1.00 mmol/L and severe hypocalcemia as iCa < 0.75 mmol/L. Azotemia was defined as a creatinine concentration $\geq 141 \mu\text{mol/L}$ (1.7 mg/dL). The primary cause of azotemia was determined by review of the medical record, and diagnostic criteria for acute kidney injury and chronic kidney disease were based on International Renal Interest Society guidelines. Data extracted included signalment, clinicopathologic variables, etiology of azotemia, and outcome. Outcome was defined as survival to discharge or non-survival.

Results: 302 cats and 646 dogs met the inclusion criteria. Ionized hypocalcemia was mild in 229/302 (75.8%) cats and 551/646 (85.3%) dogs, moderate in 69/302 (22.8%) cats and 83/646 (12.8%) dogs, and severe in 4/302 (1.3%) cats and 12/646 (1.9%) dogs. The lowest iCa occurred within 48 h of admission in 155/223 (69.5%) cats and 338/465 (72.7%) dogs hospitalized for at least 48 h. In cats, both renal (176/302, 58.3%) and postrenal (126/302, 41.7%) causes of azotemia were common, whereas renal causes predominated in dogs (602/646, 93.2%). Inflammatory/ischemic etiologies were the most common cause of acute kidney injury in both cats (54/90, 60.0%) and dogs (230/383, 60.1%). Ionized calcium showed weak correlations with blood phosphorus ($r = -0.36$ cats, -0.31 dogs), pH ($r = 0.21$ cats, 0.15 dogs), and creatinine concentrations ($r = -0.30$ cats, -0.26 dogs) (all $p < 0.001$). Clinical signs potentially attributable to hypocalcemia were reported in 9/16 (56.3%) severely hypocalcemic patients.

Conclusion: Ionized hypocalcemia in azotemic cats and dogs is generally mild and occurs early in hospitalization. It is only weakly associated with higher blood phosphorus and creatinine concentrations and lower blood pH. Routine monitoring of iCa in azotemic patients is recommended. These findings may not be generalizable to all azotemic patients, as iCa measurement was performed at the discretion of the attending clinician.

KEYWORDS

acute kidney injury, azotemia, calcium, chronic kidney disease, phosphorus, urinary obstruction

1 Introduction

Hypocalcemia has been associated with numerous adverse effects including hypotension, systolic dysfunction, arrhythmias, neurologic impairment, new-onset acute kidney injury (AKI) and overall increased mortality in both human and small animal medicine (1–6). Measurement of ionized calcium (iCa) rather than total calcium is recommended because it represents the biologically active fraction. Ionized hypocalcemia is frequently observed in association with azotemia in both cats and dogs (7) and has been reported in 10–26% of cats with chronic kidney disease (CKD) (8), 10–56% of dogs with CKD (9–11), and 34–75% of cats with urethral obstruction (12–14).

In critically ill people with AKI, ionized hypocalcemia has been independently associated with all-cause mortality (15, 16). This effect may be modified with treatment, suggesting a potential opportunity for intervention. In small animals, ionized hypocalcemia has been linked with increased mortality in dogs with AKI (17, 18) but this finding was not replicated in cats (19).

Proposed mechanisms for hypocalcemia in azotemia include hyperphosphatemia, decreased 1,25-dihydroxycholecalciferol formation, and parathyroid hormone resistance due to loss of functional nephrons and/or decreased glomerular filtration rate (20, 21). Hypocalcemia in AKI is thought more likely to be symptomatic because the degree of hyperphosphatemia is often greater than that observed in CKD (22, 23). However, the relationship between serum calcium and phosphate levels in people with AKI is inconsistent (24). The relationships between the severity of ionized hypocalcemia, underlying etiology of azotemia, and concurrent hyperphosphatemia in cats and dogs remain poorly defined, as large-scale comparisons across all causes of azotemia are lacking.

The aims of this study were to describe the severity, timing, and associated etiologies of ionized hypocalcemia, and to evaluate its relationship with serum phosphorus concentration and clinical outcome in a large population of dogs and cats with all-cause azotemia.

2 Materials and methods

A computerized search for biochemistry panels¹ from dogs and cats performed between January 1st, 2014, and December 31st, 2023 was performed using the medical records database at the University of California Davis, William R. Pritchard Veterinary Medical Teaching Hospital. Point of care blood gas and electrolyte panels² that included iCa were identified over the same time period. Animals that had both a biochemistry panel and a blood gas and electrolyte panel were included. Animals without azotemia defined by a blood creatinine concentration $\geq 141 \mu\text{mol/L}$ (1.7 mg/dL) on biochemistry with an iCa measured within 24 h of each other were excluded. The medical records for all cases with at least one instance of concurrent ionized

hypocalcemia and azotemia were reviewed. If concurrent azotemia and ionized hypocalcemia were documented in the same patient on multiple visits, only the visit with the lowest iCa was included. Animals could only be included once, and all other visits were excluded to avoid duplicate inclusion bias. For hospitalized patients (≥ 48 h) in which multiple blood gas panels were collected, both the iCa on presentation and the lowest iCa during hospitalization were recorded alongside the timing of the most profound ionized hypocalcemia. Patients with documented azotemia and ionized hypocalcemia only on samples taken more than 24 h apart were excluded. Cases with incomplete medical records sufficient to determine the cause of azotemia and outcome were excluded.

2.1 Population data

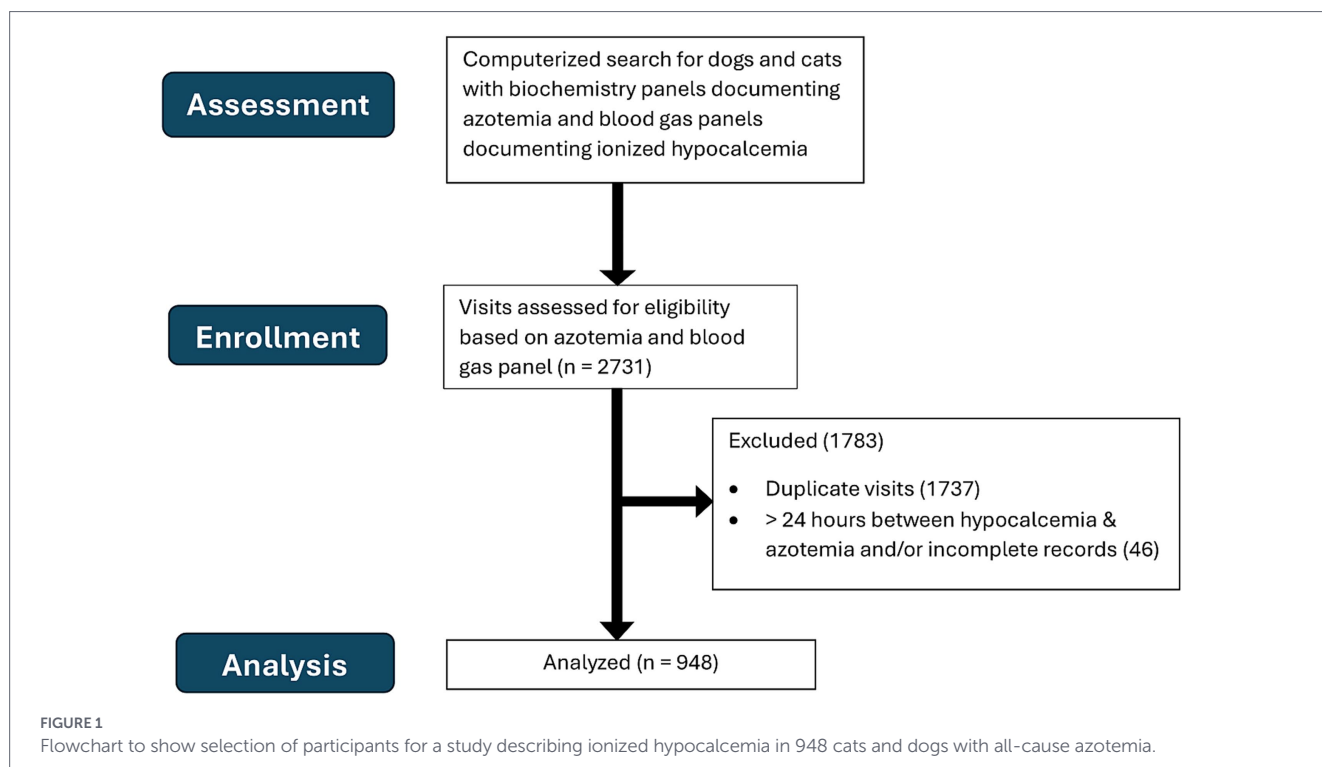
Data extracted from the electronic medical record included signalment (species, age, sex, reproductive status, bodyweight, breed), history and physical examination findings at presentation, biochemistry, blood gas analysis, ultrasonographic findings, treatments, and outcome. Outcome was defined as survival to discharge or non-survival. Non-survivors were either euthanized or died in hospital.

2.2 Etiology of azotemia

The primary cause of azotemia was determined by review of the medical record including history and diagnostic results. When 2 or more potential causes of azotemia were applicable to one patient, the disease process deemed most likely to be the cause following extensive medical record review was selected. Diagnostic criteria for AKI and CKD were based on the International Renal Interest Society (IRIS) guidelines (25). Acute on chronic kidney injury or CKD were diagnosed based on prior clinicopathologic data where available and ultrasonographic evidence consistent with CKD (decreased renal volume, irregular contour, and/or decreased corticomedullary distinction) (26). The etiology of AKI and acute on chronic kidney injury was classified as described in a previous veterinary study (27). Categories included inflammatory/ischemic, infectious, nephrotoxic, neoplastic, other, and unknown. The inflammatory/ischemic category included any cause of severe systemic inflammation or presumed ischemic renal injury. Nephrotoxicosis was defined on the basis of recent ingestion of grapes/raisins (dogs), toxic lilies (cats), or overdose of non-steroidal anti-inflammatory drugs. Pyelonephritis was diagnosed based on a positive urine culture, urine sediment findings, and ultrasonographic findings consistent with pyelonephritis (28). Leptospirosis was diagnosed in cases with a fourfold or higher increase in *Leptospira* agglutination titer at a single laboratory between acute- and convalescent-phase serum specimens (29). Neoplasia was determined as the etiology when kidney injury was caused by direct renal infiltration of neoplasia (i.e., lymphoma, carcinoma) or resulted from hypercalcemia which was confirmed to be paraneoplastic. “Other” etiology included AKI or acute on chronic kidney injury secondary to non-neoplastic hypercalcemia and glomerulopathies, diagnosed based on the presence of renal proteinuria (urine protein to creatinine ratio > 2) with or without renal biopsy (30).

¹ Chemistry analyzer, Hitachi 917, Roche Diagnostics, Indianapolis, IN.

² ABL 815, Radiometer Medical A/S, Copenhagen, Denmark.



2.3 Clinicopathological parameters

Blood gas and electrolyte analyses were conducted using a point-of-care analyzer (ABL 815, Radiometer Medical A/S, Copenhagen, Denmark) on heparinized blood samples. Institutional reference intervals for this blood gas analyzer were previously generated from 40 healthy cats (31) and 40 healthy dogs (unpublished data). The reference interval for pH was 7.266–7.422 in cats and 7.345–7.433 in dogs. The reference interval for iCa was 1.21–1.45 mmol/L in cats and 1.30–1.46 mmol/L in dogs. Therefore, ionized hypocalcemia was defined using the lower limit of these institution-specific analyzer-derived reference intervals as iCa < 1.21 mmol/L in cats and < 1.30 mmol/L in dogs. Moderate hypocalcemia was defined as iCa < 1.00 mmol/L in both dogs and cats based on previously published veterinary literature evaluating ionized hypocalcemia (7). Severe hypocalcemia was defined as iCa < 0.75 mmol/L in both dogs and cats based on the suggested treatment threshold in the IRIS consensus guidelines for AKI (32). Any reported clinical signs attributed to hypocalcemia in severely hypocalcemic patients were recorded. Serum creatinine and phosphate concentrations were recorded from a serum biochemistry profile through a reference laboratory.

2.4 Statistical methods

Statistics were performed using commercially available software.³ Numerical data were assessed for normality visually and with the Shapiro–Wilk test. Continuous parametric data were described

as mean ($\pm$ standard deviation) and non-parametric data as median (25th–75th percentile). Spearman rank correlations were calculated to assess the relationship between blood iCa and pH, serum phosphate and creatinine concentrations. A p value < 0.05 was considered significant. Correlation was considered negligible if $r = 0.0$ –0.10, weak if $r = 0.11$ –0.39, moderate if $r = 0.40$ –0.69, strong if $r = 0.70$ –0.89 and very strong if $r = 0.90$ –1.00 (33).

3 Results

3.1 Signalment

Of 2,731 visits assessed for eligibility, 1737 duplicate visits were excluded. An additional 44 visits were excluded because serum creatinine and iCa measurements were obtained more than 24 h apart, and 2 visits were excluded because of incomplete medical records. Of the remaining 994 visits (321 cats, 673 dogs) identified with azotemia and ionized hypocalcemia during the same visit, 302 cats and 646 dogs satisfied the inclusion criteria (Figure 1). Age was not recorded in 5 cats. The median age of the remaining 297 cats was 9.1 (5.5–12.5) years. In 1 cat, bodyweight was not recorded. In the remaining 301 cats, the median bodyweight was 4.5 (3.6–5.7) kg. Amongst the 302 cats, there were 120 (39.7%) females (5 intact) and 180 (59.6%) males (4 intact). In 2 cats, sex was not recorded. The most common breeds were domestic shorthair (202/302, 66.9%), domestic longhair (31/302, 10.3%), and domestic medium hair (26/302, 8.6%). In 4 dogs, age was not recorded. The median age of the remaining 642 dogs was 9.5 (6.0–12.6) years. In 5 dogs, bodyweight was not recorded. In the remaining 641 dogs, median bodyweight was 12.8 (6.2–29) kg. Amongst the 646 dogs, there were 316 (48.9%) females (51 intact) and 330/646 (51.1%) males (64 intact). The most common breeds were

³ Graph Pad Prism v 10.5, La Jolla, CA.

crossbreed (161/646, 24.9%), Labrador retriever (56/646, 8.7%), and chihuahua (22/646, 3.4%).

3.2 Etiology of azotemia

Inflammatory/ischemic causes included hypovolemic shock, pancreatitis, peritonitis, pyometra, severe gastroenteritis, pneumonia, diabetic ketoacidosis, trauma, heatstroke, anesthesia-related events, prolonged hypotension, hemorrhagic shock, cardiopulmonary arrest, systolic dysfunction, severe bradyarrhythmias, and congestive heart failure. Common inflammatory/ischemic conditions in cats included cardiogenic/thromboembolic disease (19 cases), hypovolemic shock/trauma (10 cases), diabetic ketoacidosis/hyperosmolar hyperglycemic syndrome (9 cases), pancreatitis (8 cases), septic shock/sepsis (6 cases), and gastrointestinal foreign body or perforation (5 cases). Common inflammatory/ischemic conditions in dogs included sepsis/septic shock/septic peritonitis/pyometra/bile peritonitis (45 cases), cardiogenic disease including congestive heart failure/cardiomyopathy/arrhythmias (31 cases), pancreatitis/diabetic ketoacidosis/hyperosmolar hyperglycemic syndrome (30 cases), hypovolemic or hemorrhagic shock/trauma/hypovolemia (28 cases), gastrointestinal foreign body/obstruction/perforation/gastric dilatation-volvulus (18 cases), and heat stroke (4 cases). Several patients had overlapping inflammatory or ischemic disease processes, and the reported categories reflect the predominant condition identified on medical record review. Infectious causes included pyelonephritis and leptospirosis. Postrenal causes of azotemia included ureteral obstruction, urethral obstruction and urinary tract trauma leading to uroabdomen. In cats, renal causes of azotemia were diagnosed in 176/302 (58.3%) of cats while postrenal causes were responsible for azotemia in 126/302 (41.7%) of cats (Table 1). Amongst the renal causes, AKI was diagnosed in 90/302 (29.8%) of cats. In cats with AKI, an inflammatory/ischemic cause was most common and was documented in 54/90 (60.0%) of cats. Acute on chronic kidney injury was diagnosed in 62/302 (20.5%) of which the majority (33/62, 53.2%) were of unknown cause. Chronic kidney disease was diagnosed as the cause of azotemia in 24/302 (7.9%) of cats. Amongst the 126 cats with postrenal causes of azotemia, 82/126 (65.1%) were diagnosed with ureteral obstruction and 41/126 (32.5%) with urethral obstruction. Three cases (1.0%) were diagnosed with urinary tract trauma. In dogs, renal causes of azotemia were diagnosed in 602/646 (93.2%) and postrenal causes were diagnosed in 44/646 (6.8%) of dogs. Acute kidney injury was diagnosed in 383/646 (59.3%) dogs and acute on chronic kidney injury in 169/646 (26.2%) of dogs. The most common etiologies of AKI were inflammatory/ischemic (230/383, 60.1%) and unknown (72/383, 18.8%). Acute on chronic kidney injury was diagnosed in 169/646 (26.2%) dogs with the majority of cases being of unknown etiology (95/169, 56.2%). Chronic kidney disease as the cause of azotemia was diagnosed in 50/646 (7.7%) dogs. In dogs with postrenal causes, the majority were due to urethral (18, 2.8%) or ureteral (14, 2.2%) obstruction, but trauma (10, 1.5%) and other causes including nephrolithiasis (1, 0.2%) and perineal hernia (1, 0.2%) were also identified.

3.3 Timing and severity of ionized hypocalcemia

On presentation, the median iCa amongst the 302 cats was 1.20 (1.12–1.28) mmol/L. The median lowest iCa concentration was 1.13 (1.02–1.19) mmol/L. At the time of the most profound hypocalcemia

TABLE 1 Primary etiologies of azotemia in 302 cats and 646 dogs with ionized hypocalcemia and all-cause azotemia.

Etiology	Cats, n (%)	Dogs, n (%)
Postrenal	126 (41.7)	44 (6.8)
Ureteral obstruction	82 (27.2)	14 (2.2)
Urethral obstruction	41 (13.6)	18 (2.8)
Urinary tract trauma	3 (1.0)	10 (1.5)
Nephrolithiasis	0 (0)	1 (0.2)
Perineal hernia	0 (0)	1 (0.2)
Acute kidney injury	90 (29.8)	383 (59.3)
Inflammatory/ischemic	54 (17.9)	230 (35.6)
Unknown	19 (6.3)	72 (11.1)
Neoplasia	7 (2.3)	5 (0.8)
Nephrotoxic	6 (2.0)	14 (2.2)
Infectious	4 (1.3)	46 (7.1)
Other ^a	0 (0)	16 (2.5)
Acute-on-chronic kidney injury	62 (20.5)	169 (26.2)
Unknown	33 (10.9)	95 (14.7)
Inflammatory/ischemic	16 (5.3)	38 (5.9)
Infectious	10 (3.3)	16 (2.5)
Other ^a	2 (0.7)	19 (2.9)
Neoplasia	1 (0.3)	1 (0.2)
Chronic kidney disease	24 (7.9)	50 (7.7)
Total	302	646

^aOther includes 2 cats and 35 dogs with glomerulopathy.

documented for all patients, 229 of 302 cats (75.8%) were mildly hypocalcemic, 69 (22.8%) were moderately hypocalcemic, and 4 cats (1.3%) were severely hypocalcemic. The majority of cats (223/302, 73.5%) were hospitalized for at least 48 h. Amongst this population, the lowest iCa was documented within the first 24 h of hospitalization in 99/223 (44.4%) of cats with the lowest iCa documented on Day 2 in 56/223 (25.1%) and Day 3 in 30/223 (13.5%) (Figure 2). The latest documented lowest iCa in 223 cats hospitalized for at least 2 days was on Day 21 in one cat.

On presentation, the median iCa amongst all 646 dogs was 1.22 (1.14–1.29) mmol/L. The median lowest documented iCa concentration for all dogs was 1.14 (1.04–1.21) mmol/L. At the time of the most profound hypocalcemia, 551/646 dogs (85.3%) were mildly hypocalcemic, 83 (12.8%) were moderately hypocalcemic, and 12 dogs (1.9%) were severely hypocalcemic. Of the 646 dogs, 465 (71.8%) were hospitalized for 48 h or more. Amongst this population, the most profound ionized hypocalcemia occurred within the first 24 h of hospitalization in 232/465 dogs (50.0%) with the lowest iCa documented on Day 2 in 106/465 (22.8%) and on Day 3 in 56/465 (12.0%). Clinical signs were reported in 9/16 severely hypocalcemic animals, including altered mentation ($n = 7$) and neuromuscular signs such as tremors or seizures ($n = 6$); several animals exhibited both (Table 2).

3.4 Serum phosphate concentration

On the biochemistry panel measured within 24 h of the lowest iCa, the majority of cats (211/302, 69.9%) were hyperphosphatemic with

78/302 (25.8%) normophosphatemic and 13/302 (4.3%) hypophosphatemic. Amongst the dogs, the contemporaneous biochemistry panel identified that 536/646 (83.0%) were hyperphosphatemic, 102/646 (15.8%) were normophosphatemic and 8/646 (1.2%) were hypophosphatemic at the time of the most profound ionized hypocalcemia.

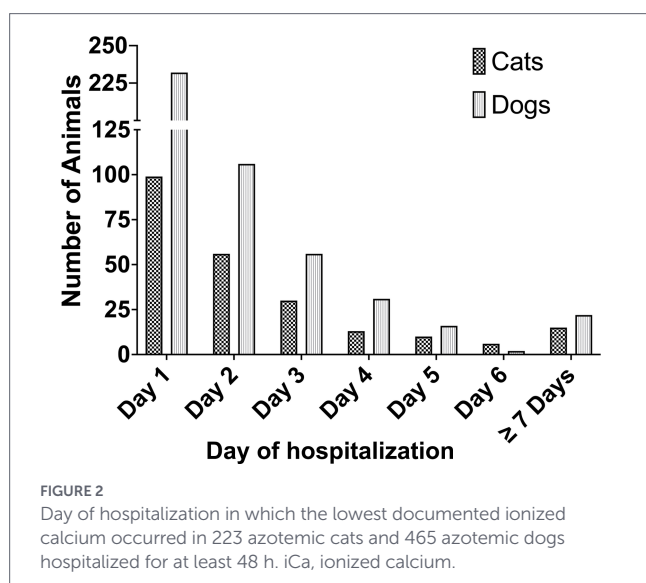
3.5 Correlation between ionized hypocalcemia, pH, phosphorus and creatinine concentration

In cats, there was a weak positive relationship between iCa and blood pH, ($r = 0.21$, $p = 0.0002$), a weak negative correlation between iCa and phosphorus ($r = -0.36$, $p < 0.0001$), and a weak negative

correlation between iCa and creatinine ($r = -0.30$, $p < 0.0001$). In dogs, there was a weak positive relationship between iCa and blood pH ($r = 0.15$, $p = 0.0002$), a weak negative relationship between iCa and phosphorus ($r = -0.31$, $p < 0.0001$), and a weak negative correlation between iCa and creatinine ($r = -0.26$, $p < 0.0001$).

3.6 Outcome

Amongst the overall population, 204/302 (67.5%) cats were discharged from the hospital while 15/302 (5.0%) died and 83/302 (27.5%) were euthanized. The majority of dogs (406/646, 62.8%) were discharged from the hospital while 40/646 (6.2%) died and 200/646 (31.0%) were euthanized. Among cats with mild, moderate, and severe ionized hypocalcemia, survival to discharge occurred in 155/229 (67.7%), 47/69 (68.1%), and 2/4 (50.0%), respectively. Among dogs, survival to discharge occurred in 350/551 (63.5%), 51/83 (61.4%), and 5/12 (41.7%), respectively. Due to the small number of severely hypocalcemic patients, formal statistical comparisons between severity groups were not performed.



4 Discussion

In a large group of azotemic cats and dogs with at least one documented episode of ionized hypocalcemia, the majority of cases were mild, with the most profound hypocalcemia typically occurring within the first 48 h of hospitalization in those admitted for two or more days.

In people, hypocalcemia has been documented in 35–43% of patients with AKI (16, 34) and in 22–30% of those with CKD (35–37). Ionized hypocalcemia has been documented in 10–56% of dogs with CKD (9–11), 10–26% of cats with CKD⁸, and 34–75% of cats with urethral obstruction (12–14). Published cohorts suggest that ionized

TABLE 2 Clinical signs and concurrent diagnoses in severely ionized hypocalcemic cats and dogs with azotemia.

Species	Lowest iCa (mmol/L)	Concurrent diagnoses	Clinical signs recorded
Feline	0.59	Urethral obstruction; severe azotemia; hyperphosphatemia; hyperkalemia	Stuporous; improved after unblocking
Feline	0.61	Acute-on-chronic kidney injury; severe azotemia; hyperphosphatemia	Obtunded
Feline	0.68	Ureteral obstruction; hyperkalemia; atrial standstill	Obtunded
Feline	0.69	Ureteral obstruction; congestive heart failure	Neurologic signs/twitching
Canine	0.57	Dialysis-dependent chronic kidney disease	Seizures
Canine	0.62	Acute kidney injury secondary to pyelonephritis	Pelvic limb twitching
Canine	0.69	Acute kidney injury; myocarditis; systemic hypertension	Obtunded
Canine	0.74	Immune-mediated glomerulonephritis; lymphoma	None documented
Canine	0.48	Heat stroke; hepatic failure; acute kidney injury	Facial twitching
Canine	0.54	Pyometra; duodenal perforation; severe shock	Periarrest/hypotension
Canine	0.65	Acute kidney injury secondary to pyelonephritis	Muscle fasciculations
Canine	0.69	Acute-on-chronic kidney injury; congestive heart failure	None documented
Canine	0.72	Linear foreign body; acute kidney injury	None documented
Canine	0.55	Ruptured pyometra; shock	Stuporous
Canine	0.73	Acute-on-chronic kidney injury	Seizures
Canine	0.74	Acute kidney injury requiring hemodialysis; pancreatitis	None documented

hypocalcemia is frequent in both dogs and cats with AKI but rarely report a prevalence (7, 18). Unfortunately, it was not possible to determine the prevalence of hypocalcemia in this study of all-cause azotemia given that not all animals had an iCa measured within 24 h of a serum biochemistry panel.

Acute kidney injury was the most common category of azotemia in dogs while postrenal azotemia was more common than AKI in cats in this study. Inflammatory/ischemic causes of AKI were the most frequently identified etiology for AKI in dogs, documented in 60% of this population. These findings are similar to those of a previous study of AKI in dogs reporting inflammatory/ischemic causes in 58% of cases (27). The category of inflammatory/ischemic causes includes a broad range of coexisting conditions which likely accounts for it being so frequent. In cats with postrenal azotemia, ureteral obstruction was more common than urethral obstruction. This may reflect the greater severity and duration of azotemia commonly associated with ureteral obstruction, potentially contributing to the development of hypocalcemia. The markedly higher proportion of postrenal azotemia overall in cats compared with dogs in this study may reflect both true species differences and referral population bias. Ureteral and urethral obstruction are common causes of severe azotemia in cats, and ionized hypocalcemia has previously been reported in 34–75% of cats with urethral obstruction (12–14). In contrast, postrenal azotemia is comparatively less frequent in dogs. In addition, the high caseload of ureteral obstruction managed at our tertiary referral institution likely contributed to the large proportion of cats with postrenal disease. In hospitalized patients, the lowest iCa was documented within the first 24 h of presentation in 50.0% of dogs and 44.4% of cats. In the remaining patients, the lowest documented iCa tended to occur within the first few days of hospitalization with the majority being within the first 48 h. This may be because the severity of azotemia is likely to be greater on initial hospitalization and in many patients, it improves with therapy with a concurrent improvement in ionized hypocalcemia. However, these findings reflect the timing of the lowest documented iCa during hospitalization and were likely influenced by differences in hospitalization duration and frequency of serial monitoring between patients. It is important to note that a significant proportion of both cats and dogs developed ionized hypocalcemia later during hospitalization. Although these animals were azotemic, other contributing factors such as concurrent disease (for example sepsis or pancreatitis), intravenous fluid diuresis, citrated blood product transfusion, and medications including loop diuretics and sodium bicarbonate likely contributed to the development of hypocalcemia (7, 20, 38, 39). These findings highlight the importance of ongoing monitoring of iCa during hospitalization for treatment of azotemia.

In both azotemic cats and dogs, iCa demonstrated statistically significant but weak correlations with blood pH, phosphorus, and creatinine concentrations. In both species the strongest association was with phosphorus ($r = -0.36$ in cats; $r = -0.31$ in dogs); yet this remained weak. Thus, while lower iCa concentrations tended to occur alongside lower blood pH and higher phosphorus and creatinine concentrations, none of these parameters alone appear clinically useful for predicting the degree of ionized hypocalcemia in azotemic cats or dogs. Traditionally, hypocalcemia in azotemia has been attributed to hyperphosphatemia, either through a mass-law effect or reduced calcitriol production, but our findings suggest that these mechanisms

alone are insufficient to explain the observed changes. Additional contributors to hypocalcemia in this population may include renal resistance to parathyroid hormone, impaired calcitriol synthesis independent of phosphate, complexing of calcium by uremic toxins, and the effects of systemic inflammation or critical illness on calcium homeostasis. Further studies are needed to explore the specific causes of hypocalcemia and its relationship with hyperphosphatemia in cats and dogs with azotemia. Until then, direct measurement of iCa should be considered in all azotemic cats and dogs, regardless of acid–base status or the degree of hyperphosphatemia or azotemia.

The treatment of hypocalcemia in azotemic patients is often complicated by concerns for increasing the calcium-phosphorus product and promoting metastatic calcification. For this reason, the optimum level at which to initiate therapy is uncertain. Recent IRIS consensus guidelines advocate treatment of ionized hypocalcemia in patients with AKI in the presence of compatible clinical signs and/or severe ionized hypocalcemia (<0.75 mmol/L), and irrespective of hyperphosphatemia (32). The recommended goal of treatment is to eliminate clinical signs rather than normalization of hypocalcemia. Severe hypocalcemia appeared relatively uncommon in this group of azotemic patients and was documented in 4 cats and 12 dogs. Amongst these patients, many had clinical signs that could be due to hypocalcemia. However, each patient had one or more comorbidities, and clinical signs cannot be solely attributed to hypocalcemia. Nonetheless, close monitoring is recommended, and calcium supplementation should be considered if compatible signs develop.

Ionized hypocalcemia has been associated with adverse outcomes in cats and dogs in various disease states (1, 5). However, there are few studies specifically evaluating the association of ionized hypocalcemia and outcome in animals with azotemia. One small study found an association between ionized hypocalcemia and non-survival in cats with urethral obstruction, although there was significant overlap in iCa between survivors and non-survivors (14). Ionized hypocalcemia has been associated with increased mortality in dogs with AKI (17, 18) but did not appear to be prognostic in a study of cats with AKI (19). Ionized hypocalcemia in azotemic and critically ill people has been linked with increased mortality and this relationship appears to be more pronounced when hypocalcemia is sustained over time rather than transient (15, 16). Some evidence has shown that correction of hypocalcemia in azotemic people improves survival (40). However, randomized trials are lacking and some sources even suggest harm in correcting hypocalcemia in certain patient populations (41). Unfortunately, small group sizes in this study precluded meaningful statistical analysis of outcome related to the severity of hypocalcemia as the majority of hypocalcemia was mild. However, 67.5% of cats and 62.8% of dogs were discharged from the hospital.

This study has several limitations, many of which are inherent to its retrospective design. The total number of cases likely underestimates the true prevalence of ionized hypocalcemia, as blood gas analysis and serum chemistry were performed at the discretion of the primary clinician. Many cases were excluded due to the lack of contemporaneous values. In addition, these blood tests are likely more frequently conducted in animals perceived to be more severely ill so the results of this study may not reflect a general population of azotemic animals. Additionally, recurrence of ionized hypocalcemia was likely undocumented in many patients due to inconsistent monitoring and varied duration of hospitalization. Prospective studies are needed to generate more objective data on the

incidence, recurrence, clinical impact, and outcomes of hypocalcemia. Such research would also help to clarify the optimal timing, duration, and potential adverse effects of treatment.

5 Conclusion

In conclusion, ionized hypocalcemia in a population of azotemic cats and dogs that had evaluation of iCa within 24 h of a biochemistry panel was generally mild and typically occurred within the first 48 h of hospitalization. There was only a weak correlation between lower iCa and lower blood pH and higher phosphorus and creatinine concentrations, suggesting other mechanisms of hypocalcemia are important in this patient population. Further studies are needed to determine the optimal management of ionized hypocalcemia in azotemic patients, but routine monitoring of iCa is recommended.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Author contributions

HP: Data curation, Investigation, Writing – original draft, Writing – review & editing. SE: Data curation, Formal analysis, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. KH: Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing.

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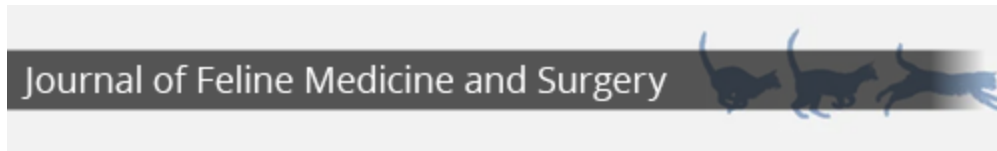
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Chronic use of maropitant for the management of vomiting and inappetence in cats with chronic kidney disease: a blinded, placebo-controlled clinical trial

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Abstract

Objectives

Maropitant is commonly used for acute vomiting. A pharmacokinetic and toxicity study in cats indicated that longer term usage appears safe. The aim of this study was to assess the efficacy of maropitant for management of chronic vomiting and inappetence associated with feline chronic kidney disease (CKD).

Methods

Forty-one cats with stable International Renal Interest Society Stage II or III CKD, no known concurrent illness, and a complaint of chronic vomiting and inappetence attributed to CKD were enrolled in a randomized, placebo-controlled, blinded clinical study. A complete blood count,

serum biochemistry, urinalysis, urine culture, T4 and blood pressure were required for entry. Maropitant was administered at a dose of 4 mg orally (median 1.1 mg/kg, range 0.6–2.9 mg/kg) daily for 2 weeks. Owners kept daily logs of vomiting incidence, appetite and activity scores. Physical examination, weight, body condition score and serum biochemistry were performed before and after the trial period. Mann–Whitney statistics were used to compare treatment groups.

Results

Thirty-three cats successfully completed the trial: 21 cats received the drug (nine Stage II cats, 12 Stage III cats) and 12 cats received placebo (seven Stage II cats, five Stage III cats). There was a statistically significant decrease in vomiting in cats with CKD that received maropitant ($P < 0.01$). Cats that received maropitant did not have statistically significant differences in appetite scores, activity scores, weight or serum creatinine compared with placebo.

Conclusions and relevance

Maropitant was demonstrated to palliate vomiting associated with CKD, and may be helpful in the nutritional management of cats with CKD.

Introduction

Chronic kidney disease (CKD) is a common condition in elderly cats, with at least 50% of the population affected.¹ As renal transplantation is often not feasible for owners, medical management is the mainstay of treatment and can help patients live with the metabolic complications of the disease and improve quality of life.^{2,3} Clinical signs of feline CKD include polyuria, polydipsia, lethargy, decreased appetite, weight loss and vomiting.⁴ In the chronically ill patient, and specifically in CKD, poor body condition has been correlated with decreased survival.^{5–7} Therefore, nutrition is important for long-term prognosis, and treatments that directly target nausea and appetite, in addition to medical therapies for metabolic complications, are of benefit to patients.⁸ Several studies have documented the therapeutic value of specially formulated diets in the management of CKD.^{9–11} However, patients that refuse to eat specially formulated diets cannot benefit from them; therefore, a key therapeutic target for these patients should be the maintenance of appetite and food intake. Additionally, poor appetite is perceived by owners as a significant quality-of-life concern, and anorexia in companion animals can cause emotional distress to owners.¹²

The exact pathophysiology of why cats with CKD suffer from anorexia and vomiting is not currently known. A recent study demonstrated that the incidence of classic uremic gastropathy lesions (gastritis and ulceration) appears to be much lower in cats than in other species.¹³ Although hypergastrinemia has been documented in cats with CKD,¹⁴ a correlation with gastric hyperacidity has not been demonstrated in cats or humans with CKD.^{13,15} Without knowing the degree to which uremic gastric lesions or gastric hyperacidity are responsible for clinical signs, other mechanisms should be considered. Uremic toxins are sensed by the chemoreceptor trigger zone of the area postrema, and ablation of this area has been shown to relieve uremic vomiting in dogs.¹⁶

Maropitant (Cerenia; Zoetis) is a selective neurokinin-1 (NK-1) receptor antagonist acting to block effectively the binding of emetic-eliciting substance P, the most potent tachykinin at the NK-1 receptor.¹⁷ Maropitant has been shown to be an effective antiemetic in cats, ameliorating xylazine-induced vomiting when given orally, subcutaneously or intravenously.¹⁷ As xylazine-induced vomiting is mediated by the area postrema in cats,¹⁸ it can be postulated that maropitant would be effective for uremic vomiting. The purpose of this clinical trial was to assess the efficacy of maropitant as an appetite stimulant and an antiemetic in cats with CKD, thus providing support for the use of this medication in the nutritional management of cats with CKD.

Materials and methods

Cats

This was a double-blind, placebo-controlled, multi-institute prospective study. Cats with stable CKD (a priori determination of serum creatinine of 177–442 $\mu\text{mol/l}$ [2.0–5.0 mg/dl]) and a history of decreased appetite, vomiting or poor body condition were enrolled. Diagnostic tests required within 1 month of enrollment included a serum biochemistry profile, complete blood count, urinalysis, urine culture, blood pressure and serum total thyroxine measurement. Exclusion criteria included a normal appetite, other systemic illnesses, complications of CKD (such as pyelonephritis or ureteral obstruction), or history of uremic crisis requiring hospitalization and intravenous fluid therapy within the month prior to study enrollment. Other concurrent therapies such as dietary management, famotidine, potassium supplementation, antihypertensive medications and subcutaneous fluids were accepted therapies if they were started more than 2 weeks before the beginning of the trial and given consistently throughout the study period. No treatment changes were allowed during the study period; if treatment changes were deemed medically necessary, the cat was removed from the study. The study was approved by the

Institutional Animal Care and Use Committee at Colorado State University (#11-2556A), and all owners reviewed and signed consent forms prior to participation in the study.

Drug preparation

To allow for a blinded study, commercially available 16 mg maropitant tablets were compounded into 4 mg doses by the pharmacy at the Colorado State University Veterinary Medical Center according to the Professional Compounding Centers of America protocol. The method used is guaranteed to produce accurate compounding to within 10% of the target dose. Analysis of random compounded capsules for maropitant content using liquid chromatography coupled to tandem mass spectrometry showed accuracies of $98.5\% \pm 1.1\%$ to the intended content and a stability of at least 6 months as formulated. Maropitant capsules were compounded within 6 months of use and stored at room temperature. An identical placebo capsule was manufactured containing lactose. The lots were coded A and B, and the pharmacy staff kept the key to the code until the completion of the study.

Study design

Predetermined randomization for order of treatment allocation, stratified by International Renal Interest Society (IRIS) Stage, was established using an online random number generator, and as cats were enrolled they were assigned consecutively to a treatment group. The clinician and owner were blinded as to the treatment group. A physical examination, body weight measurement, body condition score (BCS; Purina Body Condition System) ¹⁹ and serum biochemistry profile were performed at the beginning of the study. The 4 mg or placebo capsule was administered orally daily, followed by 3 ml of water via syringe, for 2 weeks. Owners were asked to fill out a daily log sheet regarding appetite, vomiting episodes, activity level, quality of life and occurrence of unusual behavior in the cats compared with baseline (Supplementary material). At the end of the study period, the log sheets were collected and a physical examination, body weight measurement, BCS and serum biochemistry profile were performed. For each cat enrolled, physical examinations and BCSs were performed by the same clinician. Occurrence of adverse effects was determined using incidence of unusual behaviors recorded in the daily owner log, and results of serum biochemistry performed before and after each phase of the study. All serum biochemistry profiles were performed at the Colorado State University's diagnostic laboratories.

Statistical analysis

An a priori power calculation was performed based on weight gain as a primary endpoint. Assuming a mean body weight of 3 kg for cats enrolled in the study, a gain in weight of 300 g (0.3 kg) would be regarded as clinically significant. A power calculation was performed using a statistical calculator (Lenth, R. V. [2006]; Java Applets for Power and Sample Size [<http://www.stat.uiowa.edu/~rlenth/Power>]). Using a difference in means of 0.3 kg, a SD of 0.5 and a *P* value of 0.05, the power calculation for a paired *t*-test with 24 cats in the treatment group gives a power of 0.80 for this experiment.

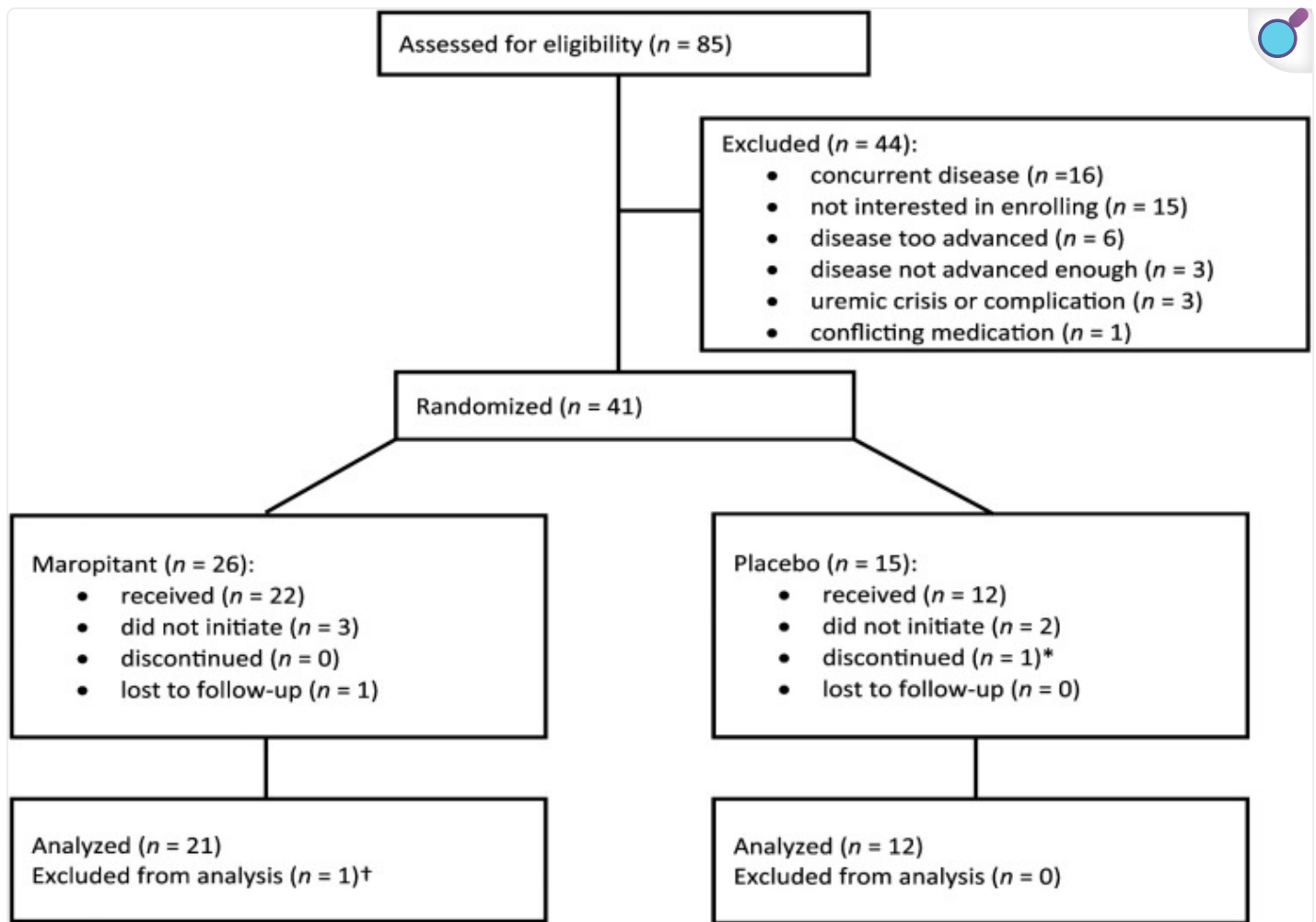
Appetite and activity data were converted to clinical scores; decreased appetite or activity were scored as -1, unchanged appetite or activity were scored as 0 and increased appetite or activity were scored as 1. Scores were then summed for the 2 week treatment period. The number of vomiting episodes over 14 days and summed clinical scores for appetite and activity were compared between placebo and maropitant treatment phases using a Mann–Whitney U-test. Weight and clinicopathologic parameters (serum creatinine, blood urea nitrogen [BUN], phosphorus, potassium) were compared between placebo and maropitant treatment phases at baseline and post-treatment using a Mann–Whitney U-test. For all analyses, a *P* value of <0.05 was considered to be statistically significant.

Results

Cats

A flow chart outlining study enrollment, allocation, outcome and analysis is depicted in [Figure 1](#). Thirty-four cats successfully completed the trial (one cat was excluded from analysis as it was subsequently discovered to be the other cat in the household that was vomiting); data were analyzed from 21 cats that received drug (nine Stage II cats, 12 Stage III cats) and 12 cats that received placebo (seven Stage II cats, five Stage III cats). The 21 cats allocated to the maropitant group included domestic shorthair (*n* = 10), domestic longhair (*n* = 6), domestic mediumhair (*n* = 1), Siamese (*n* = 3) and Birman (*n* = 1). The median age was 14 years (range 10–20 years) with 14 spayed females and seven neutered males. The 12 cats allocated to the placebo group included domestic shorthair (*n* = 8), domestic longhair (*n* = 2), Siamese (*n* = 1) and Ragdoll (*n* = 1). The median age was 14 years (range 6–18 years) with seven spayed females and five neutered males. There was no statistically significant difference in age or sex between groups. There was no statistically significant difference in weight, serum creatinine, potassium, phosphorus or BUN between maropitant and placebo treatment groups in the prestudy period.

Figure 1.



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Flow chart describing assessment for eligibility, enrollment, allocation, outcome and analysis of cats with chronic kidney disease for the maropitant study, which was a placebo-controlled, blinded clinical trial. *Medication was discontinued as the owner could no longer successfully administer it. †Subsequently discovered to be the other cat in the household that was vomiting

The 21 cats allocated to the maropitant group had been receiving a variety of medications, including no medications (n = 7), subcutaneous (SC) fluids (n = 5), benazepril (n = 4), meloxicam (n = 3), mirtazapine (n = 2), amlodipine (n = 2), famotidine (n = 1), aluminum hydroxide (n = 1), potassium supplementation (n = 1), lanthum carbonate (n = 1), polysulfated glycosaminoglycans

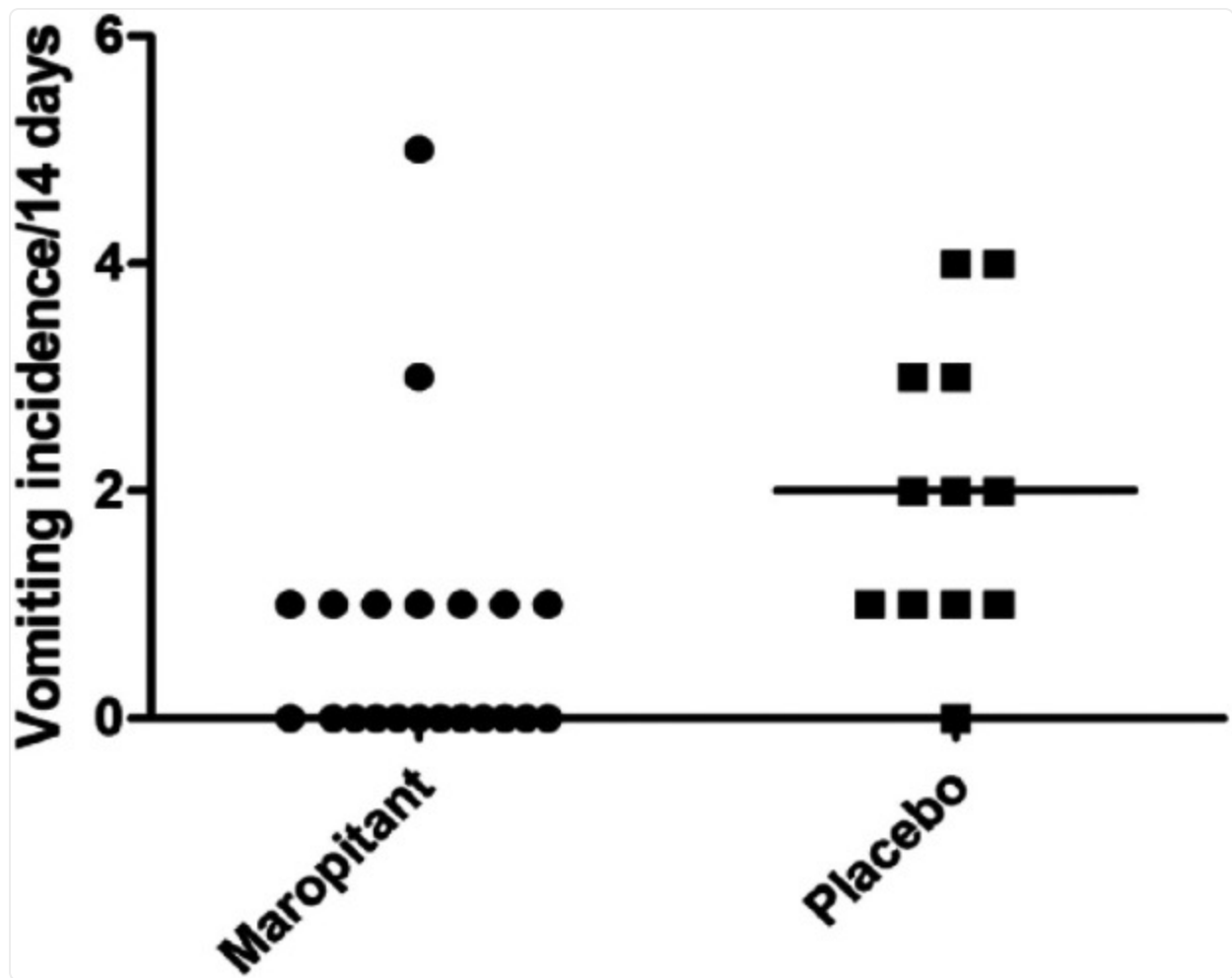
injection (n = 1), multivitamin (n = 1), ciclosporin (n = 1), butorphanol (n = 1) and tea pill supplement (n = 1). The 12 cats allocated to the placebo group had been receiving a variety of medications, including no medications (n = 6), SC fluids (n = 2), Azodyl renal support supplement (Vetoquinol) (n = 1), amlodipine (n = 1), tramadol (n = 1), amitriptyline (n = 1) and benazepril (n = 1). The 21 cats allocated to the maropitant group had been eating a renal diet (n = 4), a renal diet plus a non-renal diet (n = 9) or a non-renal diet (n = 7). The 12 cats allocated to the placebo group had been eating a renal diet (n = 2), a renal diet plus a non-renal diet (n = 5) or a non-renal diet (n = 5).

No reported changes in medication or diet were made during the study period. At initial assessment for study entry, two cats were treated for a urinary tract infection, one cat was started on amlodipine for hypertension and one cat was started on potassium supplementation. All four cats were required to wait to start the clinical trial an additional 2 weeks beyond the recheck, at which they were deemed medically stable by the attending veterinarian (antibiotics finished and urinary tract cleared as confirmed by urinalysis and culture; blood pressure within normal range; potassium within normal range).

Effect of maropitant in cats with CKD

Administration of maropitant to cats with CKD for 2 weeks resulted in a statistically significant decrease in vomiting in comparison with placebo based on a Mann–Whitney U-test ($P < 0.01$; [Figure 2](#)). The median number of vomiting episodes for cats in the maropitant group was zero (range 0–5) and the median number of vomiting episodes for cats in the placebo group was two (range 0–4). There was no statistically significant difference in appetite or activity scores between maropitant and placebo treatment groups during the study period. There was no statistically significant difference in weight, serum creatinine, potassium, phosphorus or BUN between maropitant and placebo treatment groups in the post-study period. Weight and clinicopathologic data are presented in [Table 1](#). No unusual behavior that could be considered an adverse effect was reported in the daily log by the owners.

Figure 2.



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Effect of 2 weeks of maropitant administration on vomiting in cats with chronic kidney disease (CKD). Compared with placebo, a statistically significant decrease in vomiting was seen in cats with CKD (n = 21) administered 4 mg maropitant orally daily for 2 weeks ($P < 0.01$)

Table 1.

Pre- and post-treatment comparison of weight and serum biochemistry parameters relevant to renal function in cats treated with maropitant (4 mg PO daily) or placebo for 14 days. Results are displayed as median (range)

	Maropitant				Placebo			
	Pretreatment		Post-treatment		Pretreatment		Post-treatment	
	Median	Range	Median	Range	Median	Range	Median	Range
Weight (kg)	3.6	2.3–6.0	3.6	2.3–6.1	3.6	1.4–6.3	3.7	1.3–6.3
Creatinine (0.8–2.4 mg/dl)	3.1	2.0–4.0	2.7	1.9–4.2	2.5	2.0–4.8	2.3	1.9–5.3
Creatinine (71–212 μmol/l)	274	177– 354	239	168– 371	221	177– 424	203	168– 469
BUN (18–35 mg/dl)	54	29–120	56	31–109	52	32–77	49	25–87
BUN (6.4–12.5 μmol/l)	19.3	10.4– 42.8	20	11.1– 38.9	18.6	11.4– 27.5	17.5	8.9– 31.1
Phosphorus (3–6 mg/dl)	4.7	2.8–9.0	4.8	2.9–9.0	4.9	2.8–10.2	5.2	3.0– 11.9
Phosphorus (1–1.9 μmol/l)	1.5	0.9–2.9	1.6	0.9–2.9	1.6	0.9–3.3	1.7	1.0–3.8
Potassium	4.4	3.5–5.5	4.3	3.6–5.8	4.4	3.6–6.0	4.6	3.8–5.2

	Maropitant				Placebo			
	Pretreatment		Post-treatment		Pretreatment		Post-treatment	
	Median	Range	Median	Range	Median	Range	Median	Range
(3.5–5.2 meq/l)								

[Open in a new tab](#)

BUN = blood urea nitrogen

Discussion

In this randomized, placebo-controlled, double-blind prospective clinical trial, daily administration of 4 mg maropitant orally for 2 weeks resulted in a statistically significant decrease in vomiting in cats with CKD. This study assesses the use of maropitant for antiemetic therapy in a chronic disease state. Maropitant seems a logical choice for vomiting associated with uremia as it has previously been shown to be effective against xylazine-induced vomiting in cats,¹⁷ and both xylazine and uremic toxins have been demonstrated to act via the area postrema.^{16–18}

Management of uremic vomiting in CKD cats may be helpful in preserving fluid balance and calories consumed. Similar to a previous safety study in cats, which assessed daily doses up to 5 mg/kg, no apparent adverse effects of daily maropitant administration were appreciated during the 2 week trial period.¹⁷ No statistically significant increase in appetite, weight or activity was documented during the 2 week trial period. Various explanations for this finding include a placebo effect, the length of the clinical trial, the possibility that decreasing vomiting may not have had a discernible effect on appetite or weight, and multi-cat households may affect accuracy of assessment.

Owner assessment of appetite may have been influenced by a placebo effect, thus altering the ability of this study to assess the efficacy of maropitant to improve appetite in CKD cats. Four of 12 cats in the placebo group were reported to have increased appetite for more than 6/14 days during the trial period. In a previous study looking at the effect of the appetite stimulant mirtazapine on the appetite of cats with CKD, similar appetite scores resulted in significant weight gain over 3 weeks.⁸ In the four cats in the present study a discernible change in weight was not documented. However, unlike mirtazapine, maropitant has not been described as an appetite stimulant, rather as

an antiemetic. Therefore, an increase in appetite is not necessarily expected. An effect on appetite would potentially come secondary to the palliation of nausea associated with uremia, which is an endpoint virtually impossible to measure in veterinary patients. As the effect of maropitant on appetite is potentially more subtle than a drug that acts directly as an appetite stimulant, a 2 week clinical trial may not have been long enough to appreciate weight gain as a result of administration.

During the enrollment phase of the clinical trial, every effort was made to recruit cats that did not have evidence of other systemic disease, particularly one that would cause vomiting. To the best of the attending clinician's knowledge, this was the case. Abdominal ultrasound and intestinal biopsies would have been helpful to more completely assess enrolled cats but were not possible within the scope of this project. Owing to the commonality of comorbidities in elderly cats, it is therefore possible that cats were enrolled with underlying systemic disease that could have affected their outcomes. One cat in the maropitant group had a vomiting incidence much greater than the rest of the group despite medication administration; vomiting incidence was essentially unchanged during the study period despite medication administration. As this is an unexpected finding given the apparent potency of this drug as an antiemetic, regurgitation misclassified and reported by the owner as vomiting or other unknown systemic disease should be considered as a differential for this cat. A limitation of any feline clinical trial is the likelihood of a multi-cat household making the assessment of food intake and vomiting incidence challenging. Although other members of the feline household could have been vomiting concurrently, most owners felt that they could differentiate their cats' vomit. However, in one case a cat was removed from analysis when it was subsequently discovered it was in fact the other cat in the household that was chronically vomiting and not the cat with CKD that was enrolled in the study.

One aspect of the study that could potentially confound results is the variety of concurrent therapies that cats were receiving at the time of enrollment. Cats within the maropitant group were collectively receiving more therapies than cats in the placebo group. It is also impossible to say whether a given therapy could have an unknown synergistic effect with maropitant. Unfortunately, it is virtually impossible to standardize therapy without affecting enrollment for clinical trials; thus, the results of the study need to be interpreted with this variability in mind. Another aspect of the study that could affect interpretation of the results is that it cannot entirely be ruled out that an inherent difference in vomiting incidence existed between the treatment groups at baseline. However, as the study was randomized and sufficiently powered it is most likely that the significant difference in vomiting was the effect of maropitant administration.

A limitation of this study was that it was necessary to compound the maropitant in order to blind the study. Because of the concern for accuracy of the dose with compounding, analysis of compounded capsules was performed and found to be acceptable. As owners would typically be administering a quarter of a 16 mg tablet for a cat, dose variability would likely be similar in the clinical setting. Although the dose administered was found to be accurate, there is a possibility the stability of the drug was affected and this could have an unknown effect on the efficacy of the drug.

Conclusions

Daily oral administration of maropitant was demonstrated to palliate vomiting associated with CKD. However, within the 2 week trial period it did not appear to improve significantly appetite or result in weight gain in cats with Stage II and III CKD. However, longer clinical trials and studies in late-stage disease are needed to better assess the effect of maropitant on weight gain and nutrition in cats with CKD.

Supplemental Material

Click here for Supplementary Appendix

[sj-pdf-1-jfm-10.1177_1098612X14555441.pdf](#) ^(22.7KB, pdf)

Daily log completed by owners during the study

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Footnotes

Supplementary material: Daily log completed by owners during the study.

The authors have no conflict of interest to declare.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

Click here for Supplementary Appendix

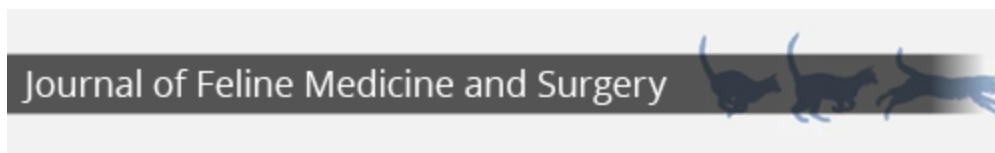
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Daily log completed by owners during the study

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Assessment of intravenous adipose-derived allogeneic mesenchymal stem cells for the treatment of feline chronic kidney disease: a randomized, placebo-controlled clinical trial in eight cats

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Abstract

Objectives

Feline chronic kidney disease (CKD) is characterized by chronic tubulointerstitial nephritis, and inflammation contributes to the progression of renal fibrosis. Mesenchymal stem cells (MSCs) have demonstrated anti-inflammatory and antifibrotic effects in rodent CKD models. However, few randomized trials evaluating the effectiveness of MSC therapy for diseases in companion animals have been reported. The purpose of this study was to evaluate the effectiveness of allogeneic MSCs for the treatment of feline CKD using a randomized, placebo-controlled trial.

Methods

MSCs were isolated from the cryopreserved adipose tissues of specific pathogen-free research cats and culture expanded. CKD cats were enrolled in a randomized, placebo-controlled, blinded one-way crossover clinical study. Four CKD cats were randomized to receive 2×10^6 MSCs/kg intravenously at 2, 4 and 6 weeks. Four CKD cats were randomized to receive placebo, with two cats crossing over to the MSC treatment group and one cat failing to complete the trial. Complete blood counts, chemistry and urinalysis were performed at weeks 0, 2, 4, 6 and 8. Glomerular filtration rate (GFR) via nuclear scintigraphy and urine protein:creatinine ratio (UPC) were determined at weeks 0 and 8.

Results

Six cats received three doses of allogeneic MSC culture expanded from cryopreserved adipose without adverse effects. No significant change in serum creatinine, blood urea nitrogen, potassium, phosphorus, GFR by nuclear scintigraphy, UPC or packed cell volume was seen in cats treated with MSCs. Individual changes in GFR were 12%, 8%, 8%, 2%, -13% and -67% in treated cats compared with 16%, 36% and 0% in placebo-treated cats.

Conclusions and relevance

While administration of MSC culture expanded from cryopreserved adipose was not associated with adverse effects, significant improvement in renal function was not observed immediately after administration. Long-term follow-up is necessary to determine whether MSC administration affects disease progression in cats with CKD.

Introduction

Chronic kidney disease (CKD) is a common medical condition in geriatric cats and is characterized histologically by tubulointerstitial inflammation and fibrosis, with subsequent progressive loss of renal function.^{1,2} Currently, renal transplant is the only definitive therapy to improve kidney function in cats with CKD. Therefore, novel and effective therapeutic options are highly sought after to provide additional treatment options for cats suffering from this disease.

Mesenchymal stem cells (MSCs) have been proposed as a novel treatment option for the management of CKD. MSCs exert potent anti-inflammatory and antifibrotic effects and may therefore indirectly improve renal function by reducing disease-associated inflammation and fibrosis through paracrine effects.³⁻⁷ Demonstrated immunomodulatory effects of MSCs include

inhibition of lymphocyte proliferation and cytokine production, suppression of dendritic cell function and suppression of interferon- γ production by natural killer cells.⁸ In vitro studies have demonstrated that MSCs can produce growth factors, cytokines and anti-inflammatory mediators, all of which could help maintain or improve renal function and suppress intra-renal inflammation.⁹⁻¹¹ The ability of MSCs to suppress inflammation appears to be mediated both by secreted factors and by direct contact with inflammatory cells.^{10,11} In the majority of studies with experimentally induced CKD in rodents, administration of both adipose-derived and bone marrow-derived MSCs has demonstrated significant renoprotective effects, including reduction of intrarenal inflammatory infiltrate, decreased fibrosis and glomerulosclerosis.^{3,4,6,7,12} Parameters of renal function and clinical health, including weight, creatinine, blood urea nitrogen (BUN), proteinuria, blood pressure and hematocrit have also been demonstrated to improve as a result of MSC therapy.^{3,4,6,7,12} Several routes of administration – intraparenchymal, subcapsular, intraperitoneal, intravenous (IV) – have been explored, and all seem to be effective. Efficacy is thought to originate from paracrine effects rather than cell engraftment into the kidney.^{4,9} Multiple repeated injections of MSCs appear to be even more effective than single injections.^{3,4} As inflammation appears to be present at all stages of CKD in cats, the immunomodulatory actions of MSC are appealing as a potential means of suppressing intrarenal inflammation and subsequent fibrosis. Thus, MSC therapy may be an effective new approach to slow the progression of CKD and improve renal function.

Previous studies in cats performed by our group have explored the use of MSCs for treatment of feline CKD, but results have been variable.^{13,14} The purpose of this study was to conduct a placebo-controlled trial to assess the efficacy of adipose-derived allogeneic MSCs to improve renal function in cats with stable CKD. We hypothesized that repeated IV administration of MSCs would be well tolerated and would result in improvement of renal function as measured by serum creatinine and glomerular filtration rate (GFR) by nuclear scintigraphy.

Materials and methods

Study cats

Cats with stable CKD serum creatinine (1.6–5.0 mg/dl) were recruited from the patient population at the Veterinary Teaching Hospital at Colorado State University (CSU). Cats were determined to have stable CKD based on two repeated biochemical evaluations performed at least 2 weeks apart and ultrasonographic evidence of CKD. Stability was defined as no more than a 15% change in serum creatinine, no uncontrolled metabolic disturbances, no change in medication regimen within 2 weeks and no clinical concerns. Pretreatment evaluation included complete blood count

(CBC), biochemistry profile, urinalysis, urine culture, blood pressure, total thyroxine, urine protein:creatinine ratio (UPC), feline leukemia/feline immunodeficiency virus serology and a renal ultrasound. Cats were excluded from the study if they had evidence of ureteroliths; pyelonephritis; ureteral obstruction (pelvic dilation >13 mm and/or dilated proximal ureters); anatomic abnormalities such as polycystic kidney disease or masses; uncontrolled hypertension; or concurrent systemic disease. Administration of concurrent supportive therapies was allowed provided there were no changes in therapy 2 weeks prior to or during the study period. The study was approved by the Institutional Animal Care and Use Committee at CSU (#11-2915A), and all owners reviewed and signed consent forms prior to participation in the study.

Predetermined randomization for order of treatment allocation was established using an online random number generator; as cats were enrolled they were assigned consecutively to a treatment group. Treatment group was blinded to the owners during the initial study phase. At the conclusion of the 8 week trial, the owners of cats that were allocated to the placebo group were given the option to crossover to the treatment group.

Adipose-derived MSC preparation and administration

Adipose-derived MSCs were isolated from adipose tissue of specific pathogen-free cats collected from a subcutaneous site on the ventral abdomen just caudal to the umbilicus, as previously described.^{[13,15](#)} For preparation of the adipose tissue for cryopreservation, the tissue was minced and divided into 1 g aliquots in 1 ml freezing medium (11% dimethyl sulfoxide, 14% MSC medium, 75% fetal bovine serum [FBS]), and stored in liquid nitrogen for no longer than 1 year prior to use. Cryopreserved adipose was then later thawed, immediately washed twice with Dulbecco's Phosphate Buffered Saline (DPBS) and then prepared for culture. For isolation of the stromal vascular fraction, the tissue was minced and digested with 1 mg/ml collagenase (Sigma-Aldrich) for 30 mins at 37°C. The sample was centrifuged, and the stromal vascular fraction was plated in MSC medium (low-glucose Dulbecco's Modified Eagle Medium [DMEM], 100 U/ml penicillin, 100 µg/ml streptomycin, 2 µM L-glutamine, 1% essential amino acids without L-glutamine, 1% non-essential amino acids, 0.075% sodium bicarbonate [Invitrogen/Gibco] plus 15% FBS [Cell Generation]). The MSCs were incubated until approximately 70% confluent, with media changes every 2–3 days. Cells were harvested by trypsinization at passage 2–3, washed and resuspended in 10 ml DPBS with 200 IU heparin sulfate, and administered as a slow IV push to the cephalic vein over 20 mins. Cats received MSCs at weeks 2, 4 and 6. As previous studies have indicated that some difference in immunomodulatory potential may exist between donors,^{[16](#)} all three injections received were from different donor cats given in a pre-randomized order. A total of three donor cats, all male, aged 2–7 years, were used to provide MSCs for the study. Placebo-control cats

received 10 ml DPBS with 200 IU heparin administered as a slow manual IV push to the cephalic vein over 20 mins at weeks 2, 4 and 6.

Characterization of MSCs

MSCs cultured from cryopreserved adipose tissue were characterized by surface marker expression using flow cytometry and a panel of cross-reactive antibodies specific for surface determinants expressed by MSC from other species. Specifically, feline MSCs were analyzed for surface expression of CD44 (antimouse/human, antibody clone IM7; eBioscience) and CD90 (antihuman, antibody clone eBio5E10; eBioscience). MSCs were also assessed for expression of CD4 (antifeline antibody clone 3-4F4; SouthernBiotech) and major histocompatibility complex (MHC) class II (antihuman antibody clone TU39; BD Biosciences). Samples were analyzed using a CyAn ADP flow cytometer (Beckman Coulter). Approximately 10,000 events were collected for analysis per sample.

In vitro differentiation assays were conducted to confirm the multipotency of feline MSCs, as assessed by their ability to differentiate into three cell lineages (osteoblasts, chondrocytes and adipocytes) that are characteristic of MSCs. For differentiation into adipocytes, MSCs at confluency were incubated with MSC medium supplemented with 0.5 μ M dexamethasone, 50 μ M indomethacin and 0.5 μ M 3-isobutyl-1-methylxanthine (all Sigma- Aldrich) for 3 weeks with media changes every 3–4 days.

Chondrogenic differentiation medium consisted of DMEM 1 $\times$ (Cellgro) supplemented with 15% FBS (Cell Generation), 10 nM dexamethasone (Sigma-Aldrich), 10 ng/ml transforming growth factor- β (R&D Systems), 50 μ g/ml ascorbic acid (Sigma-Aldrich) and 40 μ g/ml proline (Sigma-Aldrich).

Osteogenic differentiation medium consisted of MSC medium supplemented with 10 nM dexamethasone, 50 μ M ascorbic acid and 20 mM β -glycerophosphate (Sigma-Aldrich). At the end of the differentiation period, cells were fixed with 10% neutral buffered formalin and stained with Oil Red O (Sigma-Aldrich) for the presence of lipid or with Alizarin red (Sigma-Aldrich) for the presence of calcium. Cell pellets from cartilage differentiation were harvested and placed in OCT Compound (Sakura Finetek) and flash frozen prior to staining with toluidine blue (Richard-Allan Scientific) for cartilage matrix. MSCs cultured in MSC medium alone under identical conditions were used as differentiation controls.

Clinical monitoring

Each enrolled cat underwent physical examination, weighing and routine blood work consisting of CBC, serum biochemistry and urinalysis at weeks 0, 2, 4, 6 and 8. Each cat had a UPC performed at weeks 0 and 8. Cats were monitored continuously during MSC injection for any signs of reaction or adverse effects. Additionally, each cat had a GFR by nuclear scintigraphy performed at weeks 0 and 8. For the scintigraphy procedure, all enrolled cats (MSC-treated and placebo-controlled) were sedated with a standard sedation protocol (butorphanol 0.2mg/kg IV once) at a standard time before the procedure. A single nuclear technician performed all of the procedures for each particular cat. For each procedure 1.0 mCi of Tc99m-labeled diethylene triamine penta-acetic acid (Cardinal Health) was injected IV via a catheter placed in a standard location in each cat. Images were obtained using GE Millennium SPECT system applicable for small animal planar, whole body and single-photon emission CT imaging (GE Healthcare). Three blinded board-certified radiologists evaluated the GFR data, and a mean GFR value for each kidney, as well as a global value, was determined using regions of interest drawn on a summed image from the 1–3 mins postinjection time frame. The measurements calculated by the three radiologists were then averaged.

Statistical analysis

Analysis was performed using SAS Proc Mixed (SAS 9.3; SAS Institute). For each variable (GFR total, GFR left, GFR right, creatinine) the change was calculated for each animal taking week 8 – week 0. This change was used as the response variable in further analysis. A model was fit including treatment (placebo or MSC) and a random cat effect was included to account for the fact that two animals received both treatments.

Results

Study cats

Seventy-one cats were screened for entry into the clinical trial. Cats were excluded for the following reasons: concurrent illness (n = 23), end-stage or unstable disease (n = 16), extensive travel required (n = 10), owners thought the study too complex or did not want to participate in a placebo-controlled trial (n = 6), anatomic renal abnormalities (n = 4), and fractious or too stressed by visits (n = 4). Descriptive data for the eight enrolled cats are presented in [Table 1](#). One cat was enrolled and randomized into the study in the placebo group, but between week 0 and 2 the owners elected to discontinue the trial. Data for this cat are not presented.

Table 1.

Summary of demographics of cats enrolled and randomized into the intravenous allogeneic cryopreserved studies

Cat	Group	Age (years)	Sex	Breed	IRIS CKD stage	Creatinine, mg/dl (mol/l)	Treatment
1	MSC	13	FS	DSH	II	2.2 (194)	2×10^6 MSCs/kg IV, three treatments
2	MSC	19	FS	Siamese	III	3.3 (292)	2×10^6 MSCs/kg IV, three treatments
3	MSC	10	MC	DLH	III	3.5 (309)	2×10^6 MSCs/kg IV, three treatments
4	MSC	14	MC	Siamese	III	4.5 (398)	2×10^6 MSCs/kg IV, three treatments
5	Placebo MSC crossover	15	MC	DLH	III	3.8 (336)	PBS placebo, three treatments; then 2×10^6 MSCs/kg IV, three treatments
6	Placebo	11	FS	Siamese	III	3.7 (327)	PBS placebo, three treatments
7	Placebo MSC crossover	15	MC	Siamese	II	2.3 (203)	PBS placebo, three treatments; then 2×10^6 MSCs/kg IV, three treatments

Cat	Group	Age (years)	Sex	Breed	IRIS CKD stage	Creatinine, mg/dl (mol/l)	Treatment
8	Placebo	9	MC	Maine Coon	II	2.1 (186)	Owners elected not to participate after enrollment and randomization

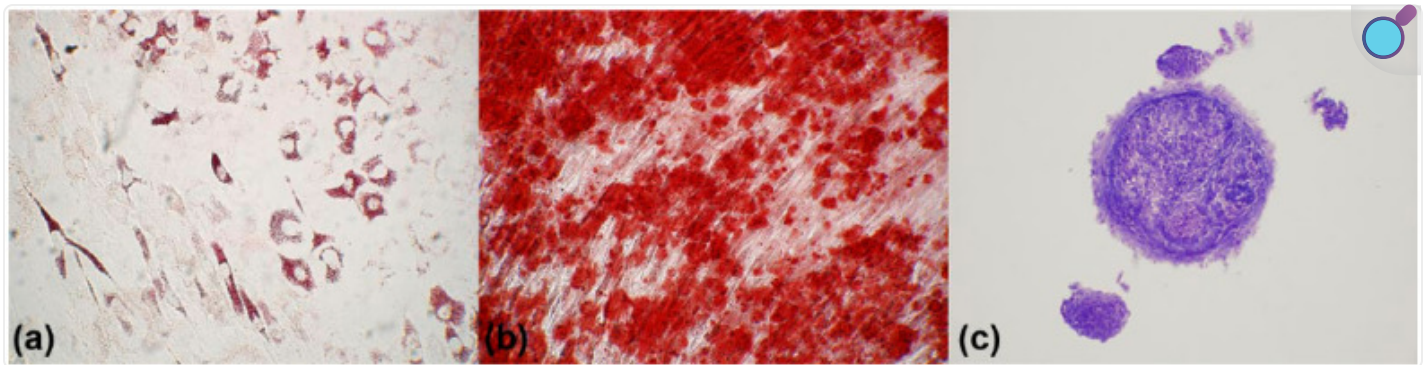
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IRIS = International Renal Interest Society; CKD = chronic kidney disease; MSC = mesenchymal stem cell; FS = female spayed; MC = male castrated; DSH = domestic shorthair; DLH = domestic longhair; IV = intravenous; PBS = phosphate-buffered saline

Characterization of MSCs

During in vitro culture, feline MSCs were observed to develop into a relatively homogeneous population of plastic-adherent cells with fibroblast-like morphology. Adipose-derived MSC expressed high levels of CD44 and CD90, and were negative for expression of CD4 and MHC class II, as previously reported.¹⁴ MSC cultured from cryopreserved fat were capable of trilineage differentiation ([Figure 1](#)).

Figure 1.



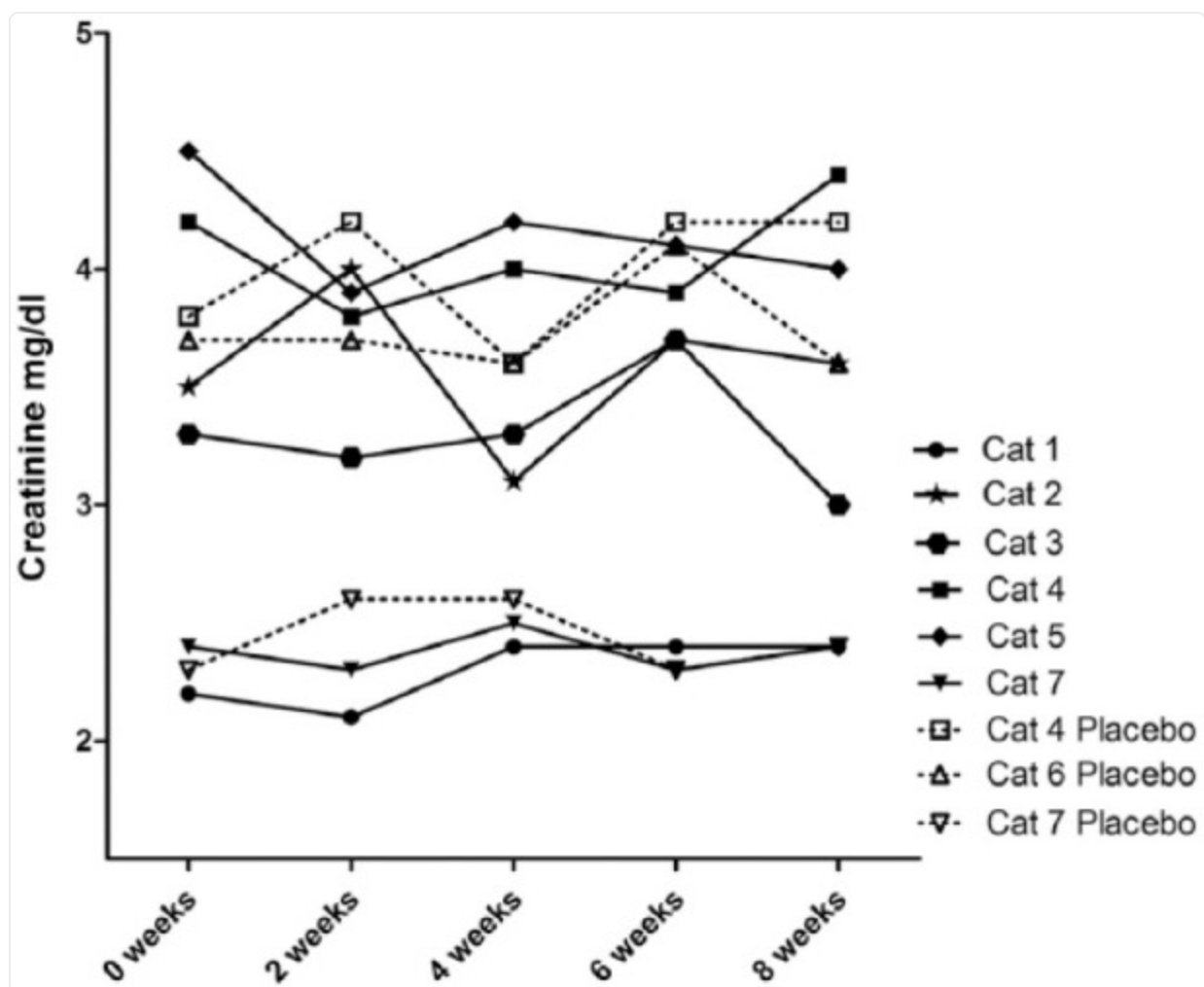
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Mesenchymal stem cells (MSCs) cultured from cryopreserved adipose were capable of trilineage differentiation. (a) MSCs formed intracellular lipid vacuoles (stained pink with Oil Red O) when incubated in adipocytic differentiation media for 21 days. (b) MSCs stained positive for calcium with alizarin red following differentiation into osteocytic phenotype after 21 days of incubation in differentiation media. (c) Cryosection of pellets of cartilage matrix (stained with toluidine blue) formed by MSCs when exposed to chondrocytic differentiation media for 21 days

Effects of MSC administration on renal function in cats with CKD

Six cats received three infusions of fresh, culture-expanded MSCs at a dose of 2×10^6 MSCs/kg body weight by slow IV infusion without developing clinically apparent adverse effects. No significant change in body weight, serum creatinine ([Figure 2](#)), BUN, phosphorus, potassium, packed cell volume, urine specific gravity or UPC was observed in cats treated with MSCs. The UPC in CKD cats treated with MSCs was a median of 0.15 (range 0.1–0.3) prior to administration of MSCs and a median of 0.2 (range 0.1–0.3) at the conclusion of the study period. The UPC in placebo-treated CKD cats was a median of 0.1 (range 0.1–0.4) at the beginning of the study and a median of 0.1 (range 0.1–0.3) at the conclusion of the study period.

Figure 2.

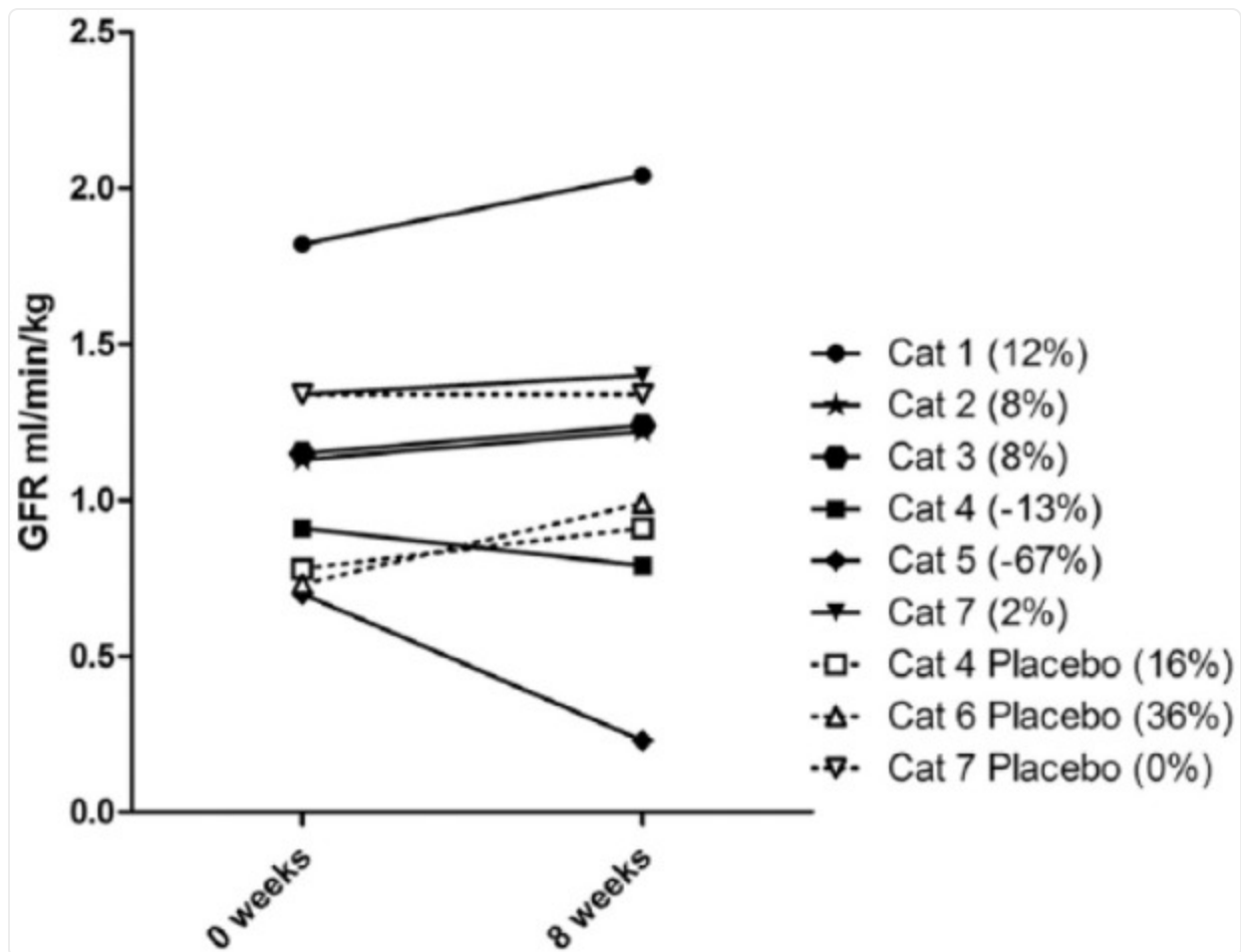


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Serum creatinine values for cats that received three doses of 2×10^6 mesenchymal stem cells/kg intravenously 2 weeks apart. No significant difference in creatinine was detected

There was no statistically significant difference in global or individual kidney GFR values as determined by nuclear scintigraphy. Individual changes in global GFR were 12%, 8%, 8% and – 67% in MSC-treated cats compared with 16%, 36% and 0% in placebo-treated cats ([Figure 3](#)). Cats that were in the placebo group and subsequently crossed over to receive MSC had changes in global GFR of 2% and –13%, respectively, during their MSC treatment period. Discordant creatinine and GFR results were observed in one animal: the MSC-treated cat that experienced the greatest decrease in creatinine also experienced the greatest decline in GFR.

Figure 3.



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Results of glomerular filtration rate (GFR) determined by nuclear scintigraphy at 0 and 8 weeks for cats that received 2×10^6 mesenchymal stem cells/kg intravenously at weeks 2, 4 and 6

Discussion

MSCs have demonstrated significant renoprotective effects in rodent models of acute and chronic kidney disease, including reduction of interstitial inflammatory infiltrate, and reduction of fibrosis and glomerulosclerosis. However, few randomized trials evaluating the effectiveness of MSC therapy for diseases in companion animals have been reported. The current study therefore evaluated the safety and efficacy of repeated IV injections of allogeneic adipose-derived MSC for

treatment of naturally occurring feline CKD using a randomized placebo-controlled study design. Key findings were that administration of three doses of 2×10^6 MSCs/kg (MSCs expanded from cryopreserved adipose to cats with CKD) was not associated with the development of clinically apparent adverse effects. Also, we did not detect a significant treatment effect (either positive or negative) on renal function in the 6 weeks after MSC treatment.

Previous clinical trials from our research group assessing MSC therapy in feline CKD have had variable results.^{13,14} For example, we found that intrarenal injections of MSCs appeared safe in CKD cats and resulted in mild improvement in renal parameters. In another study, we determined that repeated IV injections of cryopreserved MSCs in CKD cats resulted in mild improvement in renal parameters in some cats but it was questionable if the degree of improvement was clinically relevant. The clinical trials that have been performed by our research group were designed to emulate the time frame of rodent studies, which is generally short; improvements in clinicopathologic parameters are typically seen in the weeks subsequent to MSC administration, and rodents are sacrificed by 2–5 weeks for histopathologic analysis.^{3-5,7,12} Notably, none of the studies we have conducted in cats with CKD have replicated the efficacy of MSC treatment reported in rodent models of experimentally induced CKD within this time frame.^{3-5,7}

One explanation for differing results of MSC therapy in cats with CKD is that the chronic nature of feline CKD makes these patients quite different from rodents with experimentally induced disease. The 5/6 nephrectomy model in rats is a commonly used method of experimentally inducing CKD and has been shown to induce histopathologic changes consistent with CKD.¹⁷ One major difference between such models and MSC studies in cats with naturally occurring CKD is that in the majority of rodent studies the MSCs are administered a short period of time after surgical manipulation. For example, time frames for MSC administration range from immediately after nephrectomy (two studies),^{5,7} to 1 day later (two studies)^{3,18} and 2 weeks later.⁴ Two recent studies evaluated the effect of MSC administered 6 weeks after surgical manipulation; both studies demonstrated improvement in renal functional parameters and amelioration of fibrosis, even when MSCs were given after inflammation and fibrosis was already established.^{19,20} However, 6 weeks is still a much shorter time frame than is typical in feline CKD patients. In most cases, cats that present to the clinical trials service to participate in MSC studies have a several year history of CKD. Although previous studies in rodent models demonstrate impressive potential, the results of these models should be evaluated with care as administration of MSCs soon after surgical partial nephrectomy is likely not representative of long-standing, naturally occurring disease. Therefore, it needs to be considered that MSCs may not be as effective in an environment of long-standing inflammation and fibrosis, and further studies are needed to explore this question.

There is a growing body of literature supporting the theory that MSCs are adversely affected by uremia.²⁰⁻²⁴ Recent studies have documented that MSCs obtained from uremic rats have reduced proliferation in culture, loss of regenerative potential, premature senescence, decreased capacity to induce angiogenesis and an altered secretome.²⁰⁻²³ Uremic effects also have been documented in vitro as a reduced capacity of MSCs from uremic individuals to ameliorate renal damage in experimentally induced CKD in comparison with MSCs from healthy rats.^{20,21} Observations have been mixed as to whether this affects both bone marrow-derived and adipose-derived MSC. However, one study offers evidence that adipose-derived MSCs are not as susceptible to uremic effects as bone marrow-derived MSCs.²⁵ Although there is a concern regarding the effects of uremia on MSC function, clinical trials performed with allogeneic adipose-derived MSC should circumvent these concerns and give the best opportunity for efficacy. This information does imply, however, that uremic patients are not the best MSC source; a concern for autologous MSC therapy. Little data have been gathered on whether MSCs transplanted into a uremic recipient environment will become compromised. The success of MSC in palliation of acute kidney injury and CKD in rodent models argues against this being an issue. Based on the currently available data, the effect of the uremic environment seems less likely to be the explanation for lack of short-term efficacy seen in the current clinical trial; however, an adverse effect on MSCs when placed in a uremic environment cannot be entirely ruled out.

An unexpected finding in this study was that discordant serum creatinine and GFR results were observed in one cat: the MSC-treated cat that experienced the greatest improvement in creatinine also experienced the greatest decline in GFR. In addition to improved creatinine, this cat gained weight during the clinical trial and was thought by the owner to have done very well clinically, making this result unexpected and difficult to explain. Every effort was made to decrease variability between GFR studies. Studies were performed on the same day of the week, at the same time of day, with the same sedation protocol and technician. However, the potential for inherent variability in the GFR process, the analysis of the images or day-to-day changes in the physiology of the patient still exists. Daily variability in GFR by nuclear scintigraphy has been previously evaluated, but not in cats with CKD. In a previous study in normal young dogs, daily variability was as much as 25%. However, this is probably not applicable to animals with CKD due to impaired autoregulatory mechanisms.²⁶ Our group previously gained some information on variability in CKD cats by performing GFR studies 1 week apart. In these cats GFR varied by as much as 28% (average 9.6%).¹³ Taking these factors into account, GFR likely has to change by >20% to be considered a real change in response to a treatment effect. This inherent variability still does not explain the cat with a -67% change that experienced improvement based on serum creatinine, weight and clinical signs.

Over the course of our 6 year feline stem cell program, approximately 50 GFR studies have been performed, and only three technical failures have occurred in that time. On one occasion the machine broke down and took some time and effort to repair; once the isotope was not labeled by the supplier properly; and once the g camera head orientation was set wrong. In all cases it was immediately obvious that a technical problem had occurred. In the case of the cat in the present study, after careful scrutiny, no plausible explanation for the discordant result could be identified. This brings into question the value of measurement of renal function by this methodology as it requires the patient to be sedated, is expensive and is not a technique routinely used in clinical practice for patient assessment.

Although this was a randomized, blinded, placebo-controlled study, a limitation of this study was the small number of enrolled cats. Enrollment was difficult owing to the strict exclusion criteria and the study design of 50:50 ratio placebo to treatment, with placebo cats receiving the same number of procedures and sedation events, which was unpopular with owners. The latter was ameliorated by allowing placebo cats to cross over to the treatment group. Without definitive evidence of improvement in renal function during the study time frame, and the possibility that the stress of procedures outweighed therapeutic benefit, the decision was made to halt the study to move in a different direction.

Conclusions

Repeated IV injection of allogeneic MSCs expanded from cryopreserved adipose in cats with CKD was not associated with side effects; however, no discernible immediate improvement in renal function was appreciated compared with placebo. Long-term follow-up of patients that have received MSC injections will be necessary to determine if a delayed effect is appreciated. Additional studies are needed to determine if MSCs can be prepared or administered in alternative ways to increase efficacy.

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Footnotes

The authors do not have any potential conflicts of interest to declare.

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RESEARCH

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Safety and efficacy of intravenous infusion of allogeneic cryopreserved mesenchymal stem cells for treatment of chronic kidney disease in cats: results of three sequential pilot studies

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Abstract

Introduction: Administration of mesenchymal stem cells (MSCs) has been shown to improve renal function in rodent models of chronic kidney disease (CKD), in part by reducing intrarenal inflammation and suppressing fibrosis. CKD in cats is characterized by tubulointerstitial inflammation and fibrosis, and thus treatment with MSCs might improve renal function and urinary markers of inflammation in this disease. Therefore, a series of pilot studies was conducted to assess the safety and efficacy of intravenous administration of allogeneic adipose-derived MSCs (aMSCs) in cats with naturally occurring CKD.

Methods: Cats enrolled in these studies received an intravenous infusion of allogeneic aMSCs every 2 weeks collected from healthy, young, specific pathogen-free cats. Cats in pilot study 1 (six cats) received 2×10^6 cryopreserved aMSCs per infusion, cats in pilot study 2 (five cats) received 4×10^6 cryopreserved aMSCs per infusion, and cats in pilot study 3 (five cats) received 4×10^6 aMSCs cultured from cryopreserved adipose. Serum biochemistry, complete blood count, urinalysis, urine protein, glomerular filtration rate, and urinary cytokine concentrations were monitored during the treatment period. Changes in clinical parameters were compared statistically by means of repeated measures analysis of variance (ANOVA) followed by Bonferroni's correction.

Results: Cats in pilot study 1 had few adverse effects from the aMSC infusions and there was a statistically significant decrease in serum creatinine concentrations during the study period, however the degree of decrease seems unlikely to be clinically relevant. Adverse effects of the aMSC infusion in cats in pilot study 2 included vomiting (2/5 cats) during infusion and increased respiratory rate and effort (4/5 cats). Cats in pilot study 3 did not experience any adverse side effects. Serum creatinine concentrations and glomerular filtration rates did not change significantly in cats in pilot studies 2 and 3.

Conclusions: Administration of cryopreserved aMSCs was associated with significant adverse effects and no discernible clinically relevant improvement in renal functional parameters. Administration of aMSCs cultured from cryopreserved adipose was not associated with adverse effects, but was also not associated with improvement in renal functional parameters.

Keywords: Cell culture, Cytokines, Feline, Kidney, Mesenchymal stem cells, Urine

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Introduction

Chronic kidney disease (CKD) is a common condition in elderly cats and is characterized by tubulointerstitial inflammation, tubular atrophy and interstitial fibrosis with subsequent progressive loss of renal function [1,2]. Currently there is no definitive therapy short of renal transplant to improve kidney function in cats with CKD. Therefore, novel and effective therapeutic options are highly desired for treating this disease in cats.

Recently, a number of studies have suggested the use of mesenchymal stem cells (MSCs) as a novel treatment option for management of CKD, based on encouraging data from rodent studies [3-7]. There have been several rationales advanced for the use of MSCs for treatment of CKD. Rodent studies have suggested that MSCs may incorporate into the renal parenchyma and give rise to new renal tubular cells, though the process appears to be relatively inefficient [8-17]. MSCs also exert potent anti-inflammatory and antifibrotic effects and may therefore indirectly improve renal function by reducing disease-associated inflammation and fibrosis through paracrine effects [3-7]. For example, MSCs have been shown to inhibit lymphocyte proliferation and cytokine production, suppress dendritic cell function, and suppress interferon γ (IFN γ) production by natural killer (NK) cells [18].

Since inflammation appears to be present at all stages of CKD in cats, the immunomodulatory actions of MSCs are appealing as an alternative means of suppressing intrarenal inflammation long term and with fewer side effects than with conventional anti-inflammatory drugs. In the majority of experimentally-induced CKD models investigated, MSC administration resulted in improved renal function, decreased intrarenal inflammation, and reduction of renal fibrosis [3-7]. Thus, MSC therapy may be an effective new approach to slow the progression of CKD and improve renal function.

Previous studies have demonstrated that cultured MSCs can be administered intravenously to rodents as well as to humans. However, in rodent models there is a significant risk of pulmonary thrombosis when high numbers of MSCs are rapidly administered intravenously [19]. Alternative routes of delivery have included injection via the renal artery, injection directly into the renal parenchyma, and injection into the renal subcapsular space [3-7]. One potential advantage of intravenous delivery compared to other routes may be the induction of renotropic paracrine effects following intravenous administration of MSCs [20].

Based on compelling results from rodent studies, we conducted a series of pilot studies to investigate the safety and potential efficacy of adipose-derived MSC (aMSC) therapy as a treatment for cats with naturally occurring CKD. These studies were designed to test the hypothesis that allogeneic cryopreserved aMSCs could be safely administered to cats with CKD and would

result in improvement in kidney function. For this study, aMSCs were derived from allogeneic, healthy, specific pathogen-free (SPF) donor cats and cryopreserved, either as cells or adipose tissue, prior to intravenous administration to study cats. One primary study endpoint was to determine safety and potential adverse effects of repeated intravenous administration of cryopreserved aMSCs. The second major endpoint was to determine whether repeated MSC administrations were associated with improvement in renal function or urinary markers of intrarenal inflammation. These studies produced evidence of modest, but unlikely to be clinically significant, improvement in renal function but also showed evidence of significant adverse effects associated with intravenous administration of higher doses of cryopreserved aMSCs.

Methods

Study cats

Cats with stable CKD, serum creatinine 1.6 to 5.0 mg/dl, were recruited from the patient population at the Veterinary Teaching Hospital at Colorado State University. Cats were determined to have stable CKD based on two repeated biochemical evaluations performed at least 2 weeks apart and ultrasonographic evidence of CKD. Pre-treatment evaluation included complete blood count (CBC), biochemistry profile, urinalysis, urine culture, blood pressure, total T4, urine protein creatinine ratio (UPC), feline leukemia/feline immunodeficiency virus serology, and a renal ultrasound. Cats were excluded from the study if they had evidence of ureteroliths, pyelonephritis, anatomic abnormalities such as polycystic kidney disease or masses, uncontrolled hypertension, or concurrent systemic disease. Administration of concurrent supportive therapies was allowed provided there were no changes in therapy during the study period. The study was approved by the Institutional Animal Care and Use Committee at Colorado State University (#10-1603A and 11-2915A), and all owners reviewed and signed consent forms prior to participation in the study.

aMSC preparation and administration: pilot studies 1 and 2

MSCs were isolated from adipose tissue of donor SPF cats obtained from the same subcutaneous site on the ventral abdomen just caudal to the umbilicus, as previously described [21,22]. For isolation of the stromal vascular fraction, the tissue was minced and digested with 1 mg/ml collagenase (Sigma Aldrich, St Louis, MO, USA) for 30 minutes at 37°C. The sample was centrifuged, and the stromal vascular fraction was plated in MSC medium (low glucose Dulbecco's modified Eagle medium (DMEM), 100 U/ml penicillin, 100 μ g/ml streptomycin, 2 μ M L-glutamine, 1% essential amino acids without L-glutamine, 1% non-essential amino acids, 0.075% sodium bicarbonate (Invitrogen/Gibco, Carlsbad, CA, USA) plus 15% fetal bovine serum

(Cell Generation, Fort Collins, CO, USA)). The aMSCs were incubated until approximately 70% confluent with media changes every 2 to 3 days. After reaching passage 3, cells were harvested, divided into treatment aliquots in freezing medium (11% dimethyl sulfoxide, 14% MSC medium, 75% fetal bovine serum), and stored in liquid nitrogen for no greater than 1 year prior to use. For injection in pilot studies 1 and 2, cryopreserved cells were removed from liquid nitrogen, incubated in MSC medium for 10 minutes, washed three times in Dulbecco's phosphate buffered saline (DPBS; Sigma) and then counted and checked for viability using Trypan Blue (cells with less than 95% viability were not used). For cats in pilot study 1, 2×10^6 cells were resuspended in 5 ml Hank's balanced salt solution (HBSS) with 200 IU heparin sulfate and administered slowly by hand over 15 minutes; aMSCs were administered at weeks 0, 2 and 4. For cats in pilot study 2, 4×10^6 cells were initially resuspended in 10 ml HBSS with 200 IU heparin sulfate and administered as a slow intravenous push over 30 minutes. However, the first cat to receive MSCs in this manner experienced an increased respiratory rate. Subsequently, cells for the high-dose group were resuspended in 20 ml HBSS/200 IU heparin and administered over 45 to 60 minutes with a syringe pump with frequent agitation of the syringe to prevent settling of the cells. Cats in pilot study 2 received aMSCs at weeks 2, 4, and 6. All three injections received by cats treated in pilot studies 1 and 2 were from the same donor cat. A total of four donor cats were used to provide aMSCs for those studies.

MSC preparation and administration: pilot study 3

MSCs were isolated from adipose tissue of donor SPF cats as described above. For preparation of the adipose tissue for cryopreservation, the tissue was minced and divided into 1-g aliquots in 1 ml of freezing medium (11% dimethyl sulfoxide, 14% MSC medium, 75% fetal bovine serum), and stored in liquid nitrogen for no greater than 1 year prior to use. Cryopreserved adipose was then later thawed, immediately washed twice with DPBS, and then prepared for culture as described above. For injection in pilot study 3 cats, cells were harvested by trypsinization at passage 3 and resuspended in 10 ml DPBS with 200 IU heparin sulfate and administered as a slow intravenous push over 20 minutes. Cats in pilot study 3 received aMSCs at weeks 2, 4, and 6. All three injections received by cats treated in pilot study 3 were from different donor cats given in a prerandomized order. A total of three donor cats were used to provide aMSCs for the study.

Characterization of MSCs

aMSCs cultured from cryopreserved adipose tissue were characterized by surface marker expression using flow cytometry and a panel of crossreactive antibodies

specific for surface determinants expressed by MSCs from other species [23-26]. Specifically, feline aMSCs were analyzed for surface expression of CD44 (anti-mouse/human, antibody clone:IM7, eBioscience, San Diego, CA, USA) and CD90 (anti-human, antibody clone:eBio5E10, eBioscience). MSCs were also assessed for expression of CD4 (anti-feline antibody clone: 3-4F4, Southern Biotech, Birmingham, AL, USA) and major histocompatibility complex (MHC) class II (anti-feline antibody clone: TU39, BD Bioscience, San Jose, CA, USA). Samples were analyzed using a Cyan ADP flow cytometer (Beckman Coulter, Brea, CA, USA). Approximately 25,000 events were collected for analysis per sample.

In vitro differentiation assays were conducted to confirm the multipotency of feline aMSCs, as assessed by their ability to differentiate into three cell lineages (osteoblasts, chondrocytes, and adipocytes) that are characteristic of MSCs [27]. For differentiation into adipocytes, aMSCs at confluency were incubated with MSC medium supplemented with 0.5 μ M dexamethasone (Sigma Aldrich), 50 μ M indomethacin (Sigma Aldrich) and 0.5 μ M 3-isobutyl-1-methylxanthine (Sigma Aldrich) for 3 weeks with media changes every 3 to 4 days. Chondrogenic differentiation medium consisted of DMEM 1 $\times$ (Cellgro, Manassas, VA, USA) supplemented with 15% fetal bovine serum (FBS; Cell Generation), 10 nM dexamethasone (Sigma Aldrich), 10 ng/ml transforming growth factor (TGF)- β (R&D Systems, Minneapolis, MN, USA), 50 μ g/ml ascorbic acid (Sigma Aldrich), and 40 μ g/ml proline (Sigma Aldrich). Osteogenic differentiation medium consisted of MSC medium supplemented with 10 nM dexamethasone, 50 μ M ascorbic acid, and 20 mM β -glycerophosphate (Sigma Aldrich). At the end of the differentiation period, cells were fixed with 10% neutral buffered formalin and stained with Oil Red O (Sigma Aldrich) for presence of lipid or with Alizarin Red (Sigma Aldrich) for the presence of calcium [27]. Cell pellets from cartilage differentiation were harvested and placed in OCT Compound (Sakura Finetek USA, Inc., Torrance, CA, USA) and flash frozen prior to staining with toluidine blue (Richard-Allan Scientific, Kalamazoo, MI, USA) for cartilage matrix. aMSCs cultured in MSC medium alone under identical conditions were used as differentiation controls.

Clinical monitoring

Pilot study 1

Each treated cat underwent physical examination, weighing, and routine blood work consisting of CBC, serum biochemistry, urinalysis, and UPC immediately prior to aMSC injection at week 0, and at week 2, 4.

Pilot study 2

Each treated cat underwent physical examination, weighing, and routine blood work consisting of CBC,

serum biochemistry, and urinalysis at weeks 0, 2, 4, 6 and 8. Each cat had a UPC performed at weeks 0 and 8. Additionally each cat had a glomerular filtration rate (GFR) estimated by iohexol clearance performed at weeks 0 and 8. Iohexol clearance has been described as a clinically applicable alternative to estimate GFR and has previously been assessed in cats with reduced renal function [28-30]. For this method, 300 mg/kg iohexol (Omnipaque, GE Health Incorporated, Princeton, NJ, USA) was administered intravenously, and blood samples were collected at 2, 3, and 4 h after administration. Analysis is commercially available at the Michigan State University Diagnostic Center for Population and Animal Health. Assessment of GFR variability was performed by enrolling three CKD cats that did not receive MSC therapy but underwent estimation of GFR by iohexol clearance at 0 and 8 weeks.

Pilot study 3

Each treated cat underwent physical examination, weighing, and routine blood work consisting of CBC, serum biochemistry, and urinalysis at weeks 0, 2, 4, 6 and 8. Each cat had a UPC performed at weeks 0 and 8. Additionally, each cat had a GFR by nuclear scintigraphy performed at weeks 0 and 8.

For the scintigraphy procedure, cats were sedated with a standard sedation protocol (ketamine 10 mg/cat and butorphanol 0.1 mg/kg intravenously once) at a standard time before the procedure. A single nuclear technician performed all of the procedures for each particular cat. For each procedure 1.0 mCi of Tc99m-labeled diethylene triamine penta-acetic acid (DTPA) (Cardinal Health, Dublin, OH, USA) was injected intravenously via a catheter placed in a standard location in each cat. Images were obtained using GE Millennium SPECT system applicable for small animal planar, whole body and single-photon emission computed tomography (SPECT) imaging (GE Healthcare, Waukesha, WI, USA). Three independent radiologists evaluated the GFR data, and a mean GFR value for each kidney as well as a global value was determined.

Urinary cytokine analysis

Urine cytokine concentrations were determined using commercial ELISA kits, as we have described recently [31]. Briefly, urine concentrations of interleukin (IL)-6, IL-8, and IL-10 were measured using feline specific ELISA kits (R&D Systems), while monocyte chemoattractant protein 1 (MCP-1) and vascular endothelial growth factor (VEGF) were measured using commercial canine MCP-1 and VEGF kits respectively (R&D Systems). Urine transforming growth factor β 1 (TGF- β 1) concentrations were measured using an ELISA kit from Invitrogen (Camarillo, CA, USA).

Urine samples were collected by cystocentesis, and an aliquot of urine was immediately frozen at -20°C and

then stored at -80°C for cytokine measurements. Samples were thawed immediately prior to analysis and kept on ice during sample preparation. All samples were centrifuged at 2000 rpm (350 g) for 5 minutes at 4°C , and the supernatant was collected for analysis. With one exception, ELISA kits were run according to manufacturer instructions for serum samples. As one exception, urine samples were diluted 5-fold rather than 40-fold for detection of TGF- β 1.

Urine and serum creatinine concentrations were measured with a Roche Cobas Integra Chemistry Analyzer (Roche Diagnostics Limited, West Sussex, UK) at the Colorado State University Diagnostic Laboratory. All reported cytokine concentrations were normalized for differences in urine concentration as a urine cytokine-to-urine creatinine ratio. In order to maximize cytokine yield in the urine samples, additional sample processing steps were performed as previously described prior to running the IL-8, MCP-1, and VEGF ELISA kits [31].

Statistical analysis

Changes in GFR, serum creatinine, blood urea nitrogen (BUN), electrolytes, urine protein-to-creatinine (UPC) ratio, urine specific gravity, and packed cell volume (PCV) data over time in the aMSC-injected cats were compared statistically by means of repeated measures analysis of variance (ANOVA) followed by Bonferroni's correction. Values were considered statistically different for $P < 0.05$. Statistical analyses were performed using Prism5 software (GraphPad, La Jolla, CA, USA).

Results

Description of study cats

A total of 165 cats with CKD were screened for entry into this series of pilot studies. In all, 52 cats were excluded for geographical reasons (extensive travel required), 40 cats were excluded due to concurrent illness, 18 cats were excluded due to end-stage disease, 13 cats were excluded as the owners thought the study too complex, 11 cats were excluded due to unstable disease, 8 cats were excluded for anatomic abnormalities, and 2 cats were excluded due to fractious attitude in hospital. Descriptive data for the 21 cats with CKD that were enrolled in the aMSC pilot studies are presented in Table 1. Two cats that were enrolled in pilot study 3 were unable to finish the trial due to development of unrelated illnesses. A summary of cat weights throughout the study and aMSC dose by weight are presented in Table 2. No statistically significant difference in weight was seen in any pilot study as determined by a Wilcoxon sign rank test.

Characterization of MSCs

During *in vitro* culture feline aMSCs were observed to develop into a relatively homogeneous population of

Table 1 Summary of demographics of cats participating in the intravenous allogeneic cryopreserved studies

Cat no.	Group	Description	IRIS CKD stage and creatinine (mg/dl)	Treatment
1	Pilot study 1	10 yr MC DSH	IRIS CKD II; creatinine: 2.5	2 × 10 ⁶ MSCs intravenously × 3 treatments
2	Pilot study 1	15 yr FS DSH	IRIS CKD III; creatinine: 3.5	2 × 10 ⁶ MSCs intravenously × 3 treatments
3	Pilot study 1	7 yr MC Siamese	IRIS CKD III; creatinine: 4.3	2 × 10 ⁶ MSCs intravenously × 3 treatments
4	Pilot study 1	12 yr MC DLH	IRIS CKD II; creatinine: 2.4	2 × 10 ⁶ MSCs intravenously × 3 treatments
5	Pilot study 1	15 yr MC DSH	IRIS CKD II; creatinine: 2.3	2 × 10 ⁶ MSCs intravenously × 3 treatments
6	Pilot study 1	15 yr FS Siamese	IRIS CKD III; creatinine: 3.5	2 × 10 ⁶ MSCs intravenously × 3 treatments
7	Pilot study 2	11 yr MC DSH	IRIS CKD II; creatinine: 1.9	4 × 10 ⁶ MSCs intravenously × 3 treatments
8	Pilot study 2	11 yr FS DSH	IRIS CKD II; creatinine: 2.6	4 × 10 ⁶ MSCs intravenously × 3 treatments
9	Pilot study 2	18 yr FS DSH	IRIS CKD II; creatinine: 2.8	4 × 10 ⁶ MSCs intravenously × 3 treatments
10	Pilot study 2	15 yr MC DSH	IRIS CKD II; creatinine: 2.2	4 × 10 ⁶ MSCs intravenously × 3 treatments
11	Pilot study 2	7 yr MC DSH	IRIS CKD III; creatinine: 3.7	4 × 10 ⁶ MSCs intravenously × 3 treatments
12	Pilot study 2	15 yr MC DSH	IRIS CKD II; creatinine: 2.3	Iohexol GFR repeatability only
13	Pilot study 2	16 yr MC DSH	IRIS CKD II; creatinine: 2.1	Iohexol GFR repeatability only
14	Pilot study 2	16 yr MC DSH	IRIS CKD II; creatinine: 1.8	Iohexol GFR repeatability only
15	Pilot study 3	9 yr MC DSH	IRIS CKD II; creatinine: 2.6	Enrolled but decompensated before treatment initiated
16	Pilot study 3	15 yr MC Siamese	IRIS CKD II; creatinine: 2.4	4 × 10 ⁶ MSCs intravenously × 2 treatments before recurrence of previous diabetes
17	Pilot study 3	13 yr MC Siamese	IRIS CKD II; creatinine: 2.7	4 × 10 ⁶ MSCs intravenously × 3 treatments
18	Pilot study 3	8 yr MC DSH	IRIS CKD II; creatinine: 1.7	4 × 10 ⁶ MSCs intravenously × 3 treatments
19	Pilot study 3	13 yr MC DSH	IRIS CKD II; creatinine: 2.0	4 × 10 ⁶ MSCs intravenously × 3 treatments
20	Pilot study 3	15 yr MC DLH	IRIS CKD II; creatinine: 2.3	4 × 10 ⁶ MSCs intravenously × 3 treatments
21	Pilot study 3	15 yr FS DSH	IRIS CKD III; creatinine: 3.1	4 × 10 ⁶ MSCs intravenously × 3 treatments

Six cats were enrolled in pilot study 1 and received 2 × 10⁶ cryopreserved adipose-derived mesenchymal stem cells (aMSCs) per infusion. In pilot study 2 eight cats were enrolled; five received 4 × 10⁶ cryopreserved aMSCs per infusion, and three cats received only iohexol clearance studies. In pilot study 3, seven cats were enrolled; five cats received 4 × 10⁶ cryopreserved aMSCs cultured from cryopreserved adipose per infusion; two cats developed unrelated medical conditions and were removed from the trial. CKD, chronic kidney disease; GFR, glomerular filtration rate. MC = male castrated, FS = female spayed, DSH = domestic shorthair, DLH = domestic long hair, IRIS = International Renal Interest Society.

plastic-adherent cells with fibroblast-like morphology. Adipose-derived MSCs expressed high levels of CD44 and CD90 and were negative for expression of CD4 and MHC class II (Figure 1). Both cryopreserved aMSCs and aMSCs cultured from cryopreserved fat were capable of trilineage differentiation (Figure 2).

Short-term responses to MSC injection

Cats in pilot study 1 treated with three separate intravenous infusions of 2 × 10⁶ cryopreserved aMSCs and cats in pilot study 3 treated with three separate intravenous infusions of 4 × 10⁶ aMSCs cultured from cryopreserved fat tolerated their injections without apparent adverse effects. During the infusions and immediately after, the cats appeared clinically normal and unperturbed. However, cats in pilot study 2, each having received an intravenous infusion of 4 × 10⁶ cryopreserved aMSCs, experienced a number of treatment-related adverse effects. The adverse effects included vomiting (two cats) and increased respiratory rate (four cats). In two cats, vomiting and nausea occurred within 2 minutes of initiating the first aMSC infusion and were not associated with respiratory distress.

Administration of diphenhydramine (1 mg/kg subcutaneously) and maropitant (1 mg/kg subcutaneously) resulted in cessation of clinical signs of vomiting and nausea. Affected cats were premedicated with diphenhydramine and maropitant for subsequent treatments, and additional adverse gastrointestinal reactions were not noted. One cat that experienced vomiting at the time of the first injection subsequently developed increased respiratory rate at the time of the third aMSC injection.

Increased respiratory rate and effort following aMSC infusion was not pronounced in three of the four pilot study 2 cats that experienced respiratory signs but were quite pronounced in one cat. Average baseline respiratory rate was 44 breaths/minute (range 36 to 60) and average increased respiratory rate was 80 breaths/minute (range 60 to 112 breaths/minute). Increased respiratory rate and effort was generally noted after approximately two-thirds of the aMSCs had been administered (that is, after 30 minutes of infusion) in the three cats with mild respiratory signs. One cat, however, had mildly increased respiratory rate on the first infusion, a normal second infusion, and developed marked overt respiratory distress

Table 2 Weight and mesenchymal stem cell (MSC) dose for cats participating in the intravenous allogeneic cryopreserved MSC studies

Study	Stage	Weight in kg, median (range)	MSC dose in cells/kg, median (range)
Pilot study 1, n = 6	Before	4.8 (2.7 to 5.1)	4.1×10^5 (3.9×10^5 to 7.4×10^5)
	After	4.8 (2.5 to 5.2)	
Pilot study 2, n = 5	Before	4.8 (4.4 to 6.2)	8.3×10^5 (6.1×10^5 to 9.1×10^5)
	After	4.8 (4.2 to 7.1)	
Pilot study 3, n = 5	Before	4.8 (3.7 to 7.1)	8.4×10^5 (5.6×10^5 to 1.1×10^6)
	After	4.8 (3.5 to 7.1)	

on the third infusion after approximately one-third of the aMSCs had been administered. The respiratory distress was characterized by a sudden onset of increased respiratory rate that quickly led to nasal flare, open mouth breathing, and an orthopnic stance with head extended and elbows out. The injection was stopped, the cat was moved to a critical care ward and placed in an oxygen chamber, and diphenhydramine (1 mg/kg

subcutaneously) was administered. After 20 minutes no improvement was noted, and the cat became notably nauseated (foaming at the mouth and subsequently vomiting). Maropitant (1 mg/kg subcutaneously) and dexamethasone (0.05 mg/kg intravenously) were administered. Clinical signs began to abate within 20 minutes of administration of dexamethasone. Open mouth breathing and orthopnea improved within 1 h. Increased

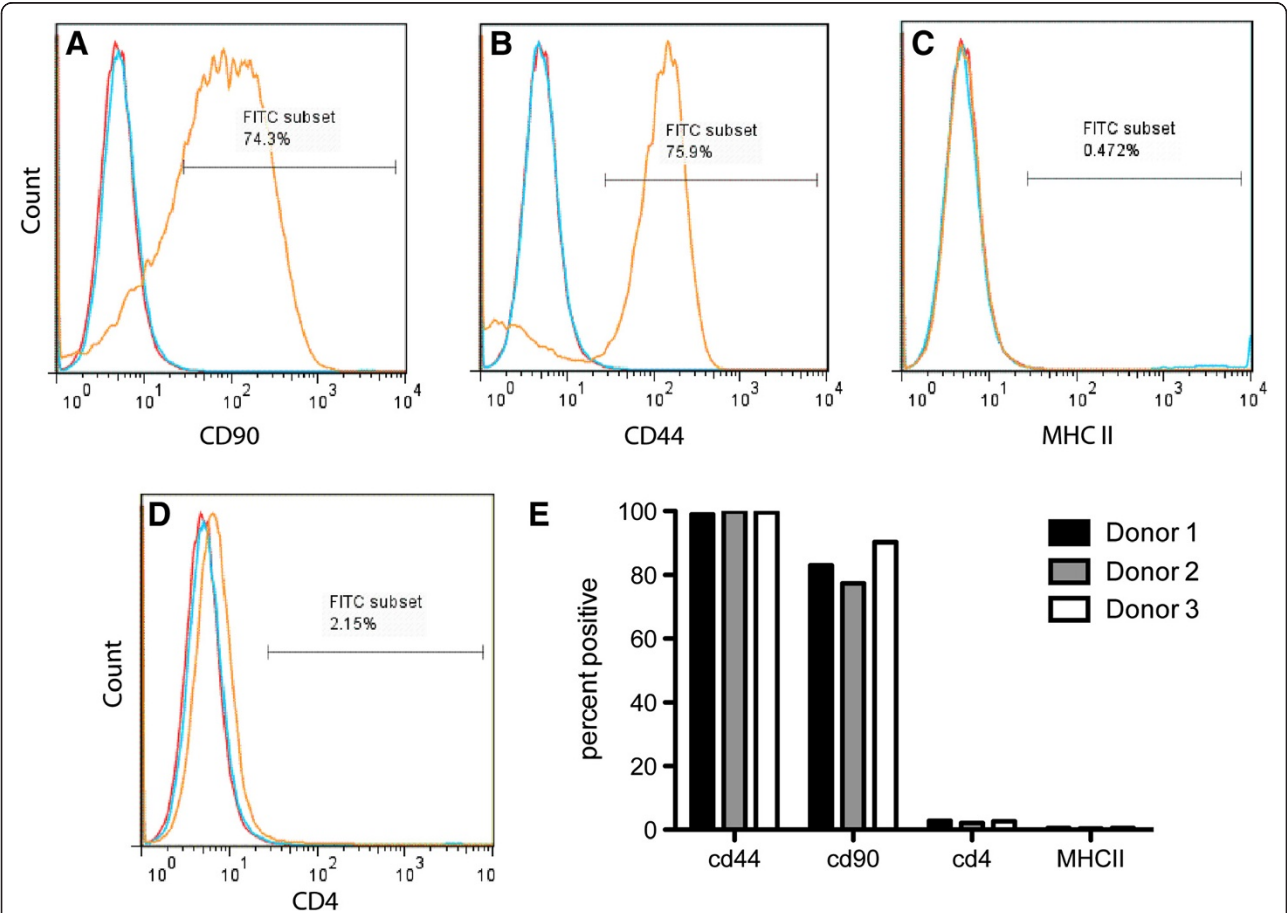


Figure 1 Expression of cell surface markers by cryopreserved feline adipose-derived mesenchymal stem cells (aMSCs). Feline aMSCs expanded from cryopreserved adipose were passaged three times in culture, then collected by trypsinization and immunostained for assessment of cell surface marker expression by flow cytometry, as described in Methods. Feline aMSCs expressed high surface levels of CD90 (A) and CD44 (B) but did not express major histocompatibility complex (MHC) class II (C) or CD4 (D). Isotype controls are represented in red and unstained MSCs are represented in blue. Similar results were obtained with aMSCs from three donor cats (E).

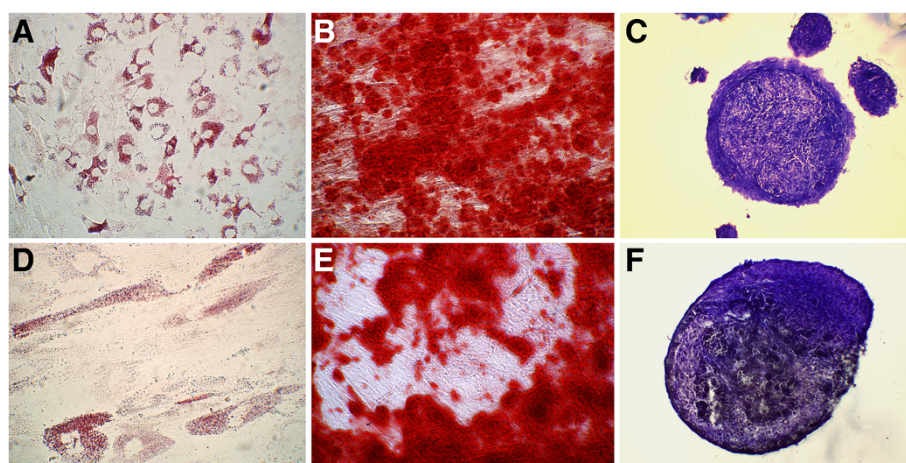


Figure 2 Trilineage differentiation of feline adipose-derived mesenchymal stem cells (aMSCs) after cryopreservation as cells or adipose. aMSCs taken directly from cryopreservation (A-C) and cultured from cryopreserved adipose (D-F) were capable of trilineage differentiation. (A,D) aMSCs formed intracellular lipid vacuoles when incubated in adipocytic differentiation media for 21 days. (B,E) aMSCs stained positive for calcium with alizarin red following differentiation into osteocytic phenotype after 21 days of incubation in differentiation media. (C,F) Cryosection of pellets of cartilage matrix (stained with toluidine blue) formed by aMSCs when exposed to chondrocytic differentiation media for 21 days.

respiratory effort continued for the next 8 h, but by 24 h post injection the cat was clinically normal and was discharged without further incident.

Effects of MSC administration on renal function clinical parameters

Serum creatinine concentrations decreased significantly ($P = 0.01$) in pilot study 1 cats over the study period but were unchanged in cats in pilot studies 2 and 3 (Figure 3A-C), however it is our opinion that the degree of observed decrease (<0.5 mg/dl) in serum creatinine concentrations is of questionable clinical significance. It should be noted that one cat from pilot study 1 was removed from analysis of serum creatinine concentration because the owners failed to administer fluids during the week prior to assessment, in violation of the study protocol. Values for BUN, serum phosphorus, serum potassium, packed cell volume, and urine specific gravity or UPC did not change significantly following administration of aMSCs as compared to pretreatment values in any of the three pilot studies (Table 3).

Effects of aMSC administration on estimated glomerular filtration rate

The GFR was not measured in pilot study 1 cats. In pilot study 2 cats, there was an overall trend towards improvement in estimated GFR by iohexol clearance values ($P = 0.056$), with individual increases of (75%, 10%, 61%, 7%, 6%) compared to untreated CKD control cats (7%, -18%, -22%) (Figure 3D). In pilot study 3 cats there was no statistically significant difference in total or individual kidney GFR values as determined by nuclear scintigraphy

however variable individual increases were seen (63%, 7%, -34%, 13%, 91%) (Figure 3E).

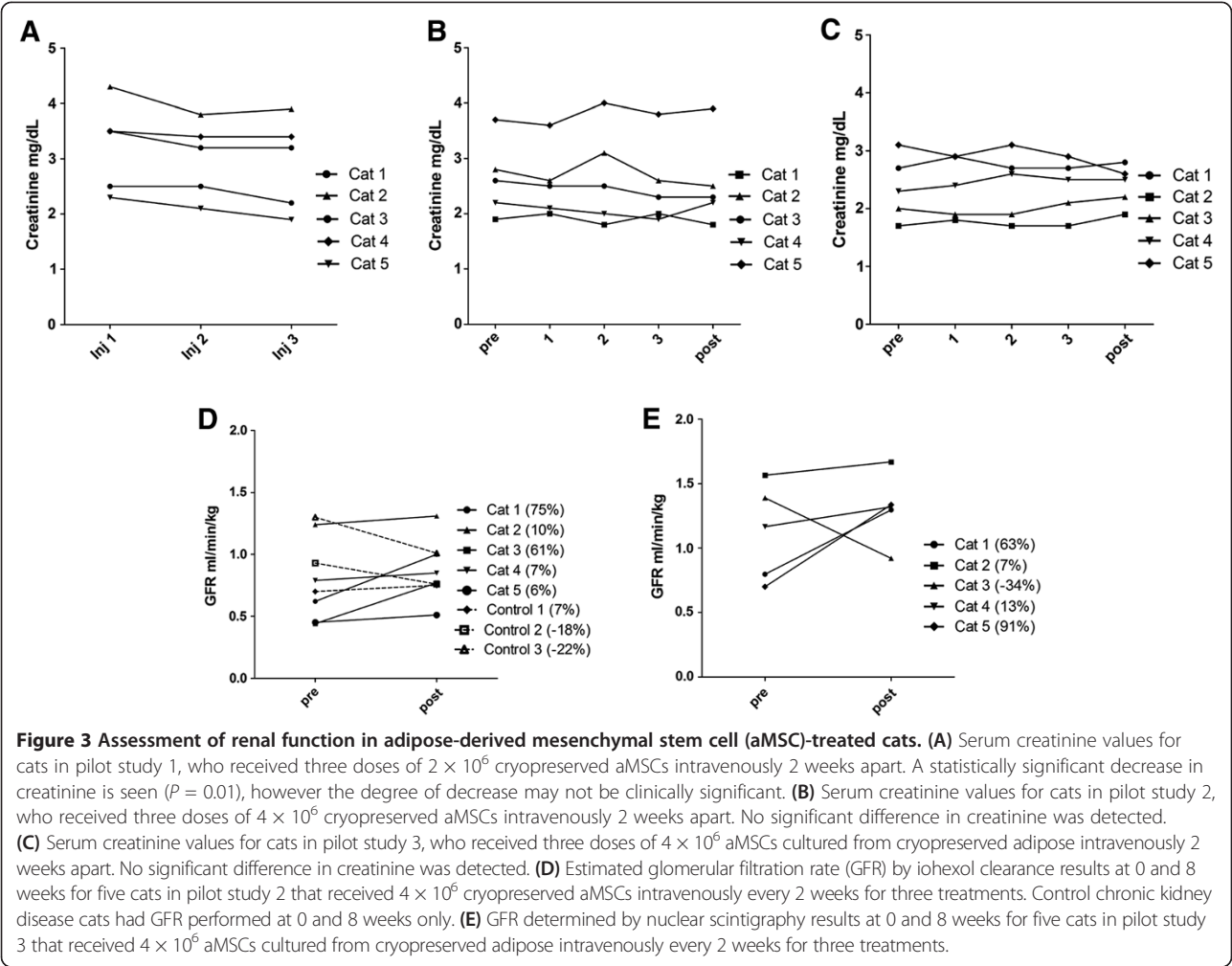
Effects of aMSC administration on urinary cytokine concentrations

In cats in pilot study 1, urine MCP-1 and IL-8 concentration data following MSC administration as measured by ELISA are depicted in Figure 4A,B. Using repeated measures ANOVA, a statistically significant decrease in MCP-1 ($P = 0.0001$) as well as in IL-8 ($P = 0.01$) was detected. However due to the variability in cytokine concentration changes (some cats decreased significantly while others increased), these results may not have clinical relevance.

In cats in pilot studies 2 and 3, urine concentrations of IL-8, MCP-1, TGF- β 1 and VEGF were assessed (Figure 2C-I). In the case of all four cytokines measured, repeated measures ANOVA failed to detect any significant changes in cytokine concentrations when pretreatment urine cytokine concentrations were compared to samples taken throughout the treatment period. In pilot study 3, MCP-1 levels were inconsistently detected in study samples.

Discussion

The safety and potential efficacy of repeated intravenous administration of allogeneic cryopreserved feline aMSCs to cats with naturally occurring CKD was assessed in this series of pilot studies. MSCs have previously been shown to be effective in suppressing some aspects of renal disease in rodent models of induced CKD. However, MSC therapy has not been evaluated in a naturally

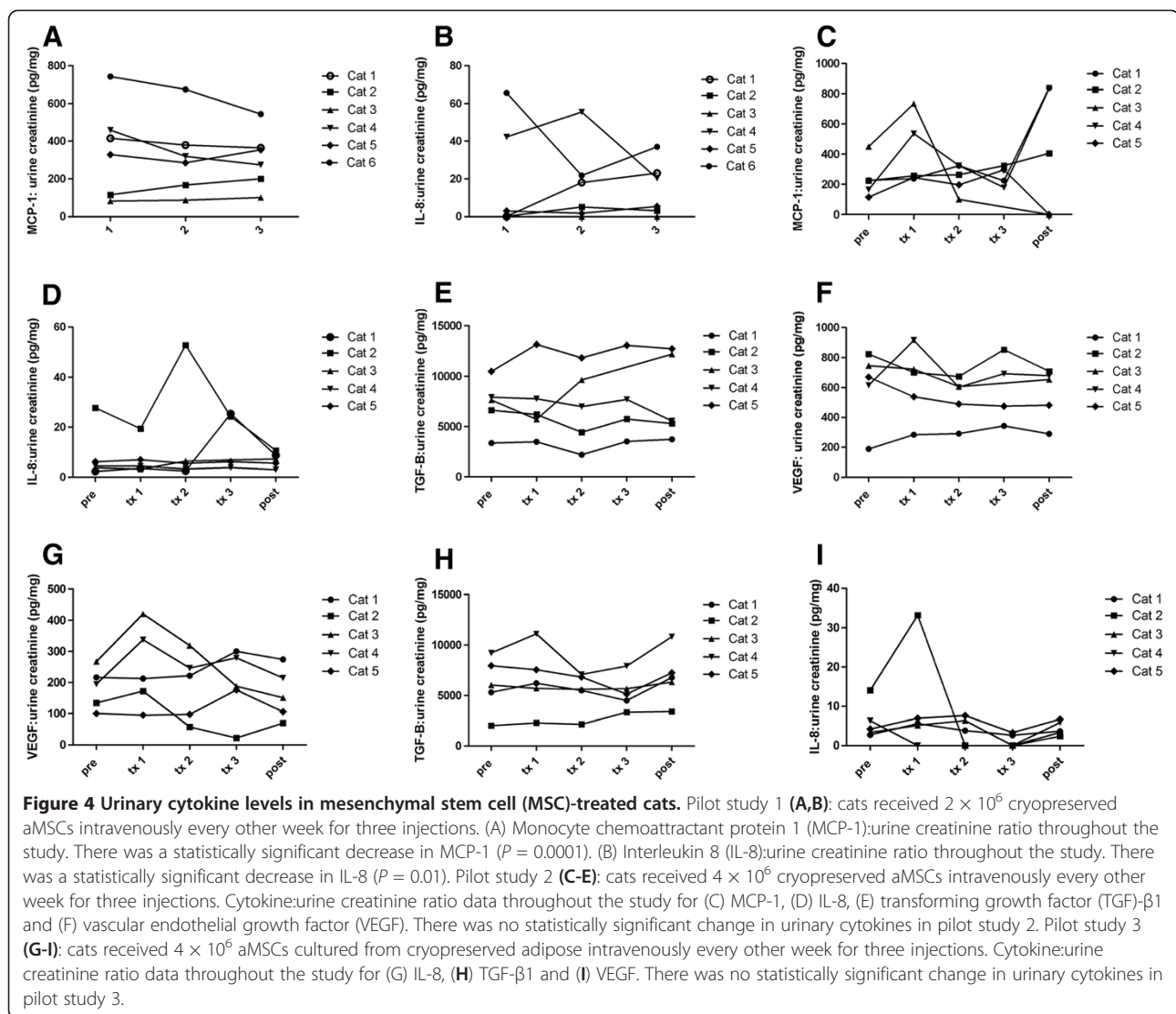


occurring animal model of CKD such as the feline CKD model. We modified existing rodent MSC administration protocols to create a clinically feasible trial in cats with CKD with translational potential. The major findings from our preliminary studies of aMSC therapy were that lower doses of allogeneic cryopreserved aMSCs were well tolerated following repeated intravenous administration and appeared to be associated with modest improvement in renal function. The level of improvement in renal function seen in the treated cats is, however, of uncertain clinical significance. The six cats in pilot study 1 that received the 2×10^6 cryopreserved aMSC dose experienced a statistically significant decrease in serum creatinine together with negligible

Table 3 Summary of clinicopathological data from cats participating in the intravenous allogeneic cryopreserved mesenchymal stem cell (MSC) studies

Study	Stage	Creatinine (mg/dl)	Blood urea nitrogen (mg/dl)	Phosphorus (mg/dl)	Potassium (meq/l)	Packed cell volume (%)	Urine protein-to-creatinine (UPC) ratio	Urine specific gravity
Pilot study 1	Before	3.5 (2.3 to 4.3)	57 (33 to 69)	3.8 (2.6 to 4.5)	4.6 (3.8 to 5.5)	31 (29 to 36)	0.1 (0.1 to 0.2)	1.016 (1.013 to 1.022)
	After	3.2 (1.9 to 3.9)	53 (36 to 58)	3.7 (3.3 to 4.5)	4.2 (3.9 to 4.7)	28 (26 to 36)	0.1 (0.1 to 0.2)	1.016 (1.013 to 1.025)
Pilot study 2	Before	2.6 (1.9 to 3.7)	42 (34 to 61)	4.0 (3.6 to 4.2)	4.6 (3.7 to 5.1)	31 (27 to 35)	0.1 (0.1 to 0.2)	1.016 (1.015 to 1.030)
	After	2.3 (1.8 to 3.9)	37 (30 to 64)	4.7 (2.4 to 6.7)	4.6 (3.7 to 5.0)	29 (27 to 34)	0.1 (0.1 to 0.3)	1.017 (1.013 to 1.036)
Pilot study 3	Before	2.3 (1.7 to 3.1)	49 (34 to 53)	4.2 (3.6 to 4.9)	4.7 (3.9 to 5.0)	34 (25 to 37)	0.1 (0.1 to 0.4)	1.017 (1.013 to 1.037)
	After	2.5 (1.9 to 2.8)	42 (36 to 53)	3.8 (3.7 to 3.9)	4.7 (4.1 to 5.0)	33 (28 to 34)	0.1 (0.1 to 0.4)	1.019 (1.013 to 1.029)

No significant changes in clinicopathological parameters were seen as a result of repeated MSC administration in any pilot group.



adverse effects from aMSC administration. However, the degree of this decrease (<0.5 mg/dl) is very small and may not translate to any change in clinical disease. In contrast, higher intravenous doses of cryopreserved aMSCs were associated with a high incidence of adverse effects and variable evidence of improvement in renal function as evidenced by GFR estimated by iohexol clearance. Most of the five cats in pilot study 2 that received 4×10^6 cryopreserved aMSCs per injection experienced side effects, including vomiting during infusion and increased respiratory rate and effort including overt dyspnea in one case. Significant changes in serum creatinine, GFR, and urinary cytokine concentrations were not observed in pilot study 2 cats. In pilot study 3, cats that received 4×10^6 aMSCs cultured from cryopreserved adipose experienced no side effects, but little evidence of clinically significant improvement in renal function. Thus, we concluded that although intravenous

administration of higher doses of cryopreserved, allogeneic aMSCs was unlikely to be an acceptable treatment option due to the high rate of complications and the lack of clinical or experimental responses; aMSCs expanded from cryopreserved fat appear a safer option although strong indication of efficacy is still lacking.

A recent study by Semedo *et al.* [4] in a rat model of CKD provided compelling evidence of the potential efficacy of repeated intravenous administration of bone marrow-derived MSCs for suppression of intrarenal inflammation and fibrosis, and this treatment protocol was the one we adapted for clinical application in CKD cats. Although aMSCs were used instead of bone marrow-derived MSCs, in the field of adipose stem cell research, the two sources are felt to be comparable in characterization and therapeutic potential in inflammatory disease states [23-27,32,33]. Why then were the results of our preliminary studies of aMSC therapy in cats with CKD so different

from the rodent studies, when similar doses of MSCs (based on body weight) were utilized? Firstly, differences in the CKD model should be taken into consideration. Most rodent models involve acute insult to the kidney with subsequent administration of MSCs within a few weeks: a relatively short time post renal insult [4,34]. Thus, the induced disease models, with their relatively short time frame, may not be representative of the changes occurring in a truly chronic, naturally occurring disease process [11]. Cats have an extended lifespan and often have CKD for months to years prior to clinical diagnosis and study enrollment, which is more similar to the disease time frame seen in humans. The possibility that this potential therapy may not be as effective in patients with more long-standing disease should be considered.

Secondly, in our study, we utilized allogeneic aMSCs, whereas autologous MSCs were used in the Semedo *et al.* study [4]. The relative efficacy of autologous vs. allogeneic cells is an area of controversy. Although allogeneic MSCs are immune privileged and are not expected to incite an immune response, according to some authors they may not be as effective as autologous cells [35]. It is argued that autologous MSCs may survive longer in the body in comparison to allogeneic cells, which could reduce efficacy of the latter. Decreased efficacy of allogeneic MSCs in comparison to autologous MSCs has been observed in one acute renal failure rodent study [35]. However, allogeneic MSCs have been widely used in experimental stem cell transfer investigations, including clinical trials in humans, with positive results [35,36].

A third major difference was the use of freshly-thawed, cryopreserved aMSCs in pilot studies 1 and 2. This decision was made based on the logistical ease afforded by use of cryopreserved rather than freshly-cultured MSCs. Successful cryopreservation of cells has been previously described [37]. However, it is unknown if cryopreservation affects the functional properties of aMSCs necessary for successful use in this model. Previous studies assessing the effects of cryopreservation on MSCs did not examine effects on potential immunogenicity and further work is needed to fully assess this subject [37,38].

Based on the results of the three pilot studies, it appears that use of higher doses of cryopreserved aMSCs was the source of the treatment-related adverse effects in pilot study 2 as similar doses of aMSCs cultured from cryopreserved adipose tissue did not result in any adverse effects. Although the specific reasons for the increased incidence of side effects is not known, it is likely related to the increase in dose of cryopreserved cells. The most likely explanation for this reaction is an instant blood mediated inflammatory reaction (IBMIR) which results in clumping of the cells as they contact the blood and potential subsequent micro pulmonary thromboembolism [39]. The IBMIR phenomenon has

been described previously in cryopreserved cells and increases in severity with dose and passage number [39]. It can result in lysis of the administered MSCs and subsequent poor efficacy. Although all cells given in pilot study 2 were of the same passage (P3) as those used in the other two pilot studies, the reaction was only seen in the pilot study group where cells were taken directly from cryopreservation and used at the higher dose. In pilot study 3 no complications during or after administration of aMSCs cultured from cryopreserved fat were appreciated. Thus, we have concluded that the administration of a higher dose of aMSCs taken directly from cryopreservation, despite careful washing, was the source of the toxic reactions observed, and this form of administration is not recommended.

Although several cats in the pilot study 1 group experienced a statistically significant improvement in serum creatinine concentrations during the study, the degree of increase is unlikely to be clinically significant. Creatinine is not the most sensitive measurement of renal function even though it is the routinely measured clinical marker of renal function. Several factors that are not directly related to renal function can influence serum creatinine values, including muscle mass and hydration status [28,40]. Decreases in weight and/or muscle mass can potentially decrease serum creatinine, while a worsening in hydration status can increase serum creatinine. In addition, changes to renal function may occur without any concomitant alteration in serum creatinine, although this is thought to be less true once an animal reaches the stage of moderate renal dysfunction [28]. Although there were no significant changes in body weight noted in these pilot studies, individual changes in body weight in the study cats could potentially have confounded serum creatinine measurements, while changes in bladder fullness at the time of weighing could potentially have compromised weight values. Therefore, in cats in pilot studies 2 and 3 we also measured estimated GFR, as a more objective way to monitor changes in renal function. Quantitating GFR allowed us to assess potential overall kidney function changes induced by the paracrine and/or autocrine effects of MSC administration. GFR results were variable in cats in pilot studies 2 and 3; two of five animals in pilot study 1 and two of five cats in pilot study 3 did experience a marked increase from pretreatment baseline; however the change in GFR was not significant overall for the group. Overall, definitive evidence of significant efficacy of intravenous administration of allogeneic aMSCs for treatment of CKD in cats is lacking at this time. Closer evaluation of the mechanism of MSC action in CKD, effect of disease stage, necessary cell dosage (for example, a cell/kg dose rather than a per cat dose as used in these pilot studies), timing of injections, and other variables remain to be determined.

Conclusions

The safety and potential efficacy of repeated intravenous administration of allogeneic cryopreserved feline aMSCs to cats with naturally occurring CKD was assessed in this preliminary study. The major findings from our pilot studies of aMSC therapy in cats with CKD were that lower doses of allogeneic cryopreserved aMSCs were well tolerated following repeated intravenous administration and appeared to be associated with modest decrease in serum creatinine. In contrast, higher intravenous doses of cryopreserved aMSCs were associated with a high incidence of adverse effects and only modest evidence of functional improvement in renal function. Thus, we concluded that intravenous administration of higher doses of cryopreserved, allogeneic aMSCs was unlikely to be an acceptable treatment option due to the high rate of complications and the lack of clinical or experimental responses.

Abbreviations

aMSCs: Adipose-derived mesenchymal stem cells; CBC: Complete blood count; CKD: Chronic kidney disease; GFR: Glomerular filtration rate; HBSS: Hank's balanced salt solution; IBMIR: Instant blood mediated inflammatory reaction; IL: interleukin; MCP-1: Monocyte chemoattractant protein 1; MSCs: Mesenchymal stem cells; SPF: Specific pathogen-free; TGF- β 1: Transforming growth factor- β 1; UPC: Urine protein-creatinine ratio; VEGF: Vascular endothelial growth factor.

Competing interests

The authors declare they have no competing interests.

Authors' contributions

JMQ conceived and designed the study, harvested adipose tissue, prepared injections, provided veterinary care and administered treatments to enrolled animals, performed data acquisition, analysis and manuscript writing and revision. TLW performed the isolation and expansion of MSCs, urine cytokine analysis, flow cytometry and differentiation assays, prepared injections, and manuscript writing and revision. LAH performed the sample preparation, data entry, and urine cytokine analysis, and manuscript writing. The study was conducted in the laboratory of SWD and he was responsible for supervision of the study and manuscript writing and revision. All authors read and approved the final manuscript.

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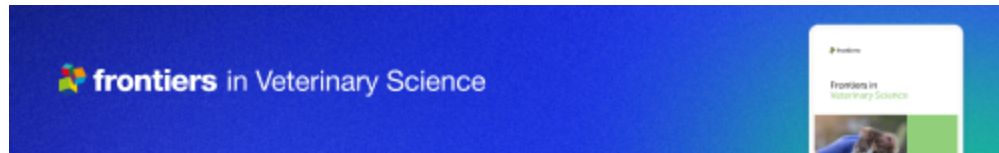
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Case Report: Allogeneic adipose-derived mesenchymal stem cells for severe feline chronic kidney disease

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Abstract

A 10-year-old neutered male Ragdoll cat presented with symptoms of anorexia, nausea, vomiting, lethargy, and progressive weight loss. Laboratory parameters and clinical signs led to a diagnosis of International Renal Interest Society (IRIS) Chronic Kidney Disease (CKD) Stage III. Following basic supportive therapy, clinical symptoms subsided, and the patient was discharged for home management. Three months later, the cat's condition worsened, and upon reassessment, it met the diagnostic criteria for IRIS CKD Stage IV. After 4 months of guideline-directed supportive care (in accordance with IRIS recommendations), there were no significant changes in clinical symptoms or laboratory markers, and the cat continued to exhibit a cachectic condition. Subsequently, allogeneic adipose-derived mesenchymal stem cell (AD-MSCs) therapy (2×10^6 cells/kg) was initiated via intravenous (IV) infusion, administered weekly for three consecutive weeks. No adverse events, such as fever or vomiting, were observed during or after therapy. Prior to AD-MSCs therapy, the cat's serum creatinine (SCr), blood urea nitrogen (BUN), and phosphorus (P) levels were all above the normal reference ranges. Three weeks after the final of the three AD-MSC

treatments, the SCr level had returned to the reference range for CKD Stage II, while BUN and P levels showed an improving trend. The cat's mental status and appetite also improved. Reevaluation according to IRIS CKD staging criteria confirmed functional enhancement from Stage IV to Stage II. During the subsequent follow-up period, the cat's physical condition improved, as shown by an increase in body weight, an improved Body Condition Score (BCS), and the normalization of mucous membrane color. Concurrently, laboratory results revealed a decrease in key renal biomarkers (SCr, BUN, and P) and an amelioration of the non-regenerative anemia. This case report suggests that allogeneic AD-MSCs have potential therapeutic efficacy in felines with end-stage CKD, offering a new possibility for the treatment of terminal chronic kidney disease.

Keywords: feline, chronic kidney disease, adipose, mesenchymal stem cells, paracrine effects

1. Introduction

Feline chronic kidney disease (CKD), characterized by persistent structural or functional renal abnormalities for more than 3 months in cats, is the most common metabolic condition in feline medicine (1). Epidemiological studies indicate that CKD is more prevalent in felines than in canines, with a particular susceptibility in older populations, although it can occur in cats of all ages (2). Ragdoll cats exhibit increased vulnerability to kidney diseases (3). A study of a randomly selected feline population revealed that the overall prevalence of CKD could be as high as 50%, rising to 80.9% in geriatric cats aged 15 to 20 years (4). In addition to its high prevalence, CKD poses a significant threat to feline survival. Furthermore, a study on feline mortality in England reported that kidney disease is the second most common cause of death in cats of all ages (12.1%) and the leading cause of mortality in cats aged 5 years and older (13.6%) (5). The principal histological features that drive the progression of CKD include glomerulosclerosis, tubulointerstitial inflammation, atrophy, and fibrosis (2, 6, 7). Using the International Renal Interest Society (IRIS) staging system, feline CKD is categorized into Stages I-IV based on serum creatinine concentrations, with further classification involving urine protein-to-creatinine ratio (UPC) and systolic blood pressure (SBP) (8). Histopathological findings confirm the increasing intensification of tubulointerstitial fibrosis and glomerulosclerosis in advanced stages, particularly during Stages III-IV (9). Currently, there are no widely recognized curative drugs for feline CKD. Therefore, management options are primarily limited to supportive and symptomatic care, aimed at improving the quality of life of affected cats and slowing disease progression (10, 11). Treatment options include conventional therapies and renal replacement therapies (9). However, the veterinary sector confronts two primary challenges: the lack of established pharmacological dosing protocols (12) and limited access to renal replacement therapies (11). These treatment constraints lead to insufficient clinical intervention, markedly diminishing the quality of life in

cats afflicted by CKD. Recent studies emphasize mesenchymal stem cells (MSCs) as promising therapeutic agents due to their complex biological features, including anti-inflammatory, immunomodulatory, anti-apoptotic, antioxidant, anti-fibrotic, pro-angiogenic, and autophagy-regulating actions ([13–15](#)). MSCs are adult stem cells characterized by their ability to self-renew and differentiate into many lineages, commonly extracted from bone marrow, umbilical cord, adipose tissue, and various other sources ([16, 17](#)). The predominant sources employed are bone marrow-derived mesenchymal stem cells (BM-MSCs), umbilical cord-derived mesenchymal stem cells (UC-MSCs), and adipose-derived mesenchymal stem cells (AD-MSCs) ([18](#)). In comparison to other MSC types, AD-MSCs exhibit superior accessibility, increased proliferative capability, reduced immunogenicity, and a lack of ethical issues ([19–21](#)). In human clinical trials, MSC-based treatments demonstrate acceptable safety profiles and tolerability, with rising evidence of efficacy in enhancing clinical parameters ([22–24](#)). Research in veterinary medicine has concentrated on the uses of MSCs across several diseases. In felines, AD-MSCs have been studied for the treatment of chronic gingivostomatitis ([25](#)), chronic enteropathy ([26, 27](#)), asthma ([28](#)), ophthalmic disorders ([29](#)), and renal illnesses ([30](#)). Recent studies further validate the safety and feasibility of MSC treatment in felines with CKD ([31–33](#)). Nonetheless, clinical evidence assessing the safety and efficacy of allogeneic AD-MSCs for feline chronic kidney disease is still scarce. This case study involved the intravenous infusion of allogeneic AD-MSCs to a feline patient with CKD who had ongoing clinical deterioration despite adherence to IRIS guideline-recommended treatment procedures.

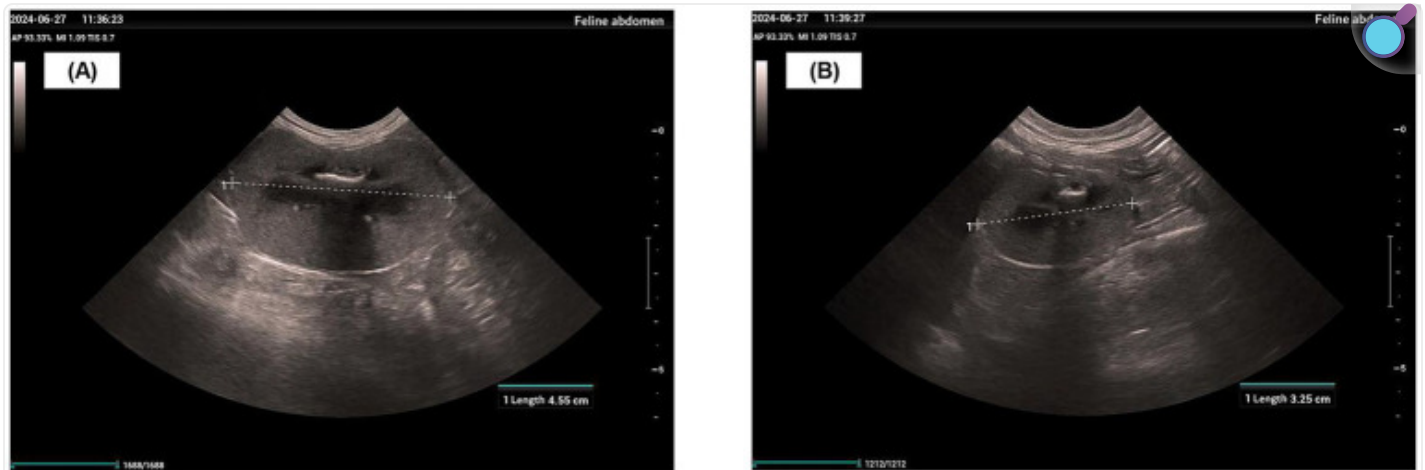
2. Case description

In August 2023, a 10-year-old neutered male Ragdoll cat presented with anorexia, nausea, vomiting, lethargy, and progressive weight loss. On physical examination, the cat weighed 6.3 kg with a body condition score (BCS) of 5/9 and had pale mucous membranes. Laboratory findings revealed a serum creatinine (SCr) of 280 $\mu\text{mol/L}$ (reference interval: 71–212 $\mu\text{mol/L}$) and a blood urea nitrogen (BUN) of 16.8 mmol/L (reference interval: 5.7–12.9 mmol/L) ([Supplementary Tables S1, S3](#)). The cat was diagnosed with IRIS CKD Stage III. Following basic supportive therapy, the cat's clinical signs subsided, and its weight stabilized at 5.7 kg. The cat was discharged for home management in mid-November.

In late June 2024, a physical examination, diagnostic imaging, and serum biochemistry were performed. The physical examination revealed a body weight of 4.6 kg, a body condition score (BCS) of 1/9, and pale mucous membranes. Diagnostic imaging identified an enlarged left kidney (4.55 cm), an atrophied right kidney (3.25 cm), thickened, hyperechoic cortices, and a loss of corticomedullary differentiation ([Figures 1A,B](#)). Serum biochemistry showed an SCr of 799 $\mu\text{mol/L}$,

BUN > 46.4 mmol/L, and phosphorus (P) > 5.20 mmol/L (reference range: 1.00–2.42 mmol/L) ([Supplementary Tables S1, S3](#)). The treatment plan for the hospitalized patient consisted of lactated Ringer's solution (200 mL SC every 24 h), maropitant (1.0 mg/kg SC every 24 h), telmisartan (1.0 mg/kg PO every 24 h), and darbepoetin alfa (1.0 µg/kg SC every 7 days), in addition to a renal prescription diet and nutraceuticals. Despite the use of IRIS guideline-recommended therapy, the cat's SCr, BUN, and P levels remained persistently above their normal physiological reference ranges. The patient's physical and mental status continued to be poor, with no improvement noted on physical examination or in biochemical markers. Therefore, with the owner's consent, MSC therapy was initiated to reduce further renal damage from chronic kidney disease, alleviate clinical signs, and improve the patient's quality of life.

Figure 1.



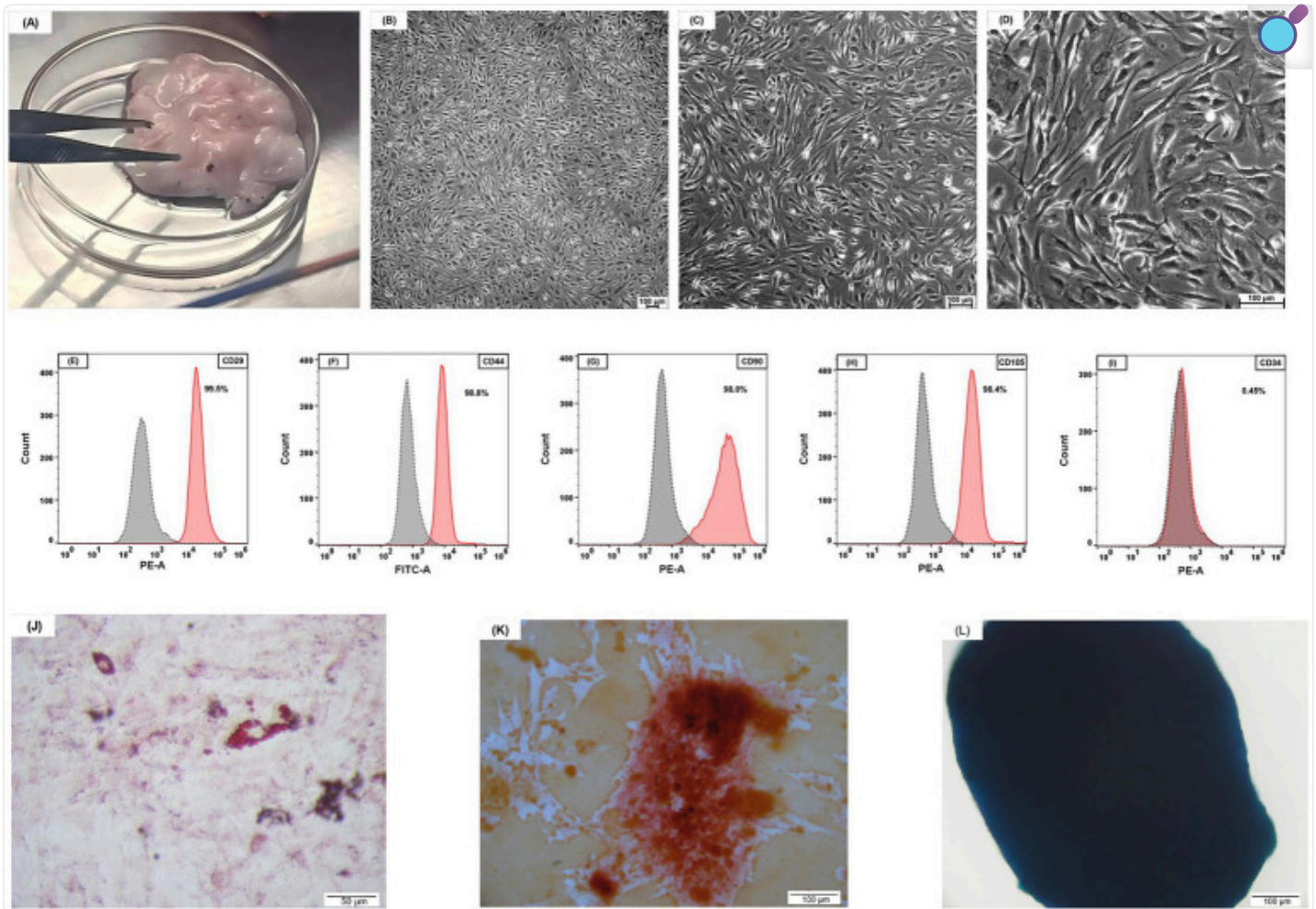
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Renal ultrasound image. **(A)** Left renal length. **(B)** Right renal length.

Adipose tissue was aseptically obtained from a 9-month-old healthy Ragdoll donor cat ([Figure 2A](#)), exhibiting normal hematological parameters and negative results for infectious disease screening. The samples were rinsed three times with phosphate-buffered saline (PBS, Servicebio, China) and then deposited into centrifuge tubes containing PBS with 1% penicillin–streptomycin (P/S, Gibco, USA) before being transferred to a laminar flow hood. The adipose tissue was consecutively submerged in three centrifuge tubes containing 75% ethanol for 3 s each, then transferred to a Petri plate containing PBS for clot removal. The treated adipose tissue was subsequently

transferred to new centrifuge tubes, to which PBS was added and gently inverted for mixing prior to discarding the PBS. The washing method was conducted for ten cycles. The adipose tissue was placed into new centrifuge tubes with sterile forceps and dissected into 0.5–1 mm³ fragments using surgical scissors. Type I collagenase (1 mg/mL; Sigma-Aldrich, USA) was then introduced, followed by incubation in a 37°C water bath with continuous agitation for 45 min to promote digestion. After digestion, an equivalent volume of complete medium was added to neutralize enzymatic activity. The complete medium was prepared using Dulbecco's modified Eagle's medium (DMEM; BasalMedia Technologies, China) as the base, supplemented with 10% fetal bovine serum (FBS; Excell, China), 1% penicillin–streptomycin (P/S), and basic fibroblast growth factor (bFGF; 5 ng/mL; Sino Biological, China). The cell suspension was subjected to extensive pipetting until the fibrous aggregates were fully dissociated and then filtered through a 100 µm cell strainer (pluriSelect, Germany) to remove any remaining undigested tissue pieces and cellular clumps. Following centrifugation, the cellular pellet was resuspended in the prepared DMEM complete medium and aseptically cultured in 6-well plates under controlled conditions (37°C, 5% CO₂ environment) for 7 days, with two medium exchanges conducted throughout the incubation period. Cell cultures were maintained until adherent cells reached 70–80% confluence, after which subculturing was performed. Cells were cryopreserved after five successive passages.

Figure 2.



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Feline adipose tissue and characterization of AD-MSCs. **(A)** Feline adipose tissue. **(B)** $\times 50$ magnification; **(C)** $\times 100$ magnification; **(D)** $\times 200$ magnification; feline AD-MSCs show a spindle-shaped morphology. **(B-D)** Scale bars, 100 μm . **(E-I)** The FCM analysis has shown that feline AD-MSCs highly expressed CD29, CD44, CD90, and CD105 but did not express CD34. **(J)** Adipogenic differentiation, evaluated using Oil Red O staining. **(K)** Osteogenic differentiation, evaluated using Alizarin Red staining. **(L)** Chondrogenic differentiation, evaluated using Alcian Blue staining. **(J-L)** Scale bars: 50 μm **(J)**; 100 μm **(K,L)**.

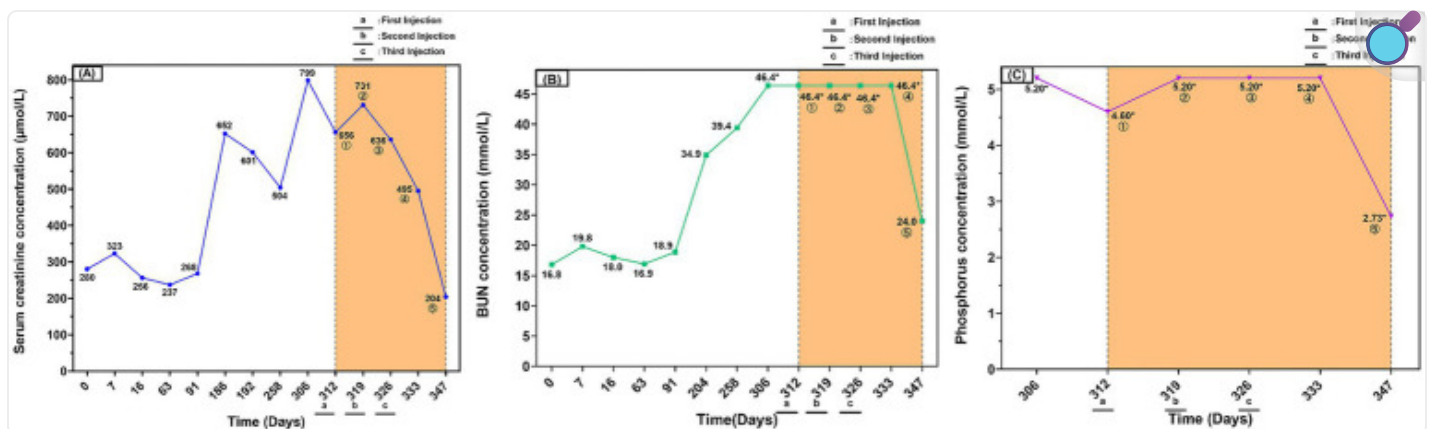
Microscopic analysis of MSCs revealed a typical spindle-shaped, fibroblast-like appearance ([Figures 2B-D](#)). The characteristics of the feline adipose-derived cells were investigated via flow

cytometry to verify their classification as mesenchymal stem cells (MSCs). The following directly conjugated antibodies were utilized: phycoerythrin (PE) conjugated against CD29 (antibody clone TS2/16, 303004, BioLegend, USA), fluorescein isothiocyanate (FITC) conjugated against CD44 (antibody clone IM7, MA1-10229, Invitrogen, USA), PE conjugated against CD90 (antibody clone 5E10, 555596, BD Biosciences, USA), and PE conjugated against CD34 (antibody clone 1H6, 559369, BD Biosciences, USA). For CD105 phenotyping, an indirect labeling technique was utilized: cells were initially treated with an unconjugated CD105 antibody (clone SN6, 14-105782, Invitrogen, USA), followed by a PE-conjugated goat anti-mouse IgG F(ab')₂ fragment secondary antibody (12-4010-82, Invitrogen, USA). The isotype controls comprised PE-conjugated Mouse IgG1, κ Isotype Control (clone MOPC-21, 555,749, BD Biosciences, USA), FITC-conjugated Rat IgG2b kappa Isotype Control (clone eB149/10H5, 11-4,031-82, Invitrogen, USA) and unconjugated Mouse IgG1 kappa Isotype Control (clone P3.6.2.8.1, 14-4,714-82, Invitrogen, USA). Cells were examined utilizing a CytoFLEX flow cytometer (Beckman Coulter, USA), and the data were assessed with FlowJo software (Tree Star, USA) ([Figures 2E–I](#)). When cultivated in certain differentiation medium, the MSCs exhibited trilineage differentiation potential into adipocytes, osteoblasts, and chondrocytes ([Figures 2J–L](#)).

Cryopreserved AD-MSCs were thawed and cultured until the adherent cells reached 80–90% confluence. The cells were then harvested by digestion with TrypLE™ Express enzyme (1X, Red; Gibco, USA) and subsequently washed with PBS to remove culture residues. Next, the cells were resuspended in 1 mL of a Cell Refrigeration Preservation Premix (Selcell, China), which is free of dimethyl sulfoxide (DMSO) and animal-derived components. The cell suspension was immediately transferred to a sterile vial and stored protected from light at 2–8°C, with the requirement for infusion within 24 h. Prior to infusion, the cell suspension was drawn and added to an infusion bottle containing 50 mL of saline. The final dose of 8.8×10^6 cells/50 mL (equivalent to 2×10^6 cells/kg) was administered as a slow intravenous infusion over 60 min via an intravenous (IV) catheter. To mitigate the risk of infusion reactions, the cat was premedicated with dexamethasone (0.5 mg/kg). Vital signs were continuously monitored throughout the infusion and for 30 min post-infusion, with observation extended to 24 h. No adverse reactions, such as fever or vomiting, were observed during or after the treatment. One week after the first stem cell injection, the SCr was 731 $\mu\text{mol/L}$. One week after the second injection, it decreased to 636 $\mu\text{mol/L}$, and 1 week after the third injection, it further reduced to 495 $\mu\text{mol/L}$. Three weeks after the final injection, the SCr level dropped to 204 $\mu\text{mol/L}$, returning to the reference range for CKD Stage II ([Figure 3A](#)). The BUN and P levels, which were consistently above the normal reference range before treatment, also showed a positive trend of improvement. Three weeks after the third stem cell injection, the BUN level decreased to 24 mmol/L, and the P level fell to 2.73 mmol/L ([Figures 3B,C](#)). The patient's mental status and appetite improved throughout the treatment period. Physical examination revealed that the cat's body weight recovered to 5 kg, the BCS improved to 3/9, and the mucous membrane color

returned to pink. Based on the laboratory data and clinical presentation, the patient was determined to have improved from CKD Stage IV to Stage II. During the subsequent follow-up period, multidimensional assessments including physical examination (PE), complete blood count (CBC), and serum biochemistry were conducted ([Supplementary Tables S1–S3](#)). The assessments revealed improvements across multiple key parameters: in physical examination, both body weight and body condition score (BCS) increased and mucous membrane color returned to normal; serum biochemistry showed a decrease in the levels of creatinine, urea nitrogen, and phosphorus; and the complete blood count reflected an amelioration of the severe renal anemia. This clinical course is consistent with a positive response to the therapy and suggests a favorable safety profile, as no significant adverse reactions were observed. Therefore, this case highlights the therapeutic potential of this regimen and provides a valuable reference for future studies on similar cases.

Figure 3.



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Serum creatinine, BUN, and phosphorus dynamics in a cat with chronic kidney disease during long-term conventional therapy and subsequent AD-MSCs treatment. **(A)** Serum creatinine (SCr) concentration over time ($\mu\text{mol/L}$). **(B)** Blood urea nitrogen (BUN) concentration over time (mmol/L). **(C)** Serum phosphorus (P) concentration over time (mmol/L). The yellow shaded area indicates the period of AD-MSCs treatment. The encircled numbers ①–⑤ on the figure represent measurements taken at the following time points: ① before the first injection (baseline); ② before the second injection; ③ before the third injection; ④ 1 week after the third injection; ⑤ 3 weeks after the third injection. Asterisks (*) indicate values that reached the upper limit of the instrument's detection range. Day 0 was defined as the day treatment commenced upon the cat's initial hospitalization in August 2023, serving as the baseline for the clinical timeline. a, b, and c indicate the time points of the first, second, and third mesenchymal stem cell infusions, respectively.

3. Discussion

Chronic kidney disease in cats is characterized by its irreversible progression and gradual onset, often evading early detection due to the absence of distinct clinical symptoms in the initial stages, resulting in diagnoses at more advanced phases (9). As a significant contributor to morbidity and mortality in older felines (5), current management strategies focus on palliative care, which includes fluid therapy (34, 35), antiemetic and orexigenic treatments (36, 37), blood pressure

control, reduction of proteinuria ([38](#)), management of anemia ([39](#)), and regulation of hyperphosphatemia through dietary phosphate and protein restriction ([40–43](#)). Although these interventions may temporarily improve the quality of life, their inability to change the disease's pathophysiology, prevent functional decline, or restore nephron structure underscores the limitations of treatment. Renal replacement therapies encompass kidney transplantation ([44, 45](#)), peritoneal dialysis ([46, 47](#)), and hemodialysis ([48, 49](#)).

Nonetheless, these methodologies present significant challenges in veterinary therapeutic practice, including ethical dilemmas, high costs, technological complexities, and the management of long-term comorbidities. Consequently, the development of innovative therapeutic strategies—particularly mechanism-targeted interventions such as stem cell-based therapies—has become a central focus in contemporary research ([50–52](#)).

MSCs show great potential in regenerative medicine for the treatment of feline diseases ([53–56](#)). For example, they have been used to treat conditions such as chronic gingivostomatitis, chronic enteropathy, asthma, ophthalmic disorders, and kidney disease. The therapeutic mechanisms of MSCs predominantly encompass two principal routes. The initial process is directed differentiation, in which a fraction of MSCs migrates to damaged renal tissues, proliferates, and differentiates into renal cells to replace the injured populations ([57, 58](#)). Nonetheless, the effectiveness of this method is limited by inadequate homing efficiency to renal locations and suboptimal post-transplant survival rates ([59–61](#)). An alternative mechanism entails MSCs providing renoprotective effects chiefly via paracrine secretion of various bioactive factors, such as vascular endothelial growth factor (VEGF), fibroblast growth factor (FGF), hepatocyte growth factor (HGF), insulin-like growth factor (IGF), transforming growth factor-beta (TGF- β), and interleukin-10 (IL-10). These mediators collectively facilitate renal epithelial cell proliferation, angiogenesis, anti-apoptotic mechanisms, anti-inflammatory responses, and anti-fibrotic activities ([62, 63](#)).

Adipose tissue is the predominant source of MSCs owing to their plentiful availability, little immunogenicity, and ethical acceptability, with AD-MSCs exhibiting distinct advantages in clinical applications ([64, 65](#)). AD-MSCs display a distinctive spindle-shaped fibroblast-like appearance and possess significant proliferation capability *in vitro* ([66, 67](#)). Immunophenotypic profiling verifies that feline AD-MSCs exhibit classical MSC surface markers (CD29, CD44, CD90, and CD105) and do not express the hematopoietic lineage marker CD34 ([21, 68, 69](#)), along with possessing multipotent differentiation capabilities. Although the therapeutic efficacy of AD-MSCs has been thoroughly established in rat disease models, its applicability for feline CKD has not been adequately explored in veterinary medicine.

The treatment protocol in this study was primarily based on key previous findings by Quimby et al. Their 2013 study demonstrated (70) that multiple intravenous infusions of culture-expanded AD-MSCs derived from cryopreserved adipose tissue have a good safety profile, whereas the direct use of cryopreserved cells may induce adverse reactions. Subsequently, their 2016 randomized, placebo-controlled trial (32) further established the clinical safety of a 2×10^6 cells/kg dose administered via multiple intravenous infusions. Although these studies did not observe significant short-term improvements in renal function, they laid the foundation for the safe application of AD-MSCs. Therefore, this study adopted a protocol of three intravenous infusions of AD-MSCs at a dose of 2×10^6 cells/kg, administered once weekly.

This case report describes clinical improvement in a cat with severe chronic kidney disease (CKD) following treatment with allogeneic mesenchymal stem cells (MSCs). No acute side effects were observed after three MSC injections. The 10-year-old male Ragdoll cat showed a decreasing trend in serum creatinine (SCr), blood urea nitrogen (BUN), and phosphorus (P) levels, along with improvements in mentation and appetite. Based on laboratory data and clinical presentation, the cat's CKD stage improved from IRIS Stage IV to Stage II. These findings suggest that MSC therapy may represent a potential therapeutic approach for end-stage CKD.

In assessing the therapeutic results of this case, essential inquiries arise: ① Considering the restricted effectiveness of traditional medication in influencing CKD indicators, which bioactive constituents of AD-MSCs primarily facilitate their renoprotective effects? ② Current research indicates that in the treatment of feline diseases, minimal dosages of mesenchymal stem cells (1×10^6 cells/kg) have restricted effectiveness, whereas elevated doses ($4\text{--}5 \times 10^6$ cells/kg) yield enhanced therapeutic results but may entail some side consequences (71). Establishing dynamic dose-adjustment techniques for MSCs based on disease stages to enhance the efficacy-safety balance remains a significant problem. ③ This example utilized intravenous MSC delivery, resulting in significant therapeutic benefits. Although intravenous injection is the predominant administration route, the “first-pass effect” considerably diminishes the percentage of cells that effectively engraft in damaged renal tissues. The suboptimal renal homing and engraftment efficiency is a significant obstacle to achieving optimal therapeutic efficacy. Would localized delivery methods—such as direct renal cortical (72), renal artery (73), or intraperitoneal (74) injection—improve target-site bioavailability and therapeutic efficacy?

This case report possesses numerous limitations. This single-case study lacks an adequate sample size to substantiate the therapeutic efficacy of MSC therapy. Secondly, although the paracrine mechanisms of MSCs were examined in light of existing research, this study did not explore the specific molecular pathways that underlie these mechanisms in the present context. Finally, a key

limitation of this study is its open-label design, in which both the evaluating veterinarian and the cat's owner were aware that stem cell therapy was being administered. This design carries an inherent risk of bias, particularly in the assessment of subjective outcomes such as appetite, mental status, and behavior. The observed improvements in these areas may have been influenced by the caregiver placebo effect. Therefore, future studies should incorporate blinded assessment to provide a more objective evaluation of these subjective parameters.

This case illustrates that AD-MSCs therapy was associated with improvement in essential renal function indicators (SCr, BUN, and P levels) in a feline CKD patient while also enhancing mental status and appetite, underscoring the therapeutic potential of stem cell-based interventions for end-stage CKD management. Future study should emphasize the following avenues: ① Increasing sample numbers with controlled trial designs that include long-term follow-up to clarify the enduring therapeutic efficacy and prolonged detrimental effects of AD-MSCs therapy in chronic renal disease. ② Recent studies demonstrate that preconditioning procedures augment the therapeutic efficacy of MSCs. Hypoxic preconditioning and melatonin pretreatment have demonstrated the ability to enhance the paracrine capacity of MSCs ([75](#), [76](#)). Future research should concentrate on examining innovative preconditioning techniques while thoroughly assessing the safety and efficacy characteristics of preconditioned MSC populations. ③ In comparison to MSCs, extracellular vesicles produced from MSCs (MSC-EVs) demonstrate reduced immunogenicity and tumorigenic potential, signifying a potentially cell-free treatment strategy. Nonetheless, their function in kidney transplantation is inadequately defined and necessitates additional research. ④ MSC therapy exhibits stage-dependent therapeutic effectiveness in CKD management, with maximum results achieved when intervention commences in the early stages of the disease. Nonetheless, their therapeutic efficacy may be limited in advanced CKD, highlighting the necessity to discover predictive biomarkers for monitoring disease progression to facilitate earlier CKD identification.

This case report illustrates that allogeneic AD-MSCs have innovative therapeutic potential for the treatment of severe CKD in feline patients. These findings warrant thorough assessment and methodical inquiry to clarify their applicability in CKD treatment strategies.

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Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author/s.

Ethics statement

The animal studies were approved by Yantai University Experimental Animal Ethics Committee. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent was obtained from the owners for the participation of their animals in this study. Written informed consent was obtained from the participants for the publication of this case report.

Author contributions

YS: Funding acquisition, Supervision, Writing – review & editing. YL: Data curation, Investigation, Writing – original draft. YY: Formal analysis, Methodology, Resources, Writing – original draft. YW: Data curation, Visualization, Writing – original draft. YM: Data curation, Visualization, Writing – original draft. SW: Data curation, Visualization, Writing – original draft. WH: Investigation, Formal analysis, Writing – original draft. HZ: Supervision, Writing – review & editing. WZ: Resources, Supervision, Validation, Writing – review & editing.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary material for this article can be found online at:

<https://www.frontiersin.org/articles/10.3389/fvets.2025.1632324/full#supplementary-material> [↗](#)

SUPPLEMENTARY VIDEO S1

A short video clip of a feline patient undergoing intravenous infusion of AD-MSCs.

[Download video file](#) (3.1MB, mp4)

SUPPLEMENTARY VIDEO S2

A short video clip documenting the resumption of voluntary feeding in a feline patient following AD-MSCs therapy.

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SUPPLEMENTARY VIDEO S3

A short video clip demonstrating locomotor performance recovery in a feline patient following AD-MSCs therapy.

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SUPPLEMENTARY VIDEO S4

A short video clip demonstrating play behavior recovery in a feline patient following AD-MSCs therapy.

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SUPPLEMENTARY TABLE S1

Physical examination.

[Data Sheet 1.pdf](#) (80KB, pdf)

SUPPLEMENTARY TABLE S2

Complete blood count (CBC).

SUPPLEMENTARY TABLE S3

Biochemistry tests.

SUPPLEMENTARY TABLE S4

Abbreviations.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

SUPPLEMENTARY VIDEO S1

A short video clip of a feline patient undergoing intravenous infusion of AD-MSCs.

[Download video file](#) (3.1MB, mp4)

SUPPLEMENTARY VIDEO S2

A short video clip documenting the resumption of voluntary feeding in a feline patient following AD-MSCs therapy.

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SUPPLEMENTARY VIDEO S3

A short video clip demonstrating locomotor performance recovery in a feline patient following AD-MSCs therapy.

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SUPPLEMENTARY VIDEO S4

A short video clip demonstrating play behavior recovery in a feline patient following AD-MSCs therapy.

[Download video file](#) (3.8MB, mp4)

SUPPLEMENTARY TABLE S1

Physical examination.

[Data Sheet 1.pdf](#) ^(80KB, pdf)

SUPPLEMENTARY TABLE S2

Complete blood count (CBC).

SUPPLEMENTARY TABLE S3

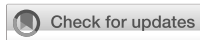
Biochemistry tests.

SUPPLEMENTARY TABLE S4

Abbreviations.

Data Availability Statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author/s.



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Advancing mesenchymal stem cell therapy for kidney diseases in companion animals: from mechanisms to clinical application

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Acute and chronic kidney diseases are major clinical challenges in companion animals, yet therapeutic options for reversing established injury remain limited. Mesenchymal stem cells (MSCs) have emerged as a promising therapy due to their immunomodulatory and tissue-repair properties. This review synthesizes current evidence on MSC therapy for kidney disease in cats and dogs, with a focus on mechanisms, preconditioning strategies, delivery routes, and clinical outcomes. Key findings indicate that MSCs exert renoprotective effects primarily through paracrine-mediated immunomodulation rather than direct differentiation. Intravenous administration, while simple, results in >80% pulmonary entrapment and renal homing below 5%; arterial or local injection increases homing to 15–20% but carries procedural risks and lacks standardized dosing. Preconditioning strategies (hypoxia, melatonin, ATRA) enhance MSC survival and homing in rodent models, but feline- and canine-specific validation remains limited. Clinical data from 19 studies in cats/dogs demonstrate that MSC therapy is generally safe, with most adverse events being infusion-related and transient. In feline CKD, consistent trends toward improved glomerular filtration rate and quality of life are reported, although statistical significance is rarely achieved. In canine AKI, MSC therapy improves survival and renal function, but results are heterogeneous and predominantly derived from experimentally induced models. Major evidence gaps include lack of dose standardization, variable MSC characterization, short follow-up durations, and limited long-term safety data. Moving forward, establishing consensus on weight-based dosing, MSC characterization standards, and multicenter randomized controlled trials in naturally occurring disease is essential to translate preclinical promise into clinical practice.

KEYWORDS

AKI, CKD, companion animals, delivery, homing, mesenchymal stem cells, preconditioning

1 Introduction

Kidney disease is a major global health burden affecting human populations and ranks among the top 10 causes of mortality worldwide, with its incidence continuously rising. Annually, approximately 13.3 million new cases of acute kidney injury (AKI) occur in humans, contributing to an estimated 1.7 million human deaths (1). According to the Kidney Disease: Improving Global Outcomes consensus, renal pathologies are classified as either acute kidney injury or chronic kidney disease (CKD) based on the severity, reversibility, and temporal onset of functional impairment (2, 3). Spontaneous kidney disease is highly prevalent in companion animals, particularly cats and dogs, and closely mimics the pathological features observed in human kidney disorders (4, 5). In recent years, the global companion animal market continues to expand, driving growing demand for effective treatments for kidney disease. Yet veterinarians still face limited therapeutic options and the challenge of reversing established renal impairment.

Currently, effective disease-modifying therapies for acute kidney injury (AKI) remain lacking. Current standard management adheres to a five key principle framework, encompassing early warning, identification, screening, intervention, and prognostic assessment (6–8). These conventional interventions primarily aim to prevent further renal deterioration and capitalize on the golden window for functional recovery; however, they fail to substantially reverse established renal parenchymal injury or adequately modulate the dysregulated immune responses involved in the pathogenesis of AKI (9, 10). Similarly, the management of chronic kidney disease continues to rely on blood pressure control, glycemic management, statin therapy, and dietary modification (11, 12). Although these strategies mitigate disease progression by reducing metabolic risks, they do not fundamentally repair renal parenchymal damage (3, 13, 14). With the advancement of regenerative medicine, the emergence of cell-based therapies offers new promise for renal repair.

Mesenchymal stem cells, characterized by their self-renewal capacity, multilineage differentiation potential, and potent regenerative and immunomodulatory properties, represent a promising therapeutic approach for kidney diseases (15, 16). Their therapeutic effects are primarily mediated by anti-inflammatory activity, immunoregulation, and tissue regeneration (17–19). MSCs exert anti-inflammatory effects primarily through paracrine signaling, secreting factors such as IL-10 that suppress pro-inflammatory mediators like TNF- α and IL-6 (20–22). Concurrently, they secrete growth factors (e.g., HGF, VEGF-C) and immunomodulatory cytokines (e.g., TGF- β), which mediate immune cell regulation—notably promoting macrophage polarization toward the M2 phenotype (17, 23, 24). While MSCs can engraft into injury sites and differentiate into tissue-specific cells, this direct regenerative mechanism contributes minimally to overall efficacy compared to paracrine actions (25–27). These mechanisms have shown promise in canine and feline kidney disease models, with MSCs demonstrating favorable outcomes across multiple therapeutic programs (28–30).

However, MSC-based therapies have shown inconsistent therapeutic outcomes. To enhance post-infusion efficacy, research has

converged on three key strategies: cellular preconditioning, delivery route optimization, and carrier-assisted delivery (31–33). Cellular preconditioning involves modifying MSCs during *in vitro* culture to augment their therapeutic potential. This includes pharmacological preconditioning with bioactive compounds and genetic engineering approaches (e.g., targeted gene knockdown or overexpression) to enhance specific functional properties (34, 35). Delivery route optimization aims to maximize renal homing while minimizing off-target sequestration. Beyond standard peripheral intravenous infusion—which subjects cells to significant pulmonary first-pass effects—investigated approaches include direct renal artery infusion, intraparenchymal renal injection, and subcapsular delivery (33, 36). Carrier-assisted delivery leverages advances in biomaterial science. Engineered carriers (e.g., hydrogels, microspheres) protect MSCs during transit, enhance site-specific retention, and improve homing efficiency without compromising renal safety (31, 32, 37).

This review synthesizes research progress on mesenchymal stem cell therapy for kidney diseases, emphasizing how delivery routes and pre-treatment strategies influence cell homing and therapeutic efficacy. It examines existing challenges based on current clinical applications in cats and dogs, and outlines future research directions. By examining the literature through an integrated framework of pre-treatment enhancement, targeted delivery, and vehicle-assisted strategies (Figure 1), we assess the current evidence base. This framework aims to support clinical practice in companion animal nephrology and lays the groundwork for optimizing future veterinary trial designs.

Although this is not a formal systematic review, we conducted a structured literature search for the section on clinical applications in companion animals (Chapter 6) to ensure comprehensive coverage. We searched PubMed, Web of Science, and the Cochrane Library from January 1, 2015, to February 1, 2026, using the following keyword combinations: (“mesenchymal stem cells” OR “MSC”) AND (“acute kidney injury” OR “chronic kidney disease”) AND (“dog” OR “canine” OR “cat” OR “feline” OR “companion animal”). The search included studies examining MSC transplantation from any source in dogs and cats with experimental or naturally occurring AKI or CKD, using intravenous, intra-arterial, or parenchymal administration. Exclusion criteria were narrative reviews, observational studies, editorials, commentaries, and conference abstracts; only English-language articles were included. Retrieved records were compiled in EndNote, and duplicates were removed. Articles were then screened by title, abstract, and keywords; studies not meeting the inclusion criteria were excluded.

2 Core mechanisms of mesenchymal stem cell therapy for kidney disease

The pathogenesis and progression of kidney disease involve three interconnected pathological processes: glomerular filtration impairment, tubulointerstitial damage, and progressive renal fibrosis (13, 38–43).

MSCs have emerged as a promising therapeutic strategy for kidney disease, offering distinctive therapeutic advantages (44–46). Their multilineage differentiation potential enables them to differentiate into renal parenchymal cells—including tubular epithelial cells and podocytes—under specific microenvironmental conditions, thereby facilitating the direct structural repair of damaged renal tissues

Abbreviations: MSCs, Mesenchymal stem cells; AKI, Acute kidney injury; CKD, Chronic kidney disease; ECM, Extracellular matrix; ADMSCs, Adipose-derived Mesenchymal Stem Cells; BMSCs, Bone marrow-derived mesenchymal stem cells; ATRA, All-trans retinoic acid; IV, Intravenous; SCr, Serum creatinine; BUN, Blood urea nitrogen; EVs, Extracellular vesicles.

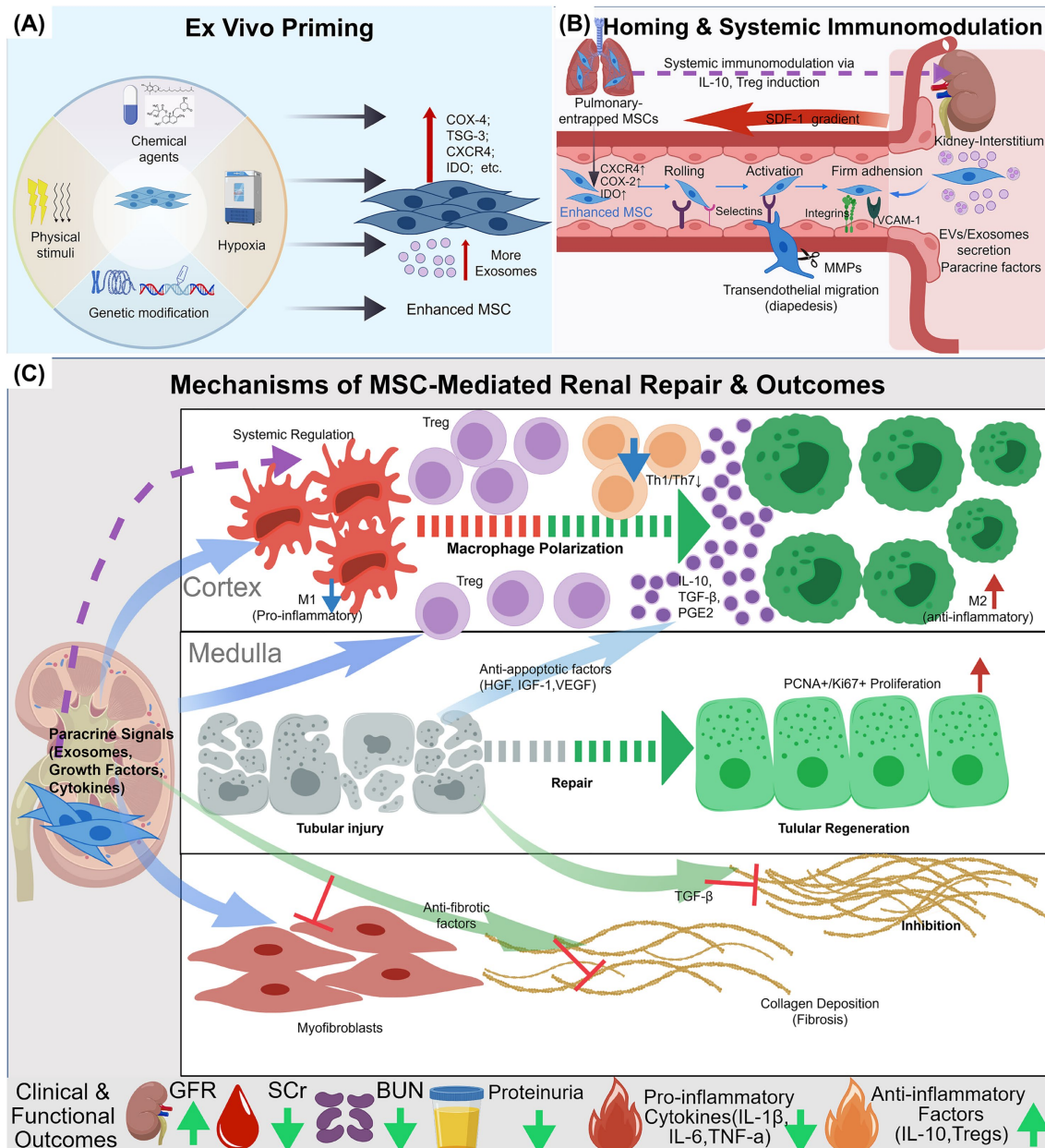


FIGURE 1

The “Prime–Target–Carrier” framework for MSC therapy in kidney disease. This figure illustrates three core strategies for optimizing MSC-based renal therapy. **(A) Ex Vivo Priming.** MSC are enhanced via four priming strategies: (1) chemical agents (ATRA, melatonin, atorvastatin); (2) physical stimuli (hypoxia, laser, electromagnetic fields); (3) genetic modification (Klotho, HGF, ISL1, CXCR4 overexpression); and (4) hypoxic preconditioning (1% O₂). Primed MSCs upregulate functional molecules (COX-4, TSG-3, CXCR4, IDO), enhance paracrine activity, increase EV/exosome production, and become “enhanced MSCs.” **(B) Homing and Systemic Immunomodulation.** Following IV infusion, >80% of MSCs are lung-entrapped, while <5% home to the kidney via SDF-1/CXCL12 gradient. Homing involves: (1) CXCR4/COX-2/IDO-mediated enhancement; (2) selectin-mediated rolling; (3) integrin (α4β1, α5β1)-VCAM-1 adhesion; and (4) MMP-mediated transendothelial migration. Lung-entrapped MSCs exert systemic effects via IL-10 and Treg induction. Renal interstitial MSCs secrete EVs/exosomes and paracrine factors (HGF, VEGF, IL-10, PGE₂) to promote repair. **(C) Renal Repair and Outcomes.** MSCs promote kidney repair via three pathways: (1) Immunomodulation: M1 → M2 macrophage polarization, Treg expansion, Th1/Th17 suppression; (2) Tubular regeneration: PCNA⁺/Ki67⁺ epithelial cell proliferation via paracrine/anti-apoptotic factors (HGF, IGF-1, VEGF); (3) Anti-fibrosis: Inhibition of α-SMA⁺ myfibroblasts and TGF-β-mediated collagen deposition. Outcomes include improved GFR, reduced SCr/BUN/proteinuria, decreased pro-inflammatory cytokines (IL-1β, IL-6, TNF-α), and increased anti-inflammatory mediators (IL-10, Tregs). Created with BioGDP.com.

(25–27). Beyond direct tissue repair, MSCs exhibit potent immunoregulatory properties by modulating of the local immune microenvironment. This suppresses excessive immune responses and attenuates renal inflammation (20, 21, 47, 48). Key mechanisms include inhibiting T-lymphocyte proliferation, downregulating pro-inflammatory cytokines (e.g., TNF-α, IL-1β), and upregulating anti-inflammatory cytokines (e.g., IL-10). These immunomodulatory effects have been

comprehensively reviewed by Han et al. (49). Furthermore, MSCs induce macrophage polarization toward the M2 phenotype, which is characterized by anti-inflammatory activity and tissue-repair capabilities. As demonstrated by Geng et al., MSC-mediated activation of trophic M2 macrophages promotes functional recovery in rhabdomyolysis-induced AKI by facilitating the transition from tubular injury to regeneration (50).

In a seminal study, Wang et al. demonstrated that MSCs enhance acute kidney injury repair through the dual mechanisms of mitochondrial regulation and ferroptosis suppression (51). Specifically, MSCs upregulated PGC-1 α expression in renal tubular epithelial cells, thereby restoring mitochondrial dynamics through balanced modulation of fusion and fission proteins, reducing reactive oxygen species production, and inhibiting NLRP3 inflammasome activation. Concurrently, MSCs administration significantly reduced pyroptosis biomarkers including IL-18 while exerting potent antifibrotic effects. These findings collectively indicate that MSCs not only facilitate acute recovery but also provide long-term protection against fibrotic progression (52).

Although most mechanistic insights are derived from rodent studies, emerging evidence in dogs and cats now supports these pathways. For example, in canine renal ischemia–reperfusion models, ADMSC-derived exosomes have been shown to reduce renal cell apoptosis, mitochondrial damage, and hydrogen peroxide production, thereby attenuating renal injury (46, 53). However, certain mechanisms warrant further investigation specifically in companion animal models.

3 Preconditioning strategies and mechanisms of mesenchymal stem cells

To augment the therapeutic efficacy of mesenchymal stem cells, contemporary research employs strategic preconditioning through optimized *in vitro* culture parameters, physical stimulation, genetic engineering, and biochemical modulation via growth factors or pharmaceutical agents (Figure 1).

3.1 Physical preconditioning

Optimizing MSC culture conditions has been shown to significantly enhance their proliferation and differentiation capacity. When cultured under hypoxia conditions, MSCs exhibit increased secretion of beneficial factors, attenuated replicative senescence, elevated proliferation rates, enhanced differentiation potential, and greater release of anti-inflammatory cytokine. These changes collectively strengthen their antioxidant and anti-inflammatory capabilities (54–57). Ishiuchi et al. demonstrated that hypoxic preconditioning of MSCs augments their paracrine activity, thereby enhancing renoprotective effects and attenuating renal fibrosis and inflammation in rats with I/R injury (58). Similarly, Zeng et al. reported that hypoxia-treated MSCs mitigate renal tubular cell apoptosis and preserve renal function during rat I/R injury by promoting tissue autophagy. Other physical preconditioning strategies have also been shown to enhance MSCs functionality. Low-intensity laser irradiation elevates proliferative activity (59), while extremely low-frequency pulsed electromagnetic fields accelerate differentiation, stimulate collagen deposition, and amplify anti-inflammatory responses in MSCs (60).

3.2 Gene modification of MSCs

Gene modification has emerged as a potent preconditioning strategy to enhance MSC function through the introduction of specific therapeutic genes such as those encoding anti-apoptotic factors or

growth factors (61). Zhang et al. transduced mouse bone marrow-derived mesenchymal stem cells (BMSCs) with Klotho-GFP adenovirus and transplanted them into mice with AKI. They demonstrated that Klotho overexpression inhibited the Wnt/ β -catenin pathway in renal tubular epithelial cells, significantly enhancing the antifibrotic effects of BMSCs while concurrently improving their proliferative capacity and immunomodulatory function (62). Similarly, Shao et al. reported that hepatocyte growth factor (HGF) modification of dental pulp stem cells markedly attenuated renal fibrosis, potentially via PI3K/AKT/GSK3 β pathway-mediated regulation of fibrosis-related genes (63). Wang et al. found that BMSCs overexpressing ISL1 attenuated renal ischemia–reperfusion injury through the paracrine secretion of haptoglobin, thereby exerting anti-apoptotic and antioxidant effects (64). Despite notable progress, the long-term safety and stability of genetically modified MSCs, particularly regarding their potential risks in clinical applications, require further rigorous investigation.

3.3 Drug preconditioning

Drug preconditioning has emerged as a promising strategy to enhance MSC therapeutic efficacy. By modulating key cellular pathways, pharmacologically preconditioned MSCs amplify beneficial processes—such as growth factor secretion, anti-inflammatory responses, macrophage polarization, and regulatory T-cell induction, while suppressing detrimental pathways, ultimately improving migration, proliferation, and renoprotection (Table 1) (65–76). Notable examples include the following:

Astilbin, a multifunctional flavonoid, protects MSCs from oxidative stress and enhances their antioxidant capacity. In models of AKI-to-CKD transition, astilbin-preconditioned MSCs attenuated AKI/CKD progression through amplified immunomodulation, mediated primarily via PTGS2-dependent macrophage polarization (70).

All-trans retinoic acid (ATRA) boosts MSC efficacy through distinct mechanisms. Zhang et al. reported that ATRA-preconditioned MSCs enhanced hyaluronic acid production and activated renal tubular PI3K/AKT signaling, thereby potentiating AKI recovery (67). Notably, Barakat et al. demonstrated that ATRA-treated Wharton's jelly MSCs upregulated Wnt/ β -catenin signaling while attenuating cell death, inflammation, and oxidative stress (66).

Erythropoietin preconditioning of bone marrow-derived MSCs accelerated AKI repair in rats, significantly reducing serum creatinine, blood urea nitrogen, and histopathological scores compared to untreated MSCs (73). Collectively, these findings confirm drug preconditioning substantially enhances MSC therapeutic outcomes. Future studies should prioritize identifying novel preconditioning agents based on mechanistic synergy with MSC biology and disease pathways.

Regarding pre-treatment-related trade-offs, Boyette et al. reported that hypoxic conditioning of human bone marrow-derived MSCs enhanced their ability to generate highly clonogenic progenitor cells but compromised their differentiation potential (77). In a related review, Hu and Li concluded that pre-treatment often enhances one functional property at the expense of others (78). To date, research on pretreated MSCs has largely focused on *in vitro* phenotypes, secreted factors, and short-term outcomes in animal models. This raises a critical question: Do pre-treatment-modified MSCs differ significantly from unmodified MSCs in terms of long-term efficacy and safety?

In feline and canine MSCs, an *in vitro* study by Park et al. demonstrated that IFN- γ pre-treatment of feline adipose-derived MSCs

TABLE 1 Effects and principles of pretreating stem cells with drugs.

Name	Principles	Efficacy	References
Iron-quercetin complex	By promoting the expression and secretion of Hepatocyte Growth Factor (HGF), activating the HGF/c-Met signaling pathway, and inhibiting apoptosis of tubular cells	Serum creatinine and blood urea nitrogen levels were significantly reduced, renal tissue structure was improved, and the scores for tubular necrosis and injury were decreased.	(65)
All-trans retinoic acid	Product hyaluronic acid, activate the PI3K/AKT pathway Potentiates the antioxidant effect, enhance its effect on the activation of the Wnt/beta catenin pathway	Enhance the anti-inflammatory, anti-apoptotic, and proliferative repair effects of MSCs	(66, 67)
Valproic acid	Activating the AKT/PI3K and SDF1/CXCR4 pathways	Significantly improved renal function and increased antioxidants	(68)
Interferon gamma	Enhance regulatory T cell induction	Suppresses inflammation, renal tubular interstitial fibrosis, and accumulation of extracellular matrix, and increases VEGF	(69)
Astilbin	Inducing of M2-like phenotype polarization in macrophages through the PTGS2-mediated pathway	Decreased the IL-6 and TNF- α levels and increased the IL-10 levels	(70)
Hyaluronic acid	Inhibit the Wnt/ β -catenin signaling pathway	Reducing inflammatory markers, apoptosis markers, and fibrosis markers	(71)
Rapamycin	Pre-activation of autophagy inhibits the mTOR/AKT signaling pathway	Enhance the survival and differentiation rates of transplanted MSCs, and improving renal tissue.	(72)
Erythropoietin	Up-regulated SIRT1 and Bcl-2 expression and down-regulated p53 expression, inhibit apoptosis of MSCs and enhance their homing and anti-inflammatory capabilities.	Lower serum IL-1 β and TNF- α levels and higher IL-10 serum level, improved tubular injury significantly	(73)
Melatonin	Reduces the expression of TGF- β 1, α -SMA, and TNF- α , increases the expression of E-cadherin. Activate the P-Erk1/2 and P-Akt signaling pathways, and increase the expression of antioxidant enzymes.	Enhance homing ability and survival rate, and improve renal regeneration outcomes	(74, 75)
Atorvastatin	Suppression of TLR4 signaling	Ameliorated oxidative stress, inhibited inflammation response, and increased the viability of implanted MSCs	(76)

enhances immunomodulatory function through the PGE2 pathway, inducing macrophage polarization and increasing regulatory T cell numbers (79). Separately, Yun, Park, and colleagues used mRNA engineering to overexpress TSG-6 in canine adipose-derived MSCs; co-culture with DH82 macrophages confirmed that TSG-6 overexpression significantly enhanced anti-inflammatory effects (80). These findings align with rodent data; however, *in vivo* studies in cats and dogs with kidney disease remain lacking (Table 2). Future research should prioritize translating pre-treatment strategies validated in rodent models to address this gap.

4 Targeting the kidney: from homing mechanisms to delivery outcomes

Cell homing refers to the process by which MSCs migrate from the site of administration to damaged tissues and subsequently reside within them, representing a critical determinant of therapeutic success (81). This process involves not only interactions between cell surface markers and target the tissue, but also regulation by various physico-chemical factors (82).

4.1 Molecular basis of renal homing: chemokine-mediated cell migration

Cell homing involves multiple molecular mechanisms, particularly chemokine signaling and receptor interactions. The SDF-1/

CXCR4 axis is central to this process, directing CXCR4-positive MSCs toward damaged renal tissue (83–85). Integrin family members are also critical for MSCs homing. By recognizing specific extracellular matrix sequences such as the RGD motif, integrins mediate both the physical coupling of MSCs to their microenvironment and the transduction of biochemical signals. Through α 4 β 1 and α 5 β 1 integrins, MSCs form cross-receptor complexes that activate the FAK–PI3K/Akt pathway and induce PDGFR- β phosphorylation, thereby enhancing their migratory capacity. Concurrently, CXCL12/CXCR4 signaling upregulates integrin α 5 β 3 and connexin expression, driving transendothelial migration and tissue colonization to complete therapeutic homing (86–89).

4.2 Regulation of homing efficiency by preconditioning

Pre-treatment strategies also play a key role in modulating homing capacity. Hypoxic preconditioning enhances MSC migratory ability by promoting chemokine receptor expression on the cell surface (90, 91). In addition to hypoxic culture, genetic modification and drug pre-treatment further enhance homing by altering the biological behavior of MSCs. For example, MSCs overexpressing CXCR4 exhibit stronger chemotaxis and migrate more effectively to damaged tissue *in vivo* (92, 93). Liu et al. demonstrated that CXCR4-overexpressing BMSCs exhibited markedly increased renal accumulation (94). In a study by Zhou et al., erythropoietin pre-treatment significantly upregulated CXCR4 expression, altered surface molecule profiles, reduced

TABLE 2 Summary of MSC preconditioning strategies tested in renal disease models and their reported effects.

Preconditioning type	Specific method	Animal model	Homing efficiency	Cell survival	Anti-fibrosis	Renal function improvement	References
Physical	Hypoxia (1% O ₂)	Rat IRI	↑ (CXCR4)	↑	↑	↑ (SCr/BUN)	(58)
Genetic modification	Klotho overexpression	Mouse AKI	–	↑	↑	↑	(62)
	HGF modification	Mouse fibrosis	–	–	↑	–	(63)
	ISL1 overexpression	Rat IRI	–	↑ anti-apoptosis	–	↑	(64)
	CXCR4 overexpression	Mouse AKI	↑↑	–	–	↑	(94)
Pharmacological	Iron-Quercetin complex	Rat AKI	–	↓ apoptosis	–	↑	(65)
	All-trans retinoic acid	Rat IRI	–	↑	–	↑	(67)
	All-trans retinoic acid	Rat AKI	–	–	–	↑	(66)
	Valproic acid	Rat IRI	↑ (CXCR4)	↑	↓	↑	(68)
	Interferon-γ	Rat IRI	–	–	↑	↑	(69)
	Astilbin	Mouse AKI -CKD	–	–	↑	↑	(70)
	Hyaluronic Acid	Rat fibrosis	–	–	↑	–	(71)
	Rapamycin	Rat AKI	–	↑	–	↑	(72)
	Erythropoietin	Rat AKI	↑ (CXCR4)	↑	–	↑	(73)
	Melatonin	Rat CKD	↑	↑	↑	↑	(74)
	Melatonin	HK-2	–	↑	–	–	(75)
	Atorvastatin	Rat IRI	–	↑	–	↑	(76)
Feline/Canine specific	IFN-γ preconditioning (feline)	<i>In vitro</i> (macrophage co-culture)	–	–	–	–	(79)
	TSG-6 overexpression (canine)	<i>In vitro</i> (macrophage co-culture)	–	–	–	–	(80)

↑, Improvement or enhancement; ↑↑, Significant enhancement; ↓, Reduction or inhibition; –, Not assessed or not reported; AKI, Acute kidney injury; CKD, Chronic kidney disease; IRI, Ischemia-reperfusion injury; CXCR4, C-X-C chemokine receptor type 4; HGF, Hepatocyte growth factor; IFN-γ, Interferon-γ; TSG-6, Tumor necrosis factor-inducible gene 6 protein.

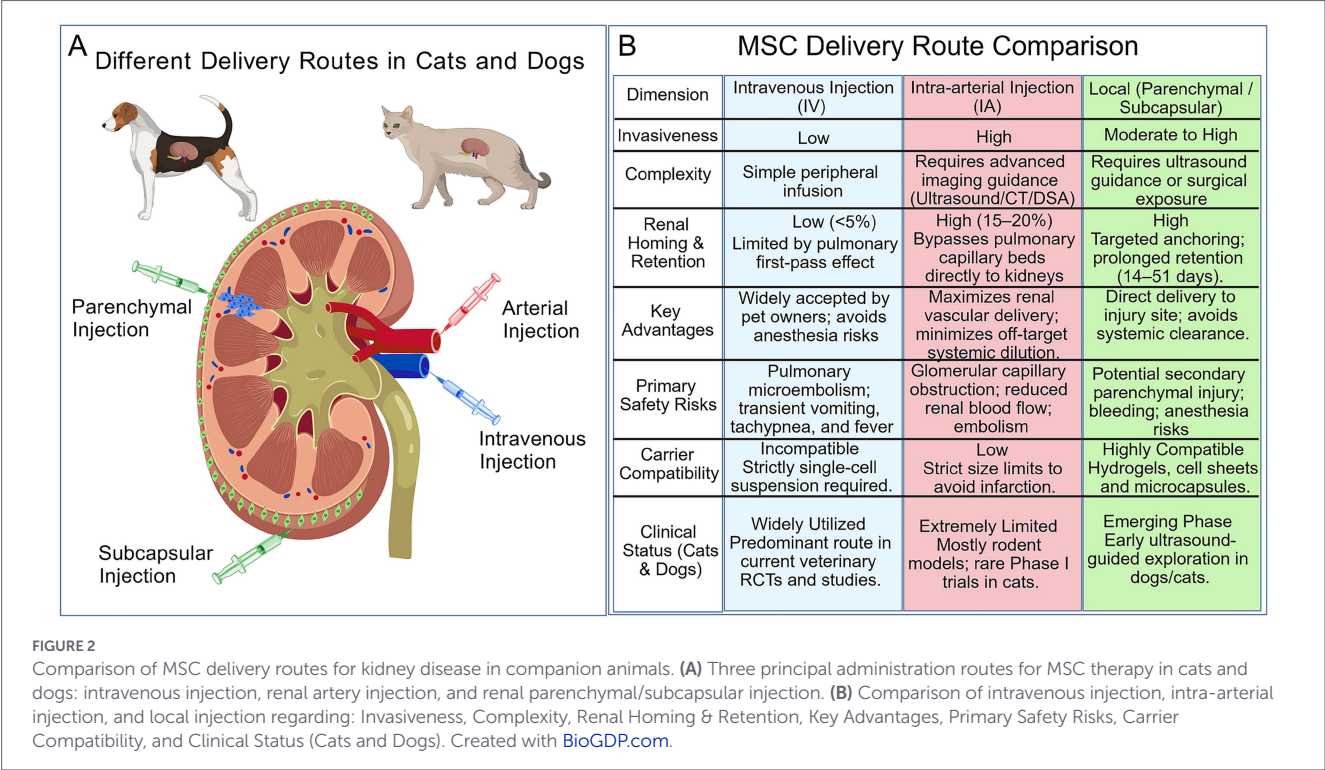
adhesion to pulmonary capillaries, and decreased pulmonary retention. As a result, renal homing of BMSCs was markedly enhanced (73).

4.3 Effect of administration route on renal homing efficiency

The route of administration plays a critical role in determining MSC homing efficiency and therapeutic outcomes (Figure 2). Common administration routes for the treatment of nephropathy include intravenous, intra-arterial, and local injection, each offering distinct characteristics and clinical applications (95, 96). Perirenal approaches, including renal artery, intrarenal, and subcapsular delivery, may offer advantages over peripheral venous infusion as techniques continue to advance (97).

Intravenous administration leads to a significant pulmonary first-pass effect, with the majority of MSCs becoming entrapped in the

lungs shortly after injection. While a small fraction distributes to the liver and spleen, only a limited number reach the kidneys or other target organs (36). This pulmonary retention is primarily attributed to several factors: the relatively large size of MSCs compared to the pulmonary capillary diameter, the extensive capillary network in the lungs, and the strong adhesive properties of culture-expanded MSCs (98, 99). In comparative kidney disease studies, untreated MSCs showed renal homing rates below 5% at 24 h post-injection, with 60–80% retained in the lungs and renal persistence lasting approximately 3–5 days. In contrast, pretreated or genetically modified MSCs achieved homing rates of 15–20% (66, 100, 101). Nevertheless, most studies using intravenous administration report that even unmodified MSCs confer measurable renal improvement, with higher homing rates correlating with better outcomes. Functional parameters including serum creatinine and blood urea nitrogen levels, decrease following treatment, and histological examination confirms renal recovery



(66, 67, 95). Cell dose also influences pulmonary retention and systemic distribution: high-dose intravenous administration ($>1 \times 10^6$ cells) has been associated with superior efficacy compared to lower doses (102).

Renal artery injection represents a more direct approach that offers high targeting specificity. By bypassing pulmonary capture, it delivers MSCs directly into the renal vasculature, theoretically enhancing homing efficiency and therapeutic outcomes. However, technical complexity and the risk of embolism limit its widespread application (33, 103–107). Yamada et al. demonstrated that administration of 5×10^5 MSCs via renal artery injection led to stable colonization of the kidneys within 1–3 days, with significantly fewer cells retained in the lungs. In a mouse model of ischemia–reperfusion injury (IRI), tracked MSCs remained detectable in the kidneys up to 42 days post-injection; in contrast, even five times the dose delivered intravenously was undetectable in the kidneys by day 7 (103). In a rat model, Cai et al. showed that renal artery injection of a low dose (1×10^5 cells) yielded superior efficacy compared to tail vein or carotid artery administration of 1×10^6 cells. The same study also revealed that high-dose arterial injection may cause glomerular capillary loop obstruction and reduce renal blood flow (105). Although renal artery injection consistently achieves high homing efficiency and shows therapeutic promise in preclinical models, concerns regarding embolization, procedural complexity, and potential secondary injury during kidney exposure have confined most studies to rodents, with limited evaluation in large animal or clinical settings. Given that the associated risks currently outweigh the potential benefits in veterinary practice, refinement of surgical techniques remains the central challenge for advancing this approach.

Local injection, primarily via the renal capsule or renal parenchyma, enables direct MSC delivery into the kidney. This approach markedly increases MSC retention within renal tissue, thereby improves therapeutic efficacy (108–111). Huang et al. compared these

two methods by transplanting umbilical cord-derived MSCs encapsulated in a collagen matrix into AKI-CKD mice via subcapsular and parenchymal routes. Both local MSC therapies reduced tubular injury, promoted tubular epithelial cell proliferation by day 3, and alleviated renal fibrosis by days 14 and 28. Using live imaging to track labeled cells, strong signals were detected in both groups on day 1, increased by day 3, and gradually diminished thereafter; however, signals remained detectable at 14 days post-transplantation (108). In a separate study, Lee et al. used ultrasound-guided renal parenchymal injection of MSCs to treat canine AKI, administering cells on days 6 and 23 after onset. Green fluorescent protein-labeled MSCs remained detectable in renal sections at day 51 (28). Xiao et al. demonstrated that minimally invasive transplantation of MSCs under the renal capsule using ultrasound guidance was safe and effective in miniature pigs; moreover, multiple transplants promoted renal functional recovery more effectively than a single transplant (111).

4.4 Correlation between homing efficiency and treatment outcomes

Current evidence indicates a clear correlation between homing efficiency and therapeutic outcomes: higher renal homing rates are associated with more pronounced reductions in biochemical markers (e.g., serum creatinine, blood urea nitrogen) and greater mitigation of renal tissue damage with accelerated repair. However, therapeutic efficacy is not solely contingent on MSC homing. In studies by Barakat et al. and Wang et al., MSCs improved renal function and histology despite limited homing to the kidney (66, 100). This suggests that MSC effects arise from combined local and systemic immunomodulation rather than homing-dependent mechanisms alone. Supporting this, Li et al. recently demonstrated that allogeneic MHC-expressing MSCs enhance immunomodulatory and anti-inflammatory effects by amplifying Treg responses and promoting IL-10

production, thereby conferring renal protection (112). The growing body of work on MSC-derived extracellular vesicles further underscores this point: although efficacy varies, acellular approaches yield therapeutic outcomes comparable to direct MSC treatment, indicating that homing represents only one component of the broader mechanism (113–121).

Several key questions remain to be addressed: (I) Is there a threshold effect—how many cells must persist within the kidney and continuously secrete paracrine factors to trigger effective treatment? (II) What is the relationship between local retention duration and therapeutic efficacy? (III) For animals with CKD, local injection may represent an optimal strategy for maximizing treatment outcomes. However, these patients frequently present with comorbidities that may elevate the risks associated with anesthesia and invasive procedures (122). Striking a balance between maximizing targeted delivery and minimizing procedural risk thus remains a critical translational challenge in veterinary medicine.

In summary, homing efficiency substantially influences therapeutic efficacy. Nevertheless, clinical treatment plans should be individualized based on disease stage, comorbidities, and procedural risk.

4.5 Veterinary clinical perspective: bridging the gap between targeted delivery and clinical application in dogs and cats

Studies in rodent and large animal models have demonstrated that renal artery and local injection can increase MSC homing rates below 5% with IV administration to 15–20% or higher. However, clinical research in companion animals has yet to translate these theoretical advantages into practice. Analysis of existing feline and canine clinical studies reveals that IV injection remains overwhelmingly dominant—whether in early exploratory studies or recent randomized controlled trials, the vast majority have employed the IV route (28–30, 53, 123–131). Only a handful of studies have attempted intrarenal injection (132, 133), while IA and subcapsular routes remain unreported in cats and dogs to date.

This disconnect between theory and practice stems from several implementation barriers: (I) Technical barriers: IA and subcapsular injections require imaging guidance (ultrasound, CT, DSA) that remains limited in veterinary clinical settings. (II) Anesthesia risks: Cats and dogs with CKD often present with comorbidities such as cardiac or endocrine disease, yet insufficient data exist to support risk assessment for interventional procedures under general anesthesia. (III) Owner acceptance: Compared to simple IV infusion, invasive procedures carry higher costs and perceived risks, which may influence treatment decisions. (IV) Dosage experience gap: Rodent studies indicate an optimal dosage range for IA administration—too low is ineffective, too high carries embolism risk. However, optimal IA doses for cats and dogs remain unknown, and determining them would require extensive, resource-intensive experimentation.

Nevertheless, rodent data offer important insights for designing feline and canine trials. If a threshold effect exists, that is, a minimum number of homing MSCs required to initiate effective repair, then exploring IA or subcapsular routes provides a strong rationale for treating refractory CKD patients with suboptimal responses to IV therapy. Future research should prioritize establishing safe dosage ranges and standardized protocols for IA and subcapsular administration in large animal models before moving to clinical trials in cats and dogs with naturally occurring disease.

5 Carrier-assisted strategy

Even when MSCs reach the kidneys via optimized administration routes, their short survival and limited retention within the injured microenvironment continue to constrain therapeutic efficacy. Carrier-assisted strategies seek to overcome this final barrier by prolonging MSC retention and sustaining functional activity through physical encapsulation or local anchoring within the kidney.

Hydrogels are three-dimensional networks formed from natural or synthetic hydrophilic polymers, offering a range of physicochemical properties (134). As common carriers for MSC delivery, hydrogels fall into two main categories: natural (collagen, gelatin, alginate, hyaluronic acid) and synthetic (PEG, GelMA) (135, 136). Natural hydrogels provide excellent biocompatibility, degradability, and extracellular matrix mimicry, whereas synthetic variants offer tunable mechanical properties, extended retention, and controlled degradation. Functionally, the three-dimensional scaffold protects encapsulated cells, enables sustained release to prolong the therapeutic window, and supports minimally invasive precision delivery via injection (137, 138). In renal applications, hydrogels can fill the subcapsular space to maximize contact area. Local enrichment is achieved through subcapsular or intrarenal injection, bypassing off-target retention in the lungs, liver, or spleen and thereby enhancing renal retention and therapeutic efficacy (139, 140). Najafi et al. used hydrogels to deliver WJ-MSCs and included control groups receiving either no carrier or nitric oxide donors combined with WJ-MSCs. The triple combination of hydrogel, nitric oxide donors, and WJ-MSCs yielded the most favorable biochemical and histological outcomes, while each monotherapy, though effective, was inferior to the combined approach (141).

Cell microcapsules offer a widely used platform for delivering MSCs. Alginate (ionically crosslinked) and chitosan (thermosensitive phase-change) are commonly selected for their biocompatibility, injectability, and controlled degradation—safety advantages over PLGA, whose acidic degradation products raise concerns. Chitosan can also be optimized by combining it with natural polysaccharides (142, 143). Functionally, microcapsules can be engineered to avoid embolization or traverse filtration barriers through size tuning. Combined with surface modifications (e.g., RGD for adhesion, PEG for immune evasion), core-shell structures, or imaging tracers, they enable single-cell encapsulation, controlled factor release, cell protection, and real-time tracking (144–147). In renal therapy, microencapsulation is frequently paired with local injection. Huang et al. delivered collagen matrix-encapsulated MSCs via local injection in an IRI model and observed tubular repair, with MSCs still detectable in renal tissue at 28 days (108). However, the study lacked an unencapsulated MSCs control group, making it difficult to determine whether therapeutic effects derived from the injection method itself or encapsulation. Future studies should include appropriate controls to clarify this.

MSCs cell sheet technology is a scaffold-free tissue engineering approach that uses thermosensitive polymers such as poly (N-isopropylacrylamide). Compared to conventional methods, this technique better preserves cell surface proteins and maintains functional potency (148–150). MSCs from various sources can be used; however, they exhibit source-dependent differences in adhesion, proliferation, and cytokine secretion. Nakao et al. reported that AD-MSCs secreted higher levels of HGF, whereas BM-MSCs and UC-MSCs showed greater PGE₂ and IL-6 expression (151). Functionally, the intact ECM network within cell sheets enables seamless integration with host tissue. In an IRI rat model, Ju et al. demonstrated that MSC

sheet transplantation significantly reduced BUN and serum creatinine levels, improved tubular injury scores, and inhibited fibrosis progression. *In vivo* imaging confirmed that, compared to single-cell suspension injection, MSC sheets exhibited markedly prolonged renal retention (>14 days) while avoiding systemic complications such as pulmonary embolism (152).

Despite the potential of carrier techniques to improve renal retention in experimental models, their application in clinical practice for companion animals encounters numerous practical challenges. First, cost and complexity: the preparation of synthetic hydrogels and associated surgical procedures raise treatment costs, potentially exceeding what some owners can afford. Second, regulatory pathways remain unclear, with no established approval standards for cell–biomaterial combination products in veterinary medicine. Third, owner acceptance remains a critical variable—for pets with advanced CKD, owners may prefer simpler, low-risk intravenous infusions over invasive procedures requiring anesthesia. Future research should therefore focus on developing injectable, minimally invasive delivery systems, while incorporating owner preference analysis and cost–benefit assessments into clinical trial study designs.

6 Clinical applications in companion animals

The MSC treatment strategy described earlier has been shown to enhance homing and amplify therapeutic effects in animal models. However, its clinical evidence in naturally occurring companion animals has yet to be systematically reviewed. Following the literature search and organization described in the introduction, this chapter presents a comprehensive review of existing clinical studies on MSC therapy for kidney diseases in cats and dogs, thereby establishing an evidence base for subsequent translational research (Table 3). A total of 208 articles were initially identified. After removing duplicates and studies that did not meet the inclusion criteria, 19 articles were retained: 11 involving cats and 8 involving dogs.

6.1 Clinical research Progress on feline MSC therapy for kidney disease

In feline kidney disease research, clinical studies have predominantly focused on chronic kidney disease (CKD), whereas investigations into acute kidney injury (AKI) remain relatively limited. In a study by Rosselli et al., 15 cats were divided into three groups ($n = 5$ per group) and received intravenous injections of 4×10^6 ADMSCs, BMSCs, or fibroblasts 1 h after renal reperfusion. Analysis 6 days post-injection revealed that none of the three treatment improved renal function parameters or histopathological outcomes in cats with acute ischemic kidney injury (130). In a study by Li et al., three cats with post-renal acute kidney injury (PR-AKI) were treated with extracellular vesicles derived from feline adipose-derived mesenchymal stem cells (ADMSC-EVs) at a dose equivalent to the amount secreted by 10 million feline ADMSCs over 24 h. After 3 days of treatment, ADMSC-EVs were found to promote rapid recovery in cats with PR-AKI, significantly accelerating the restoration of creatinine and blood urea nitrogen levels. In addition, the study identified six novel PR-AKI biomarkers (carnitine, maltose, D-glucosamine, cytidine, dihydro-lactobionic acid, and stachyose), which could serve as auxiliary

indicators for assessing disease progression and therapeutic response to ADMSC-EVs (125).

A total of nine studies have investigated MSC therapy for CKD in cats (29, 30, 104, 123, 128, 129, 131, 132, 153). In an early study, Quimby et al. performed ultrasound-guided unilateral renal injections of MSCs in six cats. Ultrasound-guided intra-renal injection of autologous adipose-derived MSCs was found to be safe and feasible in cats with CKD. Two cats with early-to-mid-stage CKD exhibited moderate improvements in glomerular filtration rate and serum creatinine levels, although these changes were not statistically significant (132). In three consecutive pilot studies, the same group administered intravenous allogeneic adipose-derived MSCs to 21 cats with naturally occurring CKD. Low-dose cryopreserved ADMSCs (2×10^6) demonstrated a favorable safety profile. In contrast, high-dose cryopreserved ADMSCs (4×10^6) were associated with a high incidence of adverse reactions: vomiting occurred in 2 of 5 cats, increased respiratory rate in 4 cats, and severe respiratory distress requiring emergency intervention in 1 cat. Cultured and expanded ADMSCs (4×10^6) also showed good safety. However, none of the treatment outcomes reached statistical significance (128). A randomized placebo-controlled trial involving intravenous infusion of allogeneic adipose-derived MSCs reported no significant improvement in serum creatinine levels, although an upward trend in quality-of-life indicators was observed (129). In a case report, intravenous administration of allogeneic adipose-derived MSCs for CKD led to notable improvements in the affected cat's body condition score and physiological parameters, with the cat's IRIS stage reduced from stage IV to stage II (29).

Recently, research has expanded to include other cell sources. One study employed amniotic-derived pluripotent cells for intravenous therapy in cats with chronic kidney disease (CKD), resulting in a significant decrease in serum creatinine ($p = 0.028$), a reduction in the urine protein-to-creatinine ratio ($p = 0.131$), and clinically relevant improvement in urine specific gravity, although the latter did not reach statistical significance (30). Another study used intravenous infusion of uterine-derived allogeneic MSCs to treat cats with nephrectomy-induced CKD, demonstrating that the rate of renal function decline was slower in the treatment group compared to controls (131).

6.2 Clinical research progress on canine MSC therapy for kidney disease

In canine acute kidney injury (AKI) research, Lim et al. administered cisplatin to induce AKI in six Beagle dogs. Three dogs received intravenous autologous BMSCs (1×10^6 cells/kg), while the control group received saline injection. After 4 days, biochemical analysis revealed that the BMSC-treated group showed no improvement in renal function and, in fact, exhibited more severe renal injury markers (IRIS stage: control group stage II vs. BMSC group stage III). Histopathological analysis demonstrated a significant reduction in mononuclear inflammatory cell infiltration and fibrosis ($p = 0.016$ and $p = 0.029$, respectively), although no significant differences were observed in other parameters (154). Lee et al. established a gentamicin- and cisplatin-induced kidney injury model in 11 dogs. Three dogs received bilateral intraparenchymal injections of 1×10^6 cells per kidney per dose on days 6 and 23, whereas eight dogs received PBS injections. By day 51, only one dog (12.5%) in the control group survived, compared to all dogs in the treatment group. Biochemical

TABLE 3 Summary of feline and canine clinical studies on MSC therapy for kidney disease.

Species	Indication	Cell source	Dose (cells)	Route	Design	Follow-up	Main outcomes	Adverse events	References
Cat	AKI model	ADMSC/Allo, BMSC/Allo	4×10^6	IV	Controlled study	6 days	Noimprovement in renal function or histopathology	Transient -fever (multiple)	(130)
Cat	AKI model	ADMSC-EVs/Allo	Equivalent to 10^7 cells/24 h secretion	IV	Case series	3 days	Significantly accelerated creatinine and BUN recovery; identified 6 novel biomarkers	None	(125)
Cat	CKD	BMSC/Auto	1×10^5 cells/kidney (3 sites)	Intrarenal	Case series	30 days	Modest GFR and creatinine improvement in 2/4 early CKD cats (not statistically significant)	Transient microscopic hematuria (in healthy cat)	(132)
Cat	CKD	ADMSC/Allo	2×10^6 (cryopreserved)	IV	Sequential study (1)	90 days	Statistically significant but clinically modest creatinine decrease	None	(128)
			4×10^6 (cryopreserved)		Sequential study (2)		No significant improvement in renal function	2/5 vomiting, 4/5 tachypnea, 1/5 severe dyspnea	
			4×10^6 (culture-expanded)		Sequential study (3)		No significant improvement in renal function	None	
Cat	CKD	ADMSC/Allo	2×10^6 /kg	IV	Randomized controlled trial	60 days	No significant creatinine decrease; Quality of life trend improvement	Transient vomiting	(129)
Cat	CKD	Amniotic MSC/Allo	2×10^6	IV	Non-randomized controlled	60 days	Significant creatinine decrease ($p = 0.028$); UPC decrease (NS)	1/9 vomiting during injection	(30)
Cat	CKD	Uterine MSC/Allo model	3×10^7	IV	Controlled study	182 days	Significant GFR increase ($p < 0.05$ at multiple time points); increased food/water intake; met primary endpoint	1 propofol overdose (unrelated)	(131)
Cat	CKD	BMSC/Auto	1×10^6	Renal artery	Phase I trial	6 months	Safe and feasible; no significant GFR improvement	No serious AEs	(153)
Cat	CKD	ADMSC/Allo	2×10^6 /kg $\times 3$	IV	Case report	180 days	IRIS stage IV $\rightarrow$ II; improved body condition and clinical parameters	None	(29)
Dog	AKI model	cUCB-MSC/Allo	1×10^6 /kidney $\times 2$	Renal parenchymal	Controlled study	51 days	100% survival (vs 12.5% control); significant BUN/SCr decrease; histological improvement	NR	(28)
Dog	AKI model	BMSC/Auto	1×10^6 /kg	IV	Controlled study	4 days	Worsened renal function (higher BUN/SCr); but reduced histological inflammation/fibrosis	NR	(154)
Dog	AKI model	BMSC/Auto vs. ADMSC/Auto	5×10^6 , 10×10^6 , 15×10^6	Renal cortical	Dose-escalation	3 months	ADSC superior to BMSC; optimal dose 10×10^6 ; faster anti-inflammatory effect with ADSC	NR	(127)
Dog	CKD	ADMSC/Allo	1×10^6 /kg $\times 3$	IV	Case series	~ 153 days	Safe; 1/5 dog creatinine decreased ($3.5 \rightarrow 2.4$ mg/dL); weight stable/increased	None	(126)
Dog	CKD model	Camel WJ-MSC/Xeno	$5 \times 10^6 \times 2$	Intrarenal	Controlled study	12 weeks	Significant BUN/SCr decrease; IRIS stage III $\rightarrow$ II; tubular regeneration on histology	NR	(133)

ADMSC, Adipose-derived mesenchymal stem cell; AKI, Acute kidney injury; Allo, Allogeneic; Auto, Autologous; BMSC, Bone marrow-derived mesenchymal stem cell; BUN, Blood urea nitrogen; CKD, Chronic kidney disease; cUCB-MSC, Canine umbilical cord blood-derived mesenchymal stem cell; EV, Extracellular vesicle; GFR, Glomerular filtration rate; IRIS, International Renal Interest Society; IV, Intravenous; MSC, Mesenchymal stem cell; NR, Not reported; NS, Not significant; eRCT, Randomized controlled trial; SCr, Serum creatinine; UPC, Urine protein-to-creatinine ratio; WJ-MSC, Wharton's jelly-derived mesenchymal stem cell; Xeno, Xenogeneic; Study design notes, Induced models (surgical/pharmacological) are specified in the "Indication" column (AKI model, CKD model). Naturally occurring cases are indicated by CKD/AKI without "model" specification.

analysis showed significant improvements in blood urea nitrogen (BUN) and serum creatinine (SCr) levels in the treatment group relative to controls ($p = 0.0485$ and $p = 0.0121$). Histopathological evaluation further confirmed more pronounced tissue-level improvements in the treatment group (28). In a more comprehensive study by Osman et al., 60 dogs were allocated into groups and treated with BMSCs or ADMSCs via renal cortical injection at low, medium, and high doses (5×10^6 , 10×10^6 , and 15×10^6 cells), with treatment durations of 2 weeks, 2 months, and 3 months. ADMSCs demonstrated superior overall efficacy compared to BMSCs. In terms of long-term renal function recovery, BMSCs achieved 8–28% recovery, whereas ADMSCs reached 20–40%. Regarding anti-inflammatory effects, ADMSCs acted more rapidly (1 month vs. 3 months). Histologically, ADMSCs initiated regenerative processes and reduced tissue damage at early time points. The optimal injection dose was identified as 10×10^6 cells (127).

Research on MSC therapy for canine chronic kidney disease (CKD) remains relatively limited but has shown promising therapeutic potential. A case series reported the treatment of five dogs with CKD via intravenous infusion of allogeneic ADMSCs, which was found to be safe and associated with improvements in selected renal function parameters (126). Another innovative study employed allogeneic MSCs derived from camel Wharton's jelly, administered via intrarenal injection in canine CKD models. Compared to the control group, significant improvements were observed in BUN and SCr levels ($p < 0.05$), along with an increased proportion of regenerated tubules and maintained normal glomerular morphology (133).

6.3 Summary of efficacy and safety

Based on the clinical studies in cats and dogs reviewed above, the efficacy and safety of MSC therapy for kidney disease can be summarized as follows:

Efficacy. Although most studies on feline chronic kidney disease (CKD) did not achieve statistical significance, they consistently demonstrated positive trends—including moderate improvements in glomerular filtration rate and serum creatinine levels, upward trends in quality-of-life indicators, and reductions in IRIS stage (29, 129, 132). Studies using alternative cell sources, such as amniotic membrane- or uterus-derived MSCs, have also reported significant decreases in serum creatinine or a slowed decline in renal function (30, 131). In canine acute kidney injury (AKI) studies, the majority of findings indicate that MSC therapy improves survival rates, reduces blood urea nitrogen and serum creatinine levels, and attenuates renal tissue damage (28, 127). However, conflicting results have also been reported (154). Notably, adipose-derived MSCs may offer superior efficacy compared to bone marrow-derived MSCs, with an optimal dose of approximately 10×10^6 cells identified in one study (127). Although research on MSC therapy for canine CKD remains limited, preliminary findings suggest potential benefits in improving renal function (126, 133).

Safety. Most studies indicate that MSC therapy is generally well tolerated. The primary adverse reactions observed are infusion related events, including vomiting, tachypnea, and fever (128, 130). High-dose cryopreserved MSCs (4×10^6 cells) may be associated with an increased risk of such reactions (128). While serious adverse events are rare, the potential risk of pulmonary microembolism warrants continued vigilance. Importantly, long-term safety data beyond 1 year

post-treatment remain entirely lacking. In addition, many studies failed to report or control for concomitant therapies (e.g., ACE inhibitors, phosphate binders, dietary management), which may confound the assessment of MSC efficacy.

In summary, existing evidence supports the short-term safety of MSC therapy in feline and canine kidney disease, with encouraging signals of efficacy that warrant further validation through larger, more rigorously designed clinical trials. Studies on feline CKD have primarily relied on naturally occurring cases, providing relatively abundant but small-sample evidence. In contrast, research on canine AKI has largely employed experimentally induced models, which offer high internal validity but require caution when extrapolating findings to clinically ill dogs.

7 Challenges and outlook

Despite the demonstrated anti-inflammatory, anti-apoptotic, and pro-reparative effects of mesenchymal stem cells in both AKI and CKD models, their clinical translation remains constrained by several key factors. To address these challenges systematically, future research should adopt a multi-layered strategy structured along temporal dimensions and problem domains, ultimately forming a closed-loop “problem–solution” framework.

7.1 Short-term bottlenecks: standardization development and data improvement

7.1.1 Lack of dose and cell standardization

Current MSC research in cats and dogs employs a wide range of cell doses (from 1×10^6 to 3×10^7 cells), with most studies using fixed cell numbers rather than weight-adjusted doses. This inconsistency precludes direct comparison of results. Furthermore, studies exhibit considerable heterogeneity in MSC identification criteria—including viability, surface marker expression, and sterility—which severely undermines research reproducibility. Consequently, establishing consensus on weight-based dosage ranges and defining minimal characterization standards for MSC products is essential. These standards should include viability $>90\%$, CD90/CD105 positivity $>90\%$, CD34/CD45 negativity $<2\%$, absence of bacterial, mycoplasma, or endotoxin contamination, and confirmed tri-lineage differentiation capacity. In addition, reporting cell passage number, culture conditions (including fetal bovine serum status), and cryopreservation and thawing protocols is critical to improving data comparability.

7.1.2 Optimizing clinical trial design

Most studies are limited by small sample sizes (<50 cases), short follow-up periods (<1 year), and a lack of standardized endpoints, making cross-study comparisons difficult (155). Therefore, clinicians should consider conducting multicenter randomized controlled trials (RCTs) that stratify enrollment by IRIS stage and compare the efficacy of MSC transplantation—administered via different infusion routes—against placebo in client-owned cats and dogs with naturally occurring nephropathy. The primary endpoint should extend beyond changes in serum creatinine to include more clinically meaningful outcomes, such as survival time and quality of life. This approach would generate

high-quality evidence on therapeutic efficacy to inform veterinary practice and establish a foundation for subsequent dose optimization studies.

7.2 Mid-term challenges: mechanism deepening and technical optimization

7.2.1 In-depth research on delivery inefficiencies and homing effects

The renal homing rate of MSCs after intravenous injection is typically low, yet improvements in renal function are still observed (33, 35). This phenomenon suggests that the therapeutic effects of MSCs may be mediated by both local actions and systemic immune regulation, although the relative contribution of each remains unclear. This ambiguity hinders the rational design of administration regimens. Quantitative correlation analysis in large animal models is required to concurrently track cellular distribution and systemic immune markers (e.g., regulatory T cells, IL-10) to clarify the mechanisms underlying homing efficiency. If systemic immune modulation proves predominant, simplified and safer intravenous administration may represent a more practical approach.

7.2.2 Cellular heterogeneity and efficacy decline

Cellular heterogeneity represents another critical bottleneck. Single-cell sequencing confirms significant differences in secretion profiles, homing capacity, and differentiation propensity between different tissue sources (UC-MSC vs. BM-MSC) and subpopulations (e.g., pro-inflammatory vs. immunoregulatory) (156, 157). Furthermore, replicative senescence induced by *in vitro* expansion diminishes MSC stemness, particularly in autologous MSCs from elderly donors, which exhibit reduced proliferative potential and accumulated DNA damage (158). Therefore, single-cell transcriptomics and functional potency assays (e.g., T-cell proliferation inhibition capacity, vascular endothelial growth factor secretion) should be incorporated as companion metrics in clinical trials. For veterinary clinical applications, establishing allogeneic MSC banks from healthy young donors should be prioritized to mitigate quality variability associated with autologous MSCs.

7.2.3 Translational gap in pre-treatment strategies

Pre-treatment strategies significantly enhance MSC homing and therapeutic efficacy in rodent models, yet *in vivo* validation in dogs and cats remains largely unexplored. The most evidence-based pre-treatment method should be selected for *in vitro* potency validation in canine and feline MSCs (assessing CXCR4 expression, VEGF secretion, and anti-apoptotic capacity), followed by small-scale proof-of-concept trials to evaluate its potential to translate into improved clinical outcomes. Concurrently, attention must be paid to potential adverse effects of pre-treatment, such as conditions that may enhance one function at the expense of others.

7.2.4 Feasibility exploration of minimally invasive local delivery

Although renal artery injection and subcapsular renal injection can elevate renal retention rates to 15–20% or higher in rodent

models, their application in veterinary clinical practice is constrained by technical barriers, anesthesia risks, and owner acceptance. While safety has been demonstrated in porcine models, clinical trials involving cats and dogs are still required to conduct safety assessments, thereby providing robust data support for the widespread adoption of this technique.

7.3 Long-term development: next-generation therapies and scientific regulation

Exosomes (EVs) have emerged as a promising “cell-free” alternative to MSCs due to their low immunogenicity, high permeability, and amenability to engineering modification (159–163). However, challenges persist in EV isolation and purification, dose standardization, long-term safety, and efficacy maintenance. Future efforts should focus on establishing GMP-grade scaled production processes for EVs, completing toxicological and pharmacokinetic studies, and conducting safety trials in client-owned cats and dogs with naturally occurring disease. Concurrently, leveraging RGD-modified hydrogel delivery systems to enhance EV retention and internalization in damaged renal tubules, in parallel with developing engineered EVs, could enable targeted delivery and optimized cargo loading (164).

7.4 Validation pathway for the “pre-treatment enhancement–targeted delivery–carrier assisted” integration strategy

Based on the hierarchical analysis outlined above, future research should systematically validate the “Prime–Target–Carrier” framework proposed in this paper:

Prime (Pre-treatment Enhancement): Screen optimal pre-treatment protocols (e.g., hypoxia, melatonin, ATRA) in canine and feline MSCs, establish potency assay methods (CXCR4 expression, VEGF secretion, anti-apoptotic capacity), and conduct small-sample proof-of-concept trials.

Target (Targeted Delivery): Develop operational protocols and safe dosage ranges for ultrasound-guided subcapsular renal injection in large animal models, then apply the optimized protocol to animals with naturally occurring nephropathy to compare the efficacy of local versus intravenous administration.

Carrier (Vehicle-Assisted): Prioritize injectable, minimally invasive carriers (e.g., degradable hydrogels, microcapsules), conduct safety trials in dogs and cats, and evaluate owner acceptance and cost-effectiveness.

In summary, only through this problem-oriented, tiered research strategy—combined with systematic validation using the “Prime–Target–Carrier” framework—can MSC therapy transition from experimental treatment to routine clinical practice for companion animals.

8 Conclusion

This review summarizes research progress on mesenchymal stem cell therapy for both acute kidney injury and chronic kidney disease. Existing evidence indicates that MSC therapy is generally safe and well

tolerated, with the potential to improve renal function and reduce histopathological damage. However, its clinical application remains constrained by low delivery efficiency, insufficient cell survival, and limited quality of evidence.

To achieve efficient MSC homing and enhance therapeutic efficacy, the integrated “Prime–Target–Carrier” strategy remains the focus of current optimization efforts. However, critical challenges persist across this framework. Optimal pre-treatment conditions remain unstandardized, clinical data on the risks versus benefits of local injection are lacking, and the feasibility of cellular carriers requires further evaluation.

Future research should advance using a phased approach: In the short term, establish product standards and conduct multicenter randomized controlled trials (RCTs). In the medium term, optimize delivery pathways, validate pre-treatment strategies, and utilize single-cell technology to analyze heterogeneity. In the long term, promote the translation of cell-free therapies and establish a veterinary regulatory framework. Concurrently, leveraging natural kidney disease models in cats and dogs to accumulate real-world evidence will help advance the safe and effective application of MSC therapies in companion animal clinical settings.

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Conflict of interest

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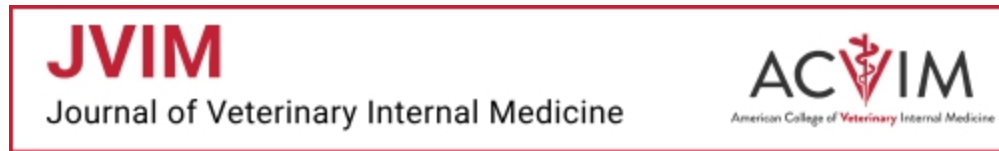
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Renoprotective effects of curcumin in cats with chronic kidney disease

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Abstract

Background

Chronic kidney disease (CKD) in cats is characterized by oxidative stress, inflammation, and apoptosis. Curcumin has shown renoprotective effects in experimental models, but clinical evidence in cats with CKD remains limited.

Hypothesis/Objectives

Evaluate whether PO curcumin supplementation (80 mg/cat/day), administered with standard treatment, favorably modulates renal, oxidative, inflammatory, and apoptotic biomarkers in cats with International Renal Interest Society (IRIS) stage 3 CKD.

Animals

Sixteen client-owned cats with IRIS stage 3 CKD.

Methods

Prospective, randomized trial. Cats were allocated to medical treatment alone (MT; $n = 8$) or medical treatment plus curcumin (MT + C; $n = 8$). Curcumin was administered PO at 80 mg/cat/day in 2 divided doses for 60 days.

Results

Baseline renal biomarkers did not differ between groups. The primary outcome, serum symmetric dimethylarginine (SDMA) concentration, was significantly lower in the MT + C group than in the MT group on days 30, 45, and 60 ($P < .01$). Urinary protein-to-creatinine ratio decreased over time in the MT + C group and was lower than in the MT group at later time points. Total antioxidant status increased, whereas total oxidant status decreased in the MT + C group, with a significant between-group difference in total oxidant status on day 45. Findings for nuclear factor kappa B and caspase-3 were also consistent with decreased inflammatory and apoptotic activity in curcumin-treated cats.

Conclusions and clinical importance

Curcumin supplementation was associated with improvement in SDMA and favorable changes in mechanistic biomarkers in cats with IRIS stage 3 CKD.

Keywords: apoptosis, cat, chronic kidney disease, curcumin, oxidative stress, SDMA

Introduction

Chronic kidney disease (CKD) is a progressive and irreversible condition frequently diagnosed in aging cats, characterized by glomerulosclerosis, tubular atrophy, interstitial fibrosis, and chronic inflammation.^{[1,2](#)} The International Renal Interest Society (IRIS) classifies CKD into 4 stages based on serum creatinine and symmetric dimethylarginine (SDMA) concentrations, with stage 1 representing early disease and stage 4 reflecting severe azotemia and advanced renal dysfunction.^{[3](#)} Despite advances in veterinary nephrology, current treatments remain primarily palliative, aiming to slow disease progression and alleviate clinical signs.^{[1,4](#)} However, these approaches do not directly target the molecular mechanisms underlying renal injury.^{[1](#)}

The diagnosis of CKD in cats is based on clinical signs, laboratory findings, and diagnostic imaging. Persistent azotemia, inadequately concentrated urine (urine specific gravity [USG] < 1.035), and increased SDMA concentrations are key markers for early detection and staging.^{5,6} According to IRIS guidelines, proteinuria and systemic blood pressure also are used for substaging.^{1,3} Whereas serum creatinine concentration traditionally has been used to evaluate renal function, SDMA may provide an earlier indication of decreased glomerular filtration rate (GFR) in some cats.⁷ Treatment strategies include renal diets, SC fluids, antihypertensives such as amlodipine, and angiotensin receptor blockers such as telmisartan.^{8,9} Management of anemia and gastrointestinal signs further improves quality of life.¹⁰

Oxidative stress, inflammation, and apoptosis are key contributors to CKD pathogenesis.¹¹ Reactive oxygen species (ROS) generated in diseased renal tissue cause lipid peroxidation, DNA damage, and protein oxidation, leading to activation of pro-inflammatory cytokines and transcription factors such as nuclear factor kappa B (NF-κB). These cascades contribute to tubular epithelial cell death and progressive nephron loss, as demonstrated in experimental CKD models in rodents and in humans with CKD.¹²

Curcumin, a polyphenolic compound from *Curcuma longa*, has demonstrated antioxidant, anti-inflammatory, and antiapoptotic properties in rodent models of kidney injury, including ischemia-reperfusion, diabetic nephropathy, and toxin-induced nephropathy.¹³⁻¹⁵ These effects involve modulation of oxidative pathways, upregulation of antioxidant enzymes including superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GPx); inhibition of NF-κB and cytokines such as tumor necrosis factor-alpha (TNF-α), interleukin-6 (IL-6), and interleukin-1 beta (IL-1β); as well as regulation of apoptosis via Caspase-3 and B-cell lymphoma 2 (Bcl-2).¹⁶⁻¹⁸

Despite these promising properties, curcumin's clinical efficacy in cats with CKD remains untested. Improved bioavailability using novel formulations enhances its potential for clinical use.^{19,20} We hypothesized that daily PO administration of curcumin (80 mg/cat/day), in combination with conventional treatment, would exert renoprotective effects in cats with stage 3 CKD by modulating oxidative stress, inflammation, and apoptosis.

Materials and methods

Twenty-four client-owned cats were enrolled: 8 clinically healthy controls and 16 cats with IRIS stage 3 chronic kidney disease (CKD). Chronic kidney disease was diagnosed by persistent azotemia, USG < 1.035, compatible clinical signs, and abdominal ultrasonography. Staging was based on serum creatinine concentration (2.9-5.0 mg/dL) after stabilization; SDMA concentration

(≥ 26 $\mu\text{g}/\text{dL}$) supported chronic renal dysfunction but was not used for staging. Cats with prerenal azotemia, concurrent systemic disease, or recent antioxidant or anti-inflammatory supplementation were excluded.

Cats with CKD were randomized to medical treatment alone (MT; $n = 8$) or medical treatment plus curcumin (MT + C; $n = 8$). Healthy controls were sampled at baseline only for disease characterization and were not used to assess treatment efficacy. The primary endpoint was longitudinal change in serum SDMA concentration. Secondary endpoints were urine protein-to-creatinine ratio (UPC), total antioxidant status (TAS), total oxidant status (TOS), NF- κ B, and caspase-3; other variables were exploratory.

Both CKD groups received IRIS-guided medical management, including renal diet, fluid therapy when indicated, telmisartan for hypertension or proteinuria, lanthanum carbonate for persistent hyperphosphatemia, dietary calcium adjustment for hypercalcemia, potassium supplementation, maropitant or mirtazapine when required, and darbepoetin with iron for anemia. The MT + C cats additionally received PO curcumin (Solgar Full Spectrum Curcumin) at 80 mg/cat/day in 2 divided doses with food for 60 days. No placebo was administered to the MT group. Laboratory personnel were blinded to group allocation throughout the study.

Blood and urine were collected from cats with CKD on day 0 before treatment and on days 15, 30, 45, and 60; controls were sampled once at baseline. Additional methodologic details, including treatment protocols, sampling procedures, and analytical characteristics of the assays are provided in [Supplementary Methods S1](#). Hematology, biochemistry, SDMA, UPC, USG, systolic blood pressure (SBP), oxidative stress markers, and inflammatory and apoptotic biomarkers were evaluated as part of the study protocol.

Statistical analyses were performed using SPSS 25.0. Longitudinal CKD outcomes were analyzed using linear mixed effects models, with Bonferroni correction applied within outcomes. Two-sided $P < .05$ was considered significant for the primary endpoint, SDMA; secondary and exploratory outcomes were interpreted as hypothesis-generating. Additional statistical details are provided in [Supplementary Methods S1](#).

Results

Clinical outcomes

No significant differences were found between the MT and MT + C groups in age, sex, neuter status, or body weight ($P = .73$, $P = .68$, $P = .71$, and $P = .62$, respectively). Among cats with CKD ($n = 16$), ages ranged from 7 to 15 years, with a mean $\pm$ SD of 11.2 ± 3.4 years. The population of cats with CKD consisted predominantly of domestic shorthair cats (69%, 11/16), followed by British shorthair (6%, 1/16), Scottish fold (6%, 1/16), Persian (6%, 1/16), Siamese (6%, 1/16), and Ankara cats (6%, 1/16). Overall, 62% were male (10/16; all neutered) and 38% were female (6/16; all spayed). The most frequent clinical signs in cats with CKD were weight loss and anorexia (94%, 15/16), polyuria and polydipsia (88%, 14/16), dehydration (75%, 12/16), and lethargy (69%, 11/16). The IRIS staging was performed based on serum creatinine concentrations at baseline, with SDMA evaluated as a supportive biomarker. One cat from the MT group died on day 44, and one from the MT + C group died on day 31; both deaths were attributed to progression of CKD rather than study procedures or interventions.

The healthy control cats ($n = 8$) were 6 to 13 years of age (9.8 ± 2.6 years). The control group included 5 males (all neutered) and 3 females (all spayed). Breeds consisted of domestic shorthair (6/8), British shorthair (1/8), and Scottish fold (1/8). Body weight was 4.6 ± 0.8 kg. All control cats were clinically healthy and had an ideal body condition score.

All CKD cats received a renal diet and lanthanum carbonate as standardized baseline treatment (MT: 8/8; MT + C: 8/8). Telmisartan was administered to all hypertensive and proteinuric cats in both groups (hypertension: MT: 8/8; MT + C: 8/8; proteinuria: MT: 8/8; MT + C: 8/8). Darbepoetin treatment was required in 6 anemic cats (MT: 3/8; MT + C: 3/8), all of which also received concurrent PO iron supplementation. Potassium supplementation was administered in 3 cats (MT: 2/8; MT + C: 1/8). Appetite stimulation with mirtazapine was used in twelve cats (MT: 6/8; MT + C: 6/8), and maropitant citrate was administered for vomiting in thirteen cats (MT: 7/8; MT + C: 6/8). Overall, the distribution of baseline concurrent treatments was comparable between groups, and no new medications were introduced during the study period.

Ultrasonographic findings in CKD cats showed a consistent pattern of chronic renal change, including increased cortical echogenicity (100%, 16/16), decreased corticomedullary definition (69%, 11/16), irregular renal margins (50%, 8/16), and decreased kidney size (13%, 2/16).

Hematologic findings

Healthy controls were included in baseline (day 0) cross-sectional comparisons to characterize disease-associated hematologic changes at study entry, whereas longitudinal treatment-related comparisons were restricted to the MT and MT + C groups. The white blood cell (WBC) counts did

not differ significantly between the MT and MT + C groups at baseline or on days 15 and 30 ($P > .05$), but were significantly higher in the MT + C group on days 45 and 60 ([Table 1](#)).

Table 1.

Mean white blood cell count (WBC), hemoglobin (HGB), hematocrit (HCT), and red blood cell count (RBC) measurements at different time points in control, medical treatment (MT), and MT + curcumin (MT + C) groups.

Variable	Time	Control $\bar{X} \pm SD$	MT $\bar{X} \pm SD$	MT + C $\bar{X} \pm SD$	P
WBC ($\times 10^3 / \mu\text{L}$)	0	8.51 $\pm$ 0.89 _b	11.75 $\pm$ 0.57 _{aAB}	11.18 $\pm$ 1.13 _a	<.05
	15		11.19 $\pm$ 1.58 _{abAB}	14.95 $\pm$ 2.03 _a	<.05
	30		15.10 $\pm$ 2.11 _{aA}	13.43 $\pm$ 0.94 _a	<.05
	45		10.21 $\pm$ 1.53 _{bB}	16.26 $\pm$ 2.20 _a	<.01
	60		8.69 $\pm$ 0.79 _{bB}	15.74 $\pm$ 2.06 _a	<.01
HGB (g/dL)			$P < .05$	$P > .05$	
	0	13.10 $\pm$ 0.75	10.02 $\pm$ 0.96	9.78 $\pm$ 1.21	>.05
	15		9.91 $\pm$ 1.03 _b	8.51 $\pm$ 1.08 _b	<.01
	30		7.9 $\pm$ 1.13 _b	9.45 $\pm$ 1.07 _b	<.01
	45		9.05 $\pm$ 0.94 _b	8.93 $\pm$ 0.50 _b	<.01
HCT (%)	60		9.48 $\pm$ 1.02 _b	8.20 $\pm$ 0.82 _b	<.01
			$P > .05$	$P > .05$	
	0	38.11 $\pm$ 2.29 _a	29.74 $\pm$ 1.99 _b	27.98 $\pm$ 2.73 _b	<.05
	15		29.30 $\pm$ 2.62 _b	24.40 $\pm$ 2.41 _b	<.01
	30		24.06 $\pm$ 2.98 _b	27.15 $\pm$ 2.72 _b	<.01
RBC ($\times 10^6 / \mu\text{L}$)	45		25.27 $\pm$ 3.53 _b	26.61 $\pm$ 1.49 _b	<.01
	60		29.00 $\pm$ 3.20 _b	24.56 $\pm$ 2.14 _b	<.01
			$P > .05$	$P > .05$	
	0	9.37 $\pm$ 0.62 _a	6.70 $\pm$ 0.74 _b	5.88 $\pm$ 0.78 _b	<.01
	15		7.04 $\pm$ 0.79 _b	5.58 $\pm$ 0.67 _b	<.01
	30		5.80 $\pm$ 0.80 _b	6.54 $\pm$ 0.74 _b	<.01

Variable	Time	Control $\bar{X} \pm \text{SD}$	MT $\bar{X} \pm \text{SD}$	MT + C $\bar{X} \pm \text{SD}$	<i>P</i>
	45		6.05 ± 0.89 ^b	6.22 ± 0.26 ^b	<.01
	60		6.63 ± 0.98 ^b	5.82 ± 0.61 ^b	<.01
			<i>P</i> > .05	<i>P</i> > .05	

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A, B, C: Different capital letters within the same treatment group indicate significant differences across time points.

a, b, c: Different lowercase letters within the same row indicate significant between-group differences based on baseline cross-sectional comparisons or the corresponding time-point comparisons between MT and MT + C, as appropriate.

Healthy controls were sampled only at day 0; therefore, comparisons involving the Control group were restricted to baseline analyses.

P-values represent overall between-group comparisons at each time point.

Hemoglobin concentration, hematocrit, and RBC count did not differ significantly between the MT and MT + C groups at any time point (*P* > .05, [Table 1](#)). Healthy controls were included in baseline (day 0) cross-sectional comparisons to characterize anemia-related changes at study entry, whereas longitudinal treatment-related comparisons were restricted to the MT and MT + C groups.

Biochemical findings

Serum creatinine, phosphorus, potassium, and TCO₂ concentrations did not differ significantly between the MT and MT + C groups at any time point (*P* > .05, [Table 2](#)). Serum calcium concentrations showed no consistent between-group differences between the MT and MT + C groups throughout the study period, but overall time point-specific group effects were observed at later evaluations ([Table 2](#)). In contrast, SDMA concentrations were significantly lower in the MT + C group than in the MT group on days 30, 45, and 60 (*P* < .01). At day 60, the baseline-adjusted between-group difference in SDMA (MT vs MT + C) was 15.75 µg/dL (95% CI, 10.9-20.4),

indicating a clinically meaningful treatment effect. Serum creatinine concentrations remained persistently increased in both CKD groups throughout the study, with no significant within-group or between-group differences ($P > .05$; [Table 2](#)).

Table 2.

Mean creatinine, phosphorus (P), calcium (Ca), potassium (K), total carbon dioxide (TCO₂), and symmetric dimethylarginine (SDMA) measurements at different time points in control, medical treatment (MT), and MT + curcumin (MT + C) groups.

Variable	Time	Control $\bar{X} \pm SD$	MT $\bar{X} \pm SD$	MT + C $\bar{X} \pm SD$	P
Creatinine (mg/dL)	0	1.00 ± 0.11	3.45 ± 0.18 _a	3.63 ± 0.24 _a	<.01
	15		3.70 ± 0.61 _a	3.89 ± 0.75 _a	<.01
	30		4.57 ± 1.82 _a	3.89 ± 0.57 _a	<.01
	45		3.63 ± 0.42 _a	4.23 ± 0.86 _a	<.01
	60		3.77 ± 0.22 _a	3.35 ± 0.38 _a	<.01
P (mg/dL)			$P > .05$	$P > .05$	
	0	4.21 ± 0.45	7.68 ± 1.01 _a	9.66 ± 1.65 _a	<.05
	15		7.67 ± 0.82 _a	8.55 ± 1.30 _a	<.01
	30		8.18 ± 1.20 _a	7.20 ± 1.04 _a	<.05
	45		8.42 ± 1.21 _a	7.90 ± 1.34 _a	<.05
Ca (mg/dL)	60		7.91 ± 1.06 _a	7.20 ± 1.19 _a	<.05
			$P > .05$	$P > .05$	
	0	10.51 ± 0.25	11.48 ± 0.41	9.85 ± 0.82 _B	>.05
	15		12.33 ± 2.01	12.07 ± 0.65 _A	>.05
	30		11.47 ± 0.50	11.91 ± 0.48 _A	>.05
K (mEq/L)	45		11.32 ± 0.44 _{ab}	12.10 ± 0.43 _{aA}	<.05
	60		11.69 ± 0.18 _a	11.81 ± 0.34 _{aA}	<.01
			$P > .05$	$P < .05$	
K (mEq/L)	0	3.98 ± 0.20	4.15 ± 0.35	4.03 ± 0.56	>.05
	15		4.18 ± 0.17	4.68 ± 0.41	>.05
	30		4.55 ± 0.24	4.00 ± 0.17	>.05

Variable	Time	Control $\bar{X} \pm \text{SD}$	MT $\bar{X} \pm \text{SD}$	MT + C $\bar{X} \pm \text{SD}$	<i>P</i>
TCO ₂ (mmol/L)	45	25.71 ± 0.67	4.32 ± 0.06	3.86 ± 0.18	>.05
	60		4.20 ± 0.22	4.05 ± 0.23	>.05
			<i>P</i> > .05	<i>P</i> > .05	
	0		27.27 ± 2.07	21.66 ± 1.94	>.05
	15		26.33 ± 2.44	25.70 ± 2.11	>.05
	30		25.36 ± 2.31	25.53 ± 1.85	>.05
	45		26.52 ± 2.46	26.46 ± 1.25	>.05
SDMA (µg/dL)	60	8.75 ± 0.67	30.41 ± 2.24	27.78 ± 2.54	>.05
			<i>P</i> > .05	<i>P</i> > .05	
	0		38.75 ± 4.20 _a	38.50 ± 3.25 _{aA}	<.01
	15		36.00 ± 2.80 _a	33.00 ± 2.84 _{aAB}	<.01
	30		40.37 ± 5.46 _a	29.83 ± 2.23 _{bB}	<.01
	45		39.42 ± 5.01 _a	28.62 ± 2.15 _{bB}	<.01
	60		43.00 ± 4.88 _a	27.25 ± 1.72 _{bB}	<.01
			<i>P</i> > .05	<i>P</i> < .01	

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Healthy controls were sampled only at day 0; therefore, comparisons involving the Control group were restricted to baseline analyses.

P-values represent overall between-group comparisons at each time point.

Urinalysis findings

Within-group analysis showed a significant decrease in UPC over time in the MT + C group, beginning on day 30 and persisting through days 45 and 60 ([Table 3](#)). In addition, UPC results were significantly lower in the MT + C group than in the MT group on days 30, 45, and 60. USG remained low in both treatment groups throughout the study period, with no significant temporal changes within groups ($P > .05$; [Table 3](#)).

Table 3.

Mean urine protein-to-creatinine ratio (UPC) and urine specific gravity (USG) measurements at different time points in control, medical treatment (MT), and MT + curcumin (MT + C) groups.

Variable	Time	Control $\bar{X} \pm SD$	MT $\bar{X} \pm SD$	MT + C $\bar{X} \pm SD$	P
UPC	0	0.03 ± 0.01	0.53 ± 0.18 _a	0.78 ± 0.14 _{aA}	<.01
	15	-	0.57 ± 0.20 _a	0.53 ± 0.14 _{aB}	<.05
	30	-	0.64 ± 0.20 _a	0.19 ± 0.03 _{bC}	<.01
	45	-	0.51 ± 0.17 _a	0.31 ± 0.07 _{bC}	<.05
	60	-	0.39 ± 0.06 _a	0.24 ± 0.04 _{cC}	<.01
			$P > .05$	$P < .01$	
USG (g/mL)	0	1.047 ± 0.004	1.010 ± 0.003 _b	1.011 ± 0.005 _b	<.01
	15	-	1.010 ± 0.002 _b	1.012 ± 0.003 _b	<.01
	30	-	1.011 ± 0.002 _b	1.012 ± 0.003 _b	<.01
	45	-	1.011 ± 0.004 _b	1.013 ± 0.003 _b	<.01
	60	-	1.012 ± 0.002 _b	1.015 ± 0.003 _b	<.01
			$P > .05$	$P > .05$	

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Healthy controls were sampled only at day 0; therefore, comparisons involving the Control group were restricted to baseline analyses.

P-values represent overall between-group comparisons at each time point.

Oxidative stress findings

Total antioxidant status was lower at baseline in both treatment groups than in healthy controls, and it remained unchanged through days 15 and 30. Significant increases from baseline were observed on days 45 and 60 in the MT + C group ($P < .01$) and in the MT group ($P < .05$; [Table 4](#)).

Table 4.

Mean total antioxidant status (TAS) and total oxidant status (TOS) measurements at different time points in control, medical treatment (MT), and MT + curcumin (MT + C) groups.

Variable	Time	Control $\bar{X} \pm SD$	MT $\bar{X} \pm SD$	MT + C $\bar{X} \pm SD$	P
TAS ($\mu\text{mol Trolox equivalent/L}$)	0	2.14 $\pm$ 0.05	1.70 $\pm$ 0.07 ^{bB}	1.64 $\pm$ 0.04 ^{bB}	<.01
	15	-	1.83 $\pm$ 0.07 ^{bAB}	1.75 $\pm$ 0.07 ^{bB}	<.05
	30	-	1.86 $\pm$ 0.09 ^{bAB}	1.83 $\pm$ 0.04 ^{bB}	<.01
	45	-	2.17 $\pm$ 0.14 ^A	2.04 $\pm$ 0.07 ^A	>.05
	60	-	2.14 $\pm$ 0.16 ^A	2.23 $\pm$ 0.08 ^A	>.05
TOS ($\mu\text{mol H}_2\text{O}_2$ equivalent/L)	0	3.26 $\pm$ 0.06	5.14 $\pm$ 0.19 ^a	5.31 $\pm$ 0.20 ^{aA}	<.01
	15	-	4.99 $\pm$ 0.21 ^a	4.98 $\pm$ 0.19 ^{aAB}	<.01
	30	-	4.98 $\pm$ 0.34 ^a	4.60 $\pm$ 0.21 ^{aBC}	<.01
	45	-	4.86 $\pm$ 0.34 ^a	4.28 $\pm$ 0.23 ^{bC}	<.01
	60	-	4.58 $\pm$ 0.22 ^a	4.21 $\pm$ 0.20 ^{abC}	<.01
			P < .05	P < .01	
			P > .05	P < .01	

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Healthy controls were sampled only at day 0; therefore, comparisons involving the Control group were restricted to baseline analyses.

P-values represent overall between-group comparisons at each time point.

Total oxidant status was increased at all time points in both treatment groups. In the MT group, TOS did not change significantly over time ($P > .05$). Although TOS in the MT + C group was lower than in the MT group on days 45 and 60, a significant between-group difference was observed only on day 45 ($P < .01$; [Table 4](#)).

Inflammatory and apoptotic marker findings

From day 30 onward, NF- κ B concentrations in the MT group remained significantly higher than baseline ($P < .01$), whereas no significant intra-group changes were observed in the MT + C group ($P > .05$). The NF- κ B concentrations were significantly lower in the MT + C group than in the MT group on days 30 and 60 ([Table 5](#)).

Table 5.

Mean nuclear factor κ B (NF- κ B) and Caspase-3 measurements at different time points in control, medical treatment (MT), and MT + curcumin (MT + C) groups.

Variable	Time	Control $\bar{X} \pm SD$	MT $\bar{X} \pm SD$	MT + C $\bar{X} \pm SD$	P
NF- κ B (ng/mL)	0	2.15 $\pm$ 0.12	3.42 $\pm$ 0.63 _{ab}	4.71 $\pm$ 0.65 _a	<.01
	15	-	3.48 $\pm$ 0.58 _{ab}	4.44 $\pm$ 0.80 _a	<.05
	30	-	4.65 $\pm$ 0.48 _a	3.45 $\pm$ 0.25 _b	<.01
	45	-	3.62 $\pm$ 0.30 _a	3.14 $\pm$ 0.22 _{ab}	<.01
	60	-	3.82 $\pm$ 0.22 _a	2.73 $\pm$ 0.21 _b	<.01
			$P > .05$	$P > .05$	
Caspase-3 (ng/mL)	0	1.33 $\pm$ 0.08	2.57 $\pm$ 0.30 _a	2.81 $\pm$ 0.25 _{aA}	<.01
	15	-	2.80 $\pm$ 0.36 _a	2.84 $\pm$ 0.26 _{aA}	<.01
	30	-	2.81 $\pm$ 0.32 _a	2.26 $\pm$ 0.06 _{aAB}	<.01
	45	-	2.75 $\pm$ 0.55 _a	1.96 $\pm$ 0.07 _{abB}	<.05
	60	-	2.24 $\pm$ 0.75 _a	1.66 $\pm$ 0.07 _{bC}	<.01
			$P > .05$	$P < .01$	

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Healthy controls were sampled only at day 0; therefore, comparisons involving the Control group were restricted to baseline analyses.

P-values represent overall between-group comparisons at each time point.

Caspase-3 concentrations in the MT + C group decreased progressively over time, whereas no significant intra-group changes were observed in the MT group. A significant between-group difference was observed only on day 60, with lower Caspase-3 concentrations in the MT + C group than in the MT group ([Table 5](#)).

Systolic blood pressure findings

Systolic blood pressure was increased at baseline in both treatment groups. It did not differ significantly between the MT and MT + C groups at most time points ($P > .05$). In the MT + C group, a significant decrease in SBP was observed at day 60 compared with baseline ($P < .05$). On day 60, SBP was significantly lower in the MT + C group than in the MT group ($P < .05$).

Discussion

We evaluated the renoprotective effects of curcumin supplementation in cats with CKD, with a focus on its impact on renal function biomarkers, oxidative stress, inflammation, and apoptosis. The findings highlight curcumin's therapeutic potential as adjunctive treatment in cats with CKD, indicating its ability to improve SDMA concentrations, modulate oxidative balance, suppress NF- κ B-mediated inflammation, and decrease apoptotic activity through caspase-3 regulation.

Renal function biomarkers such as serum creatinine and SDMA concentrations did not differ significantly between the MT and MT + C groups at baseline, confirming that the treatment groups were comparable before intervention. Throughout the study, serum creatinine concentrations remained stable in both groups, which is consistent with expectations in stage 3 CKD where nephron loss is typically advanced. In contrast, SDMA concentrations were significantly lower in the MT + C group compared with the MT group from day 30 onward. Symmetric dimethylarginine is commonly regarded as a sensitive indicator of changes in GFR in cats.⁶ Recent comparative studies however have demonstrated that SDMA does not consistently outperform creatinine when directly correlated with measured GFR in azotemic CKD cohorts.^{7,21} These findings suggest that SDMA should be interpreted within a broader physiological context rather than as a sole indicator of GFR. Curcumin has been reported to modulate renal vascular and microcirculatory function through enhancement of nitric oxide bioavailability, improvement of endothelial function, and support of cortical microvascular perfusion.^{22,23} Nevertheless, given that both SDMA and

creatinine are predominantly eliminated via GFR with minimal tubular handling, changes in renal hemodynamics would be expected to influence both biomarkers. Accordingly, the observed decrease in SDMA in the absence of a parallel change in serum creatinine concentration in the MT + C group should not be interpreted as evidence of selective glomerular or tubular effects of curcumin. Rather, this pattern likely reflects differential biomarker responsiveness over a short observation period in cats with CKD. Clarification of these relationships will require future studies incorporating direct measurements of GFR.

Phosphorus and calcium dysregulation are common complications in CKD, primarily resulting from impaired renal excretion and secondary hyperparathyroidism.^{24,25} In our study, both CKD groups exhibited persistently increased serum phosphorus concentrations throughout the observation period, consistent with reports that hyperphosphatemia often progresses despite multimodal medical management in advanced CKD.²⁶ The continued increase in serum phosphorus concentrations despite lanthanum use in all cats reflects the established pathophysiology of IRIS Stage 3-4 disease. In these stages, decreased functional nephron mass, increased fibroblast growth factor 23 secretion, impaired phosphate excretion, and secondary hyperparathyroidism limit the efficacy of dietary restriction and phosphate binders.²⁷ Therefore, the lack of reduction in serum phosphorus concentration in our study reflects the metabolic resistance characteristic of advanced CKD rather than differences in treatment protocols or curcumin administration.

Calcium homeostasis followed a more stable pattern. Although mild fluctuations were observed at later time points, no significant differences emerged between groups, and curcumin supplementation did not produce measurable changes in serum calcium concentrations. The relatively steady calcium profile in the MT + C group aligns with current understanding that oxidative stress, inflammation, and mineral metabolism are interdependent in CKD. Experimental studies have shown that curcumin can modulate calcium regulatory mechanisms by influencing redox balance, altering transmembrane calcium flux, and supporting intracellular homeostatic signaling pathways.²⁸ These mechanisms provide a coherent biological basis for interpreting the calcium patterns observed in our study.

Urinary protein-to-creatinine ratios decreased significantly in the MT + C group from day 30 onward and remained lower through day 60, whereas UPC in the MT group remained unchanged. Persistent proteinuria is a well-recognized prognostic indicator in CKD progression.^{29,30} The downward trend in UPC observed in the MT + C group aligns with the concept that modulation of inflammatory and redox-regulated pathways can influence glomerular permeability dynamics. Experimental research has demonstrated that curcumin can interact with key molecular networks

governing glomerular function, including transforming growth factor- β (TGF- β), NF- κ B, and oxidative stress-responsive cascades.^{17,31} These pathways are integral to maintaining endothelial stability, extracellular matrix organization, and selective permeability across the glomerular filtration barrier. Consequently, the UPC trajectory observed in the MT + C group is consistent with mechanistic patterns described in experimental nephrology and reflects a biologically plausible response within the broader framework of curcumin-mediated glomerular modulation.

Curcumin's effects in CKD extend beyond anti-inflammatory activity and include interactions with oxidative pathways. In our study, TAS increased and TOS decreased in the MT + C group from day 30 onward, reflecting a shift toward enhanced redox balance. Curcumin has been widely documented to influence endogenous antioxidant systems, including modulation of SOD, CAT, and GPx.^{18,32} These pathways play central roles in the neutralization of ROS, stabilization of cellular redox homeostasis, and protection of renal parenchymal structures from oxidative injury. In addition, curcumin can interact with ROS through non-enzymatic processes, such as chelating transition metals involved in Fenton chemistry, thereby decreasing the formation of highly reactive radicals.³³ The TAS/TOS profile observed in the MT + C group aligns with these established antioxidant mechanisms and supports the concept that curcumin may contribute to more favorable oxidative conditions in CKD. Given the interconnected nature of redox, inflammatory, and apoptotic pathways in renal disease, modulation of oxidative burden also may influence downstream cellular responses.³¹

Chronic inflammation plays a central role in the progression of CKD. NF- κ B, a redox-sensitive transcription factor, regulates genes involved in cytokine production, leukocyte recruitment, and fibrogenesis. In our study, NF- κ B concentrations remained high in the MT group, whereas NF- κ B concentrations decreased in the MT + C group in the later stages of follow-up. Curcumin has been extensively characterized as an inhibitor of NF- κ B signaling through stabilization of inhibitor of kappa B alpha (I κ B α) and attenuation of oxidative and inflammatory cues that facilitate NF- κ B activation.^{32,34} These regulatory effects influence the transcription of key cytokines (TNF- α , IL-1 β , and IL-6) and contribute to a less pro-inflammatory intrarenal environment.³¹ The NF- κ B profile observed in the MT + C group is consistent with curcumin-mediated modulation of redox-inflammatory pathways and supports its potential to modulate inflammatory signaling in CKD.

Apoptosis plays a fundamental role in renal cell loss and CKD progression.^{35,36} Caspase-3, a crucial executioner caspase in the apoptotic cascade, is widely recognized as a reliable biomarker of apoptosis.¹⁶ In our study, caspase-3 concentrations remained increased in the MT group, whereas a marked downward shift emerged in the MT + C group beginning on day 45 and continuing through day 60. Curcumin has been broadly characterized as an agent capable of

influencing apoptotic regulation through modulation of mitochondrial signaling, attenuation of cytochrome c-dependent caspase activation, and stabilization of intracellular redox conditions.^{37,38} These molecular interactions are closely tied to the balance between pro-apoptotic and anti-apoptotic signaling, including Bcl-2-associated X protein and Bcl-2 regulatory dynamics, which are central to tubular epithelial cell integrity and survival.³⁹ Accordingly, the caspase-3 profile observed in the MT + C group is compatible with the mechanistic framework described for curcumin-mediated modulation of apoptotic pathways and reflects a biochemical environment supportive of decreased apoptotic drive within renal tissue.

Our study had several limitations. Sample size was relatively small, which may limit the generalizability of the results. Moreover, although multiple markers were used to evaluate renal function and systemic responses, renal histopathology was not performed, which would have provided direct evidence of tissue level changes. Future studies should consider longer follow-up periods, include histologic assessments, and explore the pharmacokinetics of curcumin in feline models to optimize dosing strategies. The healthy control group was included to facilitate baseline physiological comparisons and to contextualize disease-associated alterations in biomarker profiles. However, the limited number of control animals precludes definitive conclusions regarding population-level reference values, and treatment-related inferences were therefore primarily derived from comparisons between the MT and MT + C groups. The small sample size in each study arm, dictated by the availability of eligible stage 3 CKD cases, limits statistical power and increases the risk of both false-negative and false-positive results. Therefore, our findings should be interpreted as preliminary and hypothesis-generating.

In conclusion, PO curcumin supplementation, administered at 80 mg/cat/day in 2 divided doses for 60 consecutive days in addition to conventional medical treatment, was associated with lower SDMA concentrations and favorable changes in several mechanistic biomarkers in cats with IRIS stage 3 CKD. The observed changes in oxidative stress, inflammatory, and apoptotic markers were consistent with potential adjunctive effects of curcumin, but these secondary findings should be interpreted as hypothesis-generating. Given the small sample size, these results should be considered preliminary and warrant confirmation in larger, longer-term studies incorporating direct renal functional and histopathologic assessments.

Supplementary Material

R1_Supplementary_Table_S1_aalag138

[r1 supplementary table s1 aalag138.docx](#) (13.9KB, docx)

Supplementary_Methods_S1_aalag138

[supplementary methods s1 aalag138.docx](#) (21.4KB, docx)

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Abbreviations

ANOVA

analysis of variance

Bcl-2

B-cell lymphoma 2

CAT

catalase

CKD

chronic kidney disease

FGF-23

fibroblast growth factor 23

GFR

glomerular filtration rate

GPx

glutathione peroxidase

I κ B α

inhibitor of kappa B alpha

IL-1 β

interleukin-1 beta

IL-6

interleukin-6

MT

medical treatment

MT + C

medical treatment plus curcumin

NO

nitric oxide

NF- κ B

nuclear factor kappa B

ROS

reactive oxygen species

SDMA

serum symmetric dimethylarginine

SOD

superoxide dismutase

SBP

systolic blood pressure

TAS

total antioxidant status

TCO₂

total carbon dioxide

TOS

total oxidant status

TNF- α

tumor necrosis factor alpha

UPC

urine protein-to-creatinine ratio

USG

urine specific gravity

Appendix

Materials and methods

Ethical approval

Our study was conducted in accordance with the ethical principles for animal experimentation and was approved by the Atatürk University Animal Experiments Local Ethics Committee (approval no. 66; approval date: April 12, 2021; meeting no. 3). Written informed consent was obtained from the owners of all participating cats before inclusion in the study.

Animals

Twenty-four client-owned domestic cats were enrolled in the study, including 8 clinically healthy controls and 16 cats with IRIS stage 3 CKD. Chronic kidney disease diagnosis was based on persistent azotemia and clinical findings, and IRIS staging was performed exclusively according to serum creatinine concentrations (2.9-5.0 mg/dL for Stage 3). Concentrations of SDMA (≥ 26 $\mu\text{g/dL}$) were evaluated as supportive biomarkers to confirm chronic renal dysfunction but were not used for staging. Cats with prerenal azotemia were stabilized before assessment, and findings inconsistent with CKD (eg, normal SDMA with persistent isosthenuria) were excluded. Diagnostic criteria also included persistent isosthenuria (USG < 1.035) and clinical signs such as polyuria, polydipsia, weight loss, and decreased appetite. All CKD cats were clinically stable and not dehydrated at the time of staging, and IRIS classification was performed after stabilization of renal variables. Cats presented with prerenal azotemia or dehydration were initially treated using fluid therapy and re-evaluated after 72 h before inclusion. Most CKD cases were previously diagnosed within the previous 3 months and had been maintained on a commercial renal diet before study enrollment. Cats with concurrent systemic diseases (eg, hyperthyroidism, diabetes mellitus, neoplasia) or recent use of antioxidant or anti-inflammatory supplements were excluded. Before enrollment, all animals underwent comprehensive physical examination, CBC, serum biochemistry, urinalysis, and abdominal ultrasonography.

Study design and group allocation

The primary outcome of the study was the longitudinal change in serum SDMA concentrations between the MT and MT + C groups, selected because of its clinical relevance as a sensitive marker of renal function in cats with CKD. Secondary outcomes included UPC, oxidative stress markers (TAS, TOS), and inflammatory and apoptotic biomarkers (NF- κ B, Caspase-3), which were assessed to provide mechanistic context for the primary endpoint.

Twenty-four cats were assigned into 3 equal groups ($n = 8$ per group) as follows:

Control Group (Control): Clinically healthy cats without evidence of CKD, as confirmed by normal physical examination findings, normal hematologic and biochemical test results, USG > 1.035 , and normal renal ultrasonographic appearance. The control group was included to facilitate baseline physiological comparisons with the CKD groups and to contextualize disease-associated alterations in biomarker profiles.

Medical treatment group (MT): Cats diagnosed with IRIS stage 3 CKD received medical treatment only, including a renal prescription diet, fluid therapy, phosphate binders, and antihypertensive agents, without any additional supplementation.

Curcumin Group (MT + C): Cats with stage 3 CKD received the same medical treatment as the MT group, and PO curcumin supplementation (Solgar® Full Spectrum Curcumin, Istanbul, Turkey) at a dosage of 80 mg/cat/day (2 divided doses) for 60 consecutive days.⁴⁰ This formulation, designed for humans, contained micronized curcumin in liquid form (40 mg curcumin per capsule) and was administered with food to enhance absorption.

Cats were treated and monitored at a Pethome Veterinary Polyclinic. Randomization was conducted using a computer-generated sequence. No placebo was administered to the MT group, but all investigators and laboratory personnel were blinded to group allocation throughout the study.

The healthy control group was included to permit baseline cross-sectional comparisons with the CKD groups and to characterize disease-related alterations in oxidative, inflammatory, and apoptotic biomarkers at study entry, especially because standardized reference ranges for several of these biomarkers are not well established in cats. However, the healthy control group was not intended to serve as a comparator for treatment efficacy. Therefore, evaluation of therapeutic effects was based primarily on longitudinal comparisons between the MT and MT + C groups.

Treatment protocols

Throughout the study, cats in the medical treatment group (MT, $n = 8$) diagnosed with stage 3 CKD received comprehensive medical management in accordance with IRIS guidelines. All treatments were performed at a Pethome Veterinary Polyclinic. All cats were fed a renal prescription diet (Unique Renal® Diet, France) beginning on day 0. In dehydrated cats, Ringer's lactate solution was

administered IV based on body weight, followed by SC fluid administration every other day for maintenance. Metabolic acidosis was determined by measuring total carbon dioxide (TCO₂) concentrations using an automated chemistry analyzer (Fuji NX 500 I, Fujifilm Corporation, Tokyo, Japan), and cats with TCO₂ concentrations < 15 mmol/L were considered acidotic.¹ SBP was measured on days 0, 15, 30, 45, and 60 using a high-definition oscillometric device (Hasvet 838 PM, Hasvet Medical, Istanbul, Turkey). Measurements were obtained with the cat in sternal recumbency, using an appropriately sized cuff (40% of limb circumference) placed on the median portion of the forelimb. All measurements were performed in a quiet room after a 5-min acclimation period to minimize stress-induced variability. For each time point, 5 consecutive readings were recorded, and the mean value of the last 3 measurements was used for analysis. Cats with SBP ≥ 160 mmHg received telmisartan (Semintra® 4 mg/mL, Boehringer Ingelheim Vetmedica, Ingelheim, Germany) at a dosage of 2 mg/kg/day to decrease systolic pressure below 160 mmHg.^{41,42} Serum creatinine concentration and UPC ratios were evaluated on days 0, 15, 30, 45, and 60. Serum creatinine concentrations were measured using a biochemistry analyzer (Fuji NX 500 I, Fujifilm Corporation, Tokyo, Japan) and UPC ratios were determined using an automated chemistry analyzer (Catalyst One, IDEXX Laboratories, Westbrook, ME, USA). Proteinuric cats were treated with telmisartan at a dosage of 1 mg/kg/day.⁴¹ In cats with persistent hyperphosphatemia (serum phosphorus concentration > 4.6 mg/dL) despite dietary restriction, lanthanum carbonate (Antax® 500 mg, Vilsan Pharmaceuticals, Ankara, Turkey) was administered PO at a dosage of 12.5 mg/kg/day, mixed with food.⁴³ Lanthanum treatment had been initiated at least 4 weeks before study enrollment and was continued unchanged throughout the 60-day study. For hypercalcemia (serum total calcium concentration > 12 mg/dL), the renal diet was mixed 1:1 with a maintenance diet to decrease calcium intake.⁴⁴ Hypokalemic cats received potassium gluconate or potassium citrate at a dosage of 1-2 mmol/kg/day PO.²⁴ In cases of anorexia or weight loss, mirtazapine (Zestat® 15 mg, Sanofi Aventis, Istanbul, Turkey) was administered at a dosage of 1.88 mg/cat every 48 h.⁴⁵ For vomiting, maropitant citrate (Cerenia®, Zoetis Inc., Kalamazoo, MI, USA) was administered at a dosage of 1 mg/kg.⁴⁶ Cats diagnosed with anemia (hematocrit < 20%) on day 0 received darbepoetin alfa (Aranesp®, Amgen İlaç Tic. Ltd. Şti., Istanbul, Turkey) at a dosage of 0.5 µg/kg SC once weekly. All cats treated with darbepoetin also received iron supplementation (ferrous sulfate, 10 mg/kg PO q48h) to prevent iron deficiency and optimize erythropoietic response. The need for ongoing treatment was evaluated on days 15, 30, 45, and 60.⁴⁷

Cats in the MT + C Group (*n* = 8) received the same medical treatment described above, along with an PO curcumin supplementation (Solgar® Full Spectrum Curcumin, Istanbul, Turkey) at a dosage of 80 mg/cat/day (2 divided doses) for 60 consecutive days.⁴⁰ This formulation designed for humans contains micronized curcumin in liquid form (40 mg curcumin per capsule), which provides enhanced gastrointestinal absorption compared with conventional powder forms. The

supplement was administered with food to ensure co-ingestion with dietary fat, thereby improving bioavailability. Cats in the Control Group ($n = 8$) did not receive any medical treatment.

Sample collection

Blood and urine samples were collected from all cats for eligibility assessment and longitudinal follow-up. In the control group, single blood and urine samples were obtained only at baseline (day 0) to provide physiologic results for contextual comparison. In the MT and MT + C groups, samples were collected at 5 time points: day 0 (before treatment) and on days 15, 30, 45, and 60. Approximately 5 mL of blood was drawn from the jugular vein at each time point and distributed as follows: 0.5 mL into EDTA tubes for hematologic analysis, 0.5 mL into lithium-heparin tubes for plasma-based assays, and 4 mL into serum tubes for biochemical and molecular analyses. Serum was separated by centrifugation at $1006 \times g$ (equivalent to 3000 rpm) for 10 minutes and stored at -80°C until testing (Esco Lexicon® ULT Freezer, Korea). Urine samples were obtained at the same time points (days 0, 15, 30, 45, and 60) by ultrasound-guided cystocentesis using a 22G needle and 10 mL syringe.

Routine laboratory analyses

Hematologic variables including WBC, red blood cell count (RBC), hemoglobin concentration, and hematocrit were measured from EDTA-anticoagulated blood samples using a veterinary hematology analyzer (Mindray BC-500, Mindray Bio-Medical Electronics Co., Shenzhen, China). Serum biochemical variables such as creatinine, phosphorus (P), calcium (Ca), sodium (Na), potassium (K), and total carbon dioxide (TCO_2) were analyzed using a fully automated biochemistry analyzer (Fuji NX 500 I, Fujifilm Corporation, Tokyo, Japan). Serum symmetric dimethylarginine (SDMA) and urine protein-to-creatinine ratio (UPC) were measured using the IDEXX Catalyst One chemistry analyzer (IDEXX Laboratories, Westbrook, ME, USA). Urine specific gravity was assessed using a veterinary refractometer, which was calibrated with distilled water before each use. Azotemic cats with $\text{USG} > 1.035$ were classified as having prerenal azotemia.

Oxidative stress analyses

Serum samples were analyzed to assess oxidative status using commercially available colorimetric assay kits for TAS and TOS (Rel Assay Diagnostics, Gaziantep, Turkey; REF: RL0017 for TAS, REF: RL0024 for TOS). All measurements were performed in the Department of Biochemistry, Faculty of Veterinary Medicine, Atatürk University, using an ELISA microplate reader (BioTek Instruments,

Winooski, VT, USA). The TAS assay was calibrated using Trolox, and results were expressed in mmol Trolox equivalents per liter (mmol Trolox equiv/L), as previously described.⁴⁸ The TOS assay was using hydrogen peroxide, and results were expressed in mmol hydrogen peroxide equivalents per liter (mmol H₂O₂ equiv/L), as described previously.⁴⁹ All analyses were conducted in duplicate according to the manufacturer's instructions.

Inflammatory and apoptotic marker analyses

Serum concentrations of NF- κ B and caspase-3 were measured using feline-specific ELISA kits (Cat NF- κ B ELISA Kit, REF: DZE201280411; Cat Caspase-3 ELISA Kit, REF: DZE201281041; Sunred Biological Technology Co., Ltd., Shanghai, China). The assays were conducted at the Department of Biochemistry, Faculty of Veterinary Medicine, Atatürk University, using a microplate reader (BioTek Instruments, Winooski, VT, USA). All procedures were performed in strict accordance with the manufacturer's instructions, and each sample was analyzed in duplicate to ensure accuracy and reproducibility.

All ELISA kits used in the study (TAS, TOS, NF- κ B, and caspase-3) were manufactured under good laboratory practice standards and validated for veterinary or research use by their respective producers. Therefore, no additional in-house validation was required beyond standard calibration and duplicate verification. Detailed analytical characteristics, including sensitivity, assay range, and intra and inter-assay coefficients of variation, are provided in [Table S1](#).

Statistical analysis

All statistical analyses were performed using IBM SPSS Statistics version 25.0 (IBM Corp., Armonk, NY, USA). Data initially were assessed for normality using the Shapiro–Wilk test. Parametric data were expressed as mean $\pm$ SD, whereas non-parametric data were expressed as median (minimum–maximum). Baseline comparisons among the 3 groups (Control, MT, and MT + C) at day 0 were performed using one-way analysis of variance (ANOVA) for normally distributed variables, followed by Tukey's post hoc test. Non-normally distributed variables were analyzed using the Kruskal–Wallis test, and pairwise comparisons were performed using the Mann–Whitney U test with Bonferroni correction. Healthy controls were sampled only at baseline (day 0). Therefore, all comparisons involving the healthy control group were restricted to baseline analyses aimed at characterizing disease-associated differences at study entry. Longitudinal treatment effects were evaluated only in the 2 CKD groups (MT and MT + C). For these repeated measurements, longitudinal outcomes were analyzed using linear mixed effects models to account for within-cat correlation and missing observations caused by loss to follow-up. Cat identity was included as a

random intercept. Fixed effects included treatment group (MT vs MT + C), time (day 0, 15, 30, 45, 60), and the group $\times$ time interaction. Model assumptions were assessed by inspection of residual plots; if needed, outcomes were log-transformed to improve normality and homoscedasticity.

Multiplicity adjustment was applied separately for baseline cross-sectional comparisons and for longitudinal post hoc comparisons. The primary endpoint was SDMA. Secondary prespecified endpoints were UPC, TAS, TOS, NF- κ B, and caspase-3; all additional laboratory variables and SBP were treated as exploratory. Baseline comparisons involving the healthy control group were considered supportive analyses for disease characterization, whereas efficacy analyses were based on longitudinal comparisons between the MT and MT + C groups. For each outcome, post hoc pairwise comparisons across time points or between groups at specific time points in the longitudinal analyses were adjusted using Bonferroni correction within that outcome. Two-sided *P*-values $< .05$ were considered statistically significant for the primary endpoint; secondary and exploratory analyses were interpreted as hypothesis-generating.

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Conflicts of interest

Authors declare no conflicts of interest.

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Off-label antimicrobial declaration

Authors declare no off-label use of antimicrobials.

Institutional animal care and use committee or other approval declaration

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Human ethics approval declaration

Authors declare human ethics approval was not needed.

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Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Supplementary Materials

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[r1 supplementary table s1 aalag138.docx](#) (13.9KB, docx)

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RESEARCH

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N-acetylcysteine reduces serum creatinine, blood urea nitrogen, symmetric dimethylarginine and urine protein to creatinine ratio in cats with chronic kidney disease: a double-blind, placebo-controlled clinical trial

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Abstract

Background Oxidative stress is considered a significant contributing factor of chronic kidney disease (CKD). To date, there is a paucity of clinical data in the literature regarding the effect of N-Acetylcysteine (NAC) in cats with naturally developing CKD. The aim of the study is to evaluate whether the addition of NAC in the treatment of cats with acute exacerbations of CKD could improve kidney function biomarkers over the use of intravenous fluid therapy alone.

Methods A total of 50 client-owned cats were included in the study. The inclusion criteria comprised cats previously diagnosed with azotemic CKD (IRIS stage 2–4) in addition to ultrasonographic evidence of bilaterally decreased renal mass, rough surface contours, and alteration of renal cortical echogenicity. All cats were examined using standard clinical procedures, including clinical examination, blood analyses, abdominal ultrasonography, dipstick urinalysis and urine culture. Computer-generated randomisation was utilised to assign the cats into the following groups: NAC ($n:40$): N-acetylcysteine (70 mg/kg, diluted in 50 ml 0.9% saline solution, administered intravenously over a period of seven days, and a placebo group ($n:10$) 50 ml 0.9% saline solution, IV for 7 days. Blood analyses and dipstick urinalysis were repeated on the eighth day of treatment. Between-group differences in baseline age and weight were assessed using the Student's t-test, while sex distribution was evaluated with the Fisher's exact test. Treatment effects across time were analysed using a two-way mixed-design ANOVA, with "Group" and "Time" entered as fixed factors and their interaction term included in the model.

Results SDMA and creatinine concentrations decreased significantly in both groups, but the concentrations of both were significantly lower in the NAC group after treatment (Day 8 values: SDMA NAC 16.5 ± 1.21 µg/dl versus placebo

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$27 \pm 3.89 \mu\text{g/dl}$; $P=0.04$ and Creatinine NAC $4.01 \pm 0.25 \text{ mg/dl}$ versus placebo $6.44 \pm 0.9 \text{ mg/dl}$; $P < 0.001$). UPC and BUN decreased significantly in the NAC group, but no change was observed in the placebo group.

Conclusion The incorporation of NAC into treatment regimens demonstrates potential as a treatment strategy for cats with acute-on-chronic kidney disease.

Keywords Acetylcysteine, Cat, Kidney disease, Oxidative stress

Background

Chronic kidney disease (CKD) has been reported as a prevalent clinical condition in geriatric cats [1]. While the precise pathophysiology of CKD remains to be fully elucidated, the presence of cellular injury has been established, thereby initiating a process in which oxidative stress emerges as a potential pivotal factor in the progression of CKD [2]. Oxidative stress is characterised by an imbalance between prooxidative processes and antioxidant defense mechanisms. It is also the result of the production of reactive oxygen species (ROS). The deterioration of antioxidant defense mechanisms is known to result in damage to biological macromolecules, as well as the disruption of metabolism and physiology of the tissues [3]. Kidney tissue cells are characterised by an abundance of mitochondria, rendering them particularly susceptible to oxidative stress. In particular, tubular cells are characterised by their high metabolic activity and exposure to elevated levels of ROS [4]. In human patients, increased ROS production has been observed to be associated with proteinuria and the progression of CKD [5]. Furthermore, cats with CKD demonstrate the presence of oxidative stress [6]. The primary function of N-Acetylcysteine (NAC) is associated with its antioxidant and anti-inflammatory properties. NAC has been demonstrated to have a plethora of biological functions, including the scavenging of oxygen free radicals, the regulation of cell metabolism by preventing DNA damage and the regulation of gene expression [7, 8]. It is hypothesised that NAC supplementation may reduce nerve damage, most likely due to the antioxidant, anti-inflammatory, and anti-apoptotic effects of NAC [9]. The renoprotective effects of NAC in the prevention of contrast-induced nephropathy and the improvement of kidney function have been reported in human patients [10]. A number of studies have reported a decrease in serum creatinine after exposure to NAC. In experimental and human studies exploring the potential for kidney damage reduction, NAC has been observed to curtail the escalation of serum creatinine, BUN, and proteinuria [11–13]. In the field of animal studies, the administration of NAC has been documented to attenuate the effects of ischemic kidney failure in glomerular mesangial cells [14]. In rat models of kidney failure induced by the administration of NSAIDs NAC has been demonstrated to effectively alleviate the impairment of glomerular function by means

of kidney vasodilatory induction, whilst concurrently suppressing oxidative stress through the obstruction of intrarenal reactive oxygen storage [15]. A review of the literature reveals a paucity of clinical data concerning the use of NAC in cats with CKD. The objective of the present study is to evaluate whether the addition of NAC in the treatment of cats with acute exacerbations of CKD could improve kidney function biomarkers over the use of intravenous fluid therapy alone.

Materials and methods

Animals

Non-pregnant and normotensive client-owned cats were referred with signs of chronic kidney disease (CKD), including vomiting, polyuria, polydipsia, anorexia, halitosis, lethargy, and small kidneys on palpation. The inclusion criteria encompassed the evaluation of IRIS stage 2, 3, and 4 cats [16] in conjunction with confirmed ultrasonographic evidence of bilaterally decreased renal mass, rough surface contours, and alteration of renal cortical echogenicity. Cats were excluded from the study if they had pre-renal (i.e. heart disease, hepatic disorders, neoplastic conditions, acute dehydration, diabetes mellitus and hyperthyroidism) or post-renal pathologies (obstructive uropathies) affecting kidney function. Cats with renal cysts, nephrolith and pyelonephritis were excluded from the study. Cases with cystitis were also excluded due to its possible effects on UPC. Furthermore, cats diagnosed with feline infectious peritonitis, feline immunodeficiency virus or feline leukemia virus by rapid tests were excluded from the study. It is also noteworthy that none of the cats had received medical or surgical treatment at the time of referral or within the preceding three months prior to admission. The final total of 50 cats included in the study was determined based on these inclusion and exclusion criteria. The cats enrolled in the study had an IRIS classification performed in the same hospital in the stable phase of CKD. All cats were being fed commercial renal dry food before admission and had undergone regular antiparasitic therapy and routine vaccinations. Written consent was also obtained from the owners. The present study was carried out in accordance with the principles of the European Convention for the Protection of Vertebrate Animals Used for Experimental and Other Scientific Purposes and the requirements of European

Union Directive 2010/63 on the protection of animals used for experimental and other scientific purposes.

Study design

In this prospective, randomised, placebo-controlled trial, all cats were examined using a series of standard clinical procedures. These included a complete clinical examination, as well as blood analyses (CBC, glucose, creatinine, blood urea nitrogen, total protein, albumin, inorganic phosphorus, alanine aminotransferase, alkaline phosphatase, aspartate aminotransferase, symmetric dimethylarginine-SDMA, total thyroxine, sodium, potassium, calcium, chloride [IDEXX procyte hematology and catalyst one Blood Chemistry Analyzer]), abdominal ultrasonography, dipstick urinalysis, urine culture and urine protein to creatinine ratio (UPC). Thyroid hormone levels were measured in all cats included in the study, and those diagnosed with hyperthyroidism were excluded. The mean of five to seven consecutive consistent indirect blood pressure measurements was obtained with an oscillometric device (Vet-Dop2, Veterinary Doppler blood pressure system, Vmed technology) [17]. Computer-generated randomisation was utilised to assign the cats into the following groups: NAC ($n:40$): N-acetylcysteine (70 mg/kg, diluted in 50 ml 0.9% saline solution, administered intravenously over a period of 7 days [18, 19], and a placebo group ($n:10$) 50 ml 0.9% saline solution, IV for 7 days). Fluid therapy with ringer solution was also provided intravenously to all cats, with the dosage calculated based on the patient's body weight (kg) and their dehydration status (as a decimal), as determined by physical examination parameters [20]. All cats in study had a dehydration status as 6%–8% [21]. All therapeutic interventions were administered twice daily for a period of one week at the hospital. All cats in groups received no other medications during the study, despite the study protocol allowing the administration of ancillary medications. A 7-day course of treatment was deemed adequate for cats that refused voluntary food intake and showed standardized dehydration levels of 6–8%. Water and dry food were offered ad libitum to all patients throughout the hospitalization period but none of the cats consumed them voluntarily. Force oral feeding was not also preferred due to risk of inducing vomiting. Blood analyses were repeated on the eighth day of treatment. The treatment protocol utilised in the groups was presented in a uniform physical manner to both researchers and owners. Treatment was also continued for CKD cats that still required treatment after seven days. Subsequent to the 7-day treatment period, a special prescription dry or wet kidney diet were continued in all CKD cats that commenced oral intake.

Statistical analysis

The descriptive statistics for each variable were calculated and presented as “Mean $\pm$ Standard Error of Mean”. Prior to conducting the statistical analysis, the data underwent scrutiny using the Shapiro-Wilk test for normality and the Levene test for homogeneity of variances, as these are fundamental to the validity of parametric test assumptions. The Student t-test was employed to evaluate the differences between the study groups with respect to age and weight. The Fisher exact test was used to assess the frequency distribution of sex among study groups. The collected data were subjected to two-factor mixed-design ANOVA (analysis of variance) using General Linear Model procedure for repeated measurements. The model incorporated “Group” and “Time” as the primary effects, along with their two-way interaction terms. Post hoc testing was performed using simple effect analysis with Bonferroni adjustment. To align scientific rigor with clinical responsibility, we implemented a computer-generated randomization scheme that allowed for unbalanced allocation. Unbalanced randomization is accepted in both human and veterinary clinical trials, particularly when greater information is needed on the efficacy or safety of the new treatment, or ethical/practical constraints limit placebo allocation [22, 23]. From a statistical perspective, our design is supported by power analysis. Based on using Cohen's $f = 0.25$, α (type 1 error rate) = 0.05 and 80% power in a two way mixed design ANOVA, the required total sample size was approximately 34 subjects [24]. A probability value of less than 0.05 was considered to be significant, unless otherwise indicated. The statistical analysis was conducted using SPSS 21.

Results

A total 50 of 58 client-owned cats were included in the study, comprising Chinchila ($n:1$), domestic short hair ($n:19$), Persian ($n:5$), Ankara cat ($n:4$), Siamese ($n:9$), Scottish fold cat ($n:7$), British short hair ($n:3$), Himalayan ($n:1$) and Van cat ($n:1$). Eight cats were excluded from the study for the following reasons: previous drug therapy ($n:3$), concurrent renal mass ($n:2$), renal cysts ($n:1$) and renal calculi ($n:2$). The mean age, weight and sex distributions of cats in the groups are shown in Table 1. Prior to treatment, serum creatinine, SDMA, BUN and UPC levels were higher in the groups (Table 2). SDMA and creatinine concentrations decreased significantly in both groups, but the concentrations of both were significantly lower in the NAC group after treatment (Day 8 values: SDMA NAC 16.5 ± 1.21 $\mu\text{g/dl}$ versus placebo 27 ± 3.89 $\mu\text{g/dl}$; $P=0.04$ and Creatinine NAC 4.01 ± 0.25 mg/dl versus placebo 6.44 ± 0.9 mg/dl ; $P<0.001$). UPC and BUN decreased significantly in the NAC group, but no change was observed in the placebo group (Table 2). Furthermore, the tables present the mean values of RBC, HCT,

Table 1 Characteristics of cats in groups

	NAC Group (n:40)	Placebo Group (n:10)	P
Age (years)	10.74 ± 0.53	11.5 ± 0.75	0,501
Weight (kg)	3.97 ± 0.14	4.21 ± 0.25	0,428
Sex (n, %)	♀: 24 (60) ♂: 16 (40)	♀: 6 (60) ♂: 4 (40)	
Breeds (n)	Chinchila (n:1) Domestic short hair (n:17) Persian (n:4) Ankara cat (n:2) Siamese (n:7) Scottish fold cat (n:5) Himalayan (n:1) British short hair (n:2) Van cat (n:1)	Domestic short hair (n:2), Persian (n:1), Ankara cat (n:2), Siamese (n:2), Scottish fold cat (n:2), British short hair (n:1)	
IRIS Stage 2	9	3	
IRIS Stage 3	18	5	
IRIS Stage 4	13	2	

NAC N-acetylcysteine, IRIS International Renal Interest Society

HGB, PHOS, TP and ALB. A comparative dipstick urinalysis of urine gravity between the NAC and control groups on day 8 revealed a range from 1.026 to 1.029 and from 1.024 to 1.032, respectively, with no significant relationship observed ($P > 0.05$). Culture results revealed no pathogens in all cats. All cats survived to discharge from

the hospital. Discharge times ranged from 8 to 12 days for the NAC group and 9–14 days for the placebo group. No adverse effects associated with NAC administration were also observed.

Discussion

The antioxidative and anti-inflammatory properties of NAC render it a promising therapeutic agent against the destructive effects of oxidative stress. Indeed, it is one of the most widely used antioxidants in human medicine today [25]. Numerous studies have demonstrated the efficacy of NAC in the treatment of a wide range of diseases, including diabetes, obesity, cancer, cardiovascular diseases, cardiovascular problems and autoimmunity [26]. A substantial number of clinical research articles and studies have emphasised the beneficial therapeutic effects of NAC in the veterinary field, including the following: investigation of the pharmacokinetic effects of NAC in cats and chickens [27, 28], protective effects in dogs exposed to acetaminophen toxicity [29], healing of infectious keratitis wounds in cats [30], proven effects of NAC on strengthening the intestinal mucosal barrier in experimental intestinal diseases of pigs [31], optimal therapeutic effects in mastitis and metritis of cows [32],

Table 2 Blood analyses in groups

	Group	t0		t8		P	Group (G)	Time (T)	G x T
		Mean ± SEM	SD	Mean ± SEM	SD				
RBC	NAC Group	8.16 ± 0.23 ^{a, A}	1,46	6.68 ± 0.24 ^{b, B}	1,54	0,004	< 0.001	< 0.001	0,435
	Placebo Group	9.75 ± 0.48 ^{b, A}	1,53	7.81 ± 0.56 ^{a, B}	1,78				
HCT	NAC Group	34.34 ± 1.13 ^{b, A}	7,12	27.58 ± 1.01 ^{b, B}	6,39	0,002	< 0.001	< 0.001	0,349
	Placebo Group	41.53 ± 2.21 ^{a, A}	6,99	31.78 ± 2.51 ^{a, B}	7,95				
HGB	NAC Group	11.44 ± 0.31 ^{b, A}	1,97	9.94 ± 0.36 ^{b, B}	2,29	0,027	< 0.001	< 0.001	0,442
	Placebo Group	13.11 ± 0.56 ^{a, A}	1,78	10.83 ± 0.74 ^{a, B}	2,35				
SDMA	NAC Group	30.65 ± 2.48 ^{a, A}	15,7	16.5 ± 1.21 ^{b, B}	7,66	0,123	< 0.001	< 0.001	0,04
	Placebo Group	31.9 ± 3.6 ^{a, A}	11,39	27 ± 3.89 ^{a, B}	15,46				
BUN	NAC Group	103.5 ± 4.79 ^{a, A}	30,3	51.45 ± 3.93 ^{b, B}	24,83	0,018	< 0.001	< 0.001	0,001
	Placebo Group	104.6 ± 8.33 ^{a, A}	26,33	91.4 ± 12.17 ^{a, A}	38,49				
CREA*	NAC Group	8.04 ± 0.35 ^{a, A}	2,19	4.01 ± 0.25 ^{b, B}	1,55	0,119	< 0.001	< 0.001	< 0.001
	Placebo Group	7.59 ± 0.58 ^{a, A}	1,82	6.44 ± 0.9 ^{a, B}	2,83				
PHOS	NAC Group	9.62 ± 0.6 ^A	3,77	6.28 ± 0.31 ^B	1,99	0,945	< 0.001	< 0.001	0,399
	Placebo Group	8.94 ± 0.98 ^A	3,11	6.85 ± 0.62 ^B	1,95				
TP	NAC Group	8.16 ± 0.15 ^A	0,94	7.74 ± 0.09 ^B	0,58	0,119	0,005	0,005	0,921
	Placebo Group	8.54 ± 0.25 ^A	0,79	8.08 ± 0.2 ^B	0,63				
ALB	NAC Group	3.11 ± 0.07	0,43	2.99 ± 0.05	0,35	0,864	0,218	0,218	0,51
	Placebo Group	2.87 ± 0.06	0,2	3.01 ± 0.11	0,34				
UPC	NAC Group	0.96 ± 0.07 ^{b, A}	0,41	0.52 ± 0.04 ^{b, B}	0,25	< 0.001	< 0.001	< 0.001	0,004
	Placebo Group	1.44 ± 0.18 ^{a, A}	0,57	1.33 ± 0.18 ^{a, A}	0,55				

RBC Red blood cell (10⁹/l), HCT Hematocrit (%), HGB Hemoglobine (g/dl), SDMA Symmetric Dimethylarginine (μg/dl), CREA Creatinine (mg/dl), BUN Blood urea nitrogen (mg/dl), PHOS phosphorus (mg/dl), TP Total protein (g/dl), ALB Albumine (g/dl), UPC Urine protein/creatinine ratio. *All cats in groups have >= 1.8 mg/dl creatinine levels

SEM Standard error of mean, SD Standard deviation

t0 Before treatment, t8 after treatment

^{a, b} Different letters in the same column for each variable indicate a statistically significant difference ($p < 0.05$)^{A, B} Different letters in the same row for each variable indicate a statistically significant difference ($p < 0.05$)

helping to stabilize glucose levels in diabetic cats [13], helping to increase the total leukocyte count in parvovirus-infected dogs [33], and the ability of NAC to disrupt biofilm formation in dogs with otitis media as positive therapeutic effects [34]. Notwithstanding the plethora of studies conducted in the domain of veterinary medicine, the preponderance of research assessing the effects of NAC on kidney function has been confined to experimental animal models [35–37]. To the best of the author's knowledge, there have been no clinical studies conducted to evaluate the renoprotective effect of NAC in cats with naturally occurring chronic kidney disease. The objective of the present study was to evaluate the therapeutic effects of NAC in cats with acute-on-chronic kidney disease.

A significant corpus of human research has clearly demonstrated the relationship between chronic kidney dysfunction and oxidative stress and lipid peroxidation [38, 39]. In a study of rats that had undergone nephrectomy, a reduction in dietary antioxidant levels was found to accelerate the progression of kidney disease [40]. The precise mechanisms by which oxidative stress contributes to the development of chronic kidney dysfunction remain to be fully elucidated. However, an experimental study has demonstrated that levels of malondialdehyde (MDA) and lipid peroxidation products have been observed to increase in rats following 5/6 nephrectomy [1]. The administration of NAC has been demonstrated to reduce MDA levels and improve ischaemic kidney injury in haemodialysis patients [41]. It has been demonstrated that NAC administration engenders a renal protective effect by increasing glomerular filtration rate (GFR) and reducing proteinuria [42]. In addition, NAC application in rats has been shown to cause a progressive increase in GFR [42]. As demonstrated in the research, SDMA reflects early GFR declines before creatinine increases in cats with chronic kidney disease [43]. In this study, it was observed that the decrease in SDMA in cats with chronic kidney disease treated with NAC was statistically more significant when compared to the placebo group. This finding suggests that the addition of NAC to the intravenous fluid treatment resulted in a bigger improvement in GFR biomarkers than fluid therapy alone. Conversely, a study conducted on humans who naturally developed chronic kidney dysfunction found that NAC use reduced the likelihood of developing end-stage kidney disease by 18% [12]. The administration of NAC to patients suffering from chronic kidney dysfunction was conducted both prior to and following exposure to radiocontrast material. This resulted in a 19% reduction in contrast-induced nephropathy, as evidenced by a decline in serum creatinine levels [14]. Furthermore, NAC administration was found to decrease creatinine and BUN levels, in addition to proteinuria [44]. These data are consistent with

the findings of the present study. In this study, statistically significant decreases were observed in BUN, creatinine and UPC values in patients who received NAC compared to those who received a placebo. Despite the absence of a comprehensive understanding of the pathophysiological mechanisms through which NAC prevents the progression of chronic kidney dysfunction by reducing serum creatinine and BUN levels, it is hypothesised that the renoprotective property of NAC is related to the increase in glutathione synthesis capacity [45]. NAC has been demonstrated to offer a degree of protection to kidney tubules, in part due to its antioxidant properties, and has been observed to reduce kidney cell apoptosis and support cell repair [46]. In certain studies, NAC has been shown to specifically prevent mitochondrial dysfunction in cases of acute kidney injury, with other studies demonstrating its capacity to reduce mitochondrial damage caused by oxidative stress [47]. It can be extrapolated from the evidence presented that NAC administration may be beneficial in cats with acute kidney failure or acute exacerbations of chronic kidney disease [48]. Regulation of inflammatory and mitochondrial processes by NAC has also been demonstrated to be effective in preventing kidney fibrosis [13]. It is imperative to acknowledge the pivotal role of chronic inflammatory reaction in the progression of CKD. A study has demonstrated that NAC is capable of reducing inflammatory cell infiltration and pathological damage to kidney tissue by significantly decreasing IL6 [49]. The kidney-protective effect of NAC use in rats with renal tissue damage has also been proven through histopathological analysis [35]. The current literature reveals that antioxidant capacity is reduced in felines afflicted with chronic kidney disease when contrasted with those of healthy animals [32, 50]. A study found measuring oxidative damage markers like MDA more sensitive than other plasma markers for evaluating patients with chronic kidney dysfunction [3]. The determination of oxidative stress markers and antioxidant capacity in cats with chronic kidney disease was not among the objectives of the present study. Consistent with prior reports [51, 52], the provision of nutritional support is important in the management of cats with CKD and should be addressed in future prospective studies to ensure that cats receive adequate enteral nutritional whilst hospitalized. The absence of kidney histopathology and the inclusion of cats at different stages of chronic kidney disease in the study can be considered as limiting factors. Larger number of cats with CKD could be useful to further evaluation.

Conclusion

In conclusion, the findings of this study demonstrate that NAC was well tolerated and led to a significant reduction in creatinine, BUN, SDMA, and proteinuria within one

week compared to the placebo group. The incorporation of NAC into treatment regimens demonstrates potential as a treatment strategy in cats with acute-on-chronic kidney disease.

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Authors' contributions

Hadi Alihosseini: Conceptualization (lead); data curation (equal); investigation (equal); methodology (lead); writing – original draft (lead), validation (equal) Ekrem Çağatay Çolakoğlu: Conceptualization (lead); data curation (equal); investigation (equal); methodology (lead); writing – original draft (lead), validation (equal) Ali Evren Haydardeğözü: Data curation (equal); investigation (equal); methodology (lead); validation (equal). Doğan Özgen: Writing– review & editing, Methodology, Investigation, Formal analysis.

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Data availability

The datasets generated and/or analysed during the current study are not publicly available but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

Informed consent was obtained from all owners included in the study. The present study was carried out in accordance with the principles of the European Convention for the Protection of Vertebrate Animals Used for Experimental and Other Scientific Purposes and the requirements of European Union Directive 2010/63 on the protection of animals used for experimental and other scientific purposes. The authors confirm that the ethical policies of the journal, as noted on the journal's author guidelines page, have been adhered to and the appropriate ethical review committee approval has been received. The authors confirm that they have adhered to either US or European standards for the protection of animals used for scientific purposes.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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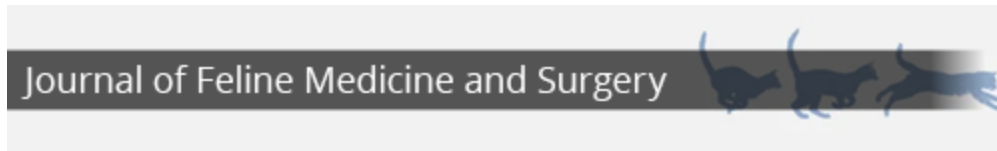
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Evaluation of the effect of Renaltec in cats with IRIS stage 2 and 3 chronic kidney disease

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Abstract

Objectives

Chronic kidney disease (CKD) is associated with accumulation of uremic toxins such as indoxyl sulfate (IS) in blood and tubular epithelial cells, which contribute to the progression of CKD. The aim of this prospective, randomised, two-centre clinical pilot study was to evaluate the effect of oral spherical carbonaceous adsorbent Renaltec (Porus One) on IS plasma concentration, progression of CKD, parameters of calcium/phosphate homeostasis (total calcium [tCa], phosphorus, fibroblast growth factor 23 [FGF-23]), urine protein:creatinine ratio (UPC) and faecal dysbiosis index (FDI) in cats with stable CKD International Renal Interest Society (IRIS) stage 2 and 3 over a 6-month period.

Methods

A total of 19 cats with CKD were randomly assigned to the Renaltec group (n = 10, eight stage 2, two stage 3) or control group (n = 9, eight stage 2, one stage 3). All cats received standard of care treatment according to IRIS guidelines. Cats in the Renaltec group received an additional 500 mg q24h of Renaltec administered mixed with food or a liquid cat treat. All cats were examined on day 0 (t0), after 3 months (t1) and after 6 months (t2). Statistical analysis was performed using generalised linear mixed models.

Results

At t0, the Renaltec group cats had significantly lower tCa ($P = 0.026$) than the control group cats. At t1, there were no differences between the groups, while at t2, the Renaltec group cats had significantly lower IS ($P = 0.045$) and UPC ($P = 0.001$) than cats in the control group. Progression of CKD was noted in two cats from each group. The Renaltec group cats had a significantly lower FDI at t2 compared with t1 ($P = 0.023$) and t0 ($P = 0.009$).

Conclusions and relevance

Administration of Renaltec over 6 months was associated with the mitigation of parameters linked to CKD progression. Given the pilot nature of the study, these findings should be interpreted with caution and warrant further investigation.

Keywords: Chronic kidney disease, indoxyl sulfate, proteinuria, faecal dysbiosis index, Renaltec

Plain language summary

Study on the effect of Renaltec in cats with chronic kidney disease

Chronic kidney disease (CKD) is a common condition in older cats. It causes waste products such as indoxyl sulfate (IS) to rise in the blood. These substances can further damage the kidneys and accelerate the progression of the disease. The aim of this study was to find out whether the dietary supplement Renaltec (Porus One) can lower IS levels and whether it has other positive effects on the health of cats with CKD. A total of 19 cats with stable stage 2 or 3 CKD were randomly divided into two groups. All cats received the usual treatment in accordance with international guidelines. Ten cats were also given Renaltec once a day, while the other nine were not. The animals were examined at the start of the study, after 3 months and after 6 months. Various blood and urine values were measured, and the state of gut bacteria was examined. After 6 months, the cats that

received Renaltec had lower levels of IS in their blood and less protein in their urine than the cats that did not receive Renaltec. Both are important factors that can influence the progression of the disease. The state of the intestinal bacteria also improved in the Renaltec group. In two cats in each group, the kidney disease worsened despite treatment. The results show that Renaltec could help to slow the progression of CKD in cats. However, as this was only a small pilot study, further research with more cats is needed to confirm the results.

Introduction

In recent years, there has been increasing interest in the role of uremic toxins, particularly of indoxyl sulfate (IS), in the progression of chronic kidney disease (CKD). IS belongs to the group of protein-bound indole compounds that are produced by intestinal microbes during the degradation of dietary L-tryptophan to indole, which is absorbed by the gut,^{[1,2](#)} metabolised to IS by the liver and excreted in the urine.^{[3-5](#)}

IS accumulates with increasing severity of CKD in cats.^{[6-9](#)} The accumulation of IS in renal tubular cells leads to inflammation and damage to these cells in vitro.^{[10](#)} In rats, IS accumulation disrupts the functionality of antioxidative defence mechanisms^{[11](#)} and leads to suppressed Klotho expression through the production of reactive oxygen species and activation of nuclear factor-kappa B.^{[12](#)} It accelerates glomerular sclerosis^{[13,14](#)} and is associated with renal fibrosis.^{[14,15](#)} IS induces intestinal epithelial cell damage in mice by disrupting mitophagy flow, leading to intestinal barrier injury.^{[16](#)}

In humans with CKD, increased serum IS concentrations activate immune cells, raise inflammatory markers and indicate oxidative stress.^{[17](#)} In these patients, elevated IS blood levels are associated with heart disease, including aortic calcification, vascular stiffness,^{[18](#)} endothelial and vascular smooth muscle cell dysfunction,^{[18,19](#)} coronary atherosclerosis^{[20](#)} and left ventricular diastolic dysfunction,^{[21](#)} all leading to a higher mortality risk.

Knowledge of these negative effects of IS led to experimental and clinical studies on the carbon-based intestinal adsorber AST-120 and its impact on plasma IS concentration and progression of CKD. In rats, oral administration of AST-120 for 10 weeks decreased IS plasma concentration, which was associated with a reduction in signs of oxidative stress.^{[22](#)} In mice, AST-120 reduced IS-induced renal fibrosis.^{[23](#)} In rats, a decrease in urine protein:creatinine ratio (UPC) was observed after treatment with AST-120 over 12–19 weeks, suggesting improved renal function.^{[14,15](#)} In humans with CKD, AST-120 significantly reduced IS blood concentrations after 12 weeks^{[24](#)} and lowered creatinine levels in a 1-year study compared with a control group.^{[25](#)}

Renaltec (Porus One; Dechra Veterinary Products) is a selective intestinal adsorber for precursors of uremic toxins developed for cats with CKD. Similar to AST-120, it is carbon based and consists of small spherical particles (0.1–0.3 mm) with micropores that enhance the molecule's adsorption capacity. A preliminary study showed that Renaltec reduces IS blood concentrations by 50% in geriatric cats.²⁶ A recent study also demonstrated that Renaltac lowered IS in cats with experimentally induced CKD.²⁷

Therefore, the aims of this prospective, randomised pilot study were to evaluate the effect of Renaltec in cats with International Renal Interest Society (IRIS) stage 2 and 3 CKD on plasma IS concentrations and disease progression. Furthermore, the effect on clinical parameters, serum electrolytes, fibroblast growth factor 23 (FGF-23) concentrations and the faecal dysbiosis index (FDI) was evaluated. It was hypothesised that Renaltec would lower IS plasma concentrations and positively influence parameters associated with CKD progression.

Materials and methods

Study design

The prospective, randomised, two-centre clinical pilot study included 19 cats with IRIS stage 2 and 3 CKD. All cats received standard-of-care treatment according to IRIS guidelines. Before inclusion, cats were randomly allocated to the Renaltec group or the control group. Cats in the Renaltec group received an additional 500 mg q24h of Renaltec with Add One (a liquid treat, Porus One; Dechra Veterinary Products) directly before or with their food. If further medication was indicated, an interval of 2 h was maintained between the administration of the drug and Renaltec. Between March 2019 and January 2023, 17 cats were treated at the LMU Small Animal Clinic, Ludwig Maximilian University Munich and two cats were treated at Tierklinik Oberhaching.

The study was approved by the ethical committee of the Centre for Clinical Veterinary Medicine, Ludwig Maximilian University Munich (reference number 154-14-01-2019). The study was conducted in accordance with the German Animal Welfare Act. All owners gave their written consent to participate before enrolling the cats in the study.

Patients

CKD was diagnosed based on history, clinical signs and persistent elevations of creatinine and symmetric dimethylarginine (SDMA) combined with a urine specific gravity (USG) <1035 at two

time points at least 3 weeks apart. Sonographic assessment of the urinary tract was performed to exclude postrenal azotaemia due to upper urinary tract obstruction. CKD staging was performed according to IRIS guidelines. Exclusion criteria included proteinuria (UPC >0.4), hypercalcaemia, steroid treatment or uncontrolled systemic extrarenal diseases (chronic intestinal diseases, hepatic disease, pancreatic disease, diabetes mellitus or cancer). Screening for these comorbidities was based on detailed medical history, physical examination and laboratory investigations (complete blood count [CBC], serum chemistry, total thyroxine in cats aged >7 years). A complete abdominal ultrasound was performed in all cats with gastrointestinal signs or laboratory abnormalities that could not be explained with CKD. Cats receiving treatment for CKD-associated hypertension (amlodipine) were also included as well as cats with well-controlled hyperthyroidism. All cats received a commercial renal diet or a renal diet calculated by a veterinary nutritionist throughout the study period.

The following investigations were performed at the time of inclusion into the study (t0), after 3 months (t1) and after 6 months (t2): physical examination; systolic blood pressure measurement; CBC; serum chemistry profile; SDMA; urinalysis including USG, dipstick, urine sediment and UPC; and, if indicated, an aerobic bacterial culture (performed in cases of active sediment). EDTA plasma samples were collected at each time point for the measurement of plasma IS concentrations, and faecal samples were collected to determine the FDI. At t0, cats were randomly allocated to the Renaltec group or the control group.

Diet

Cats included in the study had been on a kidney diet for at least 3 weeks before t0. Cats were fed commercial renal diets (Hill's Prescription Diet k/d wet and dry; Royal Canin early renal dry and renal wet and dry) or home-cooked meals with receipts calculated by Diet Check Munich or by nutritional specialists from the nutritional consulting service at the LMU Small Animal Clinic, Munich. At the beginning of the study, owners reported the types of food their cats were eating. A combination of dry and wet food was permitted according to owner preference, provided that all diets met renal diet criteria. Nutrient intake based on each cat's body weight was calculated at the start of the study. On average, cats consumed 60 mg/kg body weight phosphorus (244.36 mg/MJ metabolisable energy [ME], 102 mg/100 kcal ME) with an average crude protein content of 3.99 g/kg body weight (16.14 g/MJ ME, 6.75 g/100 kcal ME) per day.

Assessment of treatment effect

Parameters to evaluate the treatment effect were the plasma IS concentration and the development of renal function assessed by serum creatinine and SDMA as well as phosphorus, total calcium (tCa) and potassium concentrations after 3 and 6 months. CKD progression was defined as a change to a higher IRIS stage and/or an increase in creatinine of more than 25% from baseline.

CBC and routine serum chemistry

Blood samples were collected from fasted cats (≥ 12 h) by puncture of a peripheral vein. K-EDTA tubes were used for blood count analysis, processed immediately and analysed using fluorescence flow cytometry (Sysmex XT-2001; Sysmex). The remaining EDTA blood was processed for IS analysis.

Whole-blood samples were collected in serum tubes, centrifuged (5 mins at 2540 *g*) and analysed in house. Sodium, potassium and chloride were measured with the ion-selective electrode module, while the remaining serum parameters were analysed using absorbance and photometry (Cobas Integra 400 Plus; Roche). SDMA was measured by an external commercial laboratory (IDEXX Kornwestheim); serum was sent by courier on the day of sampling.

Urinalysis

At t0, urine was collected by ultrasound-guided cystocentesis. At t1 and t2, urine samples were collected either by cystocentesis or by the owners at home using non-absorbent litter. USG was measured using refractometry. A dipstick test and a sediment examination were performed (SediVue DX; IDEXX). Samples with questionable or abnormal findings were also evaluated using light microscopy. If infection was suspected (in the presence of active sediment), an aerobic bacterial culture was obtained. UPC was measured by Cobas Integra 400 Plus (Roche). UPC was only performed in samples with an inactive sediment.

Plasma sample preparation for IS

Whole-blood samples were collected in K2-EDTA tubes and processed within 2 h. Samples were centrifuged for 5 mins at 630 *g*, and plasma was transferred to Eppendorf tubes and stored at -80°C until measurement. IS was measured using an API5500 liquid chromatography with tandem mass spectrometry system as described in previous studies.⁹

Serum preparation for FGF-23

Serum samples for FGF-23 analysis were stored at -80°C until measurement. The measurement was performed by a commercial laboratory (LABOKLIN, Bad Kissingen) using a previously validated quantitative ELISA assay kit.²⁸

Faecal sample preparation for FDI

Owners collected fresh faecal samples at all time points and kept them refrigerated for up to 12 h before storage at -80°C in the hospital. Samples were shipped frozen on dry ice to the Gastrointestinal Laboratory, Department of Small Animal Clinical Sciences, College of Veterinary Medicine and Biomedical Sciences (Texas A&M University, College Station, TX, USA), as described in previous studies.²⁹ One cat that had received antibiotics before enrolment was excluded from the FDI analysis. In addition, two cats that received antibiotics from t1 onwards and one cat from t2 onwards were excluded from FDI evaluation from the respective time point and all subsequent sampling time points. None of the cats included in the FDI analysis at any time point received medications known to affect the intestinal microbiota, such as antibiotics, proton pump inhibitors or H_2 receptor antagonists, during the study period.

Statistical analysis

A power analysis was conducted based on the previously published mean ($\pm\text{SD}$) plasma IS concentration of 15 ± 5 mg/l in cats with IRIS stage 2 and 3 CKD⁶ and the assumption of achieving a 50% reduction of plasma IS within 6 months of Renaltec administration, as demonstrated in an earlier study of geriatric cats.²⁶ For a power of 80% and a P value of 0.05, at least eight cats per group were required. Consequently, 10 cats were included in the Renaltec group and nine cats in the control group. Based on these assumptions, the minimum detectable effect for the study was an absolute difference of 7.5 mg/l in mean plasma IS concentration between groups, corresponding to a large effect size (Cohen's $d \approx 1.5$). The study was therefore powered to detect substantial treatment-related reductions in IS concentration.

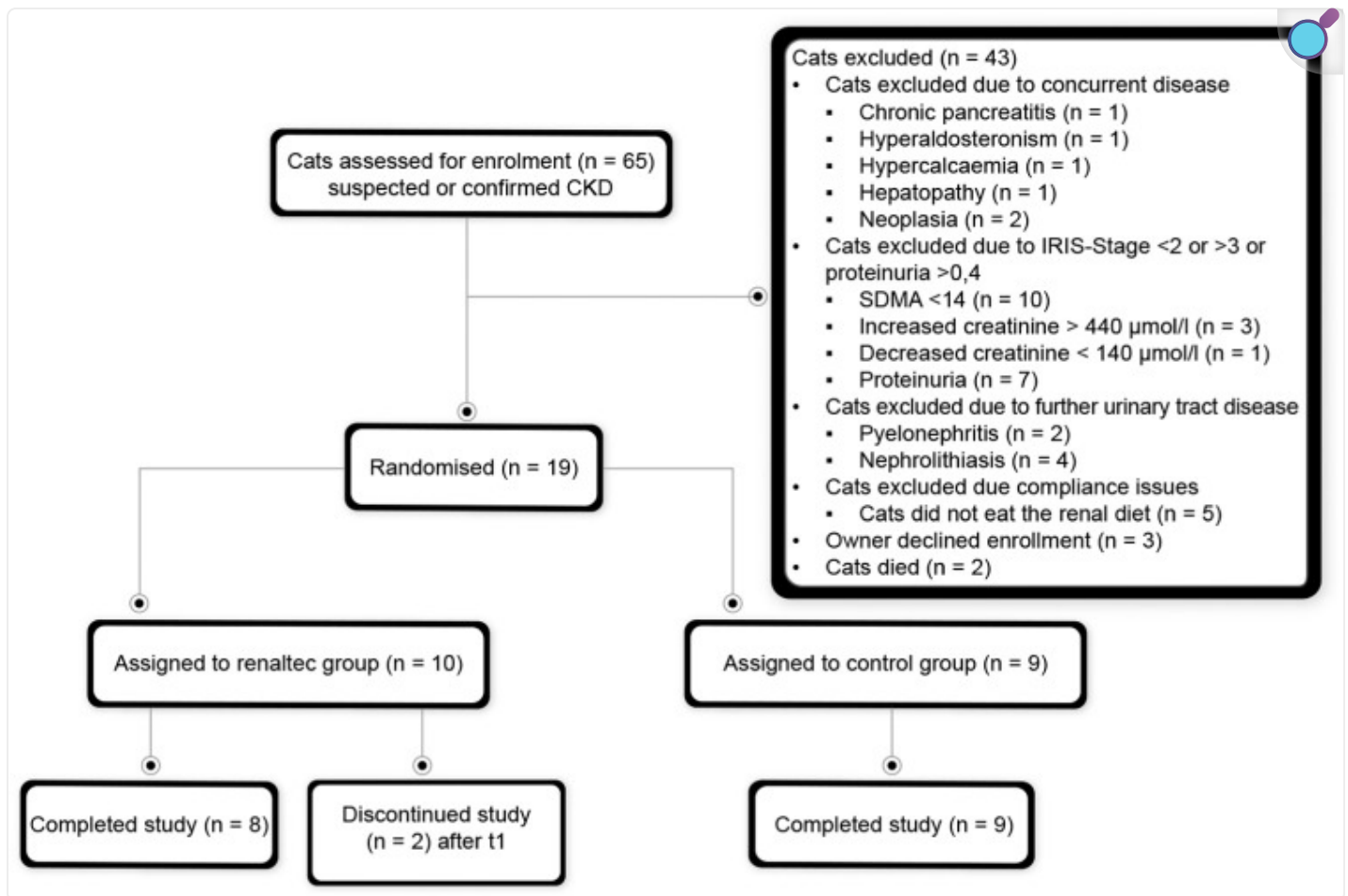
Collected data were investigated using the statistical software R version 2023.06.1. Descriptive statistics included the calculation of the continuous variables as mean, SD and median with interquartile range and the categorical variables as counts and percentages. The normal distribution of continuous variables was assessed using the Shapiro–Wilk test. The difference between body weight in the Renaltec and control groups was evaluated using the Mann–Whitney U-test, while the difference in age was evaluated using the Student's t -test. As a result of the presence of repeated measures among the time points, generalised linear mixed-effects models with individual animal ID as a random effect were chosen for analysis. The following model

assumptions were always checked: (1) the normality of residuals was checked using the Shapiro–Wilk normality test; (2) the homogeneity of variances between groups was checked using the Bartlett test; and (3) the heteroscedasticity (constancy of error variance) was checked using the Breusch–Pagan test. In case assumptions were satisfied, generalised linear mixed-effects models were used (R package – lmer). In case assumptions were violated, robust linear mixed-effects models were applied (R package – robustlmm). The robust model was used for IS, creatinine, urea, total protein, phosphate, calcium, leukocytes, haematocrit, USG, UPC, FDI, FGF-23 and body weight. The generalised linear mixed model was used for SDMA, albumin, potassium, sodium chloride, sodium and blood pressure. $P < 0.05$ was considered significant.

Results

Of the 65 cats with clinical signs compatible with CKD and assessed for enrolment, 19 with stable CKD were eligible for inclusion and randomly assigned to the Renaltec group or control group ([Figure 1](#)). The Renaltec group included 10 cats (eight at IRIS stage 2 and two at stage 3), while the control group included nine cats (eight at IRIS stage 2 and one at stage 3).

Figure 1.



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Flowchart showing the inclusion and exclusion of cats in the study. CKD = chronic kidney disease; IRIS = International Renal Interest Society; SDMA = symmetric dimethylarginine

The signalment of the cats is illustrated in [Table 1](#). The Renaltec group included significantly more castrated male cats than the control group ($P = 0.013$). There were no other significant differences between the groups.

Table 1.

Signalment of the study cohort (age, breed, sex, body weight)

Characteristics	Renaltec group (n = 10)	Control group (n = 9)	P value
Age (years)	11.90 ± 3.69	14.22 ± 2.91	0.149
Breed			
Domestic shorthair	5	4	–
Sacred Birman	4	–	–
Other breeds	1 [*]	5 [†]	–
Sex			0.013
Castrated male	7	1	
Spayed female	3	8	
	n = 10	n = 8	–
Body weight (kg)	4.75 (0.71)	4.06 (0.96)	0.146

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Data are n, mean ± SD or median (interquartile range). Differences in age between groups were evaluated using Student's *t*-test, as the assumptions of normal distribution and homogeneity of variances were met. Differences in body weight were assessed using the Mann–Whitney U-test, as the assumption of normality was not met. *P* values for group comparisons are reported. Values in bold indicate statistically significant differences (*P* < 0.05)

* One Chartreux

† One Chartreux, one Tonkinese, one Burmese, one Ragdoll, one Thai-mixed breed

At t0, laboratory results showed tCa concentrations within the reference interval for all cats; however, the cats in the Renaltec group had significantly lower tCa than those in the control group (*P* = 0.026) ([Table 2](#)). USG was significantly higher in the Renaltec group (*P* = 0.012). None of the cats in either group had an active urine sediment and none were proteinuric. One cat in the

Renaltec group and two cats in the control group were in the borderline proteinuric range. No other significant differences were observed in CBC, serum chemistry, FGF-23, FDI or IS at t0.

Table 2.

Results of serum chemistry, haematology, urine analysis and faecal dysbiosis index (FDI) of the cats examined at t0 (baseline), t1 (after 3 months) and t2 (after 6 months), analysed using mixed effects models with group, time point and their interaction as fixed effects and individual cat ID as random effect

Parameter	Time point	Renaltec group	Control group	P value
Indoxyl sulfate (µg/ml)	t0	2.32 (1.21–3.43)	2.49 (1.39–3.60)	0.833
	t1	2.21 (1.10–3.32)	2.95 (1.85–4.06)	0.352
	t2	1.77 (0.61–2.93)	3.41 (2.30–4.52)	0.045
SDMA (µg/dl)	t0	17.60 (14.50–20.60)	18.70 (15.70–21.60)	0.601
	t1	16.10 (13.20–19.00)	17.80 (14.80–20.70)	0.418
	t2	16.80 (13.70–19.90)	17.10 (14.10–20.10)	0.906
Creatinine (µmol/l)	t0	200.00 (173.00–226.00)	213.00 (186.00–239.00)	0.498
	t1	194.00 (168.00–221.00)	215.00 (189.00–242.00)	0.277
	t2	184.00 (156.00–212.00)	213.00 (186.00–240.00)	0.137
Urea (mmol/l)	t0	11.60 (8.85–14.30)	13.70 (11.02–16.30)	0.280
	t1	11.50 (8.75–14.20)	15.20 (12.59–17.90)	0.052
	t2	11.90 (9.12–14.70)	14.50 (11.80–17.10)	0.195
Total protein (g/l)	t0	72.30 (69.00–75.60)	70.50 (67.30–73.60)	0.426
	t1	71.40 (68.20–74.70)	71.00 (67.90–74.20)	0.870
	t2	72.50 (69.20–75.90)	71.20 (68.00–74.40)	0.569
Albumin (g/l)	t0	35.50 (33.40–37.50)	38.10 (36.00–40.10)	0.075
	t1	35.50 (33.40–37.50)	37.10 (35.10–39.10)	0.243

Parameter	Time point	Renaltec group	Control group	P value
Phosphate (mmol/l)	t2	36.30 (34.20–38.40)	37.30 (35.30–39.30)	0.488
	t0	1.26 (1.09–1.42)	1.10 (0.93–1.28)	0.209
	t1	1.38 (1.21–1.54)	1.22 (1.04–1.40)	0.224
FGF-23 (pg/ml)	t2	1.36 (1.18–1.54)	1.20 (1.02–1.38)	0.223
	t0	231.00 (150.00–312.00)	244.00 (163.00–325.00)	0.824
	t1	282.00 (201.00–363.00)	277.00 (196.00–358.00)	0.935
Calcium (mmol/l)	t2	255.00 (170.00–340.00)	325.00 (244.00–406.00)	0.241
	t0	2.53 (2.45–2.61)	2.66 (2.58–2.74)	0.026
	t1	2.53 (2.45–2.61)	2.59 (2.51–2.67)	0.262
Potassium (mmol/l)	t2	2.59 (2.50–2.67)	2.61 (2.53–2.69)	0.662
	t0	4.54 (4.28–4.80)	4.19 (3.92–4.46)	0.064
	t1	4.44 (4.18–4.70)	4.12 (3.85–4.39)	0.090
Sodium (mmol/l)	t2	4.46 (4.17–4.74)	4.23 (3.96–4.50)	0.242
	t0	150.00 (149.00–151.00)	151.00 (150.00–152.00)	0.302
	t1	149.00 (148.00–151.00)	151.00 (149.00–152.00)	0.427
Chloride (mmol/l)	t2	151.00 (149.00–152.00)	150.00 (149.00–152.00)	0.531
	t0	117.00 (115.00–120.00)	116.00 (113.00–118.00)	0.365
	t1	117.00 (115.00–120.00)	116.00 (114.00–118.00)	0.418
Leukocytes (10 ⁹ /l)	t2	118.00 (115.00–120.00)	116.00 (114.00–119.00)	0.392
	t0	8.97 (6.34–11.59)	6.81 (4.28–9.33)	0.245

Parameter	Time point	Renaltec group	Control group	P value
Haematocrit (l/l)	t1	9.08 (6.46–11.70)	7.20 (4.68–9.72)	0.311
	t2	8.46 (5.82–11.11)	7.02 (4.50–9.55)	0.441
	t0	0.39 (0.36–0.42)	0.40 (0.38–0.43)	0.541
	t1	0.39 (0.36–0.41)	0.40 (0.37–0.43)	0.545
	t2	0.41 (0.38–0.44)	0.37 (0.35–0.40)	0.121
USG [*]	t0	1.025 (1.021–1.030)	1.017 (1.012–1.021)	0.012
	t1	1.024 (1.020–1.029)	1.019 (1.014–1.023)	0.098
	t2	1.021 (1.016–1.025)	1.018 (1.013–1.022)	0.377
UPC	t0	0.10 (0.06–0.13)	0.12 (0.09–0.16)	0.264
	t1	0.10 (0.06–0.14)	0.14 (0.10–0.18)	0.112
	t2	0.08 (0.04–0.12)	0.17 (0.13–0.20)	0.001
FDI	t0	0.87 (–0.05 to 1.80)	1.02 (–0.02 to 2.05)	0.841
	t1	0.73 (–0.20 to 1.66)	0.43 (–0.61 to 1.46)	0.668
	t2	–0.45 (–1.47 to 0.57)	0.09 (–1.00 to 1.17)	0.482
BP [†] (mmHg)	t0	150.00 (142.00–159.00)	144.00 (135.00–153.00)	0.299
	t1	145.00 (137.00–153.00)	148.00 (140.00–157.00)	0.550
	t2	151.00 (142.00–160.00)	157.00 (149.00–166.00)	0.318

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Data are estimates (95% confidence interval) for comparisons between groups within each time point. *P* values for between-group comparisons within each time point were obtained from estimated marginal means, with Tukey adjustment for multiple comparisons. *P* values <0.05 were considered statistically significant and are shown in bold

^{*}USG was determined using a hand refractometer

[†]BP was measured using high-definition oscillometry

BP = blood pressure; FGF-23 = fibroblast growth factor 23; SDMA = symmetric dimethylarginine; UPC = urine protein:creatinine ratio; USG = urine specific gravity

Concurrent medication

In total, seven cats required additional long-term medication. In the Renaltec group, one cat received amlodipine for CKD-associated hypertension and another cat was treated with inhaled fluticasone for feline asthma. In the control group, four cats received amlodipine for hypertension, and one cat with well-controlled hyperthyroidism was treated with thiamazole throughout the study.

Administration of Renaltec

Questionnaires revealed that all cats in the Renaltec group reliably received the supplement once daily, either mixed with Add One or with their regular food. Cats that initially refused Renaltec mixed with Add One readily accepted it when offered with their regular food. Owners reported no refusals after this adjustment and no side effects or changes in acceptance throughout the study.

Treatment outcome

At t1, data from 19 cats were analysed, showing no statistically significant differences between the groups for any evaluated parameters. At t2, data from eight cats in the Renaltec group and nine in the control group were available. Two cats in the Renaltec group were excluded before t2: one on day 174 because of suppurative triaditis and another was euthanased shortly before t2 because of a forelimb neoplasm (exact date unknown).

At t2, cats in the Renaltec group had significantly lower plasma IS concentrations ($P = 0.045$) than cats in the control group. No significant differences were observed in CBC parameters, creatinine, SDMA, FGF-23, FDI, body weight or blood pressure between the groups. UPC values of eight cats of each group were available for analysis. UPC was not performed in one cat from the control group because of an active urine sediment. UPC was significantly lower in cats in the Renaltec group than in cats in the control group ($P = 0.001$) although values in both groups remained within the non-proteinuric range.

There were no statistically significant differences within the groups between t0 and t1. At t2, cats in the control group had a significantly lower haematocrit ($P = 0.026$) compared with t0, and a significantly lower body weight than at t0 ($P = 0.015$) and t1 ($P = 0.017$) ([Table 3](#)). Cats in the Renaltec group showed a significantly lower FDI at t2 compared with t1 ($P = 0.023$) and t0 ($P = 0.009$).

Table 3.

Results of the statistical analysis for serum chemistry, haematology, urinalysis, faecal dysbiosis index (FDI), blood pressure and body weight at the time intervals t0–t1, t0–t2 and t1–t2, analysed using mixed-effects models with group, time point and their interaction as fixed effects and cat ID as random effect

Parameter	Time intervals	Renaltec		Control	
		Estimate	<i>P</i> value	Estimate	<i>P</i> value
Indoxyl sulfate (µg/ml)	t0–t1	0.11 (–0.85 to 1.08)	0.959	–0.46 (–1.55 to 0.63)	0.583
	t0–t2	0.55 (–0.50 to 1.60)	0.436	–0.92 (–1.97 to 0.14)	0.103
	t1–t2	0.44 (–0.61 to 1.49)	0.592	–0.46 (–1.51 to 0.60)	0.568
SDMA (µg/dl)	t0–t1	1.47 (–1.69 to 4.61)	0.493	0.88 (–2.52 to 4.29)	0.801
	t0–t2	0.73 (–2.69 to 4.15)	0.861	1.57 (–1.73 to 4.87)	0.478
	t1–t2	–0.74 (–4.03 to 2.55)	0.846	0.69 (–2.61 to 3.99)	0.865
Creatinine (µmol/l)	t0–t1	5.27 (–21.4 to 32.0)	0.889	–2.585 (–32.5 to 27.3)	0.978
	t0–t2	15.93 (–13.0 to 44.9)	0.400	–0.47 (–29.5 to 28.5)	0.999
	t1–t2	10.66 (–18.3 to 39.6)	0.663	2.11 (–26.9 to 31.1)	0.984
Urea (mmol/l)	t0–t1	0.105 (–1.55 to 1.76)	0.988	–1.57 (–3.46 to 0.32)	0.125
	t0–t2	–0.33 (–2.13 to 1.47)	0.902	–0.78 (–2.59 to 1.03)	0.571

Parameter	Time intervals	Renaltec		Control	
		Estimate	P value	Estimate	P value
Total protein (g/l)	t1-t2	-0.44 (-2.24 to 1.360)	0.836	0.79 (-1.02 to 2.60)	0.564
	t0-t1	0.88 (-1.24 to 3.00)	0.593	-0.59 (-3.01 to 1.83)	0.835
	t0-t2	-0.20 (-2.51 to 2.10)	0.977	-0.71 (-3.03 to 1.62)	0.755
Albumin (g/l)	t1-t2	-1.08 (-3.39 to 1.22)	0.514	-0.12 (-2.44 to 2.21)	0.992
	t0-t1	0 (-1.37 to 1.365)	1.000	0.92 (-0.64 to 2.5)	0.327
	t0-t2	-0.83 (-2.31 to 0.66)	0.368	0.76 (-0.73 to 2.26)	0.431
Phosphate (mmol/l)	t1-t2	-0.83 (-2.31 to 0.66)	0.368	-0.16 (-1.65 to 1.34)	0.964
	t0-t1	-0.12 (-0.34 to 0.11)	0.430	-0.12 (-0.37 to 0.14)	0.519
	t0-t2	-0.10 (-0.35 to 0.14)	0.561	-0.10 (-0.35 to 0.15)	0.625
FGF-23 (pg/ml)	t1-t2	0.01 (-0.23 to 0.25)	0.991	0.02 (-0.23 to 0.28)	0.981
	t0-t1	-51.00 (-119.7 to 17.74)	0.191	-33.2 (-110.9 to 44.52)	0.576
	t0-t2	-23.7 (-98.3 to 50.91)	0.737	-80.9 (-155.9 to -5.91)	0.031
Calcium (mmol/l)	t1-t2	27.3 (-47.3 to 101.87)	0.667	-47.7 (-122.7 to 27.26)	0.295
	t0-t1	0.00 (-0.07 to 0.08)	0.995	0.07 (-0.02 to 0.15)	0.147
	t0-t2	-0.06 (-0.14 to 0.02)	0.235	0.05 (-0.04 to 0.13)	0.365

Parameter	Time intervals	Renaltec		Control	
		Estimate	<i>P</i> value	Estimate	<i>P</i> value
Potassium (mmol/l)	t1-t2	-0.06 (-0.14 to 0.02)	0.200	-0.02 (-0.10 to 0.0614)	0.831
	t0-t1	0.10 (-0.24 to 0.44)	0.746	0.07 (-0.30 to 0.44)	0.893
	t0-t2	0.09 (-0.28 to 0.45)	0.831	-0.04 (-0.40 to 0.33)	0.969
Sodium (mmol/l)	t1-t2	-0.01 (-0.38 to 0.35)	0.995	-0.11 (-0.47 to 0.26)	0.758
	t0-t1	0.54 (-1.76 to 2.84)	0.833	0.12 (-2.30 to 2.54)	0.992
	t0-t2	-1.04 (-3.48 to 1.41)	0.558	0.79 (-1.63 to 3.21)	0.707
Chloride (mmol/l)	t1-t2	-1.58 (-4.02 to 0.87)	0.267	0.67 (-1.76 to 3.09)	0.780
	t0-t1	-0.2 (-2.84 to 2.44)	0.981	-0.37 (-3.32 to 2.59)	0.951
	t0-t2	-0.54 (-3.40 to 2.32)	0.889	-0.57 (-3.44 to 2.30)	0.877
Leukocytes (10 ⁹ /l)	t1-t2	-0.34 (-3.20 to 2.52)	0.955	-0.21 (-3.07 to 2.66)	0.983
	t0-t1	-0.12 (-1.09 to 0.86)	0.958	-0.39 (-1.51 to 0.73)	0.688
	t0-t2	0.51 (-0.56 to 1.57)	0.504	-0.22 (-1.29 to 0.85)	0.883
Haematocrit (l/l)	t1-t2	0.62 (-0.44 to 1.68)	0.355	0.18 (-0.90 to 1.25)	0.921
	t0-t1	0.01 (-0.02 to 0.03)	0.816	0.01 (-0.02 to 0.03)	0.847
	t0-t2	-0.02 (-0.04 to 0.01)	0.314	0.03 (0.00-0.06)	0.026

Parameter	Time intervals	Renaltec		Control	
		Estimate	P value	Estimate	P value
USG [*]	t1-t2	-0.02 (-0.05 to 0.00)	0.109	0.02 (-0.00 to 0.05)	0.108
	t0-t1	0 (-0.00 to 0.01)	0.892	0 (-0.01 to 0.00)	0.625
	t0-t2	0 (-0.00 to 0.01)	0.070	0 (-0.01 to 0.00)	0.920
UPC	t1-t2	0 (-0.00 to 0.01)	0.173	0 (-0.00 to 0.01)	0.838
	t0-t1	0 (-0.05 to 0.04)	0.971	-0.02 (-0.06 to 0.03)	0.673
	t0-t2	0.02 (-0.03 to 0.06)	0.556	-0.04 (-0.09 to 0.00)	0.077
FDI	t1-t2	0.02 (-0.02 to 0.07)	0.425	-0.02 (-0.07 to 0.02)	0.398
	t0-t1	0.14 (-0.78 to 1.06)	0.932	0.59 (-0.62 to 1.80)	0.492
	t0-t2	1.32 (0.27-2.37)	0.009	0.93 (-0.36 to 2.22)	0.210
BP [‡] (mmHg)	t1-t2	1.18 (0.13-2.23)	0.023	0.34 (-0.89 to 1.57)	0.791
	t0-t1	5.63 (-8.12 to 19.38)	0.602	-4.43 (-19.38 to 10.51)	0.766
	t0-t2	-0.67 (-15.26 to 13.91)	0.994	-13.51 (-28.45 to 1.44)	0.086
Body weight (kg)	t1-t2	-6.3 (-20.89 to 8.28)	0.569	-9.07 (-23.57 to 5.43)	0.301
	t0-t1	0.09 (-0.10 to 0.28)	0.491	0 (-0.23 to 0.23)	0.999
	t0-t2	-0.02 (-0.23 to 0.18)	0.962	0.26 (0.04-0.48)	0.015
	t1-t2	-0.11 (-0.32 to 0.09)	0.389	0.26 (0.04-0.48)	0.017

Values in parentheses are 95% confidence interval (CI). Reported are the pairwise estimates of difference (estimate [95% CI of difference]) between time points for each group, with Tukey-adjusted *P* values for multiple comparisons. *P* values <0.05 were considered statistically significant and are shown in bold

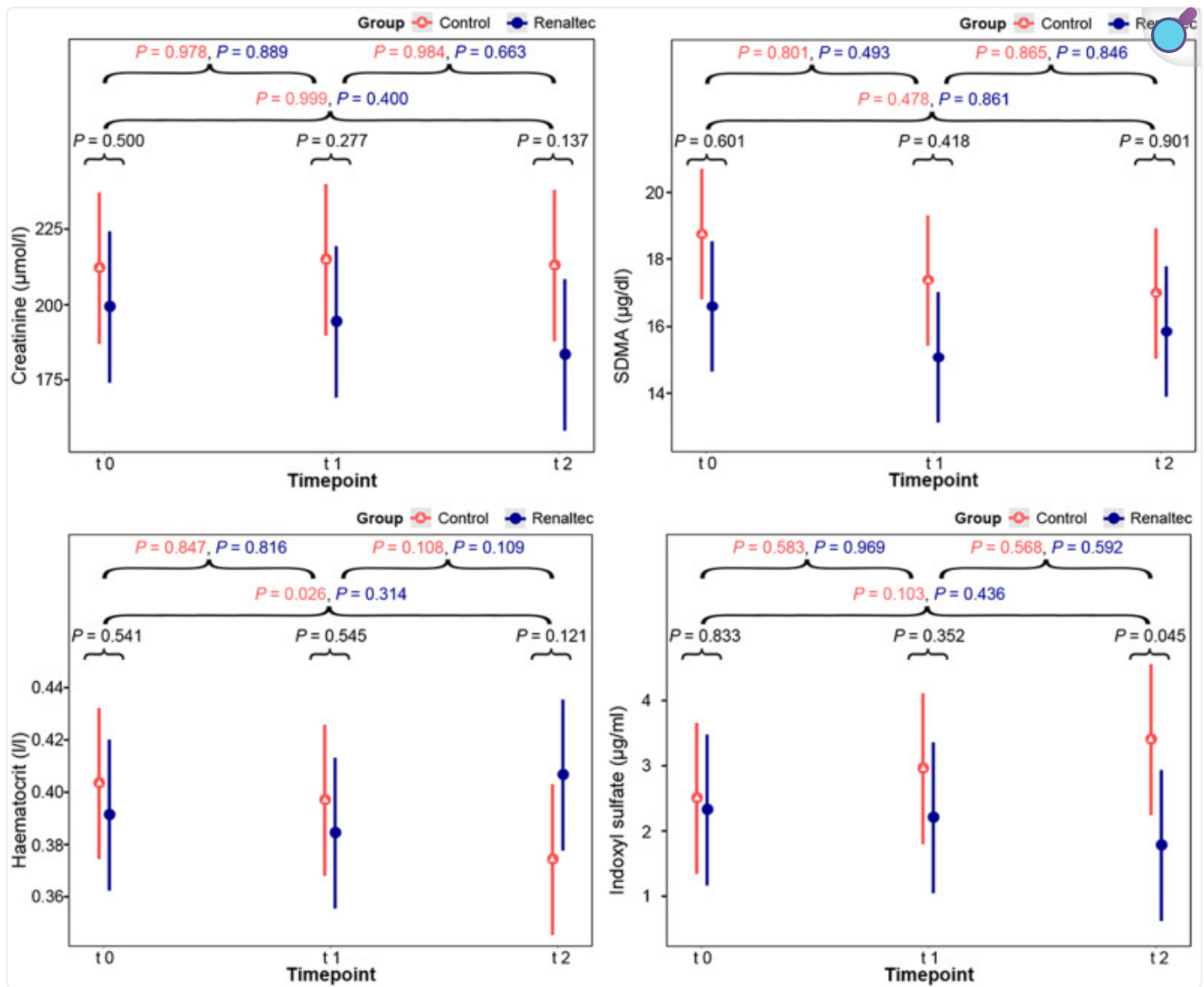
*USG was determined using a hand refractometer

†BP was measured using high-definition oscillometry

BP = blood pressure; FGF-23 = fibroblast growth factor 23; SDMA = symmetric dimethylarginine; UPC = urine protein:creatinine ratio; USG = urine specific gravity

One cat in each group progressed from IRIS stage 2 to stage 3, and one cat in each group showed an increase of creatinine of more than 25% from baseline (32% in the control group and 35% in the Renaltec group) without a change in IRIS stage ([Figures 2](#) and [3](#)).

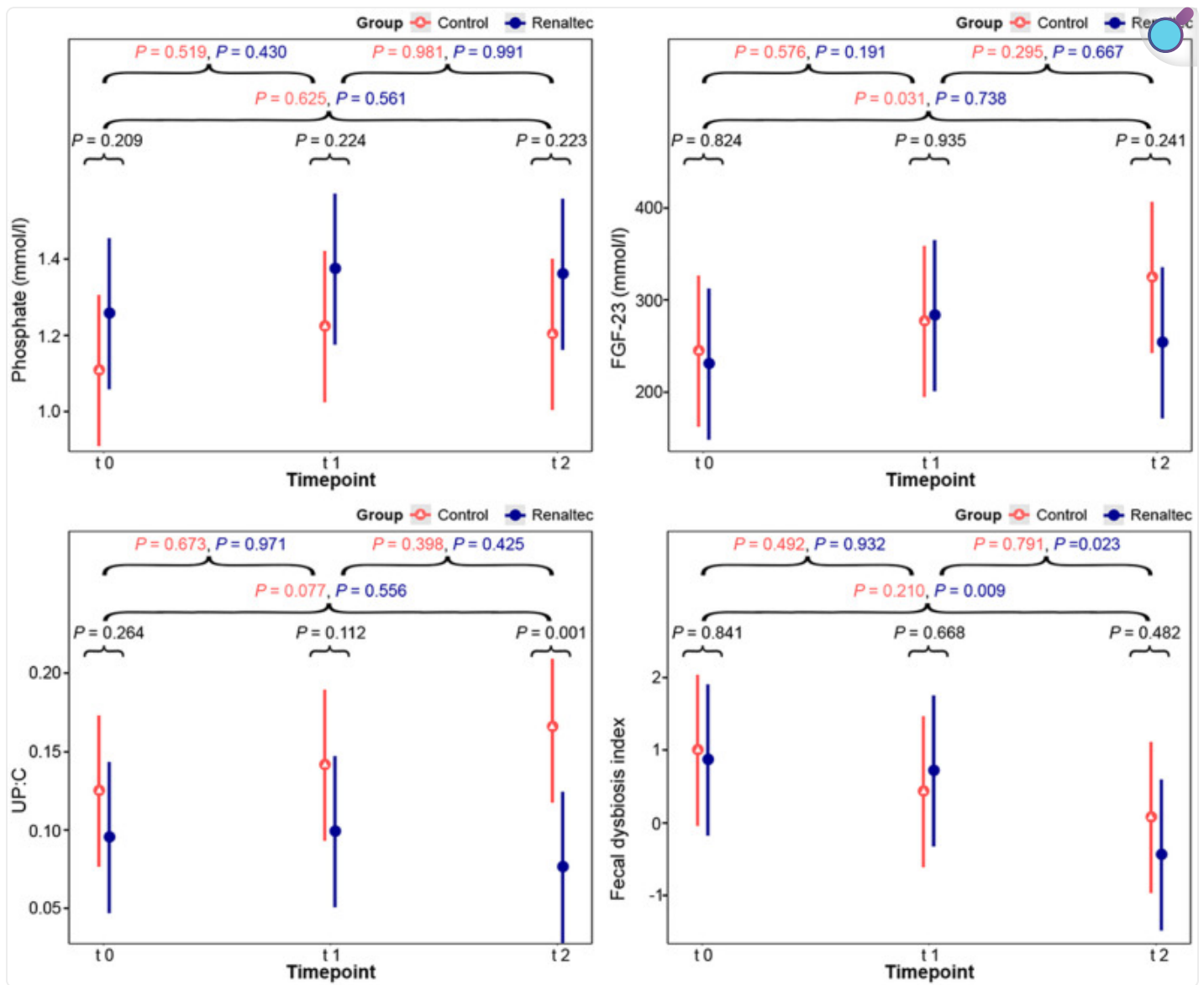
Figure 2.



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Changes in creatinine, symmetric dimethylarginine (SDMA), haematocrit and indoxyl sulfate over a period of 6 months (t0 = baseline, t1 = after 3 months, t2 = after 6 months) in the Renaltec and control groups. Estimated marginal means, 95% confidence intervals and P values originate from mixed-effects regressions

Figure 3.



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Changes in phosphate, fibroblast growth factor 23 (FGF-23), urine protein:creatinine ratio (UPC) and faecal dysbiosis index over a period of 6 months (t0 = baseline, t1 = after 3 months, t2 = after 6 months) in the Renaltec and control groups. Estimated marginal means, 95% confidence intervals and *P* values originate from mixed-effects regressions

Discussion

This pilot study investigated the effects of oral administration of Renaltec in a population of client-owned cats with naturally occurring IRIS stage 2 and 3 CKD over a period of 6 months. The main findings were that cats in the Renaltec group had significantly lower plasma IS concentrations and a significantly lower UPC compared with those in the control group after 6 months. This was attributed to an increase in plasma IS concentrations in cats from the control group and a decrease in IS levels in the Renaltec group, with neither change being statistically significant on its own.

Marked reductions in IS concentrations have been reported in geriatric cats in a previous short-term study.²⁶ Given that these animals did not suffer from CKD and were consequently expected to have a lower initial level of uremic toxins, such findings are not directly comparable with cats with naturally occurring renal disease. Moreover, biological variability in uremic toxin concentrations has been demonstrated,³⁰ indicating that more moderate reductions in IS may also be meaningful, particularly when they are achieved under routine clinical conditions in cats with naturally occurring CKD and maintained over an extended period.

Previous studies on AST-120 have demonstrated its IS-lowering effect across various species under experimental conditions. In rats, significant reductions in IS levels were observed after 10²² and 19¹⁴ weeks at dosages of 0.40–0.45 g/100 g body weight, which are substantially higher than the dose used in the present study (0.5 g/cat/day, mean weight 5.12 ± 1.82 kg). In mice, findings have been less conclusive. A 12-week study using standard food containing 5% AST-120 in mice with experimentally induced renal dysfunction showed a non-significant trend towards lower IS levels, although body weight conversion was inconclusive.³¹ Similarly, a study using a Western diet (diet with high amount of saturated fatty acids, carbohydrates and low fibre content) mixed with 8% AST-120 over 3 months in nephrectomised mice demonstrated a non-significant trend towards decreased IS levels.³¹

In dogs with experimentally induced heart failure and healthy kidneys, a 50% reduction in IS levels was observed after 4 weeks of treatment with 1 g/kg body weight AST-120 per day.³² Similarly, a multicentre, randomised, double-blind, placebo-controlled, dose-escalating study in humans with moderate to severe CKD demonstrated a dose-dependent reduction in serum IS levels after 12 weeks, with concentrations progressively decreasing as AST-120 doses increased (0.9–3 g/patient q8h).²⁴ In contrast, another study reported a significant IS decrease in CKD patients only after a minimum of 6 months of daily therapy with 6 g of AST-120.³³ In cats, a study involving 13 cats with experimentally induced CKD (IRIS stage 2 and 3), fed a standardised renal diet, reported significant serum IS reductions after 28 days of Renaltec dosing regimens (0.5 g q24h and 0.5 mg q12h). However, by day 56, significant reductions were maintained only with twice-daily dosing. The authors attributed this finding to the significant biological variability of uremic toxins and the

small sample size, which may have limited the study's statistical power.²⁷ In summary, these studies highlight the variability in IS-lowering effects across different species, dosing regimens and study durations. Notably, longer treatment durations and higher doses in both humans and other animals have demonstrated more consistent reductions in IS levels. Further research is required to optimise Renaltec dosing strategies and to clarify its long-term effects on IS levels, particularly in cats with elevated baseline IS levels and advanced stages of CKD.

Numerous studies have demonstrated that proteinuria serves as a critical prognostic factor for CKD in cats.³⁴⁻³⁷ In the present study, cats with proteinuria (UPC >0.4) were excluded. At t2, cats in the Renaltec group showed significantly lower UPC values compared with those in the control group. Similar findings have been reported in multiple studies involving rats and humans. Studies in subtotal nephrectomised rats treated with AST-120 showed significantly lower urinary protein excretion compared with the control groups.^{14,15,38} Histological evaluations from these studies suggest that AST-120 may prevent glomerular sclerosis and tubulointerstitial fibrosis, which are conditions linked to urinary protein excretion.^{14,15} In addition, a clinical study in humans demonstrated that daily administration of AST-120 over 12 months significantly reduced proteinuria levels in non-diabetic patients with CKD compared with those receiving control treatment. The authors proposed that AST-120 treatment has a beneficial effect on reducing tubular damage.³⁹ Although the present study demonstrated a statistically significant reduction in UPC, this change occurred within the non-proteinuric range and is therefore unlikely to be clinically relevant in this cohort. However, the finding may still suggest a potential effect of Renaltec on urinary protein excretion that warrants further investigation in proteinuric CKD cats. In addition, future investigations incorporating histopathological analyses are necessary to explore potential effects on glomerular sclerosis and tubulointerstitial fibrosis.

No significant differences in serum creatinine levels or CKD progression were observed between the groups at any time point during this study. In human CKD studies, a reduction in serum creatinine was observed after more than 30 weeks of AST-120 treatment²⁵ but not after 12 weeks.²⁴ In contrast, rats treated with high doses of AST-120 showed reduced creatinine levels after only 12 weeks.²² Given the limited number of studies on the effects of Renaltec in cats with CKD, it remains uncertain whether a longer treatment duration could result in significantly reduced creatinine levels in this species.

Multiple studies have shown that human patients with CKD exhibit an altered composition of gut microbiota compared with healthy controls.⁴⁰⁻⁴³ A study investigating the potential of dietary probiotics in rats found a significant reduction in serum levels of IS, among other uremic toxins, suggesting a beneficial effect in reducing the accumulation of certain uremic toxins through

modulation of the intestinal microbiome in CKD patients.⁴⁴ Studies in mice and rats treated with AST-120 demonstrated decreased tissue accumulation of IS and *p*-cresyl sulfate (*p*CS) in multiple organs,⁴⁵ a change in the composition of gut microbiota involved in the production of uremic toxin precursors⁴⁶ as well as an increase in faecal excretion of indole, *p*-cresol and phenol and the reduction of serum IS and *p*CS levels.¹⁵ A similar effect in intestinal microbiota composition was observed in humans after 3 months of AST-120 treatment; although the trend towards reduced *p*CS levels was noted, these results did not reach statistical significance.⁴⁷ A study examining the faecal microbiome and serum IS concentrations in cats with CKD revealed decreased faecal bacterial diversity and richness, alongside higher serum IS concentrations, compared with control cats. However, the study was unable to establish a correlation between the concentration of the measured uremic toxins and the types of bacteria known to be involved in their production in humans.⁹ In the present study, a significant decrease in the FDI within the Renaltec group was demonstrated at both 3 and 6 months after treatment. This improvement may be attributed to positive changes in gut microbiota composition after administration of Renaltec. Although Renaltec is a physical adsorber of indole in the intestinal lumen, the significant reduction in FDI observed in this study may suggest indirect effects on gut microbial composition, potentially mediated by altered substrate availability. These observations are in line with findings from studies on AST-120 and raise the hypothesis that intestinal toxin adsorbers could influence both systemic toxin burden and the intestinal microbiota in feline CKD. Further studies investigating the FDI in healthy cats, cats with CKD and those treated with Renaltec are warranted to better understand these effects.

In the present study, a significant decrease in haematocrit was observed in the control group between t0 and t2, although none of the cats was anaemic. The decrease, although statistically significant, was within the reference interval and therefore not considered clinically relevant in this small cohort of cats. Increased severity of CKD is associated with the development of anaemia: of cats with CKD, 5% (IRIS stage 2), 18% (IRIS stage 3) and 53% (IRIS stage 4) develop anaemia.^{48,49} A primary cause of anaemia is erythropoietin (EPO) deficiency due to impaired oxygen-sensing mechanisms and reduced EPO production by peritubular renal EPO-producing cells.⁵⁰ Studies in rats showed that exposure to increased IS levels increases oxygen consumption in renal tubular cells⁵¹ and impairs the oxygen-sensing capacity of EPO-producing cells through a hypoxia-inducible transcription factor dependent mechanism.⁵² Longitudinal studies are needed to determine whether lowering IS plasma levels could delay the progression of anaemia in CKD cats.

The present study has some limitations. Randomisation resulted in some baseline imbalances (sex, tCa, USG, concomitant medications) that could have influenced the observed treatment effects. In

addition, a few cats did not complete the 6-month study period, which further reduced the effective sample size and the overall statistical power of the study.

There are also statistical considerations. To contextualise the observed 22% reduction (3.3 mg/l absolute difference in mean IS concentration), a post hoc sensitivity analysis was performed using the same assumptions as the a priori power analysis. Detecting this smaller effect at 80% power would require 37 cats per group (total of 74 cats). With the actual sample size (mean of 9.5 per group), the achieved power for the observed effect was approximately 27%, calculated via the `pwr` package in R.⁵³ This indicates that the study was underpowered for detecting subtler changes but aligns with its exploratory design, which was based on detecting a large hypothesised effect derived from prior data.

The FDI has not been validated in cats with CKD, and its interpretation should therefore be approached with caution. Moreover, individual differences in the administration of Renaltec (with or without Add One) and in the composition of the renal diets were not standardised. Although all cats were fed exclusively renal diets, the relative proportions of dry and wet food varied between individuals and over time. In addition, cats were fed renal diets from two different brands or a calculated ration, which did not have identical protein and phosphate contents (crude protein 13.1–18.2 g/MJ ME [5.48–7.61 g/100 kcal ME]; phosphate 124.6–315.8 mg/MJ ME [52.1–132.1 mg/100 kcal ME]). Furthermore, only cats in the Renaltec group received Add One as a liquid treat vehicle, which may have influenced the intestinal microbiota and should be considered a potential confounding factor.

Conclusions

Treatment with Renaltec was associated with significantly lower IS concentrations and significantly lower UPC after 6 months compared with a control group, although UPC values remained within the non-proteinuric range. Furthermore, cats in the Renaltec group had a significant improvement of FDI over time, whereas cats in the control group showed a significant decrease of haematocrit and body weight over time. Although these findings are encouraging and suggest a potential benefit of Renaltec in addition to treatment according to IRIS guidelines, they should be interpreted with consideration of the small sample size and other study limitations. As this was a pilot study, further controlled trials with larger populations are needed to confirm these results and to explore the long-term effects of Renaltec in feline CKD.

Footnotes

Author note: The data sets analysed during the current study are available from the corresponding author on reasonable request.

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Ethical approval: The work described in this manuscript involved the use of non-experimental (owned or unowned) animals. Established internationally recognised high standards ('best practice') of veterinary clinical care for the individual patient were always followed and/or this work involved the use of cadavers. Ethical approval from committee was therefore not specifically required for publication in *JFMS*. Although not required, where ethical approval was still obtained, it is stated in the manuscript.

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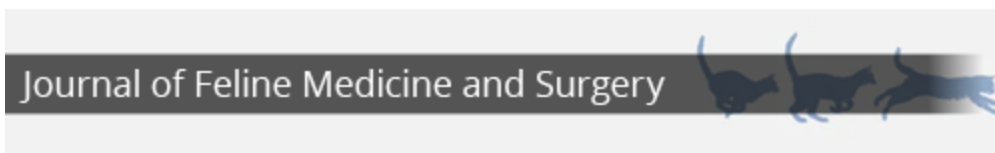
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SGLT2 inhibitor therapy in diabetic cats: first clinical experiences with non-ideal candidates

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Abstract

Case series summary

This retrospective case series describes seven diabetic cats treated with velagliflozin that were considered non-ideal candidates for this therapy. These more complicated diabetic feline cases were referred to the Small Animal Department, Faculty of Veterinary Medicine, Ghent University, Belgium. The sole inclusion criterion for this case series was treatment with velagliflozin (Senvelgo; Boehringer Ingelheim) after diagnosis of diabetes mellitus. Data on signalment, medical history, clinical findings, diagnostics, treatment, response and outcomes were available for all cats. The initial consultation for all cats took place between March 2024 and May 2025.

Relevance and novel information

Current literature on the use of sodium–glucose cotransporter-2 (SGLT2) inhibitors in feline patients remains limited, with most studies involving highly selected populations with strict exclusion criteria. This case series describes the use of velagliflozin in non-ideal candidates for SGLT2 inhibitor therapy, aiming to support clinicians managing such cases and to provide useful information for future prospective studies. In these non-ideal cases, the use of velagliflozin may be considered off label; its administration should comply with local ethical and legal regulations, with informed client consent obtained. Consultation with a veterinary specialist is recommended when clinical experience is limited. This case series includes diabetic cats with suspected and/or confirmed comorbidities that complicate diabetes management, such as hypersomatotropism and chronic kidney disease. In addition, it reports on the concurrent use of other treatments, including cabergoline and insulin. The potential interaction and possible synergistic effects of these combined therapies represent an area of interest for future research. To ensure optimal glycaemic control and enable individualised dosing, four of the presented cases were closely monitored using continuous glucose monitoring (Freestyle Libre Sensor 2; Abbott Laboratories). Furthermore, the availability of measurements of ketone concentrations in urine and/or blood provided valuable insight into the metabolic changes associated with this new treatment. The role of ketone monitoring in predicting treatment response and identifying potential adverse effects represents another important area for future research.

Keywords: Diabetes mellitus, sodium–glucose cotransporter-2 (SGLT2) inhibitor, gliflozin, velagliflozin

Plain language summary

First experiences using SGLT2 inhibitor therapy (velagliflozin) in diabetic cats with other health problems or complex conditions

Diabetes mellitus is a common disease in cats that causes high blood sugar levels and requires lifelong treatment. Traditionally, insulin injections have been the main treatment, but newer oral medications, such as velagliflozin, are now available. Velagliflozin helps lower blood sugar by allowing excess glucose to leave the body in the urine. However, there is limited information about how well this medicine works in cats with other health problems. This report describes seven diabetic cats treated with velagliflozin, even though they were non-ideal candidates for this therapy because they had additional diseases or were receiving other treatments. Some cats had conditions such as chronic kidney disease or hypersomatotropism (a hormone disorder that can make diabetes more difficult to control). The cats were monitored through regular check-ups, blood tests to measure ketones (substances that can indicate the most significant risk: diabetic

ketoacidosis) and, in some cases, continuous glucose monitoring sensors that tracked their sugar levels throughout the day. Overall, this case series provides information about using velagliflozin in more complicated diabetic cats. It shows that, with close monitoring and careful selection of patients, this medication can be a useful option beyond the strict conditions of earlier studies. These observations can help veterinarians and researchers improve the safety and effectiveness of diabetes care in cats. In situations where standard treatment is not suitable, velagliflozin may be considered outside its approved indications. This should be done only in line with local laws and ethical guidelines, and after the client has given informed consent. If the clinician has limited experience with this treatment, consultation with a veterinary specialist is advised. The findings also show that more research is needed to better understand dietary strategies, the use of combined treatments and how to interpret ketone levels in diabetic cats.

Introduction

Diabetes mellitus (DM) is a common endocrine disorder in cats, characterised by persistent hyperglycaemia and glucosuria. Clinical signs include polyuria and polydipsia (PUPD), polyphagia and weight loss. Notably, approximately one-quarter of diabetic cats have underlying hypersomatotropism (HS), which complicates DM management owing to insulin resistance.¹ Effective management is essential to prevent complications such as clinical hypoglycaemia and diabetic ketoacidosis (DKA). Although insulin therapy combined with dietary modification remains the standard of care, it poses substantial challenges for both veterinarians and pet owners. These include the need for accurate insulin dosing, twice-daily injections, frequent glucose monitoring and the risk of clinical hypoglycaemia, as well as potential strain on the owner–pet relationship. The challenges of this treatment are substantiated by a study reporting that 30% of cats or dogs with DM are euthanased within the first year of diagnosis.² These limitations highlight the need for alternative treatment options that provide effective glycaemic control while minimising adverse effects. Bexagliflozin (Bexacat; Elanco) has been available in the USA since 2022, and velagliflozin (Senvelgo; Boehringer Ingelheim) in the EU since 2023, as novel treatments for feline diabetes.

Sodium–glucose cotransporter-2 (SGLT2) inhibitors represent a new class of oral antidiabetic drugs for cats. SGLT2 is a membrane protein in the proximal tubule of nephrons, responsible for reabsorbing approximately 90% of filtered glucose, while the remaining 10% is reabsorbed by sodium–glucose cotransporter-1 (SGLT1) further along the tubule. Inhibition of SGLT2 receptors reduces glucose reabsorption, promoting urinary glucose excretion.^{3–6} In humans and mice, these drugs lower blood glucose concentrations without inducing clinical hypoglycemia.³

In feline medicine, SGLT2 inhibitors, including bexagliflozin and velagliflozin, have been evaluated as potential alternatives to insulin therapy in clinically stable diabetic cats.³⁻⁵ Previous studies applied strict exclusion and inclusion criteria, omitting cats with comorbidities, ketosis or ketonuria in the history or at presentation, and clinical signs such as vomiting, diarrhoea and anorexia. The study by Behrend et al⁵ included cats previously treated with insulin and cats with a suspicion of HS. Reported success rates, defined by improvements in glycaemic control and/or improvement in clinical signs, range from 54% to as high as 88%. None of these studies reported episodes of clinical hypoglycaemia in cats, representing a significant advantage over insulin therapy.³⁻⁵ The most common adverse effect, softer stools, was reported in 38–53% of treated cats; these episodes were generally mild and self-limiting within 2 weeks.⁶ Persistent glucosuria may contribute to ongoing PUPD in some cats, with studies reporting persistent polyuria in 12–46% of cases, indicating that this clinical sign often does not resolve despite treatment.³⁻⁵ The most serious, potentially life-threatening, adverse event associated with gliflozin therapy is DKA, which may occur even when blood glucose levels remain within normal or near-normal intervals, a condition referred to as euglycaemic diabetic ketoacidosis (eDKA). Overall, eDKA was observed in 5–7% of insulin-naïve diabetic cats treated with gliflozins.^{3,5,6} Behrend et al⁵ administered velagliflozin as monotherapy in both insulin-naïve and previously insulin-treated diabetic cats. The incidence of DKA was higher in cats with prior insulin exposure (18%) compared with newly diagnosed cats (5%). Notably, 78% of cats diagnosed with DKA were euglycaemic and 86% of cases developed within the first 2 weeks of treatment.⁵

Gliflozin therapy provides a convenient, once-daily, oral treatment option for managing stable diabetic cats, described as ‘happy diabetic cats’. These are patients that maintain normal appetite, hydration and activity levels, without clinical signs such as vomiting, diarrhoea, lethargy, cachexia or evidence of DKA, as confirmed by normal blood or urine ketone concentrations. In such cases, gliflozin has been associated with rapid glycaemic stabilisation, with measurable improvements typically seen within 7 days. In addition, the reduced need for frequent blood glucose monitoring and dose adjustments contributes to simplified long-term management.⁵

However, because of the increased risk of DKA, particularly within the first 2 weeks of treatment, close monitoring is mandatory. Owners should watch for the typical clinical signs of DM, as well as signs of DKA including anorexia, lethargy, acute vomiting, dehydration, weight loss, respiratory distress or altered mentation. Current recommendations advocate, besides monitoring of clinical signs, the measurement of ketone levels on day 2 or 3 of treatment, with prompt veterinary evaluation recommended if ketonuria or marked ketonaemia is detected. As validated reference intervals (RIs) for blood or urine ketone concentrations in cats are currently lacking, any values

exceeding the upper measurable interval of the device or showing a significant rise from baseline should be interpreted with caution and clinical context.

This case series describes second-line feline diabetes patients treated with off-label velagliflozin for various clinical indications. The purpose of this report is not to advocate off-label use of SGLT2 inhibitors, but to outline practical considerations, limitations and challenges associated with their use in complex clinical scenarios. The findings aim to contribute to a more informed and safe application of this therapeutic approach in the management of feline diabetes.

Case series description

Medical records of seven challenging second-line diabetic cats treated with velagliflozin were retrospectively reviewed. Data collected included signalment, medical history, comorbidities, diagnostic findings, treatment details, response and outcome. All cats were seen between March 2024 and May 2025. Follow-up was available for all and ongoing in some cases. Results of glucose and ketone monitoring are summarised in [Table 1](#).

Table 1.

Results of ketone and glucose measurements in seven challenging second-line diabetic cats treated with velagliflozin

Number	Reason for presentation	Days after initiating velagliflozin	Blood glucose (mg/dl)	Ketonaemia (mmol/l)	Ketonuria	F + IGF-1	Comor
1	Newly diagnosed DM with comorbidities	Baseline	300	0.1	NP	F: 649 µmol/l, IGF-1: N	CKD
		3	121 ₋ *	0.4	NP	–	
		7	104 ₋ *	0.1	NP	–	
2		Baseline	608	0.4	Neg	F: 724 µmol/l, IGF-1: 696 ng/ml	HS
		3	161–298 _±	0.2	NP	–	
		7	184–338 _±	0.2	NP	–	
3	Uncontrolled DM on insulin	Baseline	High ₋ *	<0.1	NP	–	Chronic intermittent diarrhoea
		3	122 ₋ *	0.3	NP	–	
		7	162–198 _±	0.4	NP	–	
		14	70–180 _±	<0.1	NP	–	
		112	70–180 _±	0.1	NP	–	

Number	Reason for presentation	Days after initiating velagliflozin	Blood glucose (mg/dl)	Ketonaemia (mmol/l)	Ketonuria	F + IGF-1	Comor
4	Uncontrolled DM on velagliflozin	168	324 [*] ₋	0.1	NP	F: 498 µmol/l	Chronic interm diarrhoea
		245	582 [*] ₋	<0.1	Neg	F: 678 µmol/l	
		248	130–310 [‡]	0.3	NP	–	
		252	130–310 [‡]	0.2	NP	–	
		66	>180 [‡]	1.7	Neg	IGF-1: >1000 ng/ml	
		91	255 [*] ₋	0.4	NP	F: 680 µmol/l, IGF-1: >1000 ng/ml	
		157	206 [*] ₋	0.1	NP	F: 459 µmol/l, IGF-1: >1000 ng/ml	
		232	46 [*] ₋	NP	Neg	IGF-1: >1000 ng/ml	
		237	70–180 [‡]	0.2	NP	–	
		248	70–180 [‡]	NP	NP	F: 400 µmol/l, IGF-1: >1000 ng/ml	

Number	Reason for presentation	Days after initiating velagliflozin	Blood glucose (mg/dl)	Ketonaemia (mmol/l)	Ketonuria	F + IGF-1	Comor
5	eDKA	276	100–200 [‡]	0.3	NP	–	–
		388	70–180 [‡]	0.2	NP	–	
		90	279 ^{*_}	<0.1	NP	F: 599 µmol/l, IGF-1: N	
6		Baseline	384	NP	Neg	F: 526 µmol/l	Chronic enteropathy
7		3	151	High	Pos	–	IBD, late hepatic and pancreatic involvement
		250	58 ^{*_}	0.8	NP	–	

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Blood glucose was measured by an in-clinic analyser (Catalyst one; IDEXX Laboratories) after blood draw from a peripheral vein. Ketonuria was verified by urinalysis after performing cystocentesis. All blood ketone concentrations were measured via a handheld blood ketone meter (StatStrip XPRESS 2 Glucose/Ketone Meter; Nova Biomedical)

^{*}Measured with a portable blood glucose meter (Freestyle Precision Neo Glucometer; Abbott Laboratories)

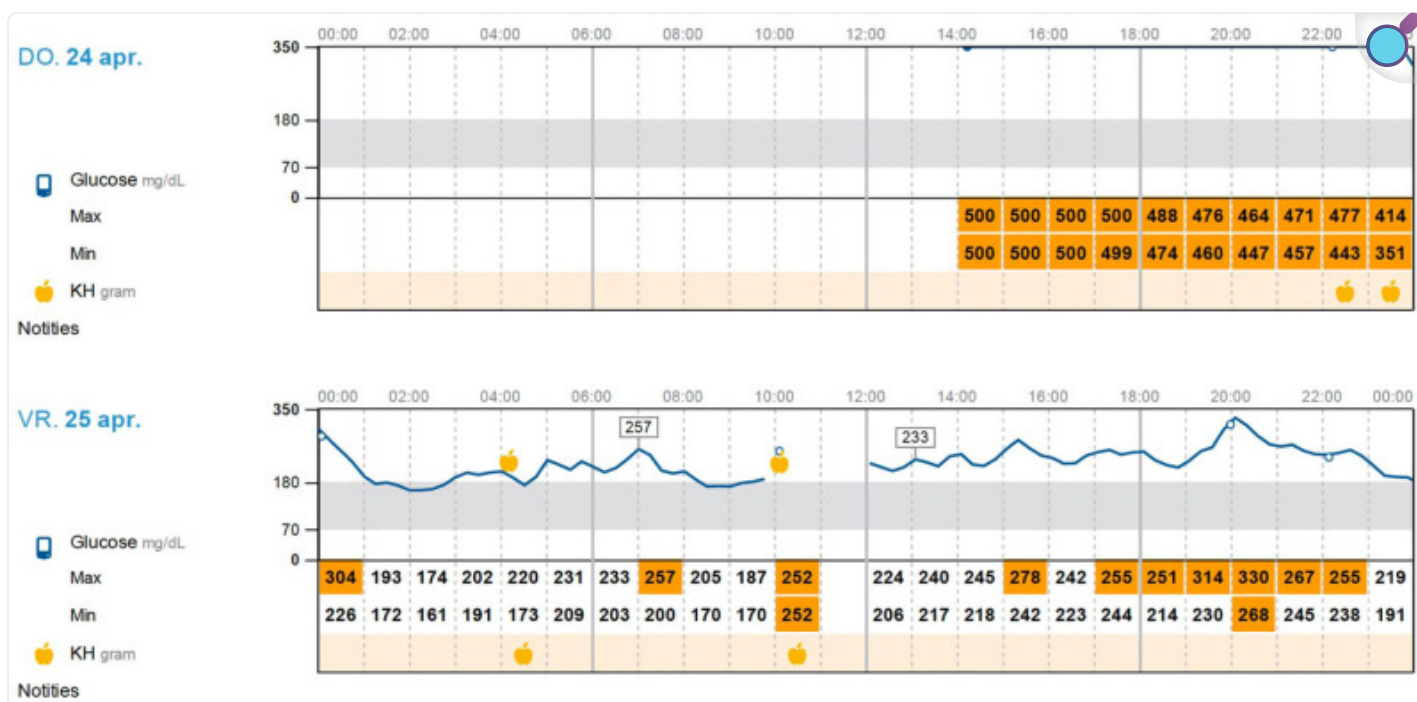
[†]Measured with continuous glucose monitoring (Freestyle Libre Sensor 2; Abbott Laboratories)

CKD = chronic kidney disease; DM = diabetes mellitus; eDKA = euglycaemic diabetic ketoacidosis; F = serum fructosamine concentration; HS = hypersomatotropism; IBD = inflammatory bowel disease; IGF-1 = insulin-like growth factor 1; N = normal; Neg = negative; NP = not performed; Pos = positive

Cat 1 was a 13-year-old spayed female Somali with newly diagnosed DM and concurrent chronic kidney disease (CKD) (International Renal Interest Society [IRIS] stage 2) that was started on velagliflozin. Glycaemic control improved within days, with mild transient ketonaemia early in treatment. Velagliflozin was discontinued after 4 months at the owner's initiative. The cat presented only 5 months after cessation of gliflozin therapy for a checkup and remission of DM was documented. Persistent clinical signs were attributed to CKD.

Cat 2 was a 15-year-old castrated male domestic shorthair (DSH) presented with newly diagnosed DM and elevated insulin-like growth factor-1 serum concentrations with pituitary microadenoma, consistent with HS. Velagliflozin was initiated, resulting in rapid glycaemic improvement confirmed by continuous glucose monitoring (CGM) ([Figure 1](#)). Cabergoline was concurrently added for HS management. Clinical control and stable glycaemia were achieved within 2 weeks.

Figure 1.

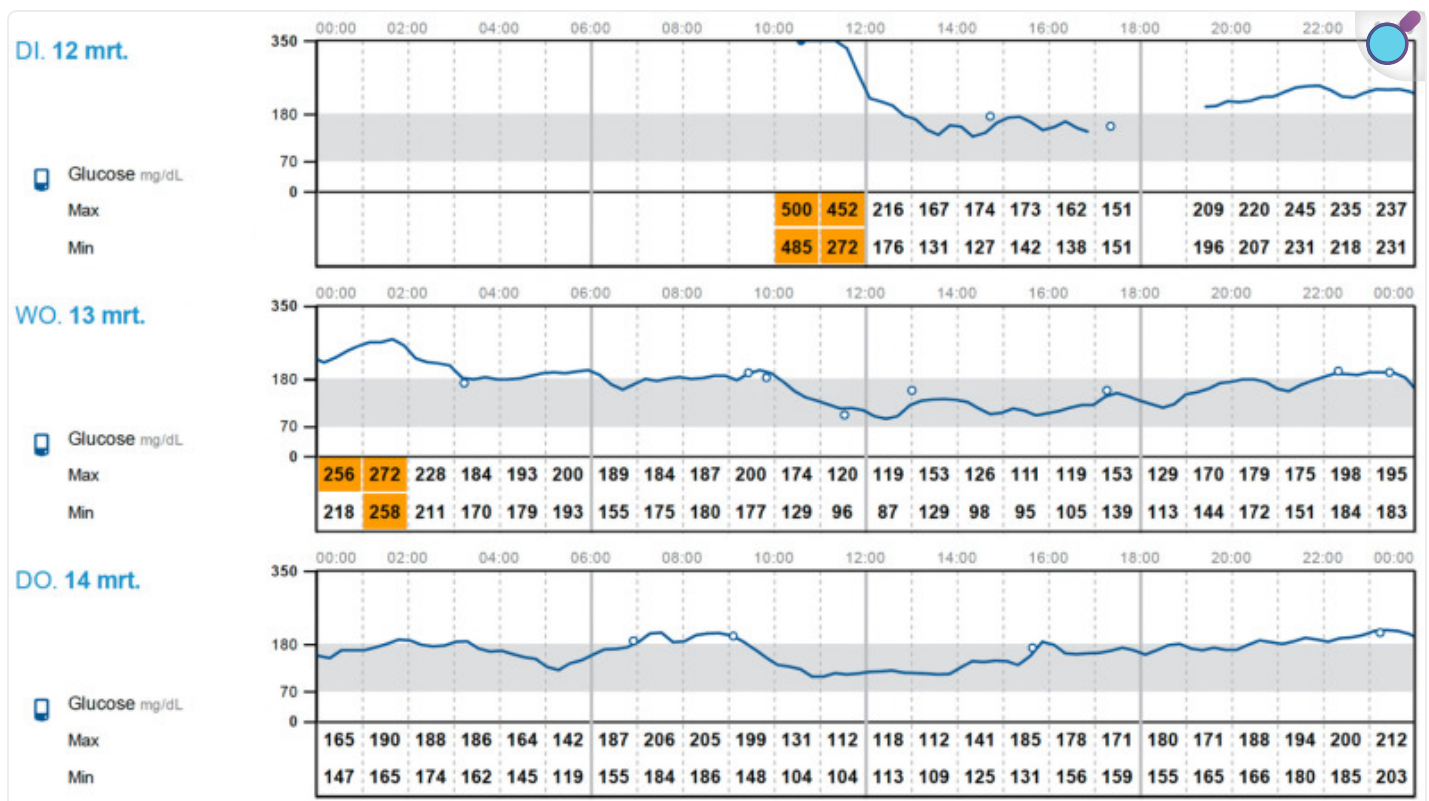


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Continuous blood glucose home monitoring (Freestyle Libre Sensor 2; Abbott Laboratories) from cat 2 diagnosed with both diabetes mellitus (DM) and hypersomatotropism. On day 1 (Thursday, 24 April 2025), the day of DM diagnosis, marked hyperglycaemia (beyond the measurable interval) was present. After administration of the first dose of velagliflozin (Senvelgo; Boehringer Ingelheim), a rapid decline in blood glucose occurred. On day 2 (Friday, 25 April 2025), the blood glucose values were just slightly above the target interval most of the time (70–180 mg/dl). Meals are represented by the apple icon

Cat 3 was a 10-year-old castrated male DSH, previously treated with insulin, that was referred for poorly controlled DM and chronic intermittent diarrhoea. The owner chose an easier treatment option and velagliflozin monotherapy was initiated after counselling on the increased risk of eDKA. After the first dose, a marked decrease in blood glucose occurred ([Figure 2](#)). Mild transient ketonaemia resolved within 2 weeks. After 3 months, remission of DM was confirmed, although relapse occurred after another 3 months. Reinitiation of velagliflozin restored glycaemic control.

Figure 2.



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Continuous blood glucose home monitoring (Freestyle Libre Sensor 2; Abbott Laboratories) from cat 3, whose owner opted for an easier treatment. On day 1 (Tuesday, 12 March 2024), marked hyperglycaemia (too high to measure) was present in the morning. After administration of velagliflozin (Senvelgo; Boehringer Ingelheim) at 10:30 am, a rapid decline in blood glucose occurred, with partial stabilisation later in the day. On day 2 (Wednesday, 13 March 2024), glycaemic variability was reduced, with most values within or slightly above the target interval (70–180 mg/dl). On day 3 (Thursday, 14 March 2024), a stable glucose profile was observed, with values largely within the target interval

Cat 4 was an 11-year-old castrated male DSH with HS and chronic intermittent diarrhoea that was referred for uncontrolled DM with persistent hyperglycaemia despite insulin and velagliflozin monotherapy. At the initial presentation, after 2 months of velagliflozin monotherapy, mild ketonaemia was observed in the absence of ketonuria or clinical evidence of eDKA. Combination therapy with insulin, velagliflozin and cabergoline, guided by CGM, showed good glycaemic

control. Blood ketones could be repeated only after 1 month and were normalised. Five months later, the cat developed a severe hypoglycaemic episode with seizures and transient blindness while under pet sitter care. Therapy was adjusted by reducing the insulin dose and levetiracetam was temporarily introduced. Cabergoline was discontinued as no positive effect was observed and it is effective in only a subpopulation of cats.⁷ Vision recovered within 7 weeks and the cat remained well controlled on the combined regimen with normal activity and glycaemia (Figure 3).

Figure 3.



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Continuous blood glucose home monitoring (Freestyle Libre Sensor 2; Abbott Laboratories) from cat 4, diagnosed with both diabetes mellitus and hypersomatotropism. Stable blood glucose concentrations were achieved with protamine zinc insulin (1 unit SC q12h, ProZinc; Boehringer Ingelheim), combined with oral velagliflozin (Senvelgo; Boehringer Ingelheim) based on ideal body weight. The green syringe symbol indicates insulin administration, the white box represents insulin dosing and the green box represents velagliflozin administration

Cat 5 was a 9-year-old castrated male DSH, earlier diagnosed with DM, that showed persistent PUPD after 3 months of velagliflozin therapy despite acceptable glycaemic parameters and absence of ketonaemia. The cat was otherwise clinically well, active and had regained weight. HS was excluded. Persistent PUPD was attributed to glucosuria-related osmotic diuresis and dietary adjustments were recommended to optimise weight and glycaemic control.

Cat 6 was a 10-year-old castrated male Devon Rex with DM and chronic enteropathy that developed lethargy and hyporexia 2 days after initiating velagliflozin therapy. Laboratory evaluation revealed eDKA. Stabilisation was achieved with intravenous fluids and short-acting insulin. Therapy was later changed to protamine zinc insulin. The cat recovered fully and velagliflozin was not reintroduced.

Cat 7 was an 11-year-old spayed female DSH known to have DM and inflammatory bowel disease that was presented with lethargy, anorexia and vomiting after treatment with velagliflozin for approximately 250 days. Although ketonaemia was only mild, DKA could not be entirely excluded given the coexistence of acidaemia, the absence of baseline beta (β)-hydroxybutyrate (BHB) data and the lack of established feline RIs. Abdominal ultrasonography revealed a large hepatic mass and pancreatitis. Given the poor prognosis, the owner elected euthanasia.

Discussion

This case series describes the clinical use, efficacy and safety of velagliflozin in cats with DM, including those with comorbidities, despite not fitting current label indications. Appropriate patient selection and clear communication with owners are and remain essential when initiating SGLT2 inhibitor therapy. Their favourable outcomes demonstrate that gliflozin therapy may be feasible beyond strictly defined candidate profiles, provided that owners are well informed and monitoring is close.

It also highlights practical aspects of glucose and ketone monitoring. Transient, mild ketonaemia was observed in several cats started on velagliflozin therapy, without clinical signs of DKA. Close observation during the first 2 weeks of treatment is essential, as eDKA can develop rapidly and may be missed owing to normal blood glucose levels, especially since this condition is still relatively new within the veterinary community. Early detection of anorexia, lethargy, vomiting or dehydration should prompt immediate ketone assessment.

Clinical hypoglycaemia is a known insulin-related adverse effect.⁸ In contrast, studies on SGLT2 inhibitors in cats report hypoglycaemia as rare or absent when used as monotherapy.³⁻⁶ However,

cat 4 experienced a severe hypoglycaemic crisis during combined insulin and velagliflozin therapy. In other cases, clinical hypoglycaemia was not reported, suggesting that the combined therapy was most likely the cause. The episode occurred 2 months after achieving glycaemic control and coincided with reduced home monitoring when the owner was absent. In hindsight, subclinical hypoglycaemia should have triggered closer follow-up. Since gliflozins cause glucosuria regardless of glycaemia, its absence cannot confirm remission of DM. As cat 4 was hyperglycaemic and glucosuric 4 days after discontinuation of the combined therapy, we concluded that remission was not the underlying cause for the hypoglycaemic crisis.

In cats, the most significant risk associated with gliflozin therapy is eDKA. Successful treatment requires sufficient endogenous insulin production, as inadequate insulin levels promote ketogenesis and subsequent ketonaemia. Although there is no definitive method to assess β -cell function at diagnosis in diabetic cats, ketone concentrations at presentation can be used as a surrogate. Persistent insulin deficiency is associated with increased production of ketone bodies, particularly BHB.⁹⁻¹¹

Healthy cats show blood BHB concentrations below 0.1 mmol/l, while BHB in insulin-treated diabetic cats without ketosis may reach up to 1.0 mmol/l.⁹ Thresholds above 2.4 mmol/l have been reported as highly sensitive for diagnosing DKA with handheld ketone meters.^{9,10} In this series, BHB measurement was used for ketone monitoring. Mild elevations (≤ 1.7 mmol/l) occurred without clinical deterioration, suggesting that low-grade ketonaemia can be benign in treated cats. These findings highlight the need for further studies to define reliable RIs and diagnostic thresholds for blood ketones in diabetic cats treated with SGLT2 inhibitors. In terms of monitoring, there is temporal distinction illustrated between cases 6 and 7. During long-term therapy, attention should shift to detecting and managing comorbidities that may cause insulin resistance.

Urine dipsticks, which detect only acetoacetate (≥ 0.5 mmol/l), were not used in this series as BHB is the predominant ketone with ketosis, but they remain common in primary care.¹² They may lag behind blood values, remain positive after resolution and are semi-quantitative, introducing subjectivity.¹³ Up to 30% of healthy cats may test positive despite being non-diabetic¹⁴ and only 55% of owners can reliably collect samples at home.¹⁵ These limitations reduce diagnostic accuracy. In contrast, handheld BHB meters provide more reliable and practical monitoring, especially in cats treated with SGLT2 inhibitors.

Conventional DKA is mostly managed with initial fluid resuscitation before starting short-acting insulin and dextrose therapy, whereas eDKA requires their simultaneous administration to

suppress ketogenesis and prevent hypoglycaemia, as blood glucose levels are often normal or only mildly elevated. [16](#)

Besides DKA, other side effects associated with gliflozin treatment were observed in this series. Persistent PUPD, as seen in cat 5, may reflect ongoing glucosuria rather than poor control and should not automatically be interpreted as treatment failure. Owners should be advised that this clinical sign may persist despite stable glycaemia. Mild gastrointestinal side effects, such as softer stools, were also observed and may be due to cross-inhibition of intestinal SGLT1 transporters, leading to increased faecal water content and mild osmotic diarrhoea secondary to incomplete glucose absorption. [6](#) However, given the non-specific nature of this sign and the transient course, alternative causes could not be excluded.

Comorbidities appear to influence both the safety and efficacy of velagliflozin. Current guidelines advise against gliflozin use in cats with reduced renal function. [3](#) Cat 1 tolerated therapy well despite IRIS stage 2 CKD, although dehydration remains a theoretical risk and reduced renal function may diminish glucose excretion. In humans, SGLT2 inhibitors may confer renal benefits, but such effects have not yet been demonstrated in cats. [17](#) HS may contribute to insulin resistance while sparing β -cell function, potentially explaining the usability of gliflozin therapy. Combination therapy with cabergoline may improve glycaemic control by reducing growth hormone secretion, but the response is inconsistent, as shown in cats 2 and 4. [7](#)

Nutritional management may affect treatment safety. Cat 5 was recommended a conventional diet. However, in humans, low-carbohydrate diets combined with SGLT2 inhibitors can enhance ketogenesis. [16](#) Because eDKA is associated with carbohydrate deficit, adequate caloric and carbohydrate intake should be maintained in treated cats. Most published feline studies kept cats on maintenance diets, suggesting that strict low-carbohydrate formulations may not always be required. [3-5](#) More research is needed to guide dietary recommendations.

Two cats in this case series achieved diabetic remission. [18](#) Although data with insulin therapy are heterogeneous, there is a general association between tighter glycaemic control, particularly early in the disease course, and an increased likelihood of remission. [8,19,20](#) As glycaemic control with SGLT2 inhibitors is generally good, remission may be expected in a subpopulation of treated cats, although current studies did not assess this. [3-5](#) These cases suggest that improved glycaemic control with SGLT2 inhibitors may support β -cell recovery and remission in selected cats, although long-term monitoring and further research are needed to clarify its role in achieving and sustaining remission.

Overall, this case series suggests that velagliflozin can be effective in managing complex feline diabetic cases, with appropriate monitoring. Although eDKA is the most serious potential complication, its risk can be minimised through owner education, early ketone testing and careful patient selection. As this was a retrospective case series, cases were not managed uniformly and some clinical findings, including chronic diarrhoea observed in several cats, were incompletely explored, although these did not appear to worsen during treatment. Other potential causes of metabolic acidosis, such as severe azotaemia, were not investigated. Moreover, the number of cases is small and heterogeneous, reflecting the off-label and exploratory use of a novel medication. These limitations indicate a need for prospective, controlled studies to further assess the safety and efficacy of velagliflozin and to identify which non-ideal candidates may be appropriate for its use. Off-label use of velagliflozin should always adhere to regional legal and ethical frameworks with informed client consent. Practitioners are encouraged to consult a specialist when considering such treatment in cases where clinical experience is still emerging.

Conclusions

This case series highlights the potential of velagliflozin in managing feline DM in cats with significant comorbidities. Glycaemic control was achieved with velaglifoizin therapy in all cases. Remission of DM was achieved in two cases and one case relapsed after 3 months. The occurrence of eDKA underscores the need for careful patient selection, early monitoring and proactive owner communication. The findings also emphasise the need for further research into nutritional strategies, combination therapies and interpretation of ketones.

Footnotes

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the use of cadavers. Ethical approval from a committee was therefore not specifically required for publication in *JFMS*. Although not required, where ethical approval was still obtained, it is stated in the manuscript.

Informed consent: Informed consent (verbal or written) was obtained from the owner or legal custodian of all animal(s) described in this work (experimental or non-experimental animals, including cadavers, tissues and samples) for all procedure(s) undertaken (prospective or retrospective studies). No animals or people are identifiable within this publication, and therefore additional informed consent for publication was not required.

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